NATIONAL STANDING ORDERS

FOR COMMUNITY HEALTH EXTENSION WORKERS

Revised By:

COMMUNITY HEALTH PRACTITIONERS' REGISTRATION BOARD OF NIGERIA IN COLLABORATION WITH NPHCDA

2024





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ACRONYMS AND ABBREVIATIONS

ANC	Antenatal Care
BHCPF	Basic Healthcare Provision Fund
CBOs	Community-Based Organizations
CE	Community Engagement
CHEW	Community Health Extension Worker
CHIPS	Community Health Influencers, Promoters and Services
CHPs	Community Health Programmes
CHO	Community Health Officer
DHIS	District Health Information System
HC	Health Clinic
HIS	Health Information System
HRH	Human Resources for Health
LGA	Local Government Area
NPHCDA	National Primary Health Care Development Agency
OIC	Officer In-charge
PHCUOR	Primary Health Care Under One Roof
PHC	Primary Health Care
SBA	Skilled Birth Attendants
SCL	Student-Centered Learning
SMOH	State Ministry of Health
TOR	Terms of Reference
UHC	Universal Health Coverage
VDC	Village Development Community
WDC	Ward Development Community
WHO	World Health Organization

FOREWORD

In today's dynamic healthcare landscape, the quest for quality and uniformity of care at the primary health care level remains paramount. It is with great pleasure that the Community Health Practitioners Registration Board of Nigeria [CHPRBN] presents the 2024 revised National Standing Orders for the Community Health Extension Workers (CHEWs). It is a comprehensive set of standard operational guidelines meticulously crafted through the collaborative efforts of the reputable regulatory establishment, principally supported by USAID/Nigeria through its Health Workforce Management Activity, the Federal Ministry of Health, the National Primary Health Care Development Agency [NPHCDA], and other Pertinent MDAs.

The significance of the National Standing Orders cannot be overstated. As a guiding framework that links Communities and the national health system, it serves as a cornerstone in ensuring consistency and excellence in healthcare practices across the nation. By providing clear protocols and procedural guidelines, it empowers Community Health Extension Workers to deliver services of the highest standard, thereby enhancing positive Community health outcomes.

Accordingly, the National Standing Orders foster a culture of accountability and professionalism within the Community Health Profession. By adhering to established guidelines, Community Health Extension Workers uphold ethical standards and demonstrate their commitment to providing safe and effective care to all individuals, regardless of their socio-economic status or geographic location.

Furthermore, this document has been painstakingly compiled to serve as a tool for capacity building and continuous improvement. Through regular updates and revisions, it reflects the latest advancements in healthcare and best practices, equipping healthcare providers with the knowledge and skills necessary to adapt to evolving healthcare needs and challenges.

In essence, the National Standing Orders epitomize our collective commitment to advancing healthcare delivery and safeguarding the well-being of our citizens. It is our fervent hope that this document will serve as a catalyst for positive change and contribute significantly to the achievement of national health objectives. Therefore, I strongly endorse the 2024 edition of the National Standing Orders to the Community Health Extension Workers, to provide quality, affordable, and uniformity of care within the national health system. Predominantly, I extend our deepest appreciation to all those who have contributed to this review, recognizing their dedication and expertise in shaping the future of healthcare in Nigeria. It is my ardent hope and trust that Community Health Extension Workers will appreciate the huge efforts put in place for this review by reciprocating with strict adherence to this national document and demonstrating a high sense of responsibility.

Dr. Tunji Alausa
Hon. Minister of State for Health

ACKNOWLEDGEMENT

The review and production of the 2024 edition were made possible by USAID/Nigeria through its Health Workforce Management Activity. The regulatory establishment [CHPRBN] reviewed this document through collaborative support from the Federal Ministry of Health, the National Primary Health Care Development Agency [NPHCDA], and other critical stakeholders. The Board acknowledged all the sources of materials used in this production.

The main objective of the National Standing Orders is to ensure quality, affordable, and uniform care, especially at the Primary Health Care level, provided by Community Health Practitioners. The review committee deserves a high level of commendation for the effort they put into the review process of this important document. Their dedication and collaboration have been instrumental in achieving the overarching objective of ensuring quality, affordable, and uniform care for individuals, families, and communities.

In light of the foregoing and the spirit of collaboration, I am pleased to extend sincere heartfelt appreciation to all contributors to this revised edition. Indeed, your invaluable contributions will undoubtedly benefit countless individuals and communities in Nigeria. Your partnership exemplifies the spirit of collective effort in advancing healthcare delivery and promoting the well-being of our dear citizens.



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INTRODUCTION

The National Standing Orders serve as a comprehensive set of specific guidelines, arranged by age group, conditions, findings (from history, physical examination, laboratory investigations), clinical judgment, and necessary actions, to standardize care for patients at Primary Health Care (PHC) Facilities. Developed through extensive collaboration between Community Health Practitioners' Registration Board of Nigeria (CHPRBN) and the National Primary Health Care Development Agency (NPHCDA), with support from the United States Agency for International Development (USAID), these comprehensive guidelines equip health workers with standardized procedures and protocols for managing a wide range of health conditions. The objectives of the Standing Orders are to:

- Ensure legal protection for health workers in their primary health care duties.
- Offer a structured approach for history taking and physical examination.
- Aid in the management of less common, easily overlooked, and more serious conditions.
- Maintain a consistently high standard of healthcare.
- Minimize unnecessary, expensive, and time-consuming laboratory investigations.
- Provide a framework for evaluating care and staff performance.

Following the guidelines will better position the Community Health Practitioners at the frontline to provide quality care that is Safe, Timely, Equitable, Effective, Effective, and Person-centered.

WHAT YOU WILL FIND

- Clear and Practical Guidelines: Organized by age groups and specific conditions, the Standing Orders provide step-by-step instructions for history taking, physical examination, treatment plans, and follow-up care. This user-friendly format ensures consistent and high-quality care for all patients.
- A framework to manage common and less frequent conditions. You'll find valuable guidance for even complex or life-threatening situations, along with clear referral pathways which will enhance clinical decision making.

More than just guidelines, these standing orders represent years of field-tested experience and continuous refinement, making them practical and relevant to the environments where Community Health Professionals operate. Each complaint or symptom that a patient may present with is addressed with a concise introduction, providing cues on important aspects, followed by a suggested history, examination, education, further investigation, treatment, and follow-up to ensure optimal patient care. Simplicity, clarity, and ease of use have been prioritized, with more common conditions addressed first and more severe conditions prioritized over moderate and mild ones.

In cases of complicated or life-threatening conditions, the Standing Orders specify referral to the next level of care or the nearest hospital as appropriate. The uniformity of care provided by these guidelines facilitates evaluation, setting a standard for care and allowing for the assessment of overall care adequacy and identification of areas needing improvement.

IMPORTANT NOTE: The Standing orders do not replace clinical judgement and critical thinking. Health workers must still gather essential history and physical examination details to make informed clinical decisions and provide appropriate treatment. These protocols delineate the necessary actions and decisions required to manage a patient's symptoms effectively. The Community Health Practitioner must ensure proper documentation of all inferences made and action taken in the management of the patient. By adhering to these standardized procedures, you can ensure your actions are well-supported and documented, minimizing legal risks and maximizing patient safety.

The effectiveness of these Standing Orders, like the broader primary health care concept, depend on well-trained, committed, and dedicated health workers. These Standing Orders are therefore complemented by the curricula developed for use in Schools of Health Technology and Community Health Officers' Training Institutions. The continued and proper utilization of these Orders, for both TRAINING AND PRACTICE, will ensure the sustenance of a high standard of health care in the community. For the 2024 revision of the Standing Orders, an illustrated version has been created to support the ease of use and improved understanding of the guidelines.

HOW TO USE THIS GUIDE

The Standing Orders are organized by complaints or problems derived by the health workers from the patient's history, an examination, and/or laboratory tests. Follow the steps below in your interaction with the patient:

1. Based on the patient's age group, identify the problems or complaints in the relevant section and refer to the appropriate page.
2. Ask all questions listed in the history box.
3. Perform all examinations listed in the examination box.
4. Record all significant information, including negative findings. For example, if examining the neck for stiffness, note the absence of stiffness as "no stiff neck."
5. Select the box containing all significant findings related to the patient's condition from the list of findings.
6. Follow the actions corresponding to the selected findings and manage the patient's complaints as instructed.

In some cases, it may be necessary to refer to more than one section of the Standing Orders to completely manage all of the patient's issues comprehensively. Take appropriate action based on your findings and record them in the home-based record (PHC Child Health card or green card for children under 5 years and Personal Health Record yellow card for those over 5 years old) or in the relevant patient record platform provided. Review the patient's condition whenever a course of medicine or injection is due to finish and based on the medications' onset of action. This is an excellent way to evaluate your work carefully. Ensure the home-based record card is returned to the mother/caregiver or patient.

ADDITIONAL INSTRUCTIONS:

- Every patient should undergo the complete examination listed under the First Visit at their initial contact with the health facility, except in emergencies, in which case the treatment should be prioritized and the 'first visit' examination should be completed at the

next visit.

- Address the health issues of all family members present in the clinic whenever possible.
- Follow up on all referred cases when patients return to ensure continuity of care and facilitate health worker learning.
- Advise all patients to return to the health facility immediately if their conditions worsen.
- Health education is crucial for patient care. Provide relevant health education to the appropriate patients on topics such as:
  - -Breastfeeding (immediate & exclusive) and complementary feeding where appropriate.
  - -Nutrition for older children & adults.
  - -Newborn care at home.
  - -Child-spacing.
  - -Personal hygiene.
  - -Immunization.
  - -Patient's presenting condition.
  - -STD/HIV/AIDS (or other current health concerns).
  - -Importance of home-based records.

This comprehensive approach ensures that health workers can provide effective and standardized care while continually improving their practice through ongoing evaluation and education.
Collaborative work: When you are in doubt about the care plan for a patient, consult with other healthcare workers at the facility or stabilize the patient and refer appropriately. When you do not have the required resources to manage a patient, promptly refer to the appropriate facility or level of care to avoid delay in the commencement of care for the patient.

FEEDBACK AND SUPPORT CONTACT

We encourage all Community Health Practitioners to actively participate in the continuous improvement of this guide. If you encounter any issues, have suggestions for improvements, or notice any outdated information, please do not hesitate to reach out to us. Your feedback is crucial in maintaining the accuracy and relevance of this essential resource and provide valuable insights for future revisions.

How To Provide Feedback

Email: Send detailed feedback, suggestions, or corrections to info@chprbn.gov.ng
Please include specific references to sections or illustrations where applicable.

Phone: Contact our support team at 08034462362 (Ibrahim Umar) for immediate assistance or to discuss your feedback in detail.

Online Form: Please fill out your structured feedback using the google form via this link

SECTION ONE
THE NEWBORN
(0 – 28 DAYS)



NATIONAL STANDING ORDERS

FOR COMMUNITY HEALTH EXTENSION WORKERS

1.1 THE NEWBORN

A newborn is an infant aged 0-28 days old who can cry and breathe unassisted when properly cared for. Crying is often a positive indicator of the newborn's breathing, as their skin will become pink once their respiratory function improves. However, newborns may experience fatigue or weakness from the strain of birth or be born with certain illnesses. Therefore, it is crucial to handle them gently with clean hands and observe them carefully without slapping any part of their body.

Newborns are susceptible to becoming cold due to being wet; thus, they must remain dry and well-wrapped at all times. It is highly recommended that mothers bring in their children for regular check-ups - monthly visits within the first two years of life, followed by every three months until age five - as well as ensure timely immunization until the child has been fully vaccinated.

1.1.1 IMMEDIATE CARE OF THE NEWBORN

It is imperative to cleanse one's hands both prior to and subsequent to tending to an infant.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. First One Hour	1. Normal Newborn		1.Wash your hands thoroughly and put on latex gloves before handling the delivery. 2.As soon as the baby is delivered and crying, check that the baby appears pink in color. 3.Dry the baby thoroughly using a clean, dry cloth. 4.Discard the wet cloth after drying and wrap the baby with another dry cloth. 5.Maintain the room temperature between 25°C - 28°C to help keep the baby warm. 6.Keep the baby away from direct sunlight. 7.Place the baby on the mother's chest and maintain this position for at least the first hour, initiating breastfeeding within 30 minutes. 8.Clean the baby's eyes with a clean cotton pad and apply either 0.5% Erythromycin ointment or 1% Tetracycline ointment. 9.Delay cord clamping for 1-3 minutes after birth. Cut the baby's cord about one hour after birth, ensuring proper care. 10.Cord care: -Always wash hands before and after cutting/ cleaning the cord. apply 4% Chlorhexidine gel (within 1 hour of birth) on the stump and ask -the mother to continue daily. -Dry soiled and wet stump after bathing and apply Chlorhexidine gel. -Do not cover the cord with a dressing or diaper, fold diaper below stump. -Do not apply methylated spirit or other medications or substances 11.To prevent hemorrhagic disease of the newborn – give IM Vitamin K1 into the anterolateral thigh (dose – 1mg for babies >1.5kg and 0.5mg for babies <1.5kg) 12.Physical examination: a.conduct a thorough physical examination, paying attention to presence of birth defects b.check breathing and count the respiratory rate (normal is 40 - 60 breaths/minute) c.measure the temperature (normal is 36.5 – 37.5°C)

			d.weigh the baby. 13.Document your findings in the baby's notes 14.Delay bathing till 24 hours after birth. -16. Give parents guidance for home care: a.Register the birth. b.Practice exclusive breastfeeding for six months. c.Recognize and manage common breast problems. d.Practice hand washing. e.Monitor the newborn closely. If neonatal jaundice or any other danger sign is observed, present baby immediately at the health facility. f.Use only 4% Chlorhexidine (7.1% Chlorhexidine digluconate) gel in 25g tube for cord care. g.Complete immunization schedule as in the child health card. At discharge give the parents the National Pictorial Newborn Discharge Guide Information Leaflet for basic instructions.
	2. Baby born with meconium-stained liquor and unable to breathe	Moderate to Severe condition Baby has asphyxia and unable to breathe (GOLDEN MINUTE)	1. If there is blood or meconium-stained liquor and the baby is not crying, 2. Put the baby on a ventilation table and immediately commence Neonatal resuscitation (Helping Babies Breathe within a minute) through the following ways: 3. Position the head and neck to open the airway. 4. Clear secretions from the airway using the mucous bulb extractor. 5. Dry newborn, stimulate and re-position. 6. Assist the baby to breathe using the bag and mask, making sure you are using the right-sized mask and bag. 7. Ensure steps 3, 4, 5, and 6 are done within 1 minute for the baby who is unable to breathe at birth. Do not handle the baby upside down, slap the feet, or pinch the chest. 8. If the baby fails to breathe, continue to ventilate and REFER to a higher-level facility for advanced care. 9. If the baby begins to breathe and cry, 10. check if the baby is fast breathing (two counts of 60 breaths or more in one minute) and 11. check temperature for high (37.5°C or more) or very low. 12. Continue with other newborn care steps under "Normal baby" above. 13. Continue to care for babies who are crying. 14. Do not remove vernix or bath baby until at least 24 hours after birth. 15. Continue to keep baby warm and initiate skin-to-skin contact with mother. 16. Encourage mother to initiate breastfeeding within 30 minutes of birth (offer help). 17. Do not give artificial teats or pre-lacteal feeds to the newborn (no water, sugar water, or local feeds). 18. Examine newborn 1 hour after delivery of placenta.

II. After 1 hour of birth	Within 60-90 minutes of life check examination findings: 1.General appearance - movement of the extremities, hypotonia, color, respiratory distress, dysmorphic features, and obvious malformations. 2.Assess breathing (baby must be calm). 3.Check for fast/difficult breathing, signs of respiratory distress, breathing pattern, respiratory rate, air entry to the lungs, presence of abnormal sounds in the lungs (listen for grunting). 4.Count the breaths (30-60 breaths per minute, repeat the count if elevated). 5.Check for chest in-drawing. 6.Check Heart rate per minute (normal rate is between 120 and 160 bpm). 7.Measure Newborn length (normal range is 48-53 cm). 8.Check for movements (normal and symmetrical). Check for jaundice - yellow palms/soles/eyes. 9.Check for presenting parts swelling, bruises, abdomen. 10.Skin - feel for warmth or cold, pallor, rash, pallor, plethora, meconium staining, cyanosis, birthmarks, etc. Check for yellow palms/soles/eyes. Temperature - check Axillary temperature (normal is between 36.5°C and 37.5°C); below 36.5°C is hypothermia.		1.After the clinical examination of the newborn, ensure DOCUMENTATION of findings: 2.Specifically, record all observed danger signs seen in the newborn: -Not feeding, poor feeding, vomiting -Lethargy -Respiratory distress, chest-in-drawing, nasal flaring -Fast breathing (breathing rate >60 per minute), grunting -Low body temperature (38°C) -Convulsions -Any jaundice in the first hour of life, or yellow palms and soles at any time AND REFER IMMEDIATELY. 3.Ensure Referral and follow-up to health facility 4.Communicate/Liaise with health facility 5.Fill out the community referral form (In triplicate) 6.Assist to mobilize for transport (Emergency transport system) 7.Keep newborn in skin-to-skin contact with the mother in KMC position 8.Express breast milk and feed baby with cup 9.Practice and maintain hygiene 10.Follow-up with the patients 11.Complete examination record in baby's notes, sign and date it. 12.Record all the findings in the newborn's registration books or chart prepared for the purpose. 13.Record any discussion or advice given to parents 14.Record any congenital abnormality in the congenital abnormality registry 15.Classify the newborn into normal or abnormal (specify problems identified) 16.Educate mothers to do the following: a.Ensure baby does not have contact of baby with wet or cold surfaces b.Ensure a room temperature of at least 25°C -28°C to help keep the baby warm but keep the baby away from direct sunlight c.Ensure continuous breastfeeding whenever the baby shows signs of readiness to feed and be sure baby is feeding and sucking well d.Ensure good positioning of the baby during breastfeeding; -Ensure baby can attach well and the mother is comfortable; -Place the baby with the head and body in a straight line; -Baby's body turns toward mother; -Baby's body is close to the mother and baby's abdomen is touching the mother's own; and -Baby's whole body is supported.

			e.Ensure good attachment of the baby during breastfeeding; -Ensure that baby's chin is touching breast; -Baby's mouth is wide open: Baby's lower lip is turned outward, and more areola is visible above than below baby's mouth. f.Ensure adequacy of feeds by feeding the baby every 2-3 hours or 8 -12 times a day; baby is getting enough milk per feed through Breasts that is soften with feeding; and Swallowing sounds are heard during feeding g.Prevent breastfeeding problems such as; -Inverted nipples by stimulating nipples before feeding and shaping the breast before attachment; -Breast engorgement (Very full, tight and shiny breasts) by feeding the baby more often and/or express milk prior to attachment -Sore or cracked nipples (usually results from poor attachment or skin infection) by ensuring good attachment and treating by applying drops of breastmilk to the skin of the nipple, and washing her breasts with water daily and avoiding soaps, medicated lotions and ointments, and continuing breastfeeding and/or expressing breastmilk to feed baby -Mastitis (occurs when there is a blocked duct or from infection -usually affects one breast with a well-defined red, sore and swollen or hardened area) by breastfeeding frequently or expressing breastmilk; -Inadequate milk volume by increasing fluid intake and frequency of breastfeeding and by applying warm compress to the breasts, massaging the back, neck, breasts and nipples. h.Always ensure baby is passing stools and urinating i.Ensure baby has received immunizations – BCG, OPV0 and hepatitis B vaccine which should have been given within 24hours of delivery if this has not been done. j.Ensure to commence postpartum family planning k.Visit the health clinic for next immunization visit at 6 weeks l.Ensure adequate home care by: a. Registering birth b. Practicing exclusive breastfeeding for six months c. Recognizing and managing common breast problems d. Practicing hand washing e. Monitoring closely for neonatal jaundice and recognizing other danger signs and visiting the health facility f. Ensure proper cord care: -Always wash hands before and after cleaning the cord. -apply only 4% Chlorhexidine (7.1% Chlorhexidine digluconate)) gel in 25g tube for cord care daily.

			-Dry soiled and wet stump after bathing and apply Chlorhexidine gel. -Do not cover the cord with a dressing or diaper, fold diaper below stump. -Do not apply methylated spirit or other medications or substances g. Completing immunization schedule as in the child health card h. Always supporting baby's head and neck and never shake or throw baby up, whether in play or in frustration.

1.1.2 FIRST POSTPARTUM HOME VISIT (2—3 DAYS AFTER THE BIRTH OF A NEWBORN)

The postnatal period begins immediately after the birth of a newborn, extending to six weeks after delivery. This period is crucial for the survival of the newborn. Proper care must be taken to avoid complications including death. The post-natal care visits days are 3, 5, 7, 9, 14, 21, 28 - If the birth weight of the baby is normal (>2.5kg) and at least three home visits are required on days 3, 7 and 14, If the birth weight of the baby is low.

HISTORY	EXAMINATION/ACTIONS
1.Ask for the frequency of newborn breastfeeding (infants eat about every 1–3 hours) 2.Ask for the number of times the Newborn wets the diaper in a day (should have about 6 wets diapers a day) 3.Ask for the number of times in a day that the newborn stools (most newborns have 3 or 4 soft bowel movements a day) 4.Ask for number of times the newborn sleeps in a day (A newborn may sleep 14 to 17 hours or more in 24 hours) 5.Ask for the number of hours that the newborn sleeps between feedings (Newborns should not sleep more than 4 hours between feedings until they have good weight gain, usually within the first few weeks). 6.Ask which of the actions below the mother has noticed in the child in the first 3-5days of the child's life: a.pay attention to faces or bright objects 8–12 inches (20–30 cm) away. b.respond to sound — they may quiet down, blink, turn head, startle, or cry. c.hold arms and legs in a flexed position d.move arms and legs equally. e.lift head briefly when on stomach (babies should be placed on the stomach only while awake and under supervision). f.have strong newborn reflexes, such as: -rooting and sucking: turns toward, then sucks breast/bottle nipple -grasp: tightly grabs hold of a finger placed within the palm -straightens arm when the head is turned to that side and bends opposite arm -throws out arms and legs, then curls them in when startled (startle response) 7.Ask mother if she has commenced Postpartum Family Planning: 8.Ask if the mother is experiencing low energy, fatigue, sleep, or appetite problems (then she may have postnatal blues) 9.Other educational needs	Perform comprehensive physical examination (PE) within 24 hours of birth, and within 72-hours of life 1.Provide essential newborn care 2.Assess for Danger signs and refer newborn to the health facility if any danger sign is seen/observed 3.Check the baby's weight, length, and head circumference and plot the measurements on a growth chart. 4.Check if Newborns seem hungry 5.Watch as the mother breastfeeds the newborn and offer help with any problems. 6.Let the mother undress the baby and conduct the following: a.an eye exam, b.listening to the i.baby's heart and feeling pulses if readings are abnormal, refer ii.inspecting the umbilical cord, If smelly or issuing pus, clean and apply Cord care daily using 4% chlorhexidine gel iii.checking the hips, Note and observe abnormalities, and refer iv.Physiologic jaundice 7.Repeat Initial examinations done for the newborn within in the first 60-90 minutes of life as shown in II above 8.Screening for: •Congenital malformations, syndromes, and associations •Congenital heart diseases •Congenital cataracts •Cryptorchidism •Developmental dysplasia of the hip 9.Discourage harmful traditional care practices and Counsel the mother against these harmful practices including the following: i.Cord care - apply only CHX and discourage the use of other substances such as toothpaste or Cow dung to the cord. ii.Colostrum is yellowish-coloured milk and should be given to the newborn because it protects the newborn from infection. 10.Promote Exclusive Breast Feeding - Exclusive breastfeeding means that a baby takes only breast milk with no additional foods or liquids, not even water (as breast milk is predominantly water), except medicine, if necessary, for the first 6 months when complementary feeding begins. Baby should be breastfed 8 – 12 times per day and on demand.

	11.Tell the mother the advantages of breast milk which include: -It is the best food for the baby as it digests easily -It protects newborn from infection -It promotes bonding between the mother and the newborn 12.Demonstrate good positioning and good attachment and discuss the importance of breastfeeding 13.Promote Immunization: Newborns are particularly vulnerable to infections. Immunization is an important way to protect an infant's health. They prevent illnesses and diseases and failure to immunize may put the infant at risk of serious illness or even death 14.Check immunization status and educate on the importance of completing all the child's immunizations. 15.Immunize newborns- applicable for ONLY oral immunization Refer to the health facility for appropriate immunization 16.Register birth (if not done already) 17.Provide information on other scheduled visits: -Second postnatal visit day 3 -Third postnatal visit: day 7 -Fourth visit: between day 14 – 21 -Fifth visit: day 28 -Sixth visit: day 42 (6 weeks) 18.Encourage caregivers to talk, sing, and play with the newborn 19.Educate and Counsel mother on Postpartum Family Planning: Family planning methods - Oral contraceptives (progesterone only.); Education on Long-Acting Methods e.g. injectables and self-injectables including DMPA-SC.; Long- acting and permanent methods (IUD/tubal ligation, implants Provide further Health Education to the Newborn mother on: A. Routine Care 1.Wash hands before handling the baby and avoid people who may be sick. 2.Keep the diaper below the umbilical cord so the stump can dry. 3.Give sponge baths with warm water until the umbilical cord falls off 4.Use fragrance-free soaps and lotions. 5.Hold the baby and be attentive to their needs. 6.Sing, talk, and read to your baby and look directly to your baby's eyes and smile when breastfeeding. Provide ways for your baby to see, hear, move arms and legs freely, and to 7.Call your Health care worker if your infant has a fever of 100.4°F (38°C), is acting sick, isn't eating, isn't peeing or pooping, looks yellow, or has increasing redness or pus around the umbilical cord or circumcision. 8.Don't give medicine to an infant younger than 2 months old without talking to the health facility 9.Breastfeed your baby 10.If you as a new mother feel intense tired, sad, moody, or anxious, call your Health worker 11.Talk to your health worker if you're worried about your living situation or don't have the things that you need to take care of your baby or have food, or a safe place to live. Your health worker can tell you about community resources or refer you to a social worker.

	12.Ensure you give your newborn six basic needs required: security, clothing, enough sleep, nutritious food, sensory stimulation, love, and attention B. Safety 1.To reduce the risk of sudden infant death syndrome (SIDS): a.Always place your baby to sleep on a firm, flat mattress on their back, and not on the stomach or side, in a crib or bassinet without any crib bumpers, blankets, quilts, pillows, or plush toys. b.Let your baby sleep in your room or crib next to your bed until your baby's first birthday, or for at least 6 months, when the risk of SIDS is highest. c.Avoid overheating by keeping the room temperature comfortable. Dress your baby for room temperature and don't cover your baby's head while they're sleeping. Watch for signs of overheating, such as sweating or feeling hot to the touch. d.Don't let your baby fall asleep on a product that isn't specifically designed for sleeping babies, such as a sitting device (like a car seat), a feeding pillow 2.Don't smoke or use cigarettes. Don't let anyone else smoke or vape around your baby. 3.Always put your baby in a rear-facing car seat in the back seat. Never leave your baby alone in a car. 4.While your baby is awake, don't leave your little one unattended, especially on high surfaces or in the bath. 5.Never shake your baby — it can cause bleeding in the brain and even death. I Call a friend, relative, or your healthcare provider for help. 6.Avoid sun exposure by keeping your baby covered and, in the shade, when possible. Counsel and provide emotional support. Reassure her that this is usually a temporary condition that happens to some women who have given birth. It sometimes helps if women know that feeling depressed following the birth of a baby is normal and many women experience these feelings. Try and talk to the woman's family and explain to them the need for extra support at this time. Verify that she and the newborn are getting the care they need.

1.2. NEWBORN WITHOUT ANY COMPLAINT- THE WELL CHILD

A well-visit, or well-child visit, is a preventative health appointment with the child's healthcare provider with the goal of helping well children to stay healthy in their earliest years, so they can grow into healthy adults. The well-child visits concentrate on the whole body, not on sickness, based on the child's age and unique needs. The well-child visit is very important and should always include screening for health problems, growth monitoring, and promotion. Health education about nutrition, immunization, and protection from illness and other conditions as relevant to the family is given. Mothers should be encouraged to take their children for visits while they are well, monthly in the first two years of life and every 3 months thereafter until 5 years old, and once a year until the child reaches 21 years. These visits also provide opportunities to monitor mothers and talk to them about postpartum depression or anxiety.

HISTORY	EXAMINATION/ACTIONS
1.Birth: a. Was pregnancy and birth normal? b. Where was the child delivered? c. Did the child cry immediately after birth? d. Was the child put to breast within 30 minutes after birth? e. Was the child jaundiced (yellow) at any time? 2.Nutrition: a. How have you been feeding the child? b. Exclusive breastfeeding? Supplementary feeding? If not exclusive breastfeeding, what else? 3.Social: a. With whom does the child live? b. Does the mother work outside the home? c. Who looks after the child when the mother is away? d. What is your source of water? e. What type of toilet do you use? f. Is your house screened against mosquitoes? g. Do you and the child sleep under a mosquito net or Insecticide-Treated Net (ITN) every day? 4.Medical: a. Has the child had any illness in the past? (e.g., yellow eyes (Jaundice), discharging eyes, fever, etc.) b. What treatment was given? Where? Any illness or chronic cough in the family? c. Does the child have any problem now?	1.General appearance 2.Obvious congenital abnormalities (e.g., mongolism, spina bifida, microcephaly, harelip) 3.Crying - active, limp, floppy 4.Skin color - normal, pale, yellow, blue 5.Weight - weigh and chart 6.Head circumference - size, swelling, color and texture of hair, fontanelle (depressed or bulging), sutures (normal or wide) 7.Eyes - discharge (sticky), yellow, red 8.Mouth - ability to suck normally, abnormality (cleft upper palate) 9.Spine - swelling, opening, curvature 10.Chest - breathing normally, respiratory rate (note if 60 per minute or more), listen to the child's lungs 11.Abdomen - condition of cord, umbilical hernia, press gently on the child's tummy to feel the organs 12.Arms and legs - normal movement, deformity, fingers, and toes (for extra or joined webbed fingers) 13.Genitalia - abnormality (check vaginal and penile opening, discharges, size of the clitoris, both testes in the scrotal sac), swelling, and congenital hernia 14.Anus - open or closed 15.Temperature - check and record 16.Developmental Milestones appropriate to age 17.Neurological tests (muscle tone, moro, suckling, rooting, and grasping reflexes) 18.Measure heart rate 19.Take blood pressure 20.Give appropriate childhood vaccines (will vary by age) 21.Give Health education: 22.Handwashing with soap and water (or use a hand sanitizer) before handling your baby. Support baby's head and neck. 23.Never shake baby, whether in play or in frustration. 24.Always fasten baby securely when using a carrier, stroller, or car seat. 25.Avoid rough play with newborns, such as jiggling them on the knee or throwing them in the air. 26.Clean around the cord stump with plain water and blot dry until the stump dries up and falls off 27.Feed babies whenever they seem hungry. Baby may show you they're hungry by crying, putting fingers in their mouth, or making sucking noises. A newborn baby needs to be fed every 2–3 hours 28.Change the position of your baby's head from night to night (first right, then left, and so on). This helps prevent a flat spot from developing on one side of the head 29.Immunization 30.Discouraging application of anything else apart from chlorhexidine gel on the cord 31.Refer for nutrition counseling and food demonstration at 5 months. 32.Remind the mother to take home-based records with her any time she takes the baby to the clinic or hospital. 33.Remind mothers about the next Well visit and information to bring for discussion including: 34.Changes in her child's health and development 35.Changes in social behaviors 36.Your child's eating habits 37.Any questions or concerns

Health Education

1. Encourage mothers to:

a. Wash hands with soap and water (or use a hand sanitizer) before handling your baby.
b. Support baby's head and neck.
c. Never shake baby, whether in play or in frustration.
d. Always fasten baby securely when using a carrier, stroller, or car seat.
e. Avoid rough play with newborns, such as jiggling them on the knee or throwing them in the air.
f. Clean around the cord stump with plain water and blot dry until the stump dries up and falls off.
g. Feed babies whenever they seem hungry. Baby may show you they're hungry by crying, putting fingers in their mouth, or making sucking noises. A newborn baby needs to be fed every 2–3 hours.
h. Change the position of your baby's head from night to night (first right, then left

2. Advice on Immunization.
3. Discourage application of anything else apart from chlorhexidine gel on the cord.
4. Refer for nutrition counseling and food demonstration at 5 months.
5. Remind the mother to take home-based records with her any time she takes the baby to the clinic or hospital.

APPENDIXES: FROM National Guidelines for Comprehensive Newborn Care
REFERRAL LEVELS November 2021 | First Edition.
Figure 5.1: Essential care for every baby in the first 90 minutes of life
Figure 5.2: Good attachment to the breast
Figure 5.3: Essential Care for Every Baby (ECEB) Chart

1.3 NEWBORN EMERGENCIES

An ill newborn can quickly become an emergency, so all newborns should be treated promptly. Immediate actions must be taken whenever these danger signs are observed in a newborn:

1.Abnormal temperature (Cold or Hot).
2.Poor sucking or inability to suckle.
3.Breathing problem – fast breathing and severe chest in-drawing.
4.Yellowness of the hands and body.
5.Poor feeding or not feeding at all.
6.Bleeding from the cord or redness around the cord stump.
7.No movement at all or only on stimulation and convulsion.
8.There should be no hesitation in initiating appropriate actions or immediately referring newborns, where necessary, to the next higher-level facility.

1.3.1. DIFFICULT BREATHING

Difficult breathing means the baby has to make some effort to breathe. Rapid breathing for a newborn is respiration rate between 40-60 per minute. Difficult breathing in children is an emergency. It may be caused by infection, cold. In babies, chest in-drawing, flaring of nose indicate difficult breathing and a danger sign to look out for.

HISTORY	EXAMINATION/ACTIONS
1.How old is the baby? 2.What is the problem with the baby? When did it start? 3.Has the baby been ill? Had fever? Or been abnormally sleepy or difficult to wake up? 4.Has the baby stopped feeding well? 5.What do you feed the baby on? 6.Was the baby given any medicine or treatment? 7.What medicines were given to the baby since he was ill? 8.Are you using methylated powder or camphor? 9.Were the baby's eyes yellow at birth? 10.Who takes care of the baby? 11.Was the pregnancy normal? If not, what were the problems? 12.Was the baby born headfirst or bottom first? 13.Was the delivery normal? 14.Was there any difficulty? 15.Was the baby normal at birth? 16.Did the baby cry immediately after birth? 17. How was the cord cut and dressed?	1.General appearance: a.Skin colour (blue, pale, yellow) b.Spontaneous movement. 2.Chest: Check respiratory rate (note if 60 or above per minute), gasping breath, heartbeat, chest in-drawing. 3.Head: a.swelling, b.anterior fontanelle –for bulging or depression, c.Face –for any obvious abnormalities. 4.Weight: weigh and chart 5.Temperature: Check with low reading or digital thermometer and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
1.Notbreathing or Gasping at birth	-Pale or bluish colour, -Gasping than 7 -Apgar score less than 7 -Limp arms or Legs No heartbeat, no respiration	Severe Condition	1.Position the baby with the back slightly extended. 2.Gently clear secretions from mouth and nose using a clean suction device or wipe (e.g., Penguin suction). 3.Stimulate breathing by gently rubbing the back once or twice. 4.If the baby is still not breathing or gasping: 5.Immediately cut the cord and move to ventilation area. 6.Ventilate with Ambu bag and appropriately sized mask for 60 seconds (first golden minute). 7.If the baby is still not breathing, improve ventilation and ventilate for another minute. 8.Check the heart rate: 9.If >100bpm and baby still not breathing or if heart rate is slow, continue ventilation. Give the first dose of antibiotics and REFER to the hospital for advanced care (e.g., IM Ceftriaxone 20-30mg/kg immediately (Stat) or Ceftazidime if available).
2. Difficult breathing after birth	-Fast breathing or slow breathing. Severe chest in-drawing	Severe Condition Emergency	1.Communicate with parents/ Caregiver: a.The need for referral. b.The referral hospital. c.Available continued care for mother (if needed) at the referral hospital. d.Obtain their consent for referral. 2.Give first dose of antibiotics (e.g. IM. Ceftriaxone 20-30mg/kg) immediately (Stat). REFER immediately to hospital

1.3.2 CONDITIONS AFFECTING HANDS AND FEET

These may be secondary to underlying medical conditions.

HISTORY	EXAMINATION/ACTIONS
1.How old is the baby? 2.What is the problem with the baby? 3.When did it start? 4.Has the baby been ill? Had fever? Or been abnormally sleepy or difficult to wake up? 5.Has the baby stopped feeding well? 6.What do you feed the baby on? 7.Was the baby given any medicine or treatment? 8.What medicines were given to the baby since he was ill? 9.Are you using methylated powder or camphor? 10.Were the baby's eyes yellow at birth? 11.Who takes care of the baby?	1. General appearance a. Skin colour (blue, pale, yellow) b. Spontaneous movement 2. Eyes: jaundice (examine under natural light) 3. Skin: rashes, dryness, jaundice 4. Temperature: check with low reading thermometer and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
Cold hands and feet	-Temperature less than 36.5oC -Cold Stress 36.0 – 36.4oC -Moderate hypothermia 32.0 – 35.9oC Severe hypothermia <32.0oC	Moderate Condition	1.Ensure baby is dry and in clean dry clothing 2.Place baby in skin to skin contact in a pre-warmed blouse/shirt opening at the front with the baby in a nappy, hat and socks 3.Wrap baby in warm blanket 4.Ensure that the room is warm (at least 25oC) and is without drafts 5.If temperature is improving but baby cannot maintain normal temperature, put baby in Kangaroo Mother Care (KMC). 6.If temperature does not improve after one hour of re-warming, REFER immediately to hospital
2.Yellow hands and feet	- Jaundice, No fever	Moderate Condition	REFER immediately to hospital where phototherapy is possible.

Health Education

1.Continue skin to skin contact between mother and baby.
2.Teach mother about Kangaroo Mother Care -KMC).
3.Exclusive breastfeeding.
4.Counsel family members to support mother go through process.

1.3.3 CONDITIONS AFFECTING UMBILICAL CORD

The umbilical cord is very significant in a newborn, any infection around the area can cause Neonatal mortality. Common indications of danger include bleeding from the umbilical cord, redness, pus, and smelly cord. These are conditions affecting umbilical cord which could endanger the baby's life.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? When did it start? 2.Is the baby having fever? Or been abnormally sleepy or difficult to wake? 3.Is umbilical cord smelly or bleeding or just wet? 4.Has the umbilical cord dropped off? What medicine have you given the baby?	1.Check umbilicus for discharge, bleeding, swelling, redness, granulation on umbilicus. 2.Temperature Weight

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Smelly umbilical cord	Discharging (pus), smelling, umbilicus swelling, redness around the umbilicus	Severe Condition	6.Wash with clean water and soap, dry with clean cloth. 7.Give antibiotic (IM. Ceftriaxone 20-30mg/ kg immediately (Stat) OR Gentamicin 5mg 1Mkg immediately (Stat)). REFER immediately to hospital.
II. Bleeding umbilical cord	Bleeding umbilicus	Moderate to severe condition	1.Re-tie the cord immediately 2.Give a third tie at the base of the cord 3.Clean with sterile water. Apply chlorhexidine gel 4.Give IM.Vitamin K, 0.5mg (preterm baby), 1.0mg (full term baby) Refer if bleeding persists.
III. Wet umbilicus	•Wet umbilicus Temperature normal (37.50C	Mild condition	1.Clean with sterile water 2.Teach mother to clean with sterile water 3.Expose umbilicus Review daily and Check immunization status
IV. Retained umbilical stump	•Granulation on umbilicus	Mild condition	1.Clean with sterile water 2.Cauterize with blue stone 3.Review daily 4.Administer immunization as necessary 5. Record in home - base card and return to mother

Health Education

1.Encourage mother/caregiver.
2.Teach mother to practice and maintain hygiene
3.Teach mother/caregiver about care of the umbilicus.
4.Counsel mother against harmful traditional practices (DISCOURAGE use of toothpaste, cow dung or any other substance on the cord).

1.3.4 OTHER BLEEDING PROBLEMS

This is bleeding from other parts of the body.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem with the baby? When did it start? 2.Has the baby been ill? Had fever? Or been abnormally sleepy or difficult to wake up? 3.Has the baby stopped feeding well? 4.What do you feed the baby on? 5.Has the baby been given any medicine or treatment? 6.What medicines were given to the baby since he has been ill? 7.Where is the bleeding coming from? a. Nose, b. Gums c. Rectum d. Ears e. From Circumcision.	Check the following: 1.Mouth 2.Ears 3.Rectum 4.Nose 5.Chest - check the breathing. 6. soloeffect- check for pallor.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Baby bleeding from mouth, rectum or other parts of the body	Bleeding from: -Mouth -Rectum -Pallor Normal breathing.	Severe Condition	1.Apply pressure on the site, where practicable. 2.Show mother/caregiver how to maintain pressure. 3.Give IM Vitamin K into the anterolateral thigh; 0.5mg (preterm baby), 1.0mg/kg (full term baby). REFER to hospital immediately.
II. Bleeding from circumcision	- Bleeding with or without pallor	Moderate condition	1.Apply pressure on site for 10 minutes. 2.Show mother how to maintain pressure (over a period of 1-2 hours). 3.Give IM Vitamin K into the anterolateral thigh; 0.5mg (preterm baby), 1.0mg/kg (full term baby). 4.If bleeding does not stop, REFER to hospital. If there is pallor, REFER to hospital.

Health Education

1.Teach mother/caregiver how to apply pressure on bleeding site.
2.Counsel/Reassure mother/caregiver on baby's condition and the need for referral.

1.3.5 CONDITIONS AFFECTING THE HEAD

There are many problems that can affect the head during the delivery process, such as injury to the head, swelling of the head, sunken or bulging fontanelles, cephalohematoma and caput succedaneum.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem with the baby? When did it start? 2.Has the baby been ill? Had fever? Or been abnormally sleepy or difficult to wake up? 3.Has the baby stopped feeding well? 4.What do you feed the baby on? 5.Has the child convulsed? 6.Was the swelling present at birth? 7.Was the head larger than normal at birth or after? 8.What part of the head is involved?	1.General Appearance: Irritable, lethargic, crying excessively, evidence of injury. 2.Head: swelling, check size (head circumference). 3.Fontanelle: depressed, bulging. 4.Evidence of attempt to drain swelling. 5.Check for lump. 6.Check the wideness of the fontanelle. 7.Check if fontanelle is pulsating.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Swelling on the head with Convulsion	-Child irritable, -Lethargic -Convulsing -Excessive crying Evidence of injury	Severe condition Suspect Birth injury	1.Reassure mother/caregiver on need for referral. 2. REFER immediately to hospital.
I. Swelling on the head not present at birth but comes up within about 12-24 hours and continues to increase	-Large swelling on head -Evidence of attempt to drain swelling Baby is blue, or infected swelling	Severe condition	1.Give IM. Ceftriaxone 20-30mg/kg immediately (Stat). 2.Reassure mother/caregiver on need for referral REFER immediately to hospital
iii.Swelling on the side of the head. No other complaint.	-Lump on the head Baby otherwise well	Moderate condition Suspect birth Injury	1.Give IM. Ceftriaxone 20-30mg/kg immediately (Stat). 2.Explain and reassure mother/ caregiver on finding. 3.Do not attempt to incise or drain swelling. 4.Ask mother/caregiver not to apply anything 5.REVIEW in 2 days, then 1 week 6. If swelling getting bigger, REFER
iv.Wide opening on front part of head	1. Fontanelle too wide. 2. Fontanelle pulsating.	Mild condition	1.Explain condition to mother/ caregiver that fontanelle should close at about 18 months. Monitor head circumference weekly for 2 month

1.36 CONDITIONS AFFECTING THE EYES

The process of delivery can be a channel for Newborn to develop eye condition when the baby sweeps through the vulva bacterial infection can set in and put the Newborn in danger. These are problems affecting the eyes which if not well managed could lead to impaired vision or blindness.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem with baby's eyes? 2.How long has the baby had the problem? 3.What's the nature of discharge? Blood, pus? 4.Has the baby had it before? 5.Did the problem come suddenly or gradually? 6.Did anything enter the baby's eyes? 7.Did the mother or anyone else put something in the baby's eyes? What was it? Has any medicine been put in baby's eyes? If so what medicine?	1.General Appearance: In pain, 2. Eyes: Redness, pus, sticky eyes

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. Discharge from baby's eye(s)	- Sticky eyes Pus in the eyes	Moderate to Severe condition Suspect Gonococcal infection	1.Wash hands with soap and water before and after treatment. 2.Gently clean eyes with cotton wool soaked in cooled boiled water 6-8 times a day 3.Use separate cotton wool for each eye. 4.Apply 1% tetracycline eye ointment in each eye 8 hourly day for 7days (ensure that the eye are clean before application) 5.Give IM. benzyl penicillin 25mg/kg immediately (stat), 12hourly for 5 days 6.Review in 2 days 7. If no improvement and eye discharge become copious, REFER to hospital.
ii. Small amount of blood in the eye	-Blood-shot in the white part of baby's eyes	Mild condition	1.Explain condition to mother (This could be due to pressure at birth). 2.No treatment is needed. Follow-up in 2 days, then 1 week. If mother anxious, REFER to hospital.

Health Education

1.Counsel and reassure mother/caregiver.
2.Teach mother/caregiver to practice and maintain hygiene.
3.Discourage mother/caregiver from applying any other substance.

1.3.7 CONDITIONS AFFECTING BABY'S BREAST

These are common problems affecting baby's breast which may be due to the effect of maternal hormones.

HISTORY	EXAMINATION/ACTIONS
1.What's the problem with the baby and when did it start? 2.Is the child crying excessively? 3.Does the child have fever? 4.When did the fever start 5.What treatment have you given at home?	1.General appearance: Ill looking, 2.Breast: check for tenderness, redness, production of little milk, warm breast 3. Body temperature

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. Swelling of breast in the newborn infant; Excessive crying; Fever	•Swollen, tender, warm breast, Rectal Temp. 38.00C	Moderate to severe condition	1.Give Syrup Paracetamol (120mg/5ml) 2.5ml immediately REFER
ii. Swelling of breast in the newborn infant	•Breast swelling, no redness, not painful, not warm, may produce little milk	Mild condition	1.Explain to mother that condition will disappear by itself 2.Advise her to leave breast alone, squeezing will increase swelling. 3.Follow-up visit until better. 4. If no improvement, REFER

1.3.8 CONDITIONS AFFECTING THE ANUS

The anus is the terminal end of the alimentary canal. Common problems affecting the anus include fissures, redness and prolapse. Some babies may be born with imperforate anus and may not be able to pass stool.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Has the child passed stools since birth? 3.Is there vomiting? If yes, how many times? 4. Has any medication been given?	4.General Appearance: Ill-looking, in pain. 5.Temperature: Check and record. 6.Anus: Confirm anal opening using thermometer.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
Newborn unable to pass stool in the first 48hours of birth.	Anal opening not present. • Confirm condition using rectal thermometer very gently.	Severe condition Imperforate anu	REFER immediately to the hospital.
Newborn unable to pass stool in the first 48hours of birth. With or without vomiting with bile-stained materials.	Anal opening present.	Severe condition Abdominal obstruction	REFER immediately to the hospital.

Health Education

1. Counsel and reassure mother about baby's condition.

1.3.9 CONDITIONS AFFECTING THE LIMB

These are common condition affecting either the arms and/or legs. Deformities of the arms and legs if present since birth often are not correctable. New deformities or swellings may be due to normal movement of the parts whenever there is history of injury, swelling or deformity.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? When did it begin? 2.Was there an accident or injury? 3.Is the baby able to move limbs? 4. Is there any pain when moving the limbs?	7.General appearance: Ill-looking. Crying excessively. 8. Limbs: Deformity - foot turned inwards, swelling, fracture, tenderness, loss of function, warm to touch (area of tenderness).

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Foot is turned inward and cannot be turned outward or upward.	•Hip is out of socket. •Foot is turned inward and cannot be turned outward or upward.	Severe condition	REFER immediately to the hospital.
II. Baby unable to move limb(s] or foot	•Less than normal movement in one or more limbs with or without tenderness	Moderate condition	1. Splint the bad limb (Upper limb) by putting the baby's palm on the chest and strap limbs down. 2. Splint the bad limb. 3. Bandage the bad limb to the good limb. 4. REFER to the hospital for advanced care.
iii.Baby unable to move upper limb pain.	•Fracture of the collar bone. • Pain on picking up baby by the armpit.	Moderate condition	1.Apply figure of 8 bandages. 2. Give Paracetamol Suppository 125mg per rectum, if not available, give Syrup Paracetamol(120mg/5ml) 2.5ml immediately (stat). 3. REFER to the Hospital.

Health Education

1.Counsel and Reassure mother/caregiver on baby's condition.
2.Teach mother about accident prevention at home.

1.3.10 CONDITIONS AFFECTING THE SKIN

These are common conditions affecting the skin. It could be due to poor hygiene and lack of safe water for bathing.

HISTORY	EXAMINATION/ACTIONS
1.How long has the postules, abcess, rash been there? 2.Does anyone in the family have a similar rash or skin problem? What is the source of your water supply?	1.General appearance: uncomfortable 2.Skin: (a) location, distribution and size of lesions (b) colour of rash or lesion Temperature, respiratory rate, wheeze: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I.Abscesses (Boil)	•Abscesses or blisters on the skin with or without fever	Moderate condition	1.Apply warm compress 3 times daily until drained 2.Daily cleaning and dressing 3.If fever is present, Give Paracetamol Suppository 125mg per rectum, if not available, give Syrup Paracetamol(120mg/5ml) 2.5ml 8hourly for 3 days Review in 2 days
II. Pustules, or blisters on the skin	•Pustules	Moderate condition	1.Wash hands with clean water and soap. 2.Dry the area with a clean towel, and paint with gentian violet. 3.Advise mother to continue the procedure. 4.Review in 2 days; if no improvement, REFER to the hospital. 5.Examine mother for skin infection and if present, treat using Adult Standing Orders. Encourage personal hygiene.
III.Skin rashes	•Heat rash	Mild condition	1.Use light clothing. 2.Bath child twice daily. 3.Apply Calamine lotion after bathing. 4.Keep baby in a well-ventilated room. 5. Ensure good hygienic practices.

Health Education

1.Proper hand washing techniques.
2.Teach mother/caregiver about personal hygiene.

1.3.11 NEONATAL TETANUS

This is a condition characterized by poor or inability to suckle, spasms and stiffness of the body.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem with the baby? 2. How long has the baby had the problem? 3. Has the baby stopped sucking well? 4. Does the baby have fever? Or cold body? 5. Does the baby have stiffness of the body? 6. Is the baby able to open mouth without difficulty? 7. Is the baby's breathing normal? Difficult?	1.General appearance: ill looking, lethargic or unconscious. 2.Mouth: locked jaw. 3.Neck: stiff neck. 4.Body: stiffness, arching of the back, spasms. 5.Limbs: stiffness. 6. Temperature: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Does not open mouth, cannot feed, "Convulsion "	•Locked jaw •Stiff abdomen •Arched back •Stiff limbs •Spasms	Severe condition (Suspect neonatal tetanus)	1. REFER immediately to the hospital perform first line management or medication before referral e.g. phenobarbitone 15 to 20mg/kg I.M stat
Incapacity to nurse at the breast. Elevated body temperature or hypothermia. Seizures. Labored respiration.	•Warm to touch or temperature 37.5°C and above. •Fever or hypothermia	Severe Condition (Suspect Sepsis)	1.Clear nose, if blocked 2. REFER immediately to the appropriate health facility hospital.

Health Education

1.Reassure mother/caregiver on baby's condition.

1.3.12 DIARRHOEA AND VOMITING IN NEWBORN

Diarrhoea is the passage of three or more loose/watery stools in a 24-hour period. Vomiting refers to the expulsion of stomach contents. Infants under six months often regurgitate small amounts of their food immediately after being fed, which is considered normal. A child who is exclusively breastfed may have bowel movements five to eight times per day, which is also within normal range.

HISTORY	EXAMINATION/ACTIONS
1.How long has the baby had diarrhea? 2.How many times has the baby passed stool? 3.Is the baby vomiting? How often? Does the baby vomit everything? 4.Is there any other complaint such as fever? 5.Is there any exclusive breastfeeding? If no, what has the baby being fed with? 6.What is the method of feeding? 7.Has any medication been given? What type? 8. Does the baby take breast eagerly, normally, or poorly?	1.General appearance: irritable, ill-looking, lethargic, unconscious, alert. 2.Head: anterior fontanelle depressed. 3.Eyes: normal, sunken, tears on crying. 4.Mouth: unable to suck/drink, drinking poorly, moist tongue. 5.Chest: check respiratory rate. 6. Skin: skin pinch goes back quickly, slowly, or very slowly.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Diarrhoea with or without vomiting	•Lethargic or unconscious •Sunken eyes •Skin pinch goes back very slowly •Not able to drink or drinking poorly	Severe condition	1.Give fluid, treat as plan C 2.Keep warm and refer immediately to hospital. 3.Encourage to continue breast-feeding. 4.Teach mother how to make Lo-ORS/Zinc 5.For Zinc: Give half tablet (10mg) every day for 10 days 6.For Lo-ORS: One half cup (50-100ml) given until diarrhea stops (500ml/day) if mother is not doing exclusive breastfeeding
II. Diarrhoea with or without vomiting	•Restlessness •Irritable •Sunken eyes •Eagerly drinks/feeds •Skin pinch goes back slowly	Moderate condition (some dehydration)	1.Treat as plan B 2. Review in 2 days . If no improvement, REFER appropriately
III. Diarrhoea with or without vomiting	•Not enough signs to classify as some or severe dehydration •Alert •Eyes normal (not sunken) •Tears on crying •Moist mouth/tongue. Child drinks normally •Skin goes back quickly when pinched	Mild condition	1.Treat as Plan A. 2.Tell mother to breast-feed frequently and for longer periods at each feed. 3.Give Lo-ORS and Zinc in addition to breast milk. 4.Teach mother how to make Lo-ORS. 5.Give 50-100ml of Lo-ORS after each loose stool in addition to regular. 6.Advise to continue giving extra fluids till diarrhea stops. 7.Teach mother about danger signs and ask her to bring the child to the clinic immediately if any of the following danger signs are present: a.Sunken eyes b.Lethargic or Unconscious. 8.Drinking poorly or not able to drink 9. Skin pinch goes back very slowly

Health Education

1.Advise on regular and complete immunization.
2.Counsel on proper personal hygiene and hand washing.
3.Advise on use of long-lasting insecticide nets (LLINs).
4.Advise on environmental hygiene.
5.Encourage exclusive breast feeding and avoid bottle feeding. Breast feed frequently and longer at each feed.
6.Keep baby warm.
7.Remind mother to take home-based records with her anytime she takes baby to clinic or hospital.

1.3.13 SEVERELY ILL BABY

HISTORY	EXAMINATION/ACTIONS
1.What is wrong with the baby? 2.When did you notice the problem? 3.Has the baby been ill? 4.Is the baby feeding well? 5.Is the baby difficult to wake? 6.What do you feed the baby on? 7.Is the baby cold or hot, how long? 8.Has the baby convulsed before? 9.What medication has been given to the baby for the problem? 10.Was the pregnancy normal? If not, what was the problem? 11.Was the delivery normal? 12.Was the baby normal at birth? 13. Did the baby cry immediately at birth?	1.General appearance: severely ill-looking, unconscious, lethargic. 2.Head: bulging fontanelle. 3.Nose: nasal flaring, grunting. 4.Ear: draining pus. 5.Chest: check respiration and record chest in-drawing. 6.Umbilical cord: redness extending to the skin. 7.Skin: many or several skin-pustules. 8.Abdomen: distension. 9.Temperature: check and record. 10.Weight: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Severely ill baby	•One or a combination of any of the following: a)Warm to touch b) Rectal temp38°C and above. c)Hypothermia (cold to touch below 36.5°C. d)Convulsion/Fits. e. Lethargic f)Unconsciousness g)Severe chest-in- drawing h)Fast breathing (60 or more per minute) i)Nasal flaring j)Grunting k)Bulging fontanelle l)Pus draining from ear m)Umbilical redness extending to the skin n)Limited movement o)Abdominal distension p) Weight- not gaining weight	Severe condition Possible serious bacterial infection	1.Give fluid, treat as plan C. 5.Keep warm and REFER immediately to the appropriate health facility hospital. 2.Encourage to continue breastfeeding. 3.Teach mother how to make Lo-ORS/Zinc. 4. For Zinc: Give half tablet (10mg) every day for 10 days. For Lo-ORS: One half cup (50-100ml) given until diarrhea stops (500ml per day) if mother is not exclusive breastfeeding

Health Education

1.Encourage exclusive breastfeeding.
2. Keep baby warm.
3.Advise on personal and environmental hygiene.
4.Advise on use of Long-Lasting Insecticidal Nets (LLINs).
5.Advise on regular and complete immunization.
6. Remind mother to take home-based records with her any time she takes baby to clinic or hospital.

1.4 NEWBORN ABNORMALITIES

Sometimes babies are born with some physical abnormalities. The health worker should examine for any abnormalities and initiate immediate action to avoid permanent disabilities.

HISTORY	EXAMINATION/ACTIONS
1.How old is the baby? 2.What is wrong with the baby? 3.When did you notice the problem? 4.How old was your pregnancy before delivery? Was the pregnancy normal, if not what was the problem? 5.Is the baby sucking well? 6. Does any other person in the family have this type of problem? What medicine or substances have been given to the baby since she was born?	1.General appearance: Any obvious abnormality, alert, ill-looking, irritable, excessive crying, sleepy. 2.Head: Swelling, size, head circumference. 3.Mouth: Cleft palate, hare-lip. 4.Eyes: Squint. 5.Ears: Positioning of the two ears. 6.Chest: Respiratory rate. 7.Limb: Abnormalities like webbed toes or fingers, extra digit. 8.Genitalia: Undescended testes, no vaginal perforation/opening, bleeding, abnormal penile opening. 9.Temperature: Check and record. 10.Weight: Weigh and chart. 11.Abdomen: Distension. 12 . Skin: Birthmarks.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. Premature /low birth weight	•Weight is 2.5kg or below •Baby sucks poorly	Severe condition	1.Exclusively feed baby with expressed breast milk with spoon/dropper, starting with breastfeeding before offering the cup. 2.Feed baby every 2-3 hours (wake baby and feed). 3.Keep baby longer at the breast, do not interrupt feeding if baby is trying. 4.Encourage mother to express milk into baby's mouth directly. 5.Keep baby warm using Kangaroo Mother Care (KMC) method. 6.Less than 1.5kg, REFER immediately to hospital. 7.Ensure warmth during referral. 1.5kg to less than 2.5kg (32-36 weeks gestational period).
ii. Abnormal appearance of genitalia	Genitalia abnormal •Undescended testes •Penile opening in wrong position •No vaginal opening.	Severe condition	1.Refer to hospital
iii. Physical abnormalities	1. Lump or bone detect at the back 2. Very big head 3. Squint eye 4. Low set ears 5. Clubfoot 6. Hare lips 7.Webbed toes and/or fingers 8. Cleft palate	Moderate to severe condition	1.Explain findings to mother. REFER to hospital

iv. Small baby	•Weight is 2.5 kg or below •Baby can suck	Moderate condition	1.Encourage exclusive breastfeeding. 2.Keep baby warm using Kangaroo Mother Care (KMC). 3.Review in 2 days. If improving, review weekly for 2 weeks. If no improvement, REFER.
v. Birth marks	•Small black or blue patches or red or bluish red	Moderate condition	1. Reassure mother
vi. Bleeding from circumcision	• Bleeding with or without pallor	Moderate	1.Put pressure on side for 10 minutes 2. Show mother how to maintain pressure 1-2 hours. 3. If bleeding does not stop, REFER appropriately.
vii. Constipation	Abdominal distention or no finding.	Mild/Moderate condition	1.Encourage exclusive breastfeeding 2.Probe rectum to stimulate defecation If no improvement, REFER
viii. Baby not feeding well, Crying always.	•No abnormality •Baby is well Irritable	Mild condition Suspect maternal causes	1.Examine mother's breasts and treat if necessary. 2.Observe how mother feeds infant and correct if necessary. Counsel mother accordingly.
ix. More than five fingers or toes	•Extra digits	Mild condition	1.If no bone present(soft) tie tightly with a piece of strong sterile cotton or silk thread 2.Advise family that finger or toe will turn black and fall off in a few days 3.Follow up in 3 days If bone is present in extra digit DO NOT TIE, REFER appropriately
x. Slight vaginal bleeding in baby girl	•Vaginal bleeding	Mild condition	1. Explain to mother that this is a passing condition, and nothing needs to be done 2. Review in 48 hours, if still bleeding. REFER appropriately

Health Education

1.Educate the mother on abnormalities of the newborn.
2.Advise on regular and completed immunization.
3.Encourage Exclusive Breastfeeding.
4.Keep Baby Warm.
5.Remind mother to take home-based records with her any time she takes the baby to the clinic or health facility.

1.5 ORAL HEALTH

This is care of oral cavity. Examination of the mouth of the newborn is an absolutely essential component of examination of the child in order to detect congenital malformations, swelling, pallor, cyanosis and infections. The areas to inspect are the lips, tongues, gum, palate, cheeks, floor and lining of the mouth.

HISTORY	EXAMINATION/ACTIONS
1.What is wrong with the baby? 2.When did you notice the problem? 3.Has the baby been ill? 4.Is the baby feeding well? 5.Was the baby born with tooth/teeth in the mouth? Or teeth came shortly after birth? 6. Has the baby been given any medication?	1.General appearance- ill-looking or not 2.Mouth- roof of mouth (formed completely) 3.Tongue- sore, fissured, smooth, dry, tied- check for band holding tongue, swelling 4.Tooth- presence or absence of white patches 5.Lips- Cleft 6.Temperature- check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. Baby is not able to feed.	Roof of mouth is not completely formed – Cleft palate. Severe condition.	Severe condition	1. Refer to the appropriate health facility.
II. Mouth: a) Baby's lip is not completely formed. b) Baby is able to feed successfully.	Hare-Lip	Moderate condition	1.Reassure parents that the condition can be corrected surgically when baby is about 6 weeks old. 2.Temporary measures can help baby feed better and look better. 3.Help mother put child to breast or feed with expressed milk using cup and spoon REFER to the appropriate health facility.
iii. Tongue Tie	• Fibrous band hold down the tongue	Mild condition	1. Explain the findings to the parents. As the baby grows older, the tie may not interfere with speech. Refer appropriate
IV. Tooth/Teeth in the mouth at birth or shortly after.	Tooth may or may not be mobile	Mild condition	1. Explain to parents that condition is normal. Reassure that baby is not evil 2. REFER appropriately.
V. Soft swelling under the tongue	Swelling contains fluid (Salivary cyst)	Mild condition Suspect salivary cyst	1.Reassure the mother 2.Give stat dose of oral antibiotic 3. Refer to the appropriate health facility

VI. White patches in mouth	White patch that rubs Off	Mild condition	1.Clean the mouth with cotton wool soaked in normal saline 2.No treatment needed, 3.Explain to mother is only milk.
VII. White patches in the mouth	White patch that does not rub off on wiping	Mild condition Suspect thrush	1. Clean mouth with cotton wool soaked in normal saline 2. Give nystatin oral suspension 0.3mls 6 hourly for 7 days 3.Review after 3 days. If no improvement, REFER appropriately

Health Education

1 Oral hygiene, emphasizing proper utilization of cleaning materials - cotton wool and salt solutions.
2 Nutrition and prevention of tooth decay, encourage exclusive breast feeding
3 Discourage uvulectomy.
4 Remind mother to take home-based records with her anytime she takes baby to the clinic or hospital.
5 Educate the mother on personal and environmental hygiene.
6 Advice on regular and completed immunization.

NB: Columns for complaint, findings on examination, clinical judgement and action as well as Health education topics have been rearranged and updated.

Reference

World Health Organization (2015)

APPENDICES

National Guidelines for Comprehensive Newborn Care REFERRAL LEVELS November 2021 | First Edition.
Figure 5.1: Essential care for every baby in the first 90 minutes of life
Figure 5.2: Good attachment to the breast
Figure 5.3: Essential Care for Every Baby (ECEB) Chart

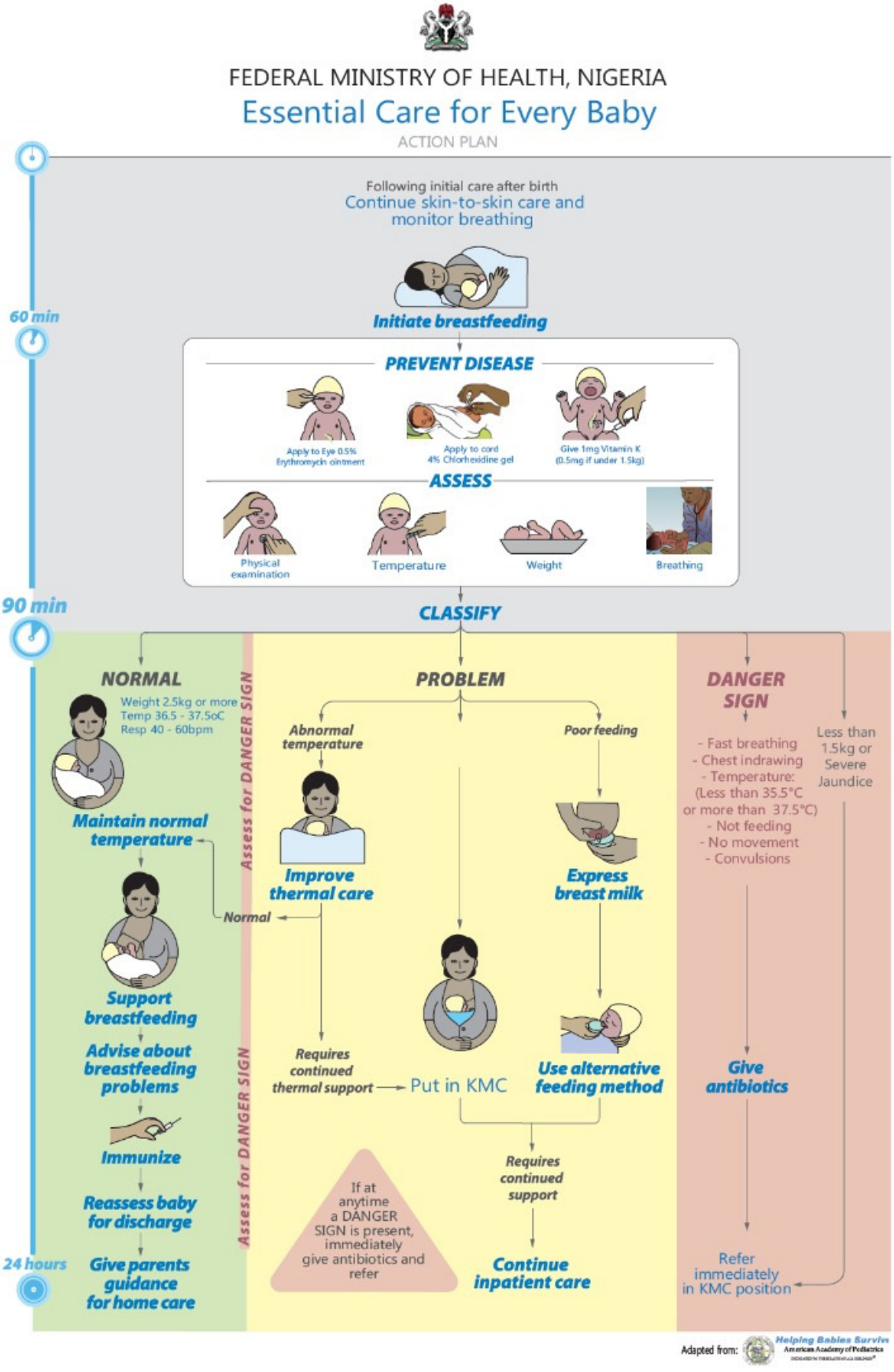


Figure 1-Essential Care for Every Baby

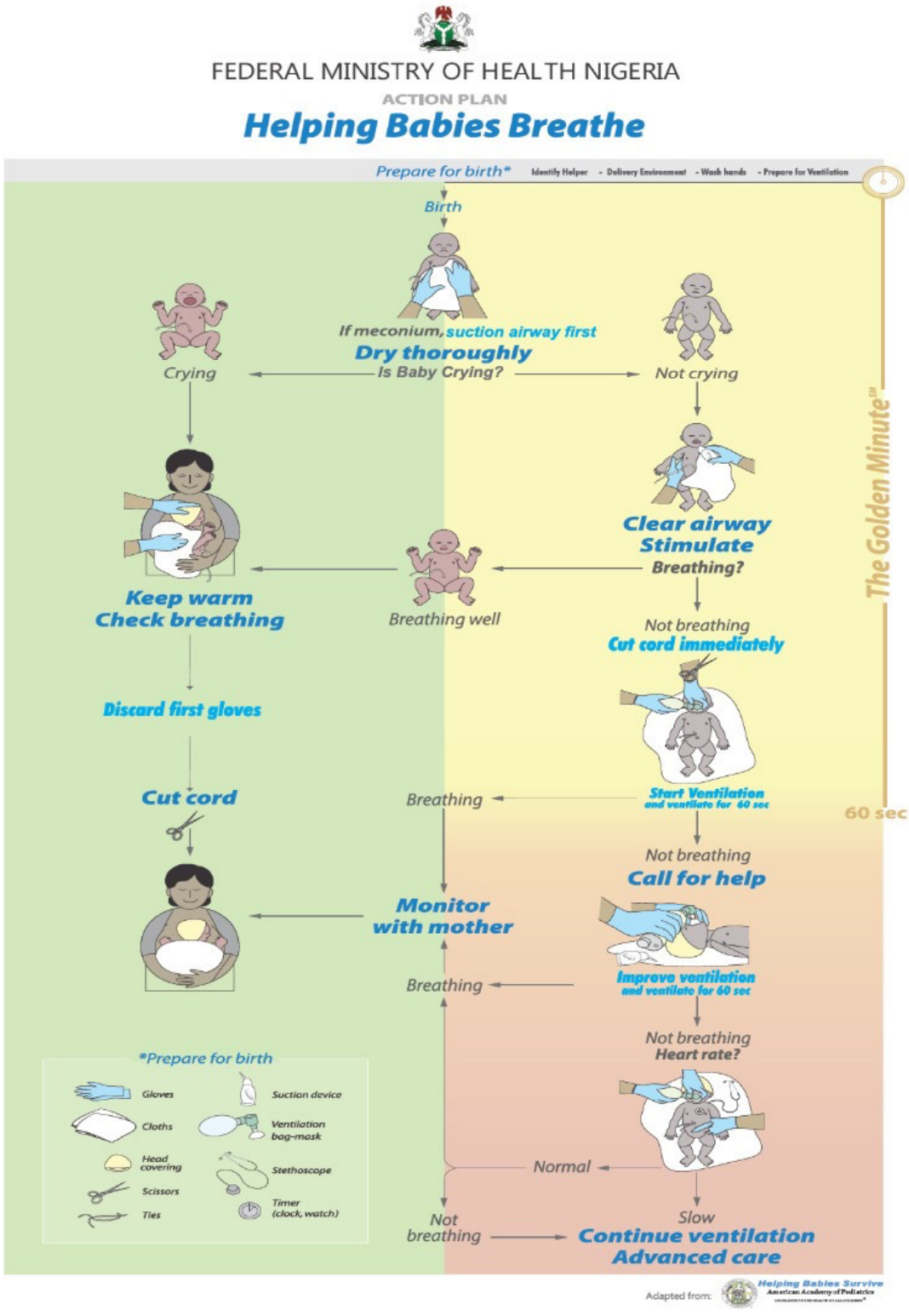


Figure 2- Helping Babies Breathe

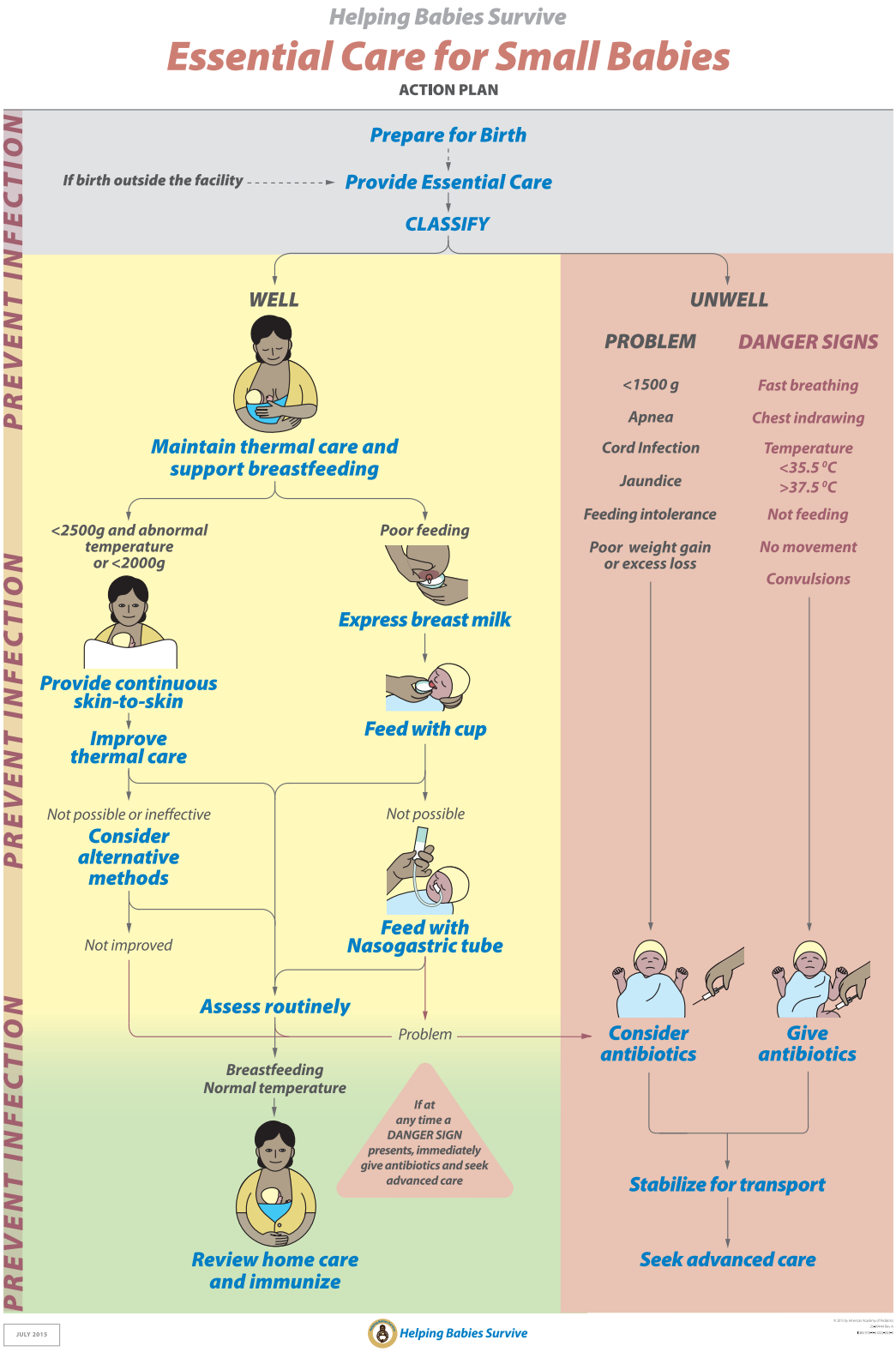


Figure 3-Essential Care for Small Babies

SECTION TWO
EARLY CHILDHOOD

2.0 EARLY CHILDHOOD

2.1. CHILD'S FIRST VISIT

The child's first visit is a very important one and should always include screening for health problems, growth monitoring and promotion, health education on nutrition, immunization and protection from illness and other conditions that are relevant to child and the family. Mothers should be encouraged to bring their children for visits while they are well monthly in the first 2 years of life, every 3 months thereafter until 5 years, and subsequent visits when necessary.

HISTORY	EXAMINATION/ACTIONS
1. Birth: Were pregnancy and birth normal? Where was the child delivered? Did the child cry immediately after birth? Was the child jaundiced (yellow) at any time? 2. Developmental Milestone: Is the child developing normally? •At about the age of 2 months-sees, responds to voice •At about the age of 4months holds head up •At about the age of 8months sits by self •At about the age of 12months walks alone 3.Immunizations: What immunization has the child had? Does the child have immunization card? 4.Nutrition: How have you been feeding the child? Exclusive breast milk if less than 6 months? Complementary feeding, supplementary feeding and breast milk after 6 months? What food is he/she eating now? If not exclusively breastfeeding, is mother using any family planning method? 5.Social: With whom does the child live? Does mother work outside the home? Who looks after the child when the mother is away? What is your source of water? What type of toilet do you use? Is your house screened against mosquitoes? Do you use Long Lasting Insecticide Net (LLIN) 6.Medical: Has the child had illnesses in the past? E.g. Measles? Mumps? What treatment was given? Where? Any illness or chronic cough in the family? 7. Does the child have any problem now?	1.General appearance: Well-looking, ill-looking 2.Colour: normal, pale, yellow, blue 3.Skin: rashes, sores, bruises 4.Head: size, swelling, colour and texture of hair, fontanelle-depressed or bulging, sutures-normal or wide 5.Eyes: discharge(sticky), yellow, red and swollen eyelids 6.Nose: discharge—blood, mucus, blockage 7.Neck: glands, masses, stiffness 8.Mouth: ability to suck normally, bleeding gums, dental caries, pallor, sores, cracks, redness of the throat (tonsillitis), abnormality (cleft upper palate) 9.Spine: swelling or opening 10.Chest: breathing normally, respiratory rate (note if 60 per minute or more) 11.Abdomen: condition of umbilical cord, umbilical hernia, rigidity, tenderness, mass and distended 12.Arms and legs: normal movement, deformity, fingers and toes (for extra or joined webbed fingers) 13.Genitalia: abnormality (check vaginal/penial opening, size of clitoris, both testis in scrotal sac), swelling, congenital hernia. 14.Anus: Redness or prolapse, irritation and imperforation. 15.Temperature and pulse: check and record. 16.Weight: check and chart 17.Haemoglobin: estimate and record 18.Developmental Milestone a.At about the age of 2 months — sees, responds to voice b.At about the age of 4 months — holds head up c.At about the age of 8 months — sits by self At about the age of 12 months — walks alone

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. No complaint (routine visit)	•No abnormality found on checking child health card for: -Immunization status -Growth monitoring -Hb (haemoglobin)	Well child	1.Encourage exclusive breast-feeding if under 6 months 2.Explain growth chart, commend mother and advise to report for regular growth monitoring 3.REFER to the appropriate health facility for food demonstration at 5 months. Mothers should attend at least once. 4.Advise mother to introduce complementary feeding at 6 months and continue breastfeeding till 2 years 5.Give Vitamin A if child has not received Vitamin A in the last 4 months: •2-6 months: 50,000 units •6-11Months: 100,000 units •12months-5years: 200,000 units 6.If not fully immunized for age, advise and immunize appropriately (see appendix) 7.Check haemoglobin every 4 months if child is less than 1 year and every 6 months thereafter. 8.Check mother's DT status and advise appropriately 9.Advise mother on family planning methods. 10.Record findings and management in child's home-based record and give to mother 11.Give appointment appropriate to child's age
Any abnormal findings			Refer to appropriate sub-section

Health Education

Encourage the mother on:

1.Exclusive breast feeding up to 6 months and breast feeding up to 2 years.
2.Immunization
3.Regular clinic attendance
4.Child spacing and good nutrition.
5.Personal and food hygiene
6.Use of long-lasting insecticidal Nets (LLINs)
7.Encourage mother to take home-based records with her anytime she brings child to the clinic or hospital.

2.2. HEALTHY CHILD

The healthy child visit is a very important one for the Community Health Practitioner and should always include screening for health problems, growth monitoring and promotion, health education about nutrition, birth registration, immunization and protection from illness and other conditions as relevant to the family. Mothers should be encouraged to take their children for visits while they are well—two years of life and 3-monthly thereafter until 5 years old. Mothers should equally be taught the importance of keeping clinic appointment(s).

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. No complaint (routine visit)	•Check child health card for: a.Last clinic appointment and current treatment (if any) b.Immunization status c.Growth monitoring d. Hb (haemoglobin)	Well child	1.Encourage exclusive breast feeding if under 6 months (feeding option if mother is HIV positive) 2.Interpret result of child's arm circumference/weight to mother and teach mother how to use arm circumference strip. 3.Advise mother to introduce complementary feeding at 6 months 4.Refer for food demonstration at 5 months. 5.Give Vitamin A if child has not received Vitamin A in the last 4 months: a. 2 - 6 months 50,000 units b. 6 - 11 months: 100,000units c. 12 months - 5 years: 200,000 units 6.Advise to report for regular growth monitoring. 7.If not fully immunized for age, advise appropriately (see appendix.) 8.Check mother's ID status and advise appropriately 9.Advise mother on family planning methods, 10.Record findings and management in child's home-based record and give it to mother 11. Give appointment appropriate to child's age

Health Education

Encourage the mother on;

1. Exclusive breast-feeding up to 6 months (feeding option if mother is HIV-positive) and breastfeeding up to 2years.
2. Regular and complete Immunization.
3. Regular clinic attendance.
4. Child spacing and good nutrition using local food to demonstrate.
5. Personal and food hygiene.
6. The use of Long-Lasting Insecticidal Nets (LLINs).
7. Importance of taking home-based records with her anytime she takes child to the clinic or hospital.

2.3. FEVER

Fever is elevation of body temperature above 37.5°C. Fever is a sign of infection or loss of body fluid. Rapid rise of temperature may lead to fits especially in young children of 6 months to 5 years (febrile convulsion). Some common causes of fever are malaria, tuberculosis, respiratory tract infections, gastrointestinal and other bacterial and viral infections. All patients with a history of fever should be treated and encouraged to drink plenty of water, other fluids and fruits. High fever should be treated as an emergency.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had the fever? 2.Is there any convulsion? 3.Has the child ever had any convulsion 4.Is the child having a running nose? Cough? Or difficult breathing? 5.Is the child coughing up blood? 6.Is the child vomiting or having diarrhea? 7.Is the child having bloody diarrhea? 8.Is the child having skin rashes? 9.Is the child having any ear pains? Sore throat? Headache? Stiffness of the neck? 10. Is the child passing urine normally? Is it painful? Is it more or less frequent? 11.Is the child bleeding inside and outside the body? From the eyes, ears and nose? 12. Has any analgesic or other medication been given?	1.General appearance - ill looking or not. 2.Mouth - roof of mouth (formed completely) 3.Tongue - sore, fissured, smooth, dry, tied - check for band holding tongue, swelling. 4.Tooth - presence or absence of white patches. 5.Lips - Cleft. 6.Temperature - check and record. 7.Ear: tenderness behind the ear, discharge, blood. 8.Throat: sores, pains or redness. 9.Neck: stiffness, nodes. 10.Abdomen: tenderness, masses, bloated. 11.Temperature and pulse: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Fever with any of the following danger signs: a.Vomiting everything b. Unable to drink, or suckle c. Convulsion or history of convulsing d. Stiff Neck	Temp.Above 37.5°C with any of the following: •Neck stiffness •Change in alertness •Convulsion •Anaemia (lack of blood) •Prostration •Bulging fontanelle •Moderate.dehydration •Lethargic (weak or no energy)	Very Severe condition Suspect febrile illness	1.Tepid sponge for children being referred with very severe febrile disease. 2.Give the first dose of intramuscular artesunate OR give the first dose of intramuscular Quinine. 3.Give the first dose of an appropriate antibiotic: 4.Susp.Amoxicillin 125mg/5ml a. 2mths-11mths: 2.5ml 8hourly for 5 days b. 12mth-5 years: 5ml 8hourly for 5days For high fever (37.5°C or above): Syrup 5.Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 6.Give breast milk (30 - 50ml) to prevent low blood sugar OR 7. sugar water if breast milk is not available.

II. Fever without any danger sign	Temp. above 38.5°C with no danger sign and stiff neck Malaria test (mRDT) positive	Moderate condition Suspect Uncomplicated Malaria	1.Treat with Oral ACT Anti-malarial Artemether/ Lumefantrine (AL 20mg/120mg): Day 1: Give 1st dose, then 2nd dose 8 hours later, then Day 2: 1.2hourly for 2 days Age: 2.1 month - 3 years, 3.Weight: 5-14kg, 4. Dose: 1 tablet at 0 hour, 8 hours later, then 12hourly for 2 days 4-5 years, Weight: 15-24kg, Dose: 2 tablets at 0hour, 8 hours later, then 12hourly for 2 days 5. Give paracetamol for high fever: a. 1 month - 3 years: Syr 2.5ml 8hourly for 3 days b. 3-5 years: Syrup 5ml 8hourly for 3 days 6.Advise the mother to use LLINs for the child 7.Advise the mother when to return immediately. 8.Follow up in 2 days if fever persists every day for more than 7 days. 9.REFER to the appropriate health facility for assessment.
III. Fever No other findings	Fever with no other findings	Mild Condition Suspect Pyrexia of Unknown Origin (PUO)	1. Give one dose of paracetamol for high fever (37.5°C or above) 2. Advise the mother to use LLINs for the child 3. Advise the mother when to return immediately 4. Follow up in 2 days 5. If fever persists ASSESS for other Conditions

IV. Fever, cough with runny nose/sore throat	Runny nose Temp. above 37.5°C Red throat No fast breathing -Respiratory rate not more than:	Mild condition Suspect Upper Respiratory Tract Infection (URTI)	1.Tepid sponge 2.Give paracetamol 120mg/5ml: a. 2 months - 2 years: 2.5mls 8hourly for 3 days b. 2 - 5 years: 5mls 8hourly for 3 days 3.If the child cannot take orally or is vomiting everything, give paracetamol suppository. 4.Give an appropriate antibiotic Suspension Amoxicillin 5ml (125mg) 8hourly for 5 days.
V. Fever and Measles rash	•Rash with any of the following: -Inability to drink or breastfeed -Lethargic -Stridor (when calm) -Severe malnutrition -Chest in-drawing -Clouding of the cornea -Deep or extensive mouth ulcer -Temp. 37.5°C or above	Severe condition Suspect complicated measles	1.Give Vitamin A treatment for Age 200,000iu a. 2-6 months 1/2 b. 6-12 months 1 c. 12 months - 5 years 2 if 200,000iu a. 2-6 months 1/4 b. 6-12 months 1/2 c. 12 months- 5 years 1 2.Give first dose of an appropriate antibiotic Susp. Amoxicillin 125mg/5ml a.Dose: 2 months–12 months: 2.5ml 8hourly for 5 days b. 12 months–5 years: 5ml 8hourly for 5 days 3.If clouding of the cornea or pus draining from the eye, apply chloramphenicol eye ointment 4.REFER immediately to the appropriate health facility.
VI. Fever with facial swelling, sore throat and bleeding from nose, eyes, ears gums	•Temperature 37.5°C or above •Headache •Joint and muscle aches •Sore throat •Weakness •Stomach pains •Loss of appetite •Vomiting or cough up blood Bloody	Severe Condition Suspect Haemorrhagic Fever	1.REFER immediately to the appropriate health facility 2.ENSURE UNIVERSAL PRECAUTIONS 3.No specific treatment for the disease is yet available Efforts to help those who are infected include giving either salt sugar solution (sss)/ORS or intravenous fluids.
VII. Fever with chills; feeling like flu, headache, muscle aches	•Temp. 37.5°C or above •Dry cough •Shortness of breath •Chest pain when coughing or breathing •Diarrhoea •Sore throat. Runny nose	Severe Condition Suspect Lower Respiratory Tract Infection (LRTI)	1.Give Susp. Amoxicillin 125mg/5ml a.2 months–12 months: 2.5ml 8hourly for 5 days b.12 months–5 years: 5ml 8hourly for 5 days 2. Review in 3 days, IF NO IMPROVEMENT REFER.

Health Education

Advise mother/caregiver on:

1.Environmental sanitation and prevention of disease transmission (vector control, drainage)
2.Proper housing condition (ventilation, lighting)
3.Use of long-lasting insecticidal net (LLIN)/ material for malaria control
4.Safe disposal of animal and human bodies infected with Lassa virus, SARS virus and Ebola
5.Wearing protective clothing when around person with the disease
6.Personal and food hygiene
7.Encourage fluid intake
8.Nutrition counseling
9.Safe water and disease prevention
10.Importance of immunization in disease prevention and control
11.Importance of taking home-based records on child's next visit to the clinic or hospital

2.4. CONVULSION

Convulsion is a jerky movement of the arms and legs and twitching of muscles. It is very common in infants and children and can be caused by high fever from any cause, head injury, tumour or sometimes infections of the brain or spinal cord. Muscle twitching due to tetanus or lock jaw should not be mistaken for convulsion. Tetanus often begins with tightness of the mouth jaw and muscle twitching.

HISTORY	EXAMINATION/ACTIONS
1.Has the child had a fever? Any accident? Head injury? 2.Did the child take any medicine, or any substance accidentally? 3.Has the child had any previous convulsion? With or without fever? 4.Has the child been given any medication? If yes, what? 5. Does any other family member suffer from convulsions?	1.General appearance: convulsing, conscious, drowsy, unconscious, smelling of cow's urine 2.Nose: flaring of alae nasi 3.Chest: respiratory rate, lower chest in-drawing 4.Head: swelling, sign of injury, fontanelle for bulging 5.Eyes: pupils for size and reaction to light 6.Neck: stiffness 7.Ears: pus, bulging drums or perforation of drums 8.Temperature and pulse: check and record 9.Weight: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Convulsion with fever	Temperature >38.7°C	Severe condition Suspect Febrile Convulsion Child convulsing now	1.Lie the child on his or her side to avoid aspiration. 2.Do not insert anything in the mouth 3.Keep the airway clear. 4.Give the first dose of i.m. Paraldehyde and REFER child urgently to the appropriate health facility a. Age or Weight Paraldehyde Dose: 1 ml = 1 gram i. 0-2 months (1.5-<4kg) 0.5 ml ii. 2 months - 4 months (4-<6kg) 1.0 ml iii. 4 months - 12 months (6-<10kg) 1.5 ml iv. 12 months - 3 years (10-<14kg) 2.0 ml v. 3 years - 5 years (14-<19kg) 2.5 ml vi. After 10 minutes, if convulsion continues, repeat IM. Paraldehyde. vii. If convulsion continues 10 minutes after the second paraldehyde, give a third dose.

II. History of one convulsion more than 24hours before coming to the clinic	•Not convulsing now •Temperature >37.5°C •Any of the following: -Bulging fontanelle -Neck stiffness -Signs of head injury	Severe condition Suspect Meningitis or head injury	1.Tepid sponge 2.Give first dose of paraldehyde: 3.Age or Weight Paraldehyde Dose: 1 ml = 1 gram i. 0-2 months (1.5-<4kg) 0.5 ml ii. 2 months - 4 months (4-<6kg) 1.0 ml iii. 4 months - 12 months (6-<10kg) 1.5 ml iv. 12 months - 3 years (10-<14kg) 2.0 ml v. 3 years - 5 years (14-<19kg) 2.5 ml 4.Explain findings to the mother. 5.REFER to the appropriate health facility
III. History of convulsion	•Not convulsing now •Temperature < 37.5°C •Conduct mRDT, and if positive, treat for uncomplicated malaria.	Moderate condition Suspect febrile illness	1.Treat malaria with ACTs (refer to section on treatment with ACTs). Dispersible Artemether/Lumefantrine 20/120mg: 2tabs at 0hour, 8 hours later, then 12hourly for 2days 2.Review in 3 days. 3.Follow up. Conduct RODT, and if positive, treat for uncomplicated malaria.

Health Education

Explain to Caregiver:

1.Causes and prevention of convulsion.

2.Management of a person during a convulsion episode.

3.Management of fever, especially in children <5 years old.

4.Use of long-lasting Insecticidal Nets (LLINs).

5.Importance of taking home-based records on the child's next visit to the clinic or hospital.

2.5 COUGH/DIFFICULT BREATHING

Cough is usually a symptom of irritation or blockage of the air passages. It is forceful pushing of air out of the lungs. Difficult breathing is when the child is making extra effort to breathe, is breathing too fast or having noisy breathing. Difficult breathing can be seen by looking for chest in-drawing below the ribs and flaring of the nose as the child breathes in lower chest in-drawing and fast breathing indicate pneumonia.

Fast breathing means a respiratory rate of 60 and above per minute for an infant less than 2months of age, respiratory rate of 50 and above per minute for an infant aged 2 -12 months; and for, a respiratory rate of 40 and above per minute for a child 12 months - 5years.

The most common causes of difficult breathing are pneumonia, asthma, mucus plug, enlarged adenoids and aspiration of a foreign body or heart disease. Aspirin or kerosene poisoning can also cause fast breathing.

HISTORY	EXAMINATION/ACTIONS
1.How old is the child? 2.How long has the child had the cough or difficult breathing? 3.Is there wheezing? Are there previous episodes? 4.Has the child been vomiting? 5.Has the child convulsed? 6. Is there a fever? For how long?	1.General Appearance: alert or abnormally sleepy or difficult to wake, severe malnutrition, respiratory distress. 2.Nose: flaring of alae nasi 3.Chest: a.Respiratory rate (count for 1 full minute in a calm child) b.Subcostal retraction lower chest in-drawing c.Stridor d.Wheeze 4.Eyes: redness, discharge 5.Throat: soreness or redness 6.Feet: swollen 7.Temperature: check and record 8. Document oxygen saturation using a pulse oximeter if available

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Cough and/or difficulty breathing with one of the following: a. Abnormally wakes b. Inability to drink c. Grunting when less than 2 months Or above 2-12months	•Fast/Difficult breathing -Severe chest in-drawing condition -Flaring of the nose •Stridor (in a calm child) •Lethargic Assessment of fast -60 calm breaths/min -50 calm breaths/min or more •Document oxygen saturation using a pulse oximeter if available	Severe condition	1. Give first dose of Susp. Amoxicillin 250mg 2. Oxygen Saturation is or less than 90% 3. REFER IMMEDIATELY to the appropriate health facility

II. Cough with fast breathing or difficult breathing	•Fast breathing •Chest In drawing	Moderate Condition	1. Susp. Amoxicillin 125mg/5ml a. 2-12 months- 2.5ml 8hourly for 5days b. 12months-2 years- 5ml 8hourly for 5days 2. Soothe the throat and relieve the cough with a safe remedy 3. Review in 3 days 4. If no improvement, REFER to the appropriate health facility.
III. Cough with fast breathing or difficult breathing	•Fast breathing •Chest In drawing	Moderate Condition	1.If less than 2hours suck nose, throat and airways with soft-tipped sunction tube immediately 2.If respiratory distress improves give, Susp. Amoxicillin 125mg/5ml a. 2-12 months- 2.5ml 8hourly for days b. 12months-2 years- 5ml 8hourly for 5days 3.Soothe the throat and relieve the cough with a safe remedy 4.Review in 3 days If no improvement, REFER to the appropriate health facility
IV. Cough with noisy Breathing	Wheezing sound	Severe condition Suspect Asthma	1. Give Sallbutamol 3 times daily (for wheezing) (not stated in the IMCI) Age: -2mg tablet -4mg tablet Syrup (2mg/5ml) -12 months-5years 1 -1h 5 2. Give Amoxicillin 250mg dispersible tablet a. 2 months up to 12months - 1 tab 2 times daily x 5days b. 12 months up to 5 years - 2 tabs 2 times daily x 5 days 3. Review the next day If no improvement, REFER to the appropriate health facility 5. For severe acute wheezing, give inhaled salbutamol for wheezing with a spacer a. Dose: Salbutamo 1 metered dose inhaler (l00microgramme / puff) give 2 puffs b. Repeat up to 3 times every 15mins NOTES: i.Oral salbutamol is not recommended for treatment of severe acute wheeze ii. No child under 5 years should be given inhaler without a spacer /nebulizer

V. Cough with difficult breathing with a whoop	•Peculiar cough with a whoop •Blood-shot eyes	Moderate to severe Suspect Whooping cough	1. Give first dose of oral Amoxicillin 250mg 2. If chest in-drawing or fast breathing, REFER immediately 3. If no chest in-drawing or fast breathing, and temp. is >37.5°C a. Give Paracetamol until fever subsides i. % 1/4 tab or 125mg qds x 5 days for 2months - 3 years ii. l/2 tabs or 250mg qds x 5 days for 3 years - 5 years 4. Review after 24 hours If no improvement, REFER immediately to the appropriate health facility
VI. Cough with measles type rash (red eyes and runny nose)			
VII. Cough with runny nose or red throat, fever (Temp. above 37.8°C)			1. See sub-section on FEVER 2. See sub-section on THROAT disorders
VIII. Cough or cold	•No signs of pneumonia •No fever •No difficulty breathing •No chest indrawing	Mild condition (No pneumonia) Suspect common cough cold	1. If coughing for more than 14days, refer immediately 2. If less than 14 days use safe remedy such as war tea and honey and breast feed if still breast feeding 3. Advise mother to return in one week

Home Care

1. Give extra fluids
2. Continue feeding/breast feeding
3. Clear the nose if it interferes with feeding
4. Do not expose child to cold or chills 5 Watch out for any of the danger-signs and if any, take the child to the health worker

Danger signs (2 months - 5years)

1.Fast breathing.
2.Difficult breathing or lower chest in drawing.
3.In-ability to suck/drink.
4.Child becomes more ill.

Health Education

Advise Mother and caregiver on:

1. Overcrowding and disease transmission control.
2. Prevention of air pollution (from firewood and kerosene smoke, and frying) and respiratory infection/ problems.
3. Role of adequate nutrition and fluid intake in prevention of infections and quick recovery
4. Use of Long Lasting Insecticidal Nets (LLINs).
5. Importance of taking home-based records on child's next visit to the clinic or hospital.

2.6. DIARRHOEA

Diarrhoea is the passage of three or more loose or watery stools in 24hours. Passing 5 to 8 soft or semi-liquid stools per day is normal in exclusively breast-feed infants. Children with diarrhoea may quickly become dehydrated and diarrhoea in children often leads to malnutrition. The main danger in diarrhoea diseases is dehydration, and it kills children more easily than adults. Replacement of fluids is the most important treatment. Persistent diarrhoea is one that lasts for two weeks or more. HIV/Aids is a major cause of persistent diarrhoea.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had diarrhoea? 2.How many times does the child usually pass stool each day? How many now? 3.Has the child passed any stools while in the clinic? (if so, look at it) 4.Are the stools watery? Is there blood or mucus? 5.Is the child vomiting? How often? 6.Has the child been active? 7.Are there other complaints such as cough, fever, draining ear, abdominal pain? 8.What has the child been fed? What is the method of feeding? 9.Has any medication been given? 10.Is anyone else in the household having diarrhoea? 11. Does the child drink normally, eagerly or poorly?	Steps for physical examination •Inspection •Percussion •Palpation •Auscultation 1.General appearance: alert, restless, irritable, thin, dehydrated, lethargic, unconscious 2.Head: anterior fontanelle depressed. 3.Eyes; normal, sunken, very sunken and dry. 4.Mouth and Tongue: moist, dry, very dry 5.Chest: respiratory rate Rapid breathing: 60 or more/minute age less than 2 months. 50 or more/minute age 2-12 months 40 or more/minute age 12 months – 5 years 6.Neck: nodes enlarged, throat inflamed 7.Ears: purulent discharge, tenderness behind ear 8.Abdomen: tenderness, distended 9.Skin: skin pinch goes back quickly, slowly or very slowly 10.Weight: weigh and chart 11.Temperature: check and record 12.Offer the child something to drink, and note if the child drinks normally, eagerly or poorly (or is not able to drink) 13. SEE APPENDIX====for chart on assessment of dehydration

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Recurrent bouts of frequent stool with or without weight loss	•Very low weight or growth faltering •Enlarged lymph nodes •Oral thrush •Parotid (salivary gland swelling) •Chronic ear discharges •Mother HIV positive	Severe condition Suspect HIV/ AIDS	1.Give Vitamin A <6months - 50,000IU 6-11months - 100,000IU 12months - 5 years - 200,000IU 2.REFER immediately to the appropriate health facility Prepare and give mother Lo-ORS/Zinc to give the child slowly on the way.

II. Frequent watery stool with weakness, with or without vomiting.	•Two or more of the following signs: -Lethargic or unconsciousness -Sunken eyes -Drinks poorly or unable to drink Skin pinch goes back very slowly.	Severe condition Suspect dehydration	1.If child has no other severe classification give low osmolarity ORS with dispersible zinc sulphate tablets for severe dehydration (see Plan C) 2.If child has another severe classification, REFER immediately 3.Advise mother to give frequent sips of low osmolarity ORS with dispersible zinc sulphate tablets Advice mother to continue breast-feeding.
III. Frequent stooling for 14 days or more	•Two or more of the following: -Restless or Irritable -Sunken eyes -Child is thirsty and drinks eagerly -Skin pinch goes back slowly -Mouth/tongue dry -Dehydration present -Child passes 3 or more watery or loose stools a day	Severe condition Suspect Some Dehydration	See Plan B. Treat SOME dehydration with Lo-ORS/ Zinc. Lo-ORS Age Dosage <2years 50–100ml(1/4to½cup) aftereachloosestool >2years 100–200ml(1/2to1cup)after each loose stool Zinc <6months 10mg once adayx10days >6months 20mg once adayx10days 2.If child has another severe classification, REFER immediately 3.Advise mother to give frequent sips of low osmolarity ORS with dispersible zinc sulphate tablets Lo-ORS Age Dosage <2years 50–100ml(1/4 to½cup) aftereachloosestool >2years 100–200ml(1/2 to1cup) after each loose stool Zinc <6months 10mgonceadayx10days >6months 20mg once adayx10days
IV. Frequent stooling of less than 14 days	•No dehydration •Well, alert. •Eyes normal. •Tears present. •Mouth/Tongue Moist. •Child drinks normally. •Skin pinch goes back quickly OR Child passes 3 or more watery or loose stools a day	Mild condition No dehydration	1.Give low osmolarity ORS, dispersible zinc sulphate tablets and fluid to treat diarrhoea at home. a.Follow Plan A 2.Review in 2 days 3.REFER if diarrhea persists

V. Frequent stooling of more than 14 days with or without vomiting,	Dehydration present	Severe condition Suspect Persistent Diarrhea	1. Give low Osmolarity ORS and dispersible zinc sulphate tablets and Vitamin A 2. Follow Plan A. 3. If child has another severe classification, 4. REFER immediately
VI. Frequent stooling of more than 14 days with or without vomiting	•No dehydration	Persistent Diarrhoea	1.Give low Osmolarity ORS and dispersible zinc sulphate tablets and Vitamin A. Advise mother on feeding child who has persistent diarrhea 2.Review in 5 days If no improvement, REFER to the appropriate health facility
VII. Diarrhea with fever, abdominal pain, cough/cold fast breathing, ear pain /sore throat	•Temperature above 37.5°C Conduct ROT, and if positive, treat for uncomplicated malaria •Fast breathing •Ear Discharge •Sore throat mRDT positive	Moderate Condition	1.Treat Diarrhoea as in Plan A 2.Treat for malaria 3.Refer to appropriate section on additional problem(s) 4. Review in 2 days
VIII. Diarrhea with vomiting	•Profuse watery diarrhoea (rice water stool) •Weakness •Dehydration	Severe condition suspect Cholera	1.ORS, 7 litres in 4 hours. 2.Give Ciprofloxacin15mg/kg/dose-2 times a day for 3 days 3.Supervise disposal of stool and vomitus 4.Notify appropriate Authority. Review Daily

<table>
<tr>
<td>IX. Frequent stool with visible blood.</td>
<td>Blood in stool with or without mucus</td>
<td>Severe condition suspect dysentry</td>
<td>1.REFER to the appropriate health facility if:
a.Child is less than12 months
b.Was dehydrated on the first visit
c.Has had measles within the last 3 months
1.Give Ciprofloxacin15mg/kg/dose-2 times a day for 3 days
<table>
<tr><td>Age</td><td>500mg</td><td>250mg Tab</td></tr>
<tr><td>2months-6months</td><td>½ tablet</td><td>½</td></tr>
<tr><td>6months-5years</td><td></td><td></td></tr>
</table>
2.Give Lo-ORS and zinc tablets.
3.Review in 3 days.
If no improvement, REFER immediately to the appropriate health facility.</td>
</tr>
<tr>
<td>X. Diarrhea with severe weight loss</td>
<td>•Weight below lower line on growth chart
•Weakness
•Dehydration
•Pallor
•Check MUAC <11.5cm (RED)
With good appetite or No appetite</td>
<td>Severe condition:

Suspect severe malnutrition</td>
<td>Give Vitamin A
a)2-6 months - 50,000IU
b)6-11 months - 100,000 IU
c)12 months - 5 years - 200,000IU
2.Give Lo-ORS and zinc
3.Give Ready-Use Therapeutic Food (RUTF) if the child has appetite
4.If no appetite refer to Stabilization Care if available
If no Outpatient Therapeutic Programme or Stabilization Care counsel mother on adequate nutritious foods (locally available food).</td>
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</table>

Danger Signs

There are signs which a mother should watch for and if present, to return to the health worker IMMEDIATELY.

1. Many watery stools.
2. Vomiting everything.
3. Child becomes more ill.
4. Breast feeding or drinking poorly.
5. Develops fever.
6. Blood in the stool (if not previously present).

Health Education

Advise mother/caregiver on:

1. Personal Hygiene.
2. Environmental sanitation.
3. Exclusive breastfeeding
4. Complementary feeding if above six months.
5. Use of Long-Lasting Insecticide Nets.
6. Remind mother to take home-based records with her anytime she takes baby to the clinic or hospital.
7. Importance of taking home-based records on child's next visit to the clinic or hospital.

2.7. VOMITING

Vomiting is the throwing up of the content of the stomach. Babies under six months often bring up little of their food immediately after being fed. This is normal. The common causes of vomiting are gastro-enteritis, meningitis, ear infection, liver disease and abdominal problems.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child been vomiting? How often? 2.Is it associated with eating? 3.Is it projectile? Does it contain blood or greenish fluid? 4.Is there diarrhea? 5.If no diarrhea, when was the last stool? (obstruction) 6.Is the child urinating normally? 7.Does the child have any other signs of illness? Fever, cough, pain in the head, abdominal pain? 8. Has the child been recently dewormed?	1.General appearance: ill, thin, jaundiced 2.Head: fontannelle depressed or bulging 3.Eyes: normal sunken, very sunken and dry 4.Neck: stiffness, enlarged and tender nodes 5.Abdomen a.Distension b.Tenderness c.board-like stiffness d.visible intestinal movement 6.Throat: redness, purulent discharge and inflamed tonsils 7.Ears: red drums, purulent discharge, tenderness behind the ear 8.Weight: weigh and chart 9.Temperature, Pulse, Respiration and Blood Pressure: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Vomiting with any of the following: a.Blood (brownish vomitus) b.Neck stiffness c.Weakness d. Forceful vomiting	•Neck stiffness •Abdominal rigidity or tenderness •Sign of severe dehydration •Lethargic	Severe condition. Suspect Meningitis	1.Give Benzyl penicillin double dose stat a.2 months to 11 months (300mg stat) b.12 months to 5 years (600mg stat) 2.REFER immediately to the appropriate health facility. 3. If referral is difficult, repeat 6 hourly till you get to hospital.
II.Vomiting, frequent watery stools	•Mild dehydration	Mild condition	1.Follow plan A for diarrhea treatment
III.Vomiting with or without diarrhoea but with fever	•Dehydration •Temperature 37.50C	Mild condition	1.Follow Plan A for diarrhea and treat fever appropriately
IV.Vomiting for more than 12 hours with no stool or fluids passed	• Baby under 3 months of age • Baby under 2.2kg • Severe dehydration • Severe abdominal pain • Abdominal distension	Severe Condition Suspect Intestinal Obstructio	1. REFER immediately to the appropriate health facility
V.Vomiting for less than 12 hours with normal stool	•Some dehydration •Reduced thirst •Reduced activity	Mild Condition Suspect Gastritis	1.Demonstrate to mother how to make and give Lo-ORS 2.Continue feeding 3.Teach mother the causes of vomiting 4.Review next day 5.If better, advise mother to continue feeding 6. If not better, REFER to the appropriate health facility.

VI. Occasional vomiting after feeding, no other complaint	•Child under 6 months Well baby	Mild condition	Advise mother to keep baby upright for 10 -15mins after feeding.

Health Education

Advise mother/caregiver on:

1. Causes of vomiting
2. Personal, food and environmental hygiene
3. Adequate fluid intake and prevention of dehydration (excessive loss of body fluids)
4. Frequent feeding with small amounts of food
5. Use of Long-Lasting Insecticidal Nets (LLINs)
6. Importance of taking home-based records on child's next visit to the clinic

2.8. PALLOR

Pallor is a major sign of anaemia. Anaemia is reduced number of red blood cells or reduced amount of haemoglobin in each red cell (below 10 grams). A child can develop anaemia as a result of not eating adequate nutritious foods, malaria, malnutrition, parasites (hook worm or whipworm), infections and sickle cell disease. Anaemia can lead to death.

HISTORY	EXAMINATION/ACTIONS
1.Is there bleeding from any source? a)Nose, gums b)Coughing or vomiting; c)In the stool or urine; d)Black stools, injury; 2.Does the child have a fever? For how long? 3.Has the child had painful swelling of the fingers, toes, the back of hands and feet? Or bone pains? 4. Does the child have a good appetite? What food does he eat? Does the child reject some foods? 5. Does the child crawl, play on the ground or go barefoot? 6. Does the child have abdominal pain? 7. Has the child been swimming or washing in the stream? 8. Does anybody in the family have similar problems? 9. Has any medication been given? 10. Is the mother using camphor?	1.General appearance: tall and thin, malnourished 2.Chest: respiratory rate, lower chest indrawing 3.Heart: rate, murmur 4.Head: bossing of the skull 5.Eyes: jaundice or conjunctival pallor 6.Abdomen: enlarged spleen, distension 7.Limbs: •Swollen and tender joints, feet and/or digits, or back of hands and feet •Nails for pallor 8.Haemoglobin: estimate and record 9.Temperature: check and record 10. Urine: colour, blood, protein

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Paleness of the body with or without tiredness	•Severe palmar pallor •Hb below 7gm/dl	Severe condition Suspect Severe Anaemia	1. REFER Immediately to the appropriate health facility.
II. Paleness of the body	•Some palmar pallor •Hb between 7gm/dl and 10gm/dl mRDT positive or negative	Moderate condition Suspect Anaemia	6.Tepid sponge 7.Give first dose of paraldehyde: 8.Age or Weight Paraldehyde Dose: 1 ml = 1 gram i. 0-2 months (1.5-<4kg) 0.5 ml ii. 2 months - 4 months (4-<6kg) 1.0 ml iii. 4 months - 12 months (6-<10kg) 1.5 ml iv. 12 months - 3 years (10-<14kg) 2.0 ml v. 3 years - 5 years (14-<19kg) 2.5 ml 9.Explain findings to the mother. 10.REFER to the appropriate health facility.
iii. Paleness of the body	No sign of palmar pallor or Hb normal	Mild condition No Anaemia	1. If child is less than 2 years old, assess child's feeding and counsel the mother on feeding.

IV. Paleness of the body	• Hb 5 -10gm with enlarged spleen • mRDT positive or negative	Severe condition Suspect Anaemia with Splenomegaly	1. Give AGT if mRDT positive, treat for uncomplicated malaria (see dosage in section 2.2) 2. Give Syr Folic Acid 5ml daily for 2weeks 3. Give Pyrantel Pamoate 1-2 tablets stat 4. Nutritional counselling 5. Review in two weeks
V. Anaemia with history of bone pains	•Hb <10gm Swollen hands and finger.	Moderate condition Suspect Sickle Cell Disease	1. Conduct mRDT 2. If positive, treat for uncomplicated malaria 3. Give Ibubrofen syrup first dose (5mg/kg body weight per dose) 4. REFER immediately to the appropriate health facility
VI. Anaemia with fever	•Hb <10gm Temp >37.0oC	Moderate Condition Suspect Malaria	1. Treat as 4 above REFER to appropriate section on fever
VII. Sickle Cell Crisis with Bone pains and Abdominal pains	•Jaundice Swollen hands and feet	Severe Condition Suspect Sickle Cell in Crisis	1. Encourage fluids intake 2. Paracetamol injection 300mg/2ml stat a. 2months - 2 year 0.5ml(75mg) b. 2years - 5years 1ml (150mg) c. REFER immediately

Health Education

Advise mother/caregiver on:

1. Personal, food and environment hygiene
2. Nutritional counseling.
3. Immunization
4. Use of Long Lasting Insecticide Nets (LLINs).
5. Regular growth monitoring.
6. Importance of taking home-based records on child's next visit to the clinic or hospital

2.9. JAUNDICE

Jaundice is the yellow discoloration of eyes (conjunctiva) and skin due to increase bile pigment in the body. Jaundice is as a result of the buildup of bilirubin - a reddish pigment resulting from haem metabolism in the body. It may or may not be as a result of liver problem. The buildup of this yellow pigment may be due to abnormalities in the formation, transportation, metabolism and excretion. At times the level may not be clinically recognizable until it is high. Jaundice occurs in liver problem, sickle cell disease, haemolysis, amoebiasis and tumour of the pancreas.

HISTORY	EXAMINATION/ACTIONS
1.Is the infant unwell? (sepsis & GIT obstruction can cause jaundice) 2.Is there yellow discolouration of eyes and skin 3.When did you notice the yellowish discolouration of the eyes? At what age? 4.Is there dehydration or poor weight gain? (both exacerbate jaundice) 5.Is there family history of haemolytic disease (ABO/G6PD, spherocytosis, sickle cell) 6.Is there dark urine or pale stools (suggest biliary obstruction) 7.What medication have you given (e.g. herbs)? 8. Birth trauma such as cephalhaematoma, significant bruising (breakdown of haem)	1.General appearance: conscious, drowsy, unconscious, 2.Chest: respiratory rate, lower chest indrawing 3.Head: swelling, sign of injury, fontannelle for bulging 4.Eyes: yellowish 5.Ears: pus, bulging drums or perforation of drums 6.Hands: palms and nail beds for pallor and yellowish discolouration 7.Temperature and pulse: check and record 8.Weight: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. NI.Yellow eye with or without swelling of hands and feet complaints(routine visit)	•Yellow eyes	Mild condition Suspect Sickle Cell Anaemia	1.Refer for diagnosis 2.Routinely follow up and give: a. Syrup Folic acid 5ml daily for life b.Tablet Proguanil 100mg daily
II.Yellow eyes with or without bone pains, abdominal pains	•Yellow eyes •Bone pain •Abdominal pain	Severe Condition Suspect Sickle Cell Disease in Crisis	1.Encourage oral fluid intake 2.Paracetamol injection 300mg/2ml a)2 months - 2 years 0.5ml (75mg) b)2 years – 5 yr 1ml (150mg) REFER immediately to the appropriate health facility.

Health Education

Advise mother/caregiver on:

1.Personal, food and environmental hygiene.
2.Nutritional counseling.
3.Regular and complete Immunization.
4.Regular growth monitoring.
5.Genetic counselling.
6.Use of Long Lasting Insecticide Nets (LLINs).
7.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.10. MALNUTRITION

Malnutrition is a state resulting from a deficiency or excess of one or more essential nutrients. Acute malnutrition is caused by a decrease in food consumption and/or illness resulting in bilateral pitting oedema or sudden weight loss. It is characterized by the presence of bilateral pitting oedema or wasting (low Mid Upper Arm Circumference or low Weight For Height). Acute malnutrition comprises of both severe acute malnutrition (SAM) and moderate acute malnutrition (MAM). All children under 5 years of age attending clinic should be examined and have their weight charted on the growth chart. Failure to gain weight should alert the health worker to take action.

HISTORY : While taking history fill the OTP (Out Patient Therapeutic Programme) card accordingly.(See Annex for OTP Card)	EXAMINATION/ACTIONS
1.What is the problem? For how long? 2.What is the child's weight at birth? 3.Is the child still breastfeeding? 4.How is the food intake in the last few days? What kind of food has the child been eating? 5.Does the child have good appetite? 6.Does the child have diarrhoea? 7.How many times does the child pass stool in a day? 8.Is the child vomiting? 9.Is the child coughing? 10.Does the child pass adequate urine? 11.Does the child have oedema? (not necessary) 12.Is the child immunized? 13.Has the child been dewormed?	1.General appearance: ill looking, weak, wasted, stunted growth, looking puffy 2.Nutritional conditions as Kwashiorkor and Marasmus 3.Head: colour, texture of hair, bossing of the forehead 4.Face: moon-shaped, oedema 5.Eyes: sunken, discharge, conjunctiva discolouration 6.,Mouth: sores and candida 7.Neck: enlarged lymph nodes, 8.Axilla: enlarged lymph nodes 9.Chest: chest indrawing 10.Skin: scabies, rashes, loss of skin turgor 11.Limbs: oedema of the legs 12.Temperature, pulse and respiration: check and record 13.Weight and height: check and record 14.Mid Upper Arm Circumference (MUAC): measure and record 15.Appetite test: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Swollen leg, hand and face with severe weight loss, not breastfeeding	Child<6months bilateral pitting oedema with any one of the following medical complication: - Intractable vomiting - Hypothermia - High fever >39°C - Chest-indrawing - Anaemia - Severe dehydration - Failed appetite test of eating 1/3 RUFT-Ready to Use Therapeutic Food- (Plumpy Nuts).	Severe condition Suspect Complicated Severe Acute Malnutrition (SAM)	1.Give Vit A and treat the child to prevent low blood sugar using 2.10% sugar water while waiting for transfer to stabilization centre a.Preparation of sugar water (10% dilution) Quantity of Water / Quantity of Sugar / Ratio Teaspoon to Sugar Cube 100ml / 10g / 2 level teaspoons (2 sugar cubes) 200ml / 20g / 4 level teaspoons (4 sugar cubes) 3.REFER to Stabilization Centre Advise mother to keep child warm on the way to hospital.

II. Swollen leg, hand and face with severe weight loss.	with Mid Upper Arm Circumference MUAC <11.5cm or 115mm (RED) orWFH < 3zscores, Bilateral pitting oedema+++, Failed appetite to eat < 1 h RUTF with any one of the following medical complications: - Intractable vomiting, - Hypothermia-G'i'C - High fever>39°C - Chest-indrawing - Anaemia - Severe dehydration - OR - No weight gain after 4visits (lmonth) to OTP	Severe condition: Suspect Severe Acute Malnutrition(SAM)	1.Give Vit A and treat the child to prevent low blood sugar using10% sugar water while waiting for transfer to stabilization care preparation of sugar water [10% dilution)
III. Child is thin, low Weight for Height (WFH), old man's look	MUAC < 11.5cm No medical complication No weakness Good appetite, Conduct mRDT, and if positive, treat for uncomplicated malaria.	Severe condition Severe Acute Malnutrition (SAM)	1.Treat malaria with ACTs (refer to section on treatment with ACTs). Dispersible Artemether/Lumefantrine 20/120mg: 2tabs at 0hour, 8 hours later, then 12hourly for 2days 2.Review in 3 days. 3.Follow up. 4.Give routine medicines to all children admitted to OTP except for malaria treatment;

Correction on the use of RUTF:

1.RUTF is designated for very thin or swollen children only and should not be shared.
2.Sick children may have reduced appetite, so offer small, regular meals of RUTF throughout the day, aiming for around eight meals if possible.
3.For breastfeeding children, continue breastfeeding and offer breast milk before each RUTF feed.
4.RUTF is the primary food for sick and malnourished children during their time in OTP. Always prioritize RUTF over other family foods.
5.Ensure children have access to plenty of clean water or breast milk while consuming RUTF, as they may need more fluids than usual.
6.Wash the child's hands and face with soap and running water before feeding, and keep food packets clean and well stored.
7.Caregivers should also wash their hands with soap and running water before feeding the child.
8.Keep sick children covered and warm to prevent them from getting cold quickly.
9.For children with diarrhea, continue feeding them and provide extra food and water to prevent dehydration.
10.Return to the health facility if the child's condition worsens or if not eating sufficiently.

Health Education

1.Educate mothers/caregivers on personal, food, and environmental hygiene practices.
2.Provide guidance on ensuring adequate nutrition for children.
3.Stress the importance of timely immunizations for children.
4.Emphasize the need for periodic deworming to prevent parasitic infections.
5.Encourage the use of Long Lasting Insecticidal Nets (LLINs) to prevent malaria.
6.Advocate for regular monitoring of child growth and development.
7.Highlight the significance of maintaining home-based records for tracking the child's health status and upcoming clinic or hospital visits.

2.11. HIV/AIDS

HIV/AIDS is caused by a virus that can be transmitted from infected mothers to their babies during pregnancy, delivery, or breastfeeding. It's crucial for HIV-positive mothers to receive guidance on the best feeding practices to ensure their babies' survival without HIV transmission. Antiretrovirals can be administered to reduce the risk of virus transmission. Babies born to HIV-positive mothers are termed HIV-exposed babies, and a definitive HIV diagnosis can only be made after proper testing and confirmation.

HISTORY	EXAMINATION/ACTIONS
1.How old is your baby? 2.What immunizations has the child taken? 3.Are you breastfeeding your baby? If yes, how many times do you breastfeed? If no, why not? 4.What are you feeding your baby with? 5.Have you introduced complementary/supplementary diet? If yes, when? 6.How many times do you feed your baby per day? 7.How do you feed your baby? 8.Has the child been sick recently? 9.Are you on ARV? 10.Do you give your child ARV?	1.General appearance: ill looking, weak, wasted, stunted growth, 2.Collect blood sample for initial DNA PCR at 4-6 weeks of age (or earliest opportunity thereafter) and 6 weeks after complete cessation of breastfeeding 3.Monitor growth as per child-welfare clinic Growth monitoring using growth chart 4.Weight and Height: check and record 5.Mid Upper Arm Circumference (MUAC): measure and record 6.Physical examination - Head: colour, texture of hair 7.Eyes: sunken, discharge, conjunctiva discolouration 8.Mouth: sores and candida 9.Neck: enlarged lymph nodes 10.Axilla: enlarged lymph nodes 11.Chest: chest indrawing 12.Skin: scabies, rashes, loss of skin turgor 13.Limbs: thin or oedema of the legs 14.Temperature, Pulse and Respiration: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Baby born to HIV- infected mother	HIV–infected woman	HIV exposed baby	1.Give baby Nevirapine daily for first six weeks only and stop 2.Continue baby on cotrimoxazole prophylaxis until baby's status is determined through PCR. 3.Provide information on Early Infant Diagnosis (EID) for HIV exposed baby 4.Collect blood sample [Dry Blood Spot (DBS)] at six weeks and send to higher level for DNA PCR. If positive REFER 5.Advise on routine immunization 6.Encourage exclusive breastfeeding for the first 6 months of life thereafter, introduce adequate complementary foods and continue breastfeeding until 12 months with mother being on ARV 7. Maintain or refill ARV for breastfeeding HIV positive mothers

II. Baby born to HIV- infected mother	•HIV-infected woman •detected for the first time in labour	HIV exposed Foetus	1. Irritate and continue ARV for the mother 2. For the baby treat as in table 1
III. Baby born to HIV- infected mother with TB co-infection	Baby born to mother with HIV-TB co-infection	Severe Condition HIV-TB co- infection	1. REFER to the appropriate health facility 2. Baby: Continue INH prophylaxis for the first six months of life 3. Mother: Continue DOT (Director Observation Therapy)
I. Baby born to HIV- infected mother with malnutrition OR II. HIV infected infants with Malnutrition	•Age<or>6months and above with Mid Upper Arm Circumference MUAC <11.5cm or 115mm(RED) or WFH<-3 z-scores •Bilateral pitting oedema +++ •Failed appetite not able to eat 1 h 92g RUTF with any one of the following - Medical complication - Intractable vomiting - Hypothermia <35.0C - High fever >39°C - Chest in-drawing - Anaemia - Severe dehydration OR - No weight gain after 4 visit (1 month) to OTP	Severe condition: Suspect Severe Acute Malnutrition (SAM)	1. REFER to Malnutrition section 2. REEFER to HIV section 3. Give nutritional Counselling

Infant Feeding in the Context of HIV:

Malnutrition is a common condition in HIV infected children and a major contributor to mortality in this population. HIV infection can result in nutritional deficiencies, growth failure and development delay. Malnutrition itself in decreased immunity and greater susceptibility to infections. Thus accelerating disease progression. Malnutrition makes HIV infection worse and HIV infection worsens malnutrition.

Health Education

Advise mother/caregiver on:

1.Factors known to increase the risk of mother –to-child transmission and prevention.
2.The continuous use of ART (Nevirapine to babies)
3.Adequate nutrition.
4.Immunization.
5.Periodical deworming.
6.Regular growth monitoring.
7.Use of Long-Lasting Insecticidal Nets (LLINs).
8.Personal, food and environmental hygiene.
9.The importance of attending ante-natal care (ANC) in order to provide appropriate care for the mother and foetus.
10.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.12. MEASLES – TYPE RASH

Measles is an acute viral infection characterized by the appearance of rashes in small groups consisting of macules (flat spots) and papules (raised spots). It typically begins with high fever, cough, nasal congestion, redness of the eyes, and inflammation of the mucous membranes of the mouth. Later stages may include mouth sores, followed by the appearance of rashes across the entire body after approximately the fourth day of illness onset.

In addition to the characteristic rash, children with measles may also experience other serious complications such as pneumonia, diarrhea, or malnutrition, either concurrently or shortly after the onset of the disease. Measles can be particularly severe in poorly nourished children and may lead to various complications, underscoring the importance of timely intervention and appropriate medical care.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child been ill? 2.Does the child have a fever? 3.Does the child have cough? Is it a barking cough? 4.Is there runny nose? 5.What food is the child taking? Is the child refusing to feed? 6.Is there diarrhoea? Vomiting? (If so, may be force-feeding) 7.Has any medicine or treatment been given? 8.Are there any other children in the family under 2 years who have not been immunized for measles? 9.Are there children in the neighbourhood with the same complaint?	1. General appearance: alert, abnormally sleepy, or difficult to wake, unconscious, ill-looking or thin 2. Skin: rash, peeling, infected spots, haemorrhagic rashes 3. Chest: a) respiratory rate, lower chest indrawing b) wheeze, stridor 4. Eyes: red, purulent discharge, or sensive to light (eyes closed), cloudy 5. Ears: bulging, discharging pus or fluid, perforation of eardrum, redness 6. Nose: runny, discharge 7. Mouth: a) sores, Koplik's spots, thrush b) throat-inflamed tonsils, purulent exudates 8. Hydration status: bulging or depressed frontanelle, sunken eyes, dry mouth and skin 9. Weight: weigh and chart 10. Temperature, and Pulse: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Measles rash with fever and cough	•Rash with any of the following -Inability to drink or suckle at breast -Lethargic -Stridor (when child is calm) -Severe malnutrition -Chest in-drawing -Clouding of the cornea -Deep or extensive mouth ulcer -Temp 37.5oc or above -Pus drawing from eyes	Severe condition Suspected severe Complicated measles	1. Give Vitamin A treatment Age 200,000IU 100,00IU 2months-6months 1/4 ½ 6-12 months 1/2 1 12 months - 5years 1 2 2. Give first dose of an appropriate antibiotic (Suspension Amoxicillin 125mg/5ml) dose a. 2months - 12months, 2.5ml 8hourly for 5 days b. 12months - 5years, 5ml 8hourly for 5 days 3. If clouding of the cornea or pus draining from the eye, apply chloramphenicol eye ointment 4. REFER immediately to the appropriate health facility.

I. Measles type rash with fever	• Cough • Measles type rash • Red eyes without ulcer • So fast breathing	Moderate condition Suspect Measles	1. Give Susp. Amoxicillin (125mg/5ml) for 5 days a. 2 months up to 12 months - 2.5ml 8hourly for 5 days b. 12 months up to 5 years – 5ml 8hourly for 5 days 2. Clean eye with cooled boiled cotton wool 3. If cornea is cloudy or pus from eyes, apply chloramphenicol ointment 8hourly for 5 days 4. Give Vitamin A <6months 6-llmonths 12 months - 5 years 50,000IU 100,000IU 200,000IU 5. Give Paracetamol 125mg/5ml a. 2months - 2years 2.5ml 8hourly for 3 days b. 2-5years 5ml 8hourly for 3 days 6. Apply Gentian Violet to mouth ulcers 7. Review in 2 days 8. If no improvement, REFER to the appropriate health facility
III. Measles rash, with fever	Measles with no complication	Moderate condition Suspect measles	1.Give Vitamin A treatment: aAge 200,000IU 100,00IU 50,000IU 2-6 months 1/4 1/2 1 6-12 months 1/2 1 2 12months-5years 1/2 1 4 Multivitamin Syrup

Health Education

Advise mother/ caregiver on:

1. Immunization for measles prevention and control
2. Personal hygiene
3. Adequate nutrition
4. Importance of taking home-based card on child's next visit to the clinic or hospital.

2.13. SKIN DISORDERS

Skin disease is the presence of itchy or non-itchy raised lesion on the body. It may be due to poor hygiene, allergy, malnutrition, poor housing condition as well as the environment. Some rashes can easily spread to other members of the family therefore, early referral and correct treatment is important for the welfare of the whole family. Some skin diseases affect the external skin (ringworm) or affect the skin in internally (pyoderma). Others affect many parts of the body systems and are generally more important (measles, leprosy, diabetes, HIV/AIDS, Jaundice). Others are scabies, eczema, tinea vesicular, vitiligo, athletes foot, urticaria! and contact dermatitis.

HISTORY	EXAMINATION/ACTIONS
1.How long has the rash been there? 2.Does it itch? Is the child scratching? 3.Are there any other symptoms? Fever, cough, headaches, difficult breathing, sore throat? 4.Has the child eaten any unusual foods? 5.Does anyone else in the family have a similar rash or skin problem? 6.How often do you bathe the child? 7.Where and with whom does the child sleep? 8.What medicine (including injections or home treatment) has the child been given lately? 9.Has the child been passing excessive urine? 10. Has the child been having excessive thirst?	1.General appearance: thin, malnourished, distressed/uncomfortable 2.Skin: a)location, distribution and size of lesions b)colour of rash or lesions c)lesions or rash: flat (macules), raised, pus or fluid-filled, weeping, crusted, ulcerated, scratch marks or d)combination of(a)to(c) 3.Ears: red or discharging 4.Eyes: redness, discolouration of eyelids (dark - due to repeated scratching) 5.Mouth: redness, discolouration of eyelids (dark-due to repeated scratching) 6.Chest: respiratory rate, wheeze 7.Temperature: check and record 8. Urine: sugar, colour.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Itchy skin with rash	Small raised red spots or postular rash between fingers, wrist, buttocks, and ankles	Moderate condition Suspect Scabies	1. Advise bathing daily with Tetmosol soap 2. Apply Benzyl Benzoate lotion for 3 nights to all parts except face and scalp 3. Examine all members of family and TREAT EVERYONE whether any rash is found or not 4. Advise on the following: a. Boil clothes and beddings b. Dry in the sun c. Iron if possible d. If pyoderma, treats boil in findings 2 5. REFER if open sores 6. Repeat treatment if no improvement after one week
ii. Boils	• Boils or infected rash	Moderate condition Suspect Boils	1. Give Susp. Co-trimoxazole a) 2-12months 2.5ml (120mg) 12hourly for 5 days b) 1-5years 5ml (240mg)12hourly for 5 days 2. Incise and drain any abscess which is fluctuant (ripe) 3. Daily cleaning and dry dressing at clinic 4. Review in 5 days 5. If no improvement, REFER immediately to the appropriate health facility.

III. White patch with itching	• Skin with well-defined edge • Reddish, with some scaling • Head is slightly itchy • Batches of scaling and hair l Loss	Mild condition Suspect Ringworm	1. Examine rest of family for similar problem 2. Apply Benzoic acid and salicylic acid (Whitfields ointment) on affected part for a minimum of 6 weeks 3. Give Susp. Griseofulvin and take a necessary 10mg/kg/day for 6 weeks 4. If no improvement after 6 weeks, REFER.
IV. Pustular rash	. Crusted lesions . Infected insect bites	Moderate condition Suspect Pyoderma	
v. Scalp scratching with marks	Scratches or marks on scalp	Mild condition Suspect head lice	1. Examine rest of the family for similar problems 2. Apply lorexane lotion to scalp close to roots of hair 3. Wash and comb hair with fine-tooth comb after 24hours 4. Repeat treatment in 1 week Note: If infestation is severe, a hair cut may be required for effective treatment
VI. Itchy skin with red papules which from small blisters	Crusty rash appears after little or no fever mainly on trunk	Moderate condition Suspect Chicken Pox	1. Syr Paracetamol For 3 days a) 2months - 3years 2.5ml 8hourly for 3days b) 3-5years – 5ml 8hourly for 3days 2. Calamine lotion 3. Gentian Violet if lesions are infected 4. Encourage fluids 5. If fever present, conduct mRDT 6. If positive, treat as appropriate in section on FEVER.

VII. Itchy skin with red lesions	• Lesions, weals or patches which come and go very quickly	Moderate condition Suspect Hives	1. Ask if child has had it before 2. Ask what medicines have been or are being given 3. Stop all medicines unless absolutely necessary 4. Ask if the child ate any unusual food in the past 24 hours 5. Give syrup promethazine 12.5mg stat 6. Review in 2days and if no improvement, REFER
VIII.Itchy skin with rash	•Fine rash all over the body	Mild condition Suspect Heat rash	

Health Education

Advise mother/caregiver on:

1.Personal hygiene (Discourage sharing of underwear and clothes).
2.Importance of washing and ironing of clothes (emphasis on not spreading clothes on grass and other inappropriate places).
3.Adequate nutrition role of water in skin infection (Protection and treatment of water sources e.g ponds with insecticide - Abate).
4.Avoid overcrowding and transmission of skin infections.
5.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.14. OEDEMA

Oedema is the abnormal collection of fluid in the tissues resulting in swelling of the affected part of the body. It could be generalized or localized, such as one foot or finger. Generalized oedema is most noticeable on the parts of the body which are lowest at that time, e.g., legs when standing.

Causes include congestive heart failure, severe anaemia, kidney disease, liver disease, and kwashiorkor. Oedema may also result from a reaction to toxins, drugs, and insect stings. Ascites, which is fluid in the abdominal cavity, may accompany oedema.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had the swelling? 2.Has the child ever had this swelling before? 3.Has the child been well otherwise? 4.Was the child bitten by an insect? 5.Has any medication been given (for example penicillin, ATS)? 6.Is there itching? 7.Is there cough? 8.Is there blood in stool or urine? 9.Any urinary frequency or pain in micturition? 10. Does the child eat well? What foods?	1. General appearance: pale lethargic, puffy, distressed 2. Chest: a) Respiratory rate, difficulty b) Crepitation's or decreased breath sounds 3. Heart: murmur, rate 4. Head: a) Hair-colour texture b) Face-oedema, particularly around eyes 5. Eyes: pallor, jaundice 6. Abdomn: swelling, tenderness, enlarged liver and spleen 7. Limbs: swelling, pitting of oedema, thinness 8. Urine: colour, cloudiness, protein, volume, (bilharzia ova in the laboratory) 9. Weigh and chart 10. Temperature: check and record 11. Haemoglobin: estimate and record 12. BP (if possible): check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Swelling of both feet and hands, and/or yellowness of eyes	Jaundice, wheezing, stridor or fast breathing	Severe condition Suspect sickle cell disease	1.Encourage oral fluid intake 2. REFER to the appropriate health facility.
ii.Swelling of limbs and/or face with itching, with or without allergic reaction to toxins drugs, stings or some food	•Swollen legs, urticaria rash Itching	Moderate condition Suspect allergy or Kidney problem	1. If no respiratory distress, promethazine 0.25ml IM stat 6.5mg 2. If respiratory distress, give Adreneline test dose 0.01ml then promethazine a) 0-5months 6-llmonths 1-2years 3-5years b) 0.05ml 0.lml 0.15ml 0.15.ml 3. Calamine lotion 4. Observe for 4-6 hours 5. If condition improves, discharge and review daily 6. If no improvement after 3 days, REFER to the appropriate health facility.
III. Swelling of feet, arms and/or face	•Swollen feet (pitting) and/or face •Protein in urine	Moderate condition Suspect Kidney disease	1. Give Cotrimoxazole (for 5days) a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 2.Encourage fluid intake 3.Review in 5 days. If no improvement, REFER appropriately

IV. Swelling of legs and/ or face	• Pallor, swollen legs and face breathlessness • Fast heart beat	Severe condition Suspect Kidney or heart problem	1. Cotrimoxazole a.2-12 months ½ tab (240mg) bd for 5 days b.1-5 years 1 tab (480) bd for 5 days OR 2. Amoxicillin DTs 250mg a.2 months up to 12 months-1tab bd for 5 days b.12 months -5 years-2 tabs 2 bd for 5 years 3. Wash with appropriate medicated soap twice daily especially crusted lesions 4. Use Gentian Violet on open sores 5. If fever is present, treat as section on FEVER 6. Review in 5 days. If no improvement, refer appropriately

Health Education

Advise mother / caregiver on:

1. Management of oedema (postural, application of crape bandage)
2. Nutritional counselling
3. Importance of taking home-based records on child's next visit to the clinic or hospital

2.15. BURNS AND SCALDS

Burns and scalds are injuries to the skin in which there is coagulative necrosis of the tissue. A burn is caused by dry heat, for example, by an iron heat, fire, chemicals, and electricity. A scald is caused by something wet, such as hot water, hot oil or steam. Burns and scalds can range from being a minor injury to a life-threatening emergency. The chart below tells approximately what the proportion of a child's total body surface is:

S/N	Rule of nines for burns (William, 2023)	Percentage of Body Surface
1.	Head and neck	9%
2.	Chest	9%
3.	Abdomen	9%
4.	Upper back	9 %
5.	Lower back	9%
6.	Right arm	9%
7.	Left arm	9%
8.	Genital area	1%
9.	Left leg	18%
10.	Right leg	18%
	Total	100%

HISTORY	EXAMINATION/ACTIONS
1.How did the child get burnt? And with what? 2.When and where did it happen? 3.Has any medication been given or applied since then? 4.Has the child been urinating since the burn? 5. Has the child been immunized against tetanus? How many times?	1. General appearance: restless, distressed, dehydrated, unconscious 2. Skin: a) Size and area of burns, blisters, leathery skin with no sensitivity to pinprick b) Signs of infections: pus, redness, warmouths 3. Temperature: check and record 4. Pulse: check rate and volume and record 5. Chest: respiratory rate, respiratory distress, wheezing, stridor 6. Check for possible signs of dehydration

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL J UDGEMENT	ACTION
1. Burns	Yellow eyes Burns of 9% or more with any of the following involved: - Face - Hands - Groin or joints - Lethargic - Without sensation or hoarse voice - Has not passed urine since burn and sign of shock (weak, thready pulse and cold extremities) - Low B.P (if Paediatric sphygmomanometer available)	Severe condition	1. Give Paracetamol 75mg I/M stat 2. Give TT 0.5ml, if not recently immunized 3. REFER immediately 4. IF REFERRAL is difficult and child is in shock, a) Give Ringers Lactate I.V. 1/3 the total amount of 8 hourly the, Darrow's half strength with 2.5% dextrose b) Syrup Ascorbic acid (Vitamin C) 100mg/5ml a. 2mths - 2yrs: 2.5ml 8hourly for 2weeks b. 2-5 years: 5ml 8hourly for 2weeks c) Apply Vaseline gauze. NOTE: Do not give oral Analgesic, if burns above 10%

II. Burns	Less than 9%with: - Blisters, - Not lethargic - No involvement of face, hands, groin or joints	Moderate condition	1.Vaseline gauze dressing 2. Syr Phenobarbitone 15mg/5ml a) 2-5 months: 2.5ml 12hourly for 3days b)6 months - 5 years: 5ml 12hourly for 3days 3. Give Syr Paracetamol: a) 2mths - 2yrs: 2.5ml 8hourly for 3days B) 2-5 years: 5ml 8hourly for 3days 4. IM Procaine Penicillin for 5 days a) 0-5 months: 75,000IU b) 6-ll months: 150,000IU c) 1-5 years: 300,000IU 5.T.T. 0.5ml stat; immunized in last 18 months 6. Rehydrate with salt sugar solution (if dehydrated) 7. Review daily until well 8. Syrup Ascorbic acid (Vitamin C) 100mg/5ml a. 2mths - 2yrs: 2.5ml 8hourly for 2weeks b. 2-5 years: 5ml 8hourly for 2weeks
III. Burns	Less than 9%; No blisters, not lethargic; No involvement of face, hands, groin or joints	Moderate condition	1.Vaseline gauze dressing or topical wounds dressing with Zinc oxide-based cream 2. Syr Phenobarbitone 15mg/5ml a) 2-5 months: 2.5ml 12hourly for 3days b)6 months - 5 years: 5ml 12hourly for 3days 3. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 4.T.T.0.5ml, if not immunized in the last 18 months; Review daily until well. 6.Ascorbic acid 500mg 8hourly,1 month – 2 years l00mg 3-5 years 7. Do not apply lotions or ointment 8. Do not break blisters or remove any loose skin 9. Remove the cause of the burn
IV. Burns with severe pains, (superficial or deep), burns from fire, flame, hotmetal, sun, high tension, lighting and friction	Redness and swelling of the area of blisters, sign of shock	Moderate to severe	Treat as 3above

V. Scald from boiling water, steam, oil, hot tea and hot liquids	Redness and swelling of the area of blisters, sign of shock	Moderate to severe	Treat as 3above

Health Education

1.Personal hygiene. .
2.Management of burns.
3.Nutritional counseling.
4.Compliance with treatment.
5.Prevention of home accidents.
6.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.16. WOUNDS, VARIOUS BITES, SPRAINS AND FRACTURES

2.16.1 WOUNDS

A wound is any break in the continuity of the skin or mucous membrane. There may also be injury to the underlying tissues such as tendons, large blood vessels, or bones. Infection or tetanus can easily set in when the skin is broken.

HISTORY	EXAMINATION/ACTIONS
1.How did it happen? 2.When did it happen? 3.Has it bled much? 4.Has anything been put on or in the wound? 5.Can the child move the area? Has the child been immunized with triple antigen or tetanus toxoid? When?	1.General appearance: in pain 2.Signs of shock: cold, clammy pale, fast breathing, rapid pulse rate 3.Wound: size, bleeding, dirt, or foreign body, pus, swelling, redness 4.Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Injury	•Swelling •Tenderness over area of injury •Unable to move the limb •In pains	Severe condition: Suspect blunt injury	1. Immobilize limb (splint) 2. Give Paracetamol (if conscious): Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. REFER immediately to the appropriate health facility.
II.Wounds on the Chest and Back from Sharp Knife and Gun shot or any other object, pains in Chest	• Laceration • Bleeding • No torn tendons • Unable to move the limb • Wound more than 4cm • No fracture • Check HB	Moderate condition	1. Clean with available antiseptic lotion 2. Stop bleeding by pressure 3. Give T.T. 0.5ml wound is less than 24hours 4. IM Procaine Penicillin for 5 days a) 0-5months b) 6-llmonth 1-5years 0.25ml 0.5ml 1ml 75,000IU 150,000IU 300,000IU 5. Give Syrup Paracetamol(125mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 6. Review the next day then at 1 week to remove sutures 7. Ask to return in 6 weeks for 2nd dose of T.T. 8. If no improvement, REFER to the appropriate health facility
III.Wounds on the Chest and Back from Sharp Knife and Gun shot or any other object, pains in Chest	• Dyspnea, cyanosis of the mouth, naideds, and skin • Cough bright red frothy blood if lungs is injured • Blood stained fluid from the wound	Severe condition Suspect Internal Hemorrhage	1. Clean with available antiseptic lotion 2. Stop bleeding by pressure 3. Gently cover the wound with a sterile pad 4. Give Td 0.5ml 5. Manage for shock 6. REFER immediately to the appropriate health facility

IV. Abdominal Wounds External or Internal Bleeding. Abdominal Pains, Vomiting	• Bleeding from the Wounds, • Part of intestine may be visible or protruding from the wound, • Signs of shock	Severe Condition. Suspect Rapture of Internal Organs.	1. Clean with available antiseptic lotion 2. Stop bleeding by pressure 3. Gently cover the wound with a sterile pad 4. Give T.T. 0.5ml 5. Manage for shock 6. REFER immediately to the appropriate health facility
V. Bleeding from circumcision	• Bleeding with or without palor or shock	Moderate condition	1. Put pressure on side for 10 minutes 2. Show mother how to maintain pressure 1-2hours 3. If bleeding does not stop, manage the shock 4. Then REFER to the appropriate health facility
VI. Wounds with sign of shock	• More than 4cm • Child is lethargic and not alert Torn tendon (unable to move limb) or fracture	Severe condition Suspect bleeding	1. Apply pressure 2. Give TT 0.5mls immediately 3. Manage shock and 4. REFER immediately to the appropriate health facility

Health Education

1. Protection from wounds
2. Personal hygiene
3. Importance of compliance with treatment
4. Nutritional Counselling
5. Environmental Sanitation
6. Importance of taking home-based records on child's next visit to the clinic or hospital

2.16.2 VARIOUS BITES

Bites may be from human, animal (dogs and snakes) and insects (mosquitoes) or stings (scorpions, bees etc). Children are prone to bites both in the home and outside the home particularly when they go out of school.

HISTORY	EXAMINATION/ACTIONS
1. How did it happen? 2. When did it happen? 3. Has it bled much? 4. Has anything been put on or in the bite? 5. Can the child move the area? 6. Has the child been immunized with triple antigen or tetanus toxoid? When?	1. General appearance: signs of shock a) pale, cold or clammy b) Increased sweating c) Increased respiration and pulse rate d) patient agitated or restless 2. Bite: a) Size and site b) Bleeding c) swelling, redness or warmth around bite 3. Temperature, Pulse and Respiration: check and record 4. Haemoglobin: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Dog bite	• Evidence of bite • Anxious • Tenderness at the site of bite with or without bleeding	Severe condition: Suspect rabies	1. Wash wound with copious clean soapy water changes in the dog's condition 3. Give anti rabies injection (if dog found or not) 4. Stop anti-rabies injection after 10 days if dog is alive 5. AMXDOSAGE 6. Give tetanus toxoid O.5mll 7. REFER immediately to the appropriate health facility
II. Snake bite.	• Wound at the site of bite • Tenderness at the site of bite with or without bleeding	Severe condition Suspect Venom Poisoning	1. Keep client calm and clean the area thoroughly with copious soapy clean water 2. Give anti-snake venom (read and follow marker's instruction) 3. Give promethazine 6.5 mg stat 4. Refer immediately to the appropriate health facility

III. Scorpion sting	•Evidence of sting •(Painful, Swelling, • itching and may be red)	Moderate condition	Administer Plain Lignocaine Sec stat at the site. Administer Adrenaline Subcutaneous stat: a) 0.05ml for 1-5 months b) 0.1ml for 6-11 months c) 0.15ml for 1-5 years if suspecting anaphylactic reactions. Provide Syr Paracetamol 125mg/5ml: a) 2 months-2 years - 2.5ml 8hourly for 3days b) 2-5 years - 5ml 8hourly for 3days Administer syrup Ascorbic acid (Vitamin C) 100mg/5ml: a) 1 month - 2 years- 2.5ml 8hourly for 3days b) 100mg 3 - 5 years 5ml 8 hourly for 3days. Review within 24 hours. If no improvement, REFER immediately to the appropriate health facility
IV. Human bite	• Wound at the site of the bite • Anxious • With or without bleeding	Moderate Condition	1. Clean wound well 2. AMXDOSAGE 3. Dress wound daily 4. Review after 3 days 5. If no improvement, REFER to the appropriate health facility
V. Bleeding from circumcision	• Bleeding with or without palor or shock	Moderate Condition	

Health Education

Advise Mother/caregiver on:

1. Protection from bites/stings.
2. Personal hygiene.
3. Importance of compliance with treatment.
4. Adequate Nutrition.
5. Environmental Sanitation.
6. Importance of taking home-based records on child's next visit to the clinic or hospital.

2.16.3 SPRAINS, STRAINS AND FRACTURES

A sprain is an injury to a joint causing pain and stiffness depending on the degree of injury to the ligaments. The signs are rapid swellings, warmth and stiffness, and sometimes discoloration and limitation of function. A spain is an injury to a muscle or tendon caused by overstretching or tearing. The signs and symptoms include pain, swelling, muscle spasms and unlimited ability to move the affected muscle. A fracture is a sudden break in the continuity of a bone causing deformity, pain, tenderness and loss of power or movement.

HISTORY	EXAMINATION/ACTIONS
1.How did the injury occur? When? 2.Was there any bleeding? 3.Where is the pain? How bad is it? 4.Can the child move/use limb? Is there limitation to movement? 5.Which movement causes greatest pain?	1.General appearance: a) Signs of shock-increased sweating, pallor, restlessness b) Degrees of pain 2. Limb or Joint a) Tenderness or swelling b) Warmouths or redness c)Crepitus (crackling) d) Open wound 3. Temperature, Pulse and Respiration: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Severe pains around the joints following an injury	• Tenderness of joint and swelling • Loss of function of the affected part • Anxious • In obvious pain With or without bleeding	Moderate Condition Suspect Sprain	1. Apply Cold compress to reduce swelling and pain 2. Apply Ung. Methyl salicylate to the affected area P.R.N 3. Apply Crepe bandage to the affected area 4. Give Paracetamol a) 1 month - 2years 1/4 tab(125mg) qds x 3/7 b) 2-5years 1/2 tab(250mg) qds x 3/7 5. REVIEW IN 2 DAYS 6. If no improvement, REFER to the appropriate health facility
II Sudden sharp pains in muscle	• Stiffness, cramps, swellings as result of violent tearing of muscles after fall or accident	Moderate Condition Suspect Strain	1. Apply cold compress 2. Apply Ung. Methyl salicylate to the affected area P.R.N 3. Rest the affected limb 4. Give Paracetamol stat a) Imonth-2years 1,4 tab(125mg) qds x 3/7 5. 2-5years 1/2 tab(250mg) qds x 3/7 6. REVIEW IN 3 DAYS 7. If no improvement, REFER to the appropriate health facility
I. Severe pains following injury, inability to move the affected part	Crackling or crepitus with loss of function No open wound	Severe condition Suspect Simple fracture	1. Give Paracetamol stat a) Imonth-2years 1,4 tab(125mg) qds x 3/7 b) 2-5 years 1lz tab(250mg) qds x 3/7 2. Apply splint to affected limb 3. Observe for signs of shock 4. REFER immediately to the appropriate health facility

IV. Open wound with inability to move the affected part	• Exposed bone • Bleeding • Pains	Severe condition Suspect compound fracture	1. Clean the wound with normal saline 2. Observe for sign of shock 3. Give Td 0.5ml IM immediately 4. AMX DOSAGE 5. Give paracetamol stat a. 1 month -2 years ½ tab (125mg) 6 hourly for 3 days b. 2-5 years ½ tab (250mg) 6 hourly for 3 days 6. Immobilize, cover wound. REFER immediately to the appropriate health facility

Health Education

Advise Mother/caregiver on:

1. Protection from bites/stings.
2. Prevention of home accidents.
3. Personal hygiene.
4. Importance of compliance with treatment.
5. Environmental Sanitation.
6. Care of wounds and sores.
7. Nutrition and effect on healing.
8. Importance of taking home-based records on child's next visit to the clinic or hospital.

2.17. DISORDERS AFFECTING THE HEAD

The head contain the brain (with the organs for the sense of hearing, seeing, smelling and tasting). The bones of the head are very strong but sometimes a serious head injury may be caused by a fall or a blow to the head. Some signs of head injury is bleeding from the nose, mouth or ears, severe headache or unconsciousness. There also may be vomiting or the client may be confused. Any delay in referring a client with a head injury for proper medical attention may result in permanent injury or death.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. Has he fallen or hit his head? When? 3. If he had an injury, has he been drowsy? 4. Has the vomited? How many times? 5. Does he have a headache? 6. Has he convulsed? 7. Is he playing and behaving normally for his age as he usually does?	1. General appearance and behaviour: dull, abnormally, sleepy, or difficult to wake restless, delirious or nconscious 2. Head: a) Circumference, shape b) External injuries, sores, or rashes c) Swellings or depressions 3. Neck: stiffness 4. Eyes: haemorrahage, pupils size (equal or unequal), reaction to light, vision, jaundice 5. Gait: steady or unsteady 6. Temperature, Pulse and Respiration: check and record 7. Haemoglobin: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. History of trauma to head, abnormally sleepy or difficult to wake, vomiting	•Pupil abnormal in size or not Reacting to light. Fracture, vomiting •Bleeding from ear or nose •Black Eye,	Severe condition Suspect Fracture of base of the skull	1. Maintain a clear airway. 2. Lie patient on the side and REFER immediately to the appropriate health facility, in that position
II. Trauma to Head from blow, fall or vehicular accident. Nausea and Vomiting. Loss of Memory	• Partial loss of consciousness • Shallow breathing, Face may be pale, cold clammy skin, • Rapid and weak pulse,	Severe Condition Suspect Concussion	1. Maintain a clear airway. 2. Lie patient on the side REFER immediately to the appropriate health facility, in that position
III. Trauma to Head, pressure from bone in a depressed fracture, from fall or vehicular	• Noisy breathing, slow and • bounding pulse, Different size of pupils, • Weakness or paralysis of one side of body, • Poor response from patient	Severe Condition Suspect Compression	1. Maintain a clear airway. 2. Lie patient on the side 3. REFER immediately to the appropriate health facility, in that position
IV. Headache with any of the following: a) Vomiting b) Fever c) Neck stiffness	• Temperature 37.5°C or above • With or without positive mRDT	Severe condition Suspect Cerebral Malaria/ Meningitis	1. If mRDT is positive, treat for uncompleted malaria with ACT (refer to section on malaria treatment 2. Give mRDT is negative, give Benzyl penicillin stat 0-3 months: 1/2 ml: 50,000ml 4 months - 2 years: 1ml: 100,000iu 3-5 years: 2ml: 200,000iu 3. REFER immediately to the appropriate health facility

V. Bulging forehead	Bossing of forehead (swelling)	Severe Condition Suspect Sickle Cell Disease	1. REFER to the appropriate health facility for sickling test 2. Folic acid 1 daily x 2weeks 3. Conduct mRDT, if positive treat for uncomplicated malaria with ACT (refer to section on malaria treatment) 4. If Hb is <lOmg/dl, REFER 5. Review in 2weeks
VI. Injury to head with a small external swelling	• Wound or swelling • Normal behaviour • No vomiting • No drowsiness	Moderate condition	1. Paracetamol stat a. 2months - 2years 1/4 tab (125mg)/5mls 6hourlytds b. 2years - 5years 1/2 tab (250mg)/10mls 6 hourly tds 2. If open wound, a) Clean with saline water and dress wound b) Give T.T 0.5ml/M stat immediately 3. AMXDOSAGE 4. If no open wound, apply cold compress on swelling 5. Review after 24hours 6. Advise mother to return with the child if condition becomes worse i.e drowsy, start vomiting showing any strange behaviour and\ REFER immediately
VII. Head larger or smaller than normal age	Head appears larger or smaller than normal age	Severe Condition	1. if above or below normal limits, REFER 2. Normal Head Circumference: a) 1 month (33 - 37cm average) b) 6months (41- 46cm) c) 12months [45 - 50cm)

Health Education

Advise Mother/caregiver on:

1. Proper care of child.
2. Prevention of head injuries.
3. Prevention of home and road accidents.
4. Importance of protection (use of car seat belt helmet etc.).l
5. Importance of taking home-based records on child's next visit to the clinic or hospital

2.18 NECK DISORDERS

Neck problems include the presences of enlarged nodes and lumps around the neck. There could also be pain or stiffness of the neck. These are usually caused by infection of the ears, the mouth or on the scalp, tuberculosis, leukemia (cancer of the blood), meningitis, other childhood malignancies. There may also be an injury to the neck following a fall.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long has the child had the problem? 3.Is there weight loss or loss of appetite? 4.Is there fever or sore throat? 5.Is there excessive sweating at night? 6.Is there headache? Earache? 7.Is there pain or vomiting? 8.Is the child restless or hyperactive? 9.Has the child convulsed?	1.General appearance: pale, thin, or wasted 2.Head: sores or abscess 3.Mouth: a). condition of teeth b).sores c).tonsils 4.Ears: discharge, red or bulging or perforated ear drum 5.Neck: stiffness, nodes, lumps, glands 6.Chest: stridor 7.Temperature, Pulse and Respiration: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Stiffness of neck with a fall and no fever	•Stiff neck with pain •Normal temperature •Abnormal head posture on lying down	Severe condition	1. Restrict movement of head and neck 2. If in extreme pain, give paracetamol a) 2months-2years: 2.5ml stat 2-5years 5ml stat REFER immediately to the appropriate health facility
II. Neck pain with Fever, with vomiting, no neck injury	Stiff neck, Temp.37.50c or above, vomiting	Severe condition Suspect Meningitis	1.If Kernig's sign positive, a. Give Benzyl penicillin injection 0 – 3 month 4 month – 2 years 3 – 5 years 1/2ml 1ml 2ml 50,00 IU 100,000IU 200,000IU 2. REFER immediately to the appropriate health facility.
III. Neck pain without Fever	Lump and Nodes present No tenderness Deformities Swelling and Masses	Moderate condition	1. Give tab paracetamol a) >2months - 2years 1,4 tab or 12Smg qds x 3 days b) >2years - Syears 1/2 tab or 250mg qds x 3 days 2. REFER immediately to the appropriate health facility
IV. Neck pain with Fever	• Tender lumps or nodes • Temp 37.SDC or above	Moderate condition	1. Conduct mRDT for malaria, if positive treat for uncomplicated malaria with ACT (refer to section on malaria treatment) 2. Review in 2days 3. If no improvement, REFER

V. Swelling of both sides of neck (cheeks)	•Enlarged salivary gland behind the jaw under the chin. •Testes may or may not be involved (swollen and/or painful)	Mild condition Suspect mumps	1. Give Syr 120mg/5ml Paracetamol a) 2month-2years : 2.5ml 8hourly for 3days b) 2-5years:5ml 8hourly for 3days 2. Encourage fluids, Saline mouth wash, Explain finding to mother. If male, advise rest and continuous observation of tests 3. Review in 3 days 4. If no improvement REFER to the appropriate

Health Education

Advise Mother/caregiver on:

1.Prevention of neck injuries.
2.Complications of mumps in males.
3.Oral hygiene.
4.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.19. EYE DISORDERS

Eye disorders are disease conditions affecting the child's eye and are commonly caused by bacteria or viruses, injury or foreign body or lack of enough vitamin A in foods. If there conditions are not prevented or treated early, they may cause blindness. It is therefore important to prevent or eye infections.

HISTORY	EXAMINATION/ACTIONS
1.What is the matter with the child's eye? 2.Do you think the child sees? 3.How long has the child had the problem? 4.Has the child had it before? 5.Did the problem come suddenly or gradually? 6.Is there a lot of pain in the eye? 7.Does the child rub the eye frequently? 8.Did anything enter the eye? 9.Has any medication been applied to the eye? If so, what? 10.Does anyone else in the family have this problem? 11.Is the child receiving treatment for any other disease? 12. Does the child take palm oil, carrots, fresh mangoes, pawpaw, and tomatoes?	1. General appearance: in pains, ill 2. Skin: rashes, scratch marks 3. Eyes: a) Redness, b) Tearing c) Swelling around the eyes d) Pus or other discharge e) Pupils: Reaction to light, white pupils f) Protruding eye balls g) Squint h) White foamy substance at corner of eyes/cloudy or white cornea i) Vision: check if child can follow a bright object, or reach out to receive an object from you without assistance 4. Nose: discharge 5. Temperature, Pulse and Respiration: Measure and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Injury to eyeball with sharp object	Injured white part of eye (s)	Severe condition Suspect Trauma to eye	1.Apply Chloramphenicol eye drops 2.Apply eye pad. 3. REFER immediately to the appropriate health facility.
II. Injury to eyeball as a result of burns, hot water or chemicals	Painful watery eyes (chemical burns to the eye)	Severe condition Suspect conjunctivitis/ keratitis	1. Irrigate the eyes immediately with clean water 2. Apply chloramphenicol eye ointment 3. Refer immediately to the appropriate health facility

Health Education

Advise Mother/caregiver on:

1.Personal, food, and environmental hygiene are essential to prevent infections and maintain overall health.
2.Eye injuries can be prevented by taking precautions such as wearing protective eyewear during activities that pose a risk.
3.Adequate nutrition, especially foods rich in Vitamin A, is important for maintaining eye health.
4.Regular eye check-ups are recommended to detect any issues early and prevent further complications.
5.It is crucial to keep home-based records of the child's health and upcoming clinic or hospital visits.

Ear Problem

Middle ear infections are common causes of ear problems, presenting with symptoms like ear pain, redness, or dullness of the eardrum, sometimes leading to perforation and pus drainage.

Examination of the bone behind the ear is important, as tenderness may indicate mastoiditis, a serious complication.

Deafness can be a major complication, which can be tested by observing the child's response to sounds

Nose Disorders

The nose serves for breathing and the sense of smell. Diseases or injuries in the throat or sinuses can affect the nose, causing watery discharge, mucous, blood, or pus.

Nose injuries can also affect the throat, and the presence of discharge from only one side of the nose may indicate a foreign object.

Throat Disorders

A red throat may indicate infection, often accompanied by nasal congestion and middle ear infections in young children.

Enlarged tonsils during childhood are normal, but treatment is necessary only if they cause difficulty in breathing or inflammation.

HISTORY	EXAMINATION/ACTIONS
1. What is wrong with the child? 2. Does the child have hearing problems? 3. Is there any discharge form the ear? 4. Has the child had any ear surgery? 5. Is there any family history of deafness? 6. Does the child have runny nose? 7. Is there any bleeding from the nose? If yes, for how long? 8. Has the child had it before? 9. Is the child able to swallow food or fluids? 10. Has the child received any injection or medication recently? If yes, what is the name? Ototoxic drug (antibiotics (e.g gentamicin), diuretics, cytotoxic). 11. Has the child been exposed to loud noises recently? Exposure to noise (e.g pneumatic drill or shooting) 12. What treatment have you given to the child for this condition?	1. General appearance: well, ill-looking, pale 2. Nose: watery discharge, mucus, blood, pus, any visible swelling deformity, or foreign body 3. Throat: pus or redness Ear 1. Head: tenderness over bone behind the ear (mastoid) 2. Ear: a) Condition of canal, b) Pus or blood, perforate drum c) Foreign body d) Red or bulging drum 3. Test for hearing: clap 4. Neck: stiffness Throat 1. General appearance: distressed, pale 2. Head: tenderness over bone behind the ear (mastoid) 3. Throat: pus or redness, enlargement of tonsils 4. Throat: General appearance: distressed, pale, 5. Neck: nodes, swellings 6. Ear: discharge, redness or bulging drum 7. Temperature, Pulse and respiration: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Ear pain with history of foreign body with bleeding or neck stiffness	Progressive inflammatory swelling over the Mastoid region Tender, swelling	Severe condition Suspect Mastoiditis	1. Give Susp. Amoxicillin 125mg/5ml 2months - 12months 2.5ml 8hourly for 5days b) 12months - 5years : 5ml 8hourly for 5days
II. Ear pain with Discharge	•Pus is seen draining from the ear and discharge is reported for less than 14 days •Tenderness behind the ear/on touching the ear	Severe Condition Suspect Acute Ear 2. Infection	1. Give Susp. Amoxicillin 125mg/5ml A) 2months - 12months 2.5ml 8hourly for 5days b) 12months - 5years : 5ml 8hourly for 5days 2. Give first dose of Paracetamol 120mg/5ml a.2months – 3years: 2.5ml 8hourly for 3days b) 3-5years: 5ml 8hourly for 3days REFER urgently to appropriate facility, if no improvement
III. Ear pain with discharge for more than 14 days with or without fever	•Pus is seen draining from the ear and discharge is reported for 14 days or more. Ear pain;Temp > 37.5^{0}C	Severe condition Suspect Chronic ear infection	1. Susp.Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 2.Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. Conduct mRDT and if positive, treat for uncomp licated malaria with ACT (refer to section on Malaria) 4. Teach mother to wick ear at least 3 times daily until the ear is dry and discourage plugging the ear or using ear drops 5. Review in 3 days 6. If no improvement, REFER to the appropriate health facility

IV. Swollen Wound in ear	•Boil seen •Tenderness Temp > 37.50C	Moderate condition Suspect Boil	1. Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 2. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. Review in 3 days 4. If no improvement REFER to the appropriate health facility
V. Difficulty in Swallowing	•Grey and dirty membrane on tonsil Temp. > 37.5^{0}C	Severe condition Suspect diphtheria	1. GIVE IM Paracetamol (300mg/2ml) a.2 months -2 years 0.5ml (75mg) b.2 years – 5 years 1ml (150mg) 2. REFER immediately to the appropriate health facility
XIII. Boil in the nose with or without fever	• Boil can be seen • Temp less than or greater than 37.SOC	Mild Condition Suspect Boil	1. Susp.Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 2. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. Give Syrup Ascorbic acid (Vitamin C) 100mg/5ml a. 2mths - 2yrs: 2.5ml 8hourly for 2weeks b. 2-5 years: 5ml 8hourly for 2weeks 4. Warm Compress 5. Review in 3 days. d) If no improvement, REFER
XIV. Ear pain with discharge formore than 14 days with or without fever	• Pus is seen draining from the ear • Dull ear drum • Warm to touch (Temp>37.S°C)	Mild condition	REFER immediately to the appropriate health facility

Health Education

Advise Mother/caregiver on:

1. Management of throat problems (gargle and discouragement of uvulectomy)
2. Prevention of exposure to air pollutants (fumes, dust, and pollen)
3. Personal hygiene
4. Discourage home removal of foreign body
5. Adequate nutrition
6. General childcare
7. Importance of taking home-based records on child's next visit to the clinic or hospital.

2.20. EAR, NOSE AND THROAT DISORDER

Ear Problem

Ear problems are diseases affecting the ear and are commonly caused by middle ear infection; usually with pain in the ear, dullness or redness of the ear drum sometimes with perforation and drainage of pus or fluid. It is important to examine the bone behind the ear, because if this is tender to touch, it may mean that the child has mastoiditis. This is a danger sign. The other major complication is deafness which can be tested by clapping your hands behind the child to see if he will turn and look.

Nose Disorders

The nose is the organ for breathing and for the sense of smell. Disease or injury in the throat or sinuses can affect the nose. This could bring about watery discharge, mucous blood or pus from the nose. Also, injury to the nose can affect the throat. The health worker should think of a foreign object if there is discharge from only one side of the nose.

Throat Disorders

A red throat may be caused by infection. It may be combined with nasal congestion and middle ear infection especially in young children. The tonsils normally enlarge during childhood, but unless they show sign of difficult breathing or inflammation, no treatment is needed.

HISTORY	EXAMINATION/ACTIONS
1. What is wrong with the child? 2. Does the child have hearing problems? 3. Is there any discharge from the ear? 4. Has the child had any ear surgery?y? 5. Is there any bleeding from the nose? If yes, for how long? 6. Has the child had it before? 7. Is the child able to swallow food or fluids? 8. Has the child received any injection or 9 medication recently? If yes, what is the name? Ototoxic drugs (antibiotics (e.g gentamicin), diuretics, cytotoxics). 10. Has the child been exposed to loud noises recently? Exposure to noise (e.g. pneumatic drill or shooting) 11What treatment have you given to the child for this condition?	Nose 1.General appearance: well, ill-looking, pale 2.Nose: watery discharge, mucus, blood, pus, any visible swelling, deformity, or foreign body. 3.Throat: pus or redness Ear 1.Head: tenderness over bone behind the ear (mastoid) 2.Ear: a) Condition ofav canal, b) Pus or blood, perforated drum c) Foreign body, d). Red or bulging drum 3.Test for hearing: clap 4.Neck: stiffness Throat 1.General appearance: distressed, pale 2.Head: tenderness over bone behind the ear (mastoid) 3.Throat: pus or redness., enlargement of tonsils 4.Throat: General appearance: distressed, pale, 5.Neck: nodes, swellings Ear: discharge, redness or bulging drum 7.Temperature, Pulse and Respiration: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Ear pain with history of foreign body with bleeding or neck stiffness	•Progressive inflammatory swelling over the Mastoid region •Tender, swelling behind the ear. May or may not be fluctuant	Severe condition Suspect Mastoiditis	1.Susp.Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 2.Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days REFER immediately to the appropriate health facility
II. Ear pain with Discharge	•Pus is seen draining from the ear and discharge is reported for less than 14 days •Tenderness behind the ear/on touching the ear	Severe condition Suspect Acute Ear Infection	Susp.Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days REFER urgently to the appropriate health facility if no improvement.
III. Ear pain with discharge for more than 14 days with or without fever	•Pus is seen draining from the ear and discharge is reported for 14 days or more. Ear pain;Temp > 37.50C	Severe condition Suspect Chronic ear infection	1.Susp.Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 2.Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3.Conduct mRDT and if positive, treat for uncomplicated malaria with ACT (refer to section on Malaria) Teach mother to wick ear at least 3 times daily until the ear is dry and discourage

IV. Swollen Wound in ear	•Boil seen •Tenderness •Temp > 37.50C	Moderate condition Suspect Boil	1.Susp.Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 2.Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3.Review in 3 days If no improvement REFER to the appropriate health facility
V. Difficulty in Swallowing	•Grey and dirty membrane on tonsil Temp. > 37.50C	Severe condition Suspect diphtheria	1.Give I/M Paracetamol (300mg/2ml) 2.2months -2 years 0.5ml(75mg) 3.2years – 5 year 1ml (150mg) REFER immediately to the appropriate health facility
VI. Pain in the throat, Inability to swallow	•Enlarged red tonsils with pussy exudate with or without tender lymph nodes in the neck	Moderate condition Suspect tonsilitis	1.Wash mouth with Salt water 2. Give I/M Amoxicillin stat a. 2months–12months 125mg b. 12months -5years 250mg 3. Give I/M Paracetamol (300mg/2ml) a. 2months -2 years 0.5ml(75mg) b. 2years – 5 yr 1ml (150mg) 4. REFER immediately to the appropriate health facility
VII. Pain in the throat, Fever, Inability to swallow or refuses to feed	•Sore throat with or without redness •Tonsils not inflamed and no pus Temp. > 37.50C	Mild condition Suspect Pharyngitis	1.Give I/M Amoxicillin stat a.2months–12months 125mg b.12months -5years 250mg c.Continue with Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 2.Give I/M Paracetamol (300mg/2ml) a.2months -2 years 0.5ml(75mg) b.2years – 5 yr 1ml (150mg) 3.Continue with Syr. Paracetamol 120mg/5ml a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 4.Syr. Ascorbic acid (100mg/5ml) 5ml 8hrly for 7 days Encourage fluid intake 7. Review in 3 days. If no improvement, REFER to the appropriate health facility

VIII. Pain on either side of the nose with headache with or without fever	•Pus from nose •tenderness i.Smelly, purulent discharge Temp. > 37.5°C	Moderate condition Suspect acute sinusitis	1. Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 1.Syr. Paracetamol 120mg/5ml a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. Syr. Chlorpheniramine (2mg/5ml) 2.5ml at night for 3 days 3. 4.Ascorbic acid (Vitamin C) tab 50mg daily in divided doses x 7 days 5.Review in 3 days. If no improvement, REFER to the appropriate health facility
IX. Bleeding from Nose	•Blood from the nose Anxious	Moderate condition Suspect Epistaxis	1.Teach mother to pinch side of the nose for 15 minutes 2.Apply Cold Compress to nose for 30 minutes 3.Re-examine 4.If no improvement, REFER to the appropriate health facility. If condition has happened before, REFER immediately to the appropriate health facility.
X. Foreign object or Polyp in the nose	Foreign object seen with or without nasal discharge	Moderate condition	1.Try to remove foreign body with blunt forceps If not removable, or if it's a polyp, REFER immediately to the appropriate health facility.
XI. Stuffy nose, with or without Fever, Sneezing with or without watery eyes	•Watery eye •Catarrh Temp. > 37.5°C	Moderate condition Suspect Common cold or allergy	1.Syr. Promethazine 0.25ml stat then 2. Syr. Chlorpheniramine (2mg/5ml) 2.5ml at night for 3 days 3. Give Syrup Ascorbic acid (Vitamin C) 100mg/5ml a. 2mths - 2yrs: 2.5ml 8hourly for 2weeks b. 2-5 years: 5ml 8hourly for 2weeks 4.Review in 3 days If no improvement, REFER to the appropriate health facility

XII. Stuffy nose or catarrh with or without fever, Sneezing with or without watery eyes, Lack of appetite, Difficulty in breathing	•Running nose and watery eyes Temp. > 37.50C	Mild condition Suspect Upper Respiratory Tract Infection	1.Syrup Paracetamol(120mg/5ml) 1. Syr. Paracetamol 120mg/5ml a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2.Give Syrup Ascorbic acid (Vitamin C) 100mg/5ml a. 2mths - 2yrs: 2.5ml 8hourly for 2weeks b. 2-5 years: 5ml 8hourly for 2weeks 3.Conduct mRDT test. If positive refer to section on malaria treatment Continue feeding/breastfeeding 4.Clear the nose if it interferes with breathing 5.Encourage extra fluids a.Keep the child warm 6.Watch out for danger signs (fast or difficult breathing, inability to suck or drink, or child becomes more ill)
XIII. Boil in the nose with or without fever	•Boil can be seen •Temp less than or greater than 37.50C	Mild Condition Suspect Boil	a.1. Susp.Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 2. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days c. b. 2-5 years: 5ml 8hourly for 3day 2.Give Syrup Ascorbic acid (Vitamin C) 100mg/5ml a. 2mths - 2yrs: 2.5ml 8hourly for 2weeks b. 2-5 years: 5ml 8hourly for 2weeks 3.Warm Compress 4.Review in 3 days. If no improvement, REFER to the appropriate health facility
XIV. Nasal discharge with or without fever	Running discharge from nose with or without fever	Mild Condition Suspect Acute Sinusitis	1. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. Give Syrup Ascorbic acid (Vitamin C) 100mg/5ml a. 2mths - 2yrs: 2.5ml 8hourly for 2weeks b. 2-5 years: 5ml 8hourly for 2weeks 3.Warm Compress 4. Review in 3 days. 5. If no improvement, REFER to the appropriate health facility

XVI.. Ear pain with discharge for more than 14 days with or without fever	• Pus is seen draining from the ear. • Dull ear drum • Warm to touch (Temp > 37.50C)	Moderate Condition	1. REFER immediately to the appropriate health facility

Health Education

Advise Mother/caregiver on:

1.Management of throat problems (gargle and discouragement of uvulectomy).
2.Prevention of exposure to air pollutants (fumes, dust, and pollen).
3.Personal hygiene.
4.Discourage home removal of foreign body.
5.Adequate nutrtion.
6.General childcare.
7.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.21. ORAL HEALTH DISORDERS

Oral health is a state of being free from chronic mouth and facial pain, oral and throat cancer, oral sores, birth defects such as cleft lip and palate, periodontal (gum) disease tooth decay and tooth loss, and other diseases and disorders that affect the oral cavity.

The most common complaint is pain which may or may not be accompanied with swelling inside the mouth or on the face. The condition of the mouth is also a reflection of the client's general health, so it should be inspected as part of physical examination. Early recognition of oral problems with prompt attention or referral will reduce morbidity and mortality arising from otherwise simple disease condition.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Where is the problem? 3.Is there pain from tongue, palate, mucus membrane, teeth or gum? 4.How long has it been noticed or present? 5.Are you able to eat your regular diet? 6.Does the pain increase with hot or cold drink or food? 7.Does the pain start on its own? 8.Does it keep you awake at night? 9.Is there swelling anywhere? 10.Is there any bleeding from gum or other part of the mouth? 11.Is there any bleeding from gum or other part of the mouth? 12.What do you use for cleaning your teeth? 13.How do you clean your teeth? 14. Do you notice any sore in the mouth?	1.General appearance: ill looking, facial swelling, malnourished. 2.Temperature: Check and record. 3.Mouth: a)Gum: sore, swelling, discharging sinus, bleeding, white patches. b)Tongue sore, coated, redness, fissures, smoothness, dryness. c)Lips: crack d)Chest: lining 4.Teeth: caries, discolouration, tenderness, mobility, missing, plague.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Severe pain that keeps patient awake a) Aggravated by chewing b) With or without headache	Hole in tooth/teeth	Moderate to severe condition Suspect Irreversible pulpitis	1. Syrup Paracetamol(125mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. Syr Metronidazole 100mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 3. Susp.Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 4. Review in 3 days, if no improvement, REFER immediately to the appropriate health facility

II. Sharp, short pain triggered by chewing hot/cold drinks Stops as soon as stimulus is removed	Hole on the offending tooth/ teeth	Mild condition Suspect reversible pulpitis	1. Syrup Paracetamol(125mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. Syr Metronidazole 100mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 3. Susp.Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 4. Review in 3 days, if no improvement, REFER immediately to the appropriate health facility
III. Gum soreness and bleeding, fever, foul breath	• Ulcers along gingival margins, increased salivation • Temperature greater than 37.5°C • mRDTnegative	Severe condition Suspect Acute Ulcerative Gingivitis (AUG)	1. Syrup Paracetamol(125mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. Syr Metronidazole 100mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 3. Susp.Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 4. Hydrogen Peroxide Solution 1 to 5 parts of water for mouth wash 8 times daily for 5 days 5. Review in 3 days, If no improvement, REFER to the appropriate facility
IV. Gum soreness fever, foul breath	• Mouth ulcers with vesicles on the tongue, mucous membrane • Inflamed gingivae mRDT negative	Severe condition Suspect viral gingivitis	1. Syrup Paracetamol(125mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. Syr Metronidazole 100mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 3. Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs.: 5ml 8hourly for 5days 4. Review in 3 days, If no improvement, REFER to the appropriate health facility

V. White patches in mouth, skin rash, soreness of the gum, not gaining weight	•White patches on tongue, oral mucosa that easily rubs off •Oral ulcers •Underweight Failure to thrive	Severe condition Suspect malnutrition and/ or HIV/AIDS	1. REFER immediately to the appropriate health facility.
	Hard or soft swelling anywhere in the mouth or jaws	Severe condition Suspect Oral Tumour	1. REFER immediately to the appropriate health facility.

Health Education

Advise Mother/caregiver on:

1.Brushing twice daily, in the morning and last thing before going to bed at night.
2.Use of fluoride-containing toothpaste.
3.Changing toothbrush as soon as the bristles begin to flare/scatter.
4.Visiting the dentist at least twice a year for routine cleaning (scaling and polishing).
5.Avoiding the intake of sugary foods (refined sugar) in-between meals. Eat mouth cleansing fruits instead.
6.Adequate nutrition.
7.Cultivating the habit of rinsing your mouth with water after meals.
8.Discouraging harmful traditional practices such as uvulectomy.
9.Importance of taking home-based records during visit to the clinic/hospital.

2.22. CHEST/HEART DISORDERS

A very rapid, slow or irregular heart rate is always a serious health sign. Changes in pulse and respiratory rate are often important signs of heart disease. Blueness of the lips or fingers, clubbing of the fingers and toes with difficult breathing are signs that the heart is not able to do its work well or easily. Congestive heart failure occurs when the heart becomes too weak to pump all the blood around and some of the fluid goes back into the lungs and often into the abdomen and legs. Any history of shortness of breath should make the health worker think of a heart disease.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? For how long? 2.Is there cough? 3.Is there difficulty with feeding due to shortness of breath? 4.Is there shortness of breath at night? 5.Is there chest pain? 6.Is there restlessness? (sign of lack of oxygen) 7.Does the child tire easily or tend to squat much of the time? 8.Have the lips or fingernails ever looked blue? 9.Does the child participate in sports? Is the child able to run around with his/her age mates without difficulty? 10.Any other problem?	1.General appearance: ill-looking, weak, thin, pale 2.Eyes: swollen or puffy 3.Nose: flaring 4.Mouth: blueness of lip 5.Toes/fingers: thickness of toes/fingers, blueness of the fingers 6.Chest: respiratory rate, lower chest in-drawing, grunting 7.Heart: rate and rhythm, murmur 8.Assess child's heart rate: Normal Heart rate 0-1year - 120/min 1-2years - 110/min 2-4 years - 100/min 4-8years - 90/min 9.Limbs: a). fingers blueness, pallor. Clubbing 10.Temperature: check and record 11.Haemoglobin: estimate and record 12.Weight: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Chest problem with: a. Shortness of breath Easily tired squats frequently b. Blue lips or finger nails c. Frequent coughing d. Poor weight gain e. Pain in the chest ii. Easily tired squats weakness iii. Easily tired with or without fever or infection	• Coughing • Swelling of limbs • Fast heart rate • More comfortable sitting up • Blueness of lips and finger nails • Fatigue Heart murmur • Slow or fast heart rate, cough, blueness, low Hb • Increased heart • Temp 37.5°Cor above	Severe condition Suspect congenital heart problem Moderate Clinical judgement for i, and iii should be seperated. also incert tables (three rows) in i to iii action be included too see page 105 of 2015	REFER immediately in a comfortable position to the appropriate health facility

Health Education

Advise Mother/caregiver on:

1.Personal hygiene.
2.Proper care of person with chest/heart problem (keep warm and allow for adequate rest).
3.Adequate nutrition.
4.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.23. ABDOMINAL DISORDERS

The common causes of abdominal pain are gastro-enteritis, malaria, appendicitis, rupture of the intestine, typhoid, sickle cell crisis and obstruction, e.g strangulated inguinal and umbilical hernias. Constipation, worm infestations may also cause abdominal pain. Abdominal pain may lead to hardness or rigidity of the abdominal wall.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? For how long? 2.Is there pain? If yes, is it continuous or intermittent? 3.Does the pain make the child restless at night? 4.What was the child fed on in the past 24 hours? 5.Is there vomiting? Any blood in the vomitus? 6.Is there diarrhoea? 7.Is there constipation? 8.Has the child passed worms in the stool? 9.Are stools black or is there bright red blood in stools? 10.Any pain on micturition? 11.Any fever? 12. Has any medication been given to the child?	1.General appearance: ill looking, crying(pain), thin, pale 2.Chest: respiratory rate, lower chest indrawing 3.Abdomen: a). swelling (distension), uniform or irregular b)visible peristalsis, bowel sound c)rigidity, tenderness, any palpable mass 4.Temperature and pulse: check and record\ 5.Haemoglobin: estimate and record Genotype: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i.Abdominal Pain; Not passing stool	•Child ill looking, crying and in pain, abdominal rigidity.	Severe condition Suspect Constipation	1. REFER immediately to the appropriate health facility
II.Abdominal pain with swelling on left side of abdomen (spleen) with or without history of injury.	Ill looking, pale, enlarged tender mass with painful swelling on the left upper part of the abdomen	Severe condition Suspect injury to spleen	1. REFER Immediately to the appropriate health facility.
III.Abdominal pain with Constipation;	•Well child; swelling of abdomen increased bowel sounds	Moderate condition	1.Review child's diet and fluids; 2.Advise fruits and vegetables 3.Give liquid paraffin 5 - 10mls daily for 2 days; If no improvement in 2 days, REFER
iv. Abdominal pain with rigidity; vomiting but no diarrhea	• Child in pain and very abnormal increased bowel Sound	Severe Condition Suspect injury to spleen	1. REFER immediately to the appropriate health facility.

V. Abdominal pain with bloody stools; Strains on passing stool	•Pale looking; blood in stools or Tenesmus(painful)	Moderate/Severe condition Suspect; Dysentery, Amoebiasis	1. If Dysentery give Syr. Metronidazole 100mg/5ml 1month-12months: 2.5ml 8hourly for 5 days, 3-5yrs: 5ml 8hourly for 5 days. OR Ciprofloxacin 15mg/kg/close-12hourly for 3 days if no improvement after 5 days; order stool test 2. if laboratory confirmation of amoebiasis, continue for 7days Syr. Metronidazole 100mg/5ml a. 2months-1 yrs.: 2.5ml 8hourly for 7days b. 2-5yrs.: 5ml 8hourly for 7days 3.Review in 3 days,If no improvement, REFER immediately to the appropriate health facility

Health Education

Advise Mother/caregiver on:

1. Food and personal hygiene.
2. Adequate nutrition.
3. Genetic counseling
4. Importance of taking home-based record's on children next to the clinic or hospital.

2.24. HERNIA

Hernia is when there is a protrusion of an organ or tissue through a gap or weakness of the body cavity in which it normally lies. Hernias can be found in the umbilicus, groin, or scrotum.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? For how long? 2. Is there vomiting? How many times? 3. Is there constipation? When was the last stool? 4. Is there any abdominal pain? 5. Has the child had the problem before? 6. Has any medication been given? If yes what?	1. General appearance: pallor, weakness, distress 2. Respiration: difficult or rapid 3. Abdomen: a) enlarged, distended b) masses, enlarged spleen c) visible intestinal movement d) presence of hernia, tenderness 4. Groin: presence of scrotal hernia 5. Hydration status: fontanels, eyes, mouth, tongue, skin 6. Hemoglobin: estimate and record 7. Temperature, pulse and respiratory rate: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Protrusion from the anus. Not reducible	• Child in distress, • Not dehydrated • Rectal prolapsed not reducible with or without diarrhea	Severe condition Suspect prolapsed of rectum	1. Apply warm saline, compress (sitz bath), 2. REFER immediately to the appropriate health facility
ii. Protrusion from the anus, reducible	•Child in distress, •Anal prolapsed •reducible with or without diarrhea	Moderate condition Suspect reducible prolapsed of rectum	1. Push prolapsed portion back and strap buttocks together 2. Teach mother how to strap the buttocks 3. If having diarrhea, Give Syr. Cotrimoxazole (240mg/5ml) a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 1. If anaemic a. Give Syrup Folic Acid 5ml daily for 2weeks b. Ferrous sulphate 200mg (60mg elemental iron) 8hourly for 2 weeks c. Lo-ORS Solution 2. Nutrition counseling 3. Review in 3 days, If no improvement REFER to the appropriate health facility
iii. Redness of anus with or without pain	•Anus red with sore Child uncomfortable	Moderate condition Suspect anal fissure	1. Clean with soap and water 2. Dry thoroughly 3. Apply zinc oxide topically 4. Give paracetamol tablet or Paracetamol suppository if oral medication is not possible Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 5. Dietary advice 6. Review in 3 days, If no improvement, REFER to the appropriate health facility

Health Education

Advise Mother/Caregiver on;

1. Personal, food and environmental hygiene
2. Protection of genital area
3. Early detection and treatment of infection
4. Discouragement of sharing of sponge, towel, underwear and clothing
5. Adequate nutrition
6. Bowel training
7. Importance of taking home-based records on child's next to the clinic or hospital

2.25. SEXUAL ASSAULT/ RAPE

Sexual assault is rape or any sexual contact that results from coercion, including seduction of a child through offers of affection or bribe; it is also includes being touched, grabbed, kissed or shown genitals. On the other hand, rape is oral, anal or vaginal penetration that involves threats or force against a person who is unwilling (i.e. non-consenting) or incapacitated (because of cognitive or physical disability or intoxication). Such penetration whether wanted or not is statutory rape if victims are younger than the age of consent. This may result in pregnancy (if female), transmission of STIs (e.g. HIV, Syphilis, depression, physical injuries, recurrent dreams and nightmares and other emotional problems). Victims of sexual assault/rape should be advised to report the incident to police if not against their wish.

HISTORY	EXAMINATION/ACTIONS
1.Was the baby sexually assaulted (raped)? If yes, where? How? Who did it? 2.How many times? How long did it last? When? 3.Was the baby threatened or forced? 4.Was the child made to drink anything (alcohol or medicine) prior to the assault? 5.Was the baby wearing clothes when the incident occurred? 6.Was the child physically harmed? 7.Was there any penetration of the anus, vagina or mouth? If yes, with penis, finger or object? 8.Was there ejaculation? Where? 9.Any bleeding? Where? 10.Did you bath, change clothes or clean up the child afterwards 11.Has the child been sexually assaulted before? 12.Did you report to anyone else about this? Who? 13.Have you given any treatment? When? Where? 14.Any witness? Any other problem?	1.General appearance: restless, physically exhausted, conscious, depressed, afraid, crying. 2.Check temperature, pulse, respiration, blood pressure and record. 3.Mouth: injuries 4.Skin: injuries 5.Genitalia: bleeding, injuries 6.Anus: injuries 7.VDRL: test and record. 8.Hepatitis B: test and record. 9. HS and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Rape with vaginal or anal bleeding and lacerations	Signs of penetration with bleeding, lacerations and injuries to other parts of the body.	Severe condition	1.Counsel and reassure mother. 2.Advise mother/caregiver to report the incident to police or appropriate NGO. 3.REFER to the appropriate health facility for further investigations.
II. Rape without vaginal or anal bleeding and lacerations	Signs of penetration without vaginal or anal bleeding and lacerations.	Moderate condition	1.Counsel and reassure mother. 2.Advise mother/caregiver to report the incident to police or appropriate NGO. 3.Give the child Hepatitis B vaccination (if not already immunized) 4.Conduct HIV test. If positive, REFER to appropriate Standing Order 5.If negative, commence emergency post exposure prophylaxis within 72 hrs. of rape (if trained). 6.Give: a) IM Ceftriaxone stat. b) Azithromycin orally c) Tab Metronidazole orally.

III. Sexual Assault	No signs of penetration of vagina, anus and mouth. No injury to other parts of the body.	Mild condition	1.Counsel and reassure mother. 2.Advise mother/caregiver to report the incident to police or appropriate NGO. Ensure child protection

Health Education

Advise Mother/Caregiver on;

1.Childcare and protection.
2.Appropriate dressing.

2.26. ANAL DISORDERS

The anus is the terminal end of the alimentary canal. Common problem affecting the anus include fissures. Redness and prolapse. An anal fissure is painful if near or at margin of anus. In rectal prolapsed; the rectum turns inside out and protrudes out of the anus as a pink mass. It is common in young children who are poorly nourished, have worms and diarrhoea frequently.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? How long have you had the problem? 2.Is there diarrhoea? How many times? 3.Is there constipation? When was the last stool? 4.Is there bright blood in stool? 5.Is there any abdominal pain? 6.Has the child had the problem before? Has any medication been given?	1.General appearance: pale weak, in distress 2.Hydration status: fontanels, eyes, mouth, tongue, skin 3.Anus: prolapsed, redness, fissure 4.Haemoglobin: estimate and record 5.Temperature: check and record 6.Weight: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Protrusion from the anus, not reducible	•Child in distress, •Not dehydrated Rectal prolapse not reducible with or without diarrhoea	Severe condition Suspect Prolapse of rectum	1.Apply warm saline, compress (sitz bath), REFER to the appropriate health facility.
II. Protrusion from the anus, reducible	•Child in distress Anal prolapse reducible with or without diarrhoea	Moderate condition Suspect reducible Prolapse of rectum	1.Push prolapsed portion back and strap buttocks together. 2.Teach mother how to strap the buttocks 3.If having diarrhoea, Give syr. Cotrimoxazole: a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 4.If anaemic a.Give Syr. Folic Acid 5ml daily for 2 wks. b.Ferrous sulphate (60mg elemental iron) 8hourly for 2 weeks c.Lo-ORS Solution 5.Nutrition Counseling 6.Review in 3 days if no improvement, REFER to the appropriate health facility.
III. Redness of anus with or without pain	•Anus red with sore Child uncomfortable	Moderate condition Suspect anal fissure	1.Clean with soap and water 2.Dry thoroughly 3.Apply Zinc Oxide topically 4.Give Paracetamol tablet or suppository if oral medication is not possible. Syr. Paracetamol(125mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 5.Dietary advice 6.Review in 3 days, If no improvement, REFER to the appropriate health facility.

Health Education

Advise Mother/Caregiver on;

1.Personal, food and environmental hygiene.
2.Protection of genital area.
3.Early detection and treatment of infection.
4.Discouragement of sharing of sponge, towel, underwear, and clothing.
5.Adequate nutrition.
6.Bowel training.
7.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.27. WORM INFESTATION

Worm infection is the presence of large amount of worm in the gut which causes discomfort to the patient. Most children have worms and Ascaris (round worm) is the commonest. Heavy round worm infestation produces abdominal pain, vomiting constipation and sometimes obstruction of the intestine. Hookworm infestation is more common in rural areas where heavy infestation causes blood in stool, anemia, and malnutrition. Severe (whip worm) infestation causes bloody diarrhea, abdominal pain, and anemia and prolapsed of the rectum. Thread worms also live in the intestine and cause itching around the anus. Tapeworm is a flat ribbon- like worm that can also be found in the intestine.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? When did it start? 2. Has any worms been passed? If yes, describe 3. Is there diarrhea? Blood in stool? 4. Is there vomiting? Is the child coughing? Any wheezing? 5. Is there abdominal pain and rectal prolapsed? 6. Is there any other problem? 7. Has any medication been given? If yes, what?	1. General appearance: thin, pale 2. Eyes: pallor 3. Chest: wheezing, rhonchus 4. Abdomen: tenderness, distension, visible peristalsis 5. Anus: redness, prolapsed rectum 6. Temperature pulse and respiration: check and record 7. Hemoglobin: estimate and record 8. Weight: weight and chart 9. Stool: examination: for ova of worm (if laboratory available)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Passing worms	Passing worms vomiting with abdominal tenderness	Severe Condition	1.REFER to the appropriate health facility.
Passing worms	Passing round worms with mild abdominal pain May be underweight No other problems	Moderate condition Suspect Ascariasis	1. Susp. Pyrantel pamoate stat a. 6-11mths:2.5ml stat b. 1-2yrs:5ml stat c. 3-5yrs: 5ml stat 2. Review in one week
Passing worms with paleness	• Passing worms. With distended but not tender abdomen: or Hb less than 10gm no diarrhea.	Moderate condition Suspect hookworm infestation	1. Susp. Pyrantel pamoate stat a. 6-11mths:2.5ml stat b. 1-2yrs:5ml stat c. 3-5yrs: 5ml stat 2. Give Syrup folic acid 5ml daily for 2weeks
IV.Loss of weight paleness	Presence of worm in the stool from laboratory result	Moderate condition Suspect tapeworm infestation	1. Niclosamide 200 mg/kg body weight single dose followed by chastic purgative 2hr later

Health Education

Advise Mother/Caregiver on;

1. Personal, food hygiene and environmental sanitation
2. Adequate nutrition
3. Regular Deworming
4. Protection of genital area
5. Importance of taking home-based records on child's visitation the clinic or hospital.

2.28. URINARY AND GENITAL DISORDERS

A urinary tract infection, commonly known as UTI is one of the most common urinary problems and is an infection that occurs in the urinary tract. This infection is most often caused by microbes such as fungi, bacteria, and virus and involves the kidneys, ureters, urethra, or bladder. This condition can be very painful, and the consequences can get critical if the infection spreads to the kidneys. Children may present with abnormalities of the genitalia such as undescended testes, imperforate hymen, hydrocele, hypospadias, and hermaphrodite. Children may also present with painful micturition with or without discharge. Redness or itching of the genitalia may occur in genital infections. Patients may complain of discharge, pain and burning sensation on passing urine and blood in urine may occur as a result of schistosomiasis infection. Other causes of urinary problems are kidney disease and diabetes which can cause frequent micturition.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long has it existed? 3. Is there pain on passing urine? Or does the child scream or rub his/ her legs together before passing urine? If yes, for how long? 4. When last did He urine? 5. What does the urine look like? Odour? 6. Has the child been bed wetting after a period of full bladder training? 7. Does ants cloud around urine passed by the child? 8. Is the child excessively thirsty?	1. General appearance: puffy, pale, iii -looking 2. Eye: pallor, peri oritel oedema 3. Mouth: dryness parched lips 4. Abdomen: tenderness, or masses, oedema 5. Flanks: tap both flank with hand for tenderness 6. Skin: dryness, wrinkle 7. Limbs:hands and feet for oedema, clubbing 8. External genitalia: abnormalities, redness, swelling or pus. In males, look at foreskin to see if is too tight or will not retract. 9. Temperature pulse and respiration; check and record 10. Urine; cloudiness, colour, odour, protein, blood sugar, and micros

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. No urine for more than 12 hours	Looks ill Puffy Restless Anxious Leg crossing Heal sitting	Severe condition Suspect acute renal failure	1. Insert catheter 2. Refer immediately to the appropriate health facility.
II. Excessive urination with history of ants crowding round urine. Excessive thirst	Dry mouth Wrinkled skin Parched lips Sugar in Urine Lethargic Loosing urine by accident	Severe condition Suspect juvenile diabetes	1. Screening for UTI should be done immediately 2. Refer immediately to the appropriate health facility.
III Dark yellow urine with yellow eyes with or without hotness of body	Dark yellow urine Yellow eyes with or without fever	Severe condition Suspect malaria	1. Test for Genotype 2. Syrup Folic acid 5ml daily for 2 weeks 3. Conduct mRDT if positive treat uncomplicated malaria with ACT (Refer to section on malaria treatment) 4. Encourage fluid and fruits intake 5. Review in 3 days 6. If no improvement, REFER to the appropriate health facility

IV. Blood in last stream of Urine with or without pain, fever or discharge	Blood in Urine, Schistosoma ova may be detected in urine	Severe condition Suspect Urinary Schistosomiasis	1. If painful Micturition; 2. Give Susp. Cotrimoxazole (240mg/5ml) a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 3. Encourage fluid intake 4. Review in 3 day, if no improvement, check for Bilharzia ova, if positive, give Metriphonate a. 100mg (1tab) every 2 weeks for 6weeks 3-5yrs b. If negative and no improvement, REFER to the appropriate health facility for further investigations.
V. Redness or swelling of genitalia with or without painful micturition, itching	Redness and swelling ofgenitalia, scratch marks Bruises	Moderate to severe condition	1. Assess for history of rape or sexual abuse 2. Wash the area with soap and water 3. Syrup Paracetamol(125mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 4. Apply zinc oxide and Mycostain cream or paste to area. 5. Encourage fluids 6.If male, retract foreskin 7. Give Susp. Cotrimoxazole (240mg/5ml) a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 8. Review in 3 days, If no improvement REFER to the appropriate health facility Treatment of constipation with stool softeners is effective in preventing recurrent UTI in children without Urinary tract abnormities
VI. Painful urination; No discharge	Painful urination No discharge	Moderate condition Suspect Urinary Tract Infection [UTI]	1. Syrup Paracetamol(125mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. Give Susp. Cotrimoxazole (240mg/5ml) a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 3. Encourage fluid and fruits intake 4. Review in 3 days, If no improvement REFER to the appropriate health facility.

VII. Pain in the Flank, fever	•Tenderness on tapping flank Muscle spasm •Temp 37.50c or above	Moderate condition Suspect kidney infection (pyelonephritis)	1. REFER to the appropriate health facility
VIII. Painful urination with or without frequent, cloudy urine or fever.	Temp – 27.50c protein in urine, painful micturition	Moderate condition	1. Encourage fluids 2. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. Give Susp. Cotrimoxazole (240mg/5ml) a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 4. Review in 2 days; if no improvement, REFER to the appropriate health facility
IX. Dark yellow Urine no other complain	Dark Urine No other findings	Moderate condition Suspect dehydration	1. Encourage adequate fluid intake 2. Review in 2 days, 3. Rule out severe Malaria by mRDT test
X. Painful, urination With or without frequent, cloudy urine or fever	Temp > 37.50c Protein in Urine; painful Micturition.	Moderate condition Suspects glomerulonephritis	1.Encourage fluids 2.Give Paracetamol tablet or suppository Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3.Give Syr Cotrimoxazole (240mg/5ml) a. 2-12mths- 2.5ml 12hourly for 5days b. 1-5yrs- 5ml 12hourly for 5days 4.Review in 2 days 5.If no improvement REFER to the appropriate health facility
XI. Urethra and/or Vaginal discharge with painful micturition	Urethral/vaginal discharge	Moderate condition Suspect Urethratis	1. Risk assessment to know if child has been raped or abused 2. Syrup Paracetamol(125mg/5ml) a. 2mths - 2yrs:2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. Syr. Ciprofloxacin 125mg 12hourly for 5days 4. See syndromic management 5. Syr. Doxycycline 25mg 12hourly for 7 days 6. Review in 3days, If no improvement, REFER to the appropriate health facility

XII. Difficulty in passing urine	•Penis blocked by tight foreskin not easily retracted bladder full	Moderate Condition Suspect phimosis constriction produce	1. Circumcise immediately, following standard procedure. (see appendices for procedure)
XIII.Abnormal Genitalia	•Un-descended testis •Imperforate hymen •Hermaphrodite •Large clitoris •Hydrocoele; •Abnormal (penial opening(s)	Mild or Moderate condition Suspect congenital malformation	1. Explain findings to mother and re-assure her. 2. REFER to the appropriate health facility.

Health Education

Advise Mother/Caregiver on;

1. Personal hygiene.
2. Drug compliance
3. Proper parental care and supervision of the child
4. Adequate fluid intake
5. Importance of taking home- based records on child's visitation the clinic or hospital.

2.29. ARMS AND LEGS DISORDERS

Deformities of the arms and legs, if not corrected early could lead to permanent disabilities. The deformities or swellings may be due to bruises. Fractures or a dislocation. It is important to test for normal movement of the parts whenever there is a history of injury, swelling or deformity. Swelling of a joint may be due to a sprain of the tenders surrounding the joint. Sickle cell disease may also be a cause of pain in the arms, legs and the hip bone. Tuberculosis can occur in the joints. Polio can seriously affect one or both legs of arms first causing weakness then wasting, or paralysis. Diabetes Mellitus may cause tingling sensation and numbness of the fingers and toes. Foot ulcer that fails to heal and black discoloration of toes and feet (Gangrene).

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? When did it begin? 2. Was there an accident or injury? 3. If an injury, has the child been able to use the part since the injury? 4. Is there any pain or tingling or numbness? Is it recurrent? 5. Is there fever? 6. Is the child a stickler? 7. Is the child passing excessive urine? 8. Is the child having excessive thirst?	1. General appearance: ill looking, wasted or deformed 2. Limb: a. Deformity b. Swelling c. Bleeding d. Fracture e. Tenderness f. Warmth g. Loss of function h. wasting 3. Eyes: jaundice 4. Temperature: check and record. 5. Haemoglobin; estimate and record. 6. Urine; sugar and colour

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Inability to move arm and/ or leg.	Unable to move and feet	Severe condition Suspect Polio Paralysis or Sickle cell disorder	1. Refer immediately to the appropriate health facility and follow up. 2. If polio notify the appropriate PHC authority
ii. History off all from a height, pain, swelling broken bone.	Pain and tenderness Swelling Lower limb hanging Inability to move limb	Severe condition	1. Immobilize the limb 2. Syrup Paracetamol(125mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. Review in 3 days. if no improvement, refer immediately to appropriate health facility
iii. History of sickle cell disease, severe joint and abdominal pain, swelling on digits, especially of fingers, toes, back of hands and feet	Intense pain Swollen fingers, toes, back of hands and feet. HB below 10 gms Temp > 37.5°C	Severe condition Suspect sickle cell disease in crises	REFER immediately in a comfortable position to the appropriate health facility

iv. Swelling of joints	Tenderness of the affected joint Swelling of the joint.	Moderate condition Suspect juvenile Arthritis	1. Immobilize the limb give Syrup Paracetamol(125mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. Susp Ampicillin + cloxacillin 250mg/5ml a. 2-12mths- 2.5ml 6hourly for 5days b. 1-5years- 5ml 6 hourly for 5days 3. Review in 3 days, if no improvement, REFER to the appropriate health facility for investigations
V. Deformity of leg with difficult in Walking	• Club foot, bow leg, walking with difficulty	Moderate condition Suspect birth defect	1. REFER to the appropriate health facility for assessment

Health Education

Advise Mother/Caregiver on;

1. Prevention of accidents
2. Genetic counseling
3. Adequate nutrition (Calcium rich food)
4. Importance of taking home-based records on child's visitation the clinic or hospital

2.30. CHANGES IN ALERTNESS AND BEHAVIOUR

Changes in alertness and behavior mean that the client does not act normal. He may be drowsy or unconscious or give a history of fainting. Big changes such as unconsciousness or delirium are easy to identify. Small changes such as drowsiness and decreased alertness are not easily noticed. If the clients is not alert or is drowsy, it may not be noticed easily if this happens, the client may not receive treatment and could come back later in more serious condition. Therefore, careful observation of client is important.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child been like this? 2.Did anything happen to the child? Any injury? 3.Did the child have a fit or convulsion, or is the child clumsy? 4.Has any medication been given? 5.Has the child been ill otherwise? 6.Has the child had fever, diarrhoea, vomiting or ear pain? 7.Has the child been drinking more recently??? 8.Has the child been urinating more frequently? 9.Was there any difficulty with the pregnancy or delivery? 10.Was the child jaundiced at birth or shortly after? 11. Has the child been frothing in the mouth or passing urine when he faints?	1.General appearance: (especially for alertness and irritability), sign of trauma. 2.Chest: respiratory rate, lower chest in-drawing. 3.Neck: stiffness 4.Head: fontanelle for bulging, signs of injury. 5.Ears: pus, redness or perforation of the drum. 6.Limbs: arms and legs for abnormal movements. 7.Temperature: check and record 8.Haemoglobin: estimate and record. 9. Urine: test for protein, sugar, acetone.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Unconscious or difficult to wake, Irritable, abnormal movements of arms and Legs II. Unconscious with: a) fever b) difficult to arouse c) confusion d) more irritable than normal e) abnormal movements of arms and legs III. Unconscious or: a) more irritable than normal b) abnormal movements of arms and legs IV. Unconscious or : a) more irritable than normal b) abnormal movement of arms and legs	•Irritable •No stiffness •Uncontrollable	Severe condition Suspect Head Injury Severe condition Suspect severe malaria Severe condition Suspect epilepsy Severe condition	1. REFER immediately to the appropriate health facility 2. Conduct mRDT, if POSITIVE, give pre-referral treatment for severe malaria 3. Refer immediately to the appropriate health facility 4. REFER immediately to the appropriate health facility 5. REFER immediately to the appropriate health facility
V. Child faints suddenly but is arousable	•Child disoriented •Weak child •Acetone in urine	Severe condition Suspect Hypoglycemia	1. Give child sugar or glucose at once and let him rest for 30mins 2. If no improvement, REFER to the appropriate health facility 3. If improved, advise on proper feeding 4. Review next day.

VI. Fainted (fit) once or twice with or without fever or frothing from mouth or family history of epilepsy	•Conscious. •Normal temperature •No fits		1. Explain condition to mother 2. Phenobarbitone (15mg/5ml) 0-5mths 1ml 12hourly for 7days 6mths-5yrs : 2.5ml 12hourly for 7days 3. Review in 7 days 4. Assess medication, if improved, resupply 5. If no improvement, REFER

Health Education

Advise Mother/Caregiver on;

1. Prevention of injuries
2. Compliance with prescribed drug regimen
3. Avoidance of self-medication
4. Healthy home environment
5. Proper parental care and emotional support
6. Need for therapy as appropriate
7. Importance of taking home-based records on child's next visit to the clinic or hospital

2.31. POISONING

A poison is a toxic substance which enters the body through swallowing, breathing, absorption through the skin or mucous membrane or infection. Poison may be accidental, suicidal or homicidal. An overdose of a medicinal drug may act as poison e.g. overdose of aspirin. It is common for children under 5 years of age to accidentally ingest medicine not prescribed for them or over dosage of prescribed, drugs; alcohol (gin), kerosene, petrol or corrosive substance e.g. acid, lead or bleaches.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.What was the substance taken? How much of it? When was the poison taken? 3.Has any medication been given (modern or traditional)? 4. Has the child vomited? Is there blood in the vomitus?	1.Try to examine poison container 2.General appearance: consciousness, alertness, pallor, Cyanosis 3.Eye: size and reaction of pupils to light 4.Chest: respiration rate, difficulty, lower chest in drawing, deep and sighing 5.Abdomen: tenderness, rigidity 6.Mouth: burn, bleeding, dribbling, colour of membrane 7.Pulse: check rate, volume, and record 8.Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Ingestion of any of the following, gin, medicine, not prescribed.	Irritable, Restless, No Mouth burns or swelling May or may not be warm to touch	Moderate condition Suspect poisoning	1. identify the poison 2. do not induce vomiting 3. give milk drink or Mist magnesium trisilicate 5-10mls 8hourly for 3 days 4. give activated charcoal 5. review after four hours 6. if no improvement, REFER to the appropriate health facility
ii. Ingestion of poison with history of convulsion. iii. Ingestion of petrol or kerosine or corrosive substance e.g alkali, caustic soda and acid	•irritable, restless, may or may not be •convulsing •no mouth burns and swelling •Irritable, swelling and mouth burns	Moderate condition. Moderate to severe condition Suspect substance poisoning	1. IM. Paraldehyde 1-3ml stat 2. Gastric lavage/washout 3. Tab. Phenobarbitone 15mg 12hourly for 3 days 4. Give activated charcoal 5. Review in 3days, If no improvement, REFER appropriately. 1. Do NOT induce vomiting 2. Give enema saponis 3. Give Amoxicillin 250mg stat. 4. Refer immediately to the appropriate health facility
iv. Aspirin poison with or without fever or other signs	•Temp normal or above 37.50c •Irritable •Thirsty •Restless		1. Do not induce vomiting 2. Give milk drink or mist magnesium Trisilicate 3. Give plenty of fluids 4. Give activated charcoal 5. Review daily until better

v. Ingestion of petrol or kerosene or corrosive substance e.g. acid, herbicides, insecticides, rodenticides etc	•Irritable •Swelling and mouth Burns	Severe condition Suspect lead/ corrosive poisoning	1. Do not induce vomiting 2. Give enema e.g. soap 3. Give IM Cloxacillin for 5 days 0-5mths: 0.25ml daily for 5 days 6-llmths: 0.5ml daily for 5days 1-5yrs 1ml daily for 5days 3. Mist magnesium Trisilicate 5mls 8hourly for 5days 4. Give activated charcoal 5. Review in 4 hours, If no improvement, REFER to the appropriate health facility

Health Education

Advise Mother/Caregiver on;

1.Prevention of home accidents especially by ingestion
2.Proper storage and labeling of drugs, various liquids and solvents
3.Importance of taking home-based records on child's next visitation to the clinic or hospital.

2.32. EMERGENCY CONDITIONS

Children 2 months to 5 years can be life threatening and should be treated promptly to save life.
There should be no hesitation in asking for or making referral where necessary. Severe acute fever, diarrhea, malnutrition, child poisoning, conjunctivitis, epilepsy, choking and some of the common anaphylaxis are emergency situations and should be treated as soon as possible to save the life of the child.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem with child? 2. Is there fever, vomiting, diarrhoea or abdominal pain? 3. Is there history of convulsion or unconsciousness? 4. Is there history of poisoning? 5. Is the child having difficulty in breathing or chocking? 6. Is there family history of asthma? 7. Is the child allergic to known substances or food? 8. Is there decrease in urine output?	1. General appearance: ill looking, alert, unconscious, convulsing, dehydrated. 2. Temperature: check and record 3. Eye: Sunken 4. Mouth and lips: Dryness 5. Chest: (a) Respiratory rate: check and record (b) Wheezing, crepitation 6. Skin: dryness, dehydration

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Acute fever with temperature above 38.5°C and child convulsing	High temperature above 38.5°C and convulsing	Severe condition Suspect Acute Haemorrhagic fever	1. Tepid sponge 2. Give paracetamol suppository 3. REFER immediately to the appropriate health facility
II. Acute diarrhea with severe dehydration / loose stools, unable to pass urine	Severe dehydration with symptoms as drowsiness, confusion, passing little urine	Severe condition Acute diarrhea	1. Give LoORS 2. REFER Immediately to the appropriate health facility.
III. Fatal child poisoning from medicines, iron pills, cleaning products, pesticides		Severe Condition Suspect Child poisoning	1. REFER Immediately to the appropriate health facility.
iv. Paediatric conjunctivitis with inflammation, redness, and discharge.	Child with inflamed eyes, redness, and discharge	Severe condition Suspect conjunctivitis	1.Give1dose of appropriate antibiotics 2.REFER Immediately to the appropriate health facility.
v. Severe seizure in children	Unconscious and convulsing	Severe condition Suspect epilepsy	1. REFER Immediately to the appropriate health facility
vi. Anaphylactic reaction to food, medications, allergens	• cutaneous manifestations: urticaria, pruritus, angioedema and flushing • cardiovascular: dizziness, hypotension and syncope • GI'l: Nausea, vomiting, diarrhea, abdominal pain	Severe condition Suspect Anaphylaxis	I. Give injectable epinephrine 1:1000, 0.01mg kgs, 0.01mg/kg into the lateral thigh, may repeat every 5 -10 mins if symptom persist to all children with signs and symptoms suspicious of anaphylaxis before arrival to hospital. 2. REFER Immediately to the appropriate health facility

VII. Choking with breathlessness	• Bluish skin color • Difficulty breathing - ribs and chest pull inward • Loss of consciousness if blockage is not cleared . Inability to cry or make much sound .Weak, ineffective coughing . Soft or high-pitched sounds .While inhaling	Severe condition Suspect chocking	1. If your child is not coughing forcefully or does not have a strong cry, follow these steps: 2. Lay the infant face down, along your forearm. Use your thigh or lap for support. Hold the infant's chest in your hand and jaw with your fingers. Point the infant's head downward, lower than the body. 3. Give up to 5 quick, forceful blows between the infant's shoulder blades. Use the palm of your free hand. 4. If the object does not come out of the airway after 5 blows: 5. Turn the infant face-up. Use your thigh or lap for support. Support the head. 6. Place 2 fingers on the middle of his breastbone just below the nipples. 7. Give up to 5 quick thrusts down, compressing the chest 1/3 to 1/2 the depth of the chest. 8. Continue 5 back blows followed by 5 chest 9. thrusts until the object is dislodged or tired infant loses alertness (becomes unconscious)
VIII. Acute asthma attack with wheezing, cough, breathlessness	Worsening asthma condition	Severe condition Suspect Acute asthma	1. Give 6 puffs of blue reliever inhaler via a Spacer. 2. If no improvement within 20 minutes repeat 6 puffs 3. If still no improvement within 20 minutes of 6 puffs, refer immediately to appropriate health facility

Health Education

Advise Mother/Caregiver on;

1. Compliance with prescribed drug regimen.
2. Avoidance of self-medication.
3. Healthy home environment.
4. Proper parental care and emotional support.
5. Importance of taking home-based records on child's next visit to the clinic or hospital.

2.33. HIGH RISK GROUPS (ORPHANS AND VULNERABLE CHILDREN)

An orphan is a child (0-17 years) who has lost one or both parents. A vulnerable child is one who, due to circumstances of birth or immediate environment, is prone to abuse or deprivation of basic needs, care, and protection, and thus is disadvantaged relative to his or her peers (FMWA&SD, 2008). A vulnerable child may lack access to education, health, and other social support, have a chronically ill parent, live in a household with terminally or chronically ill parent(s) or caregiver(s), live outside of family care (with extended family, in an institution, or on the street), or be infected with HIV. Additionally, a vulnerable child includes those in need of alternative family care, those who are sexually abused or neglected, children in hard-to-reach areas, children with mental disabilities, children affected by armed conflict, and children in need of legal protection. Most current interventions focus on Orphans and Vulnerable children (OVC) aged 6-17 years, often excluding children under five years old. In Nigeria, various organizations are involved in OVC work, including international NGOs (mainly USG and Global Fund implementing partners), local NGOs, faith-based organizations (FBOs), and community-based organizations (CBOs). Some of the main international NGOs involved in OVC work include the Columbia University International Center for AIDS Care and Treatment Programs (CU-ICAP), Catholic Relief Services (CRS), Center for Development and Population Activities (CEDPA), and Winrock International AIM.

HISTORY	EXAMINATION/ACTIONS
1. How old is the child? Who is the child living with? 2. Is the child sick or well 3. If sick, what kind of illness 4. If well, who is taking care of the child? 5. Is the mother or father alive both or one 6. Is the mother both parents HIV/AIDS before death 7. Has the child been sexually abused, and you are sick as result of that?	1. General appearance: well-nourished. ill. tired, clean, thin, congenital abnormality 2. Immunization status: if card available, check status 3. Weight and Height: weigh, measure and record 4. Head: lice, ringworm 5. Eyes: discharge, pallor, jaundice, cataract, test of vision 6. Ears: redness, discharge, test of hearing 7. Mouth: odour, caries, missing teeth 8. Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Baby born to HIV- Infected mother and orphaned	Baby on or not on ARV	Severe/moderate/ mild, Suspect HIV	1. REFER to the section on HIV/ AIDS and treat 2.REFER to Local and Inti' NGO's on OVC
II. Children 2 months to 5 years with any of the disease conditions		Severe moderate /mild, Suspect any disease condition	1. Refer to appropriate section and treat
iii. Well OVC	• No abnormality		1. REFER to OVC NGO's for Psychosocial care.

Health Education

1. Healthy home environment.
2. Proper parental care and emotional support.
3. Counsel on need for therapy as appropriate'.

2.34 SCHOOL HEALTH SERVICES

The school health services are essential in maintaining and improving the health status of members of the school community and should be available to everyone for as long as they are in the school community. School health services afford the opportunity for personal health care, screening of school food vendor, environmental health care, community mobilization, advocacy. Personal health care includes health screening for all new comers, regular health inspection and treatment for minor ailments.

HISTORY	EXAMINATION/ACTIONS
1 .Is there any problem? 2. Which immunizations have you had? 3. What type of food do you eat?(ask about main meals and snacks) 4. How many brothers and sisters do you have? 5. Whom do you live with? (Parents, grandparents, spouse other specify)? 6. How many times do you eat in a day? Where do you get your food? 7. What type of work does your father do? Mother? You? Spouse? 8. What is your source of water? 9. What type of toilet facility do you have in your house or hostel? 10. Have you had any serious illness in the past?	1. General appearance: well-nourished, ill, tired, clean, thin, congenital abnormality 2. Immunization status: if card available, check status 3. Weight and Height: weigh, measure and record 4. Head: lice, ringworm 5. Eyes: discharge, pallor, jaundice, cataract, 6. Nose: discharge, blockage, abnormality 7. Ears: redness, discharge, test of hearing 8. Mouth: odour, caries, missing teeth 9. Gum: healthy, bleeding 10. Throat: redness, inflammation 11. Chest: respiratory rate, wheezes, stridor, lower chest in drawing 12. Heart: rate, rhythm, murmur 13. Abdomen: soars, tenderness, rigidity, masses, palpable spleen and liver. 14. Skin: dryness, sores, rashes, multiple soars or wounds (think child abuse) 15. Limbs: deformity, paralysis, oedema 16. Pulse:, Blood pressure: check and record 11. Urine: sugar, protein 18. Haemoglobin estimate and record 19. Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. No complaints	• No abnormality	Well school child	1. Ensure weekly regular health inspection
II. Any abnormal finding		Severe moderate/mild	1. Refer to appropriate section and treat
III. If school food vendor is engaged		Severe moderate/mild condition	1. Check for the following at least every six months a. Chest X-ray b. Sputum for AFB o. Blood for widal c. Stool for Ova e. Treat as appropriate

SECTION THREE

MIDDLE CHILDHOOD

(6 – 12 Years)

3.0 MIDDLE CHILDHOOD

3.1 CHILD FIRST VISIT

This section is for children between the ages of 6 and 12 years. The health worker should be able to carry out simple sight and hearing tests to determine if the child has any visual or hearing problems that could affect school performance. They must also recognize social problems that might put the child at risk and be able to screen for malnutrition. Additionally, they should request genotype and blood group tests so that the child and parents are informed, and provide counseling if any abnormalities are detected. The health worker should also conduct health and nutrition education, immunize, and protect children from illnesses and other conditions relevant to the family. Parents and guardians should be encouraged to take their children for regular clinic visits, even if the children are well.

HISTORY	EXAMINATION/ACTIONS
Development; 1.Ask parent/caregiver if child is developing normally 2.When does the child start crawling, walking and speaking? 3.Is the child attending school? 4.What is the current class of the child? (Milestone) Family and Social 5.With whom does the child live? 6.Does mother/caregiver work outside home? 7.Who looks after the child when the mother/caregiver is away? 8.What is the source of water? What type of toilet is used? 9.Is house screened against mosquito/is insecticide treated net used? 10.What illnesses has the child had in the past? 11.Does the child have any problem now? 12.What work does the father do? How many Siblings does the child have? 13.How many children are there in the family? 14.How many wives does the man have? 15.Are there any of these disease conditions in the family: Asthma, sickle cell disease, Diabetes, Hypertension, Tuberculosis, Epilepsy, and other relevant conditions. Immunization; 1.Did the child receive all the routine immunization for age? 2.Does the child have the certificate for completing immunization? Nutrition; 3.Was the child exclusively breast-fed? 4.What type of weening food was introduced after six month? 5.What kind of food is the child currently eating? 6.What type of food does the child like most? Medical; 7.Has the child been hospitalized before? 8.What type of sickness was the child hospitalized for? 9.Has the child convulsed at any time in the past?	1.General appearance: healthy, ill, thin 2.Blood pressure: measure and record 3.Temperature, pulses: check and record 4.Eyes: pallor, jaundice 5.Mouth: conditions of mucous membranes, gums, caries, inflamed throat 6.Neck: swelling, and stiffness,or tenderness 7.Chest: a.respiration rate, difficulty b. crepitus. c c. any decrease or increase in breath sounds 8.Heart: rate, rhythm, murmurs 9.Abdomen: scars, tenderness, masses, hernias 10.Limbs: a.deformation, varicosities, oedema b.ankles: oedema 11.Weight: weigh and record 12.Urine: color, protein, sugar, and ova 13.Stool: parasites and ova 14.Hemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. No complaint about child	No abnormality	Healthy child	Explain findings to Mother Counsel as appropriate
II. Any abnormal finding			REFER to appropriate section

Health Education

Advise Parent/Caregiver on the following:

1. Adequate nutrition
2. Personal and food hygiene
3. Environmental sanitation
4. Hand washing
5. Prevention of accident
6. Proper parental care
7. Use of Long-lasting Insecticidal Nets (LLINs)
8. Importance of taking home-based records with her anytime she takes baby to the clinic or hospital.

3.2 FEVER

Fever is the elevation of the body temperature above 37.2 °C. Fever is a sign of infection or loss of body fluid. Rapid rise of temperature leads to fits (Convulsion) especially in young children. Some common causes of fever in children are malaria, tuberculosis, respiratory, gastrointestinal and other bacterial and viral infections. All patients with history of fever should be treated and encouraged to drink plenty of water, other fluids, and fruits, and should not be overdressed. High fever should be treated as an emergency.

HISTORY	EXAMINATION/ACTIONS
1. How long has the child had the fever? 2. Is there any convulsion? 3. Has the child ever had any convulsion? 4. Is the child having a runny nose? Cough? Or difficult breathing? 5. Is the child coughing up blood, 6. Is the child vomiting or having diarrhea? 7. Is the child having bloody diarrhea? 8. Is the child having skin rashes? 9. Is the child having any ear pains? Sore throat? Headache? Stiffness of the neck? 10. Is the child passing urine normally? Is it painful? Is it more or less frequent? 11. Is the child bleeding inside and outside the body, from the eyes, ears, and nose? 12. Has any analgesic or other medication been given? 13. Is anyone currently ill in the family or compound?	1. General appearance: alert, restless, listless, drowsy or difficult to wake, delirious, dehydrated, thin, weak, or convulsing. 2.Skin: rashes, septic spots, profuse perspiration. 3. Chest: a.respiratory rate counted for 1 full minute b.Lower chest indrawing/retraction c.wheeze or stridor in a calm child 4. Nose: discharge. clear or purulent. blood 5. Eyes: pus, blood 6. Ear: tenderness behind the ear, discharge, blood 7. Throat: sores, pains 8. Neck: stiffness, nodes. 9. Abdomen: tenderness, masses 10. Temperature and pulse: check and record 11. Conduct Rapid Diagnostic Test (RDT) for malaria (if positive proceed and treat for severe malaria)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Fever with any of the following at a) Convulsion/multiple convulsion b) Prostration c) Impaired consciousness d) Respiratory distress e) Abnormal bleeding. f) Persistent vomiting	Lethargic or unconscious, warm to Touch and Temp. 37.5°C or above. Pulmonary oedema Jaundice discolored urine not making urine / scanty urine Haemoglobinuria Renal failure Hypoglycemia Severe anaemia (Hb < 5gm/dl) Circulatory collapse	Severe condition Suspect severe malaria	1. In Seizure or Convulsion, give IM paraldehyde 0.2-0.4ml/kg 2. Give Intramuscular (IM) Artesunate 2.4-3.0mg/kg body weight at the time of admission (stat), then at 12h and 24h, then once a day until patient can take oral medication. OR 3. IM Artemether 3.2mg/kg given at the time of admission (0) hour, then 1.6mg/kg per day until patient can take oral medication. 4. IM Quinine 10mg/kg loading dose diluted in 10ml/kg of IVF 4.3% Dextrose saline 500mls or 5% dextrose over 4hours; then at 12h or glucose water over 2hours and repeat every 8hours to correct hypoglycemia till patient can tolerate orally. 5. Tepid sponge when necessary (PRN) 6. Review in 3 days if no improvement REFERto appropriate health facility.

II. Fever with neck pain with or without vomiting	Neck stiffness, abdominal rigidity with or without spasms. Warm to touch and Temp. 37.5°C or above. Kernig's sign positive	Severe Condition Suspect Meningitis	1. Conduct Rapid Diagnostic Test (RDT) for malaria- If negative proceed with the following treatment 2. IM Benzyl Penicillin 3cc (300,000 i/u) immediately OR IM. Ceftriaxone 0.5g immediately (stat) 3. Give IM Paraldehyde 0.2-0.4ml/kg Immediately (stat) 4. REFER immediately.
III. Fever every day for more than 30 days; skin rash, mouth ulcer, weight loss, with or without diarrhea.	Oral thrush Skin rash. Lethargy, Pallor.	Severe condition Suspect HIV AIDS	1. Conduct Rapid Diagnostic Test (RDT) for malaria (if positive proceed and treat for severe malaria) 2. In Seizure or Convulsion give IM paraldehyde 0.2-0.4ml/kg' 3. Give Intramuscular (IM) Artesunate 2.4-3,0mg/kg body weight at the time of admission (stat), then at 12h and 24h, then once a day until patient can take oral medication. OR 4. IM Artemether 3.2mg/kg given at the time of admission (0) hour, then 1.6mg/kg per day until patient can take oral medication. OR 5.. IM Quinine 10mg/kg loading dose diluted in 10ml/kg of IVF 4.3% Dextrose saline 500mls or 5% dextrose over 4hours; then at 12h or glucose water over 2hours and repeat every 8hours to correct hypoglycemia till patient can tolerate orally. 6. Tepid sponge when necessary (PRN)

IV. Fever with cough and runny nose	• Centralized rash • Red eyes, • Moderate runny nose • Warm to touch. • Temp. 37.5°C and above, • sticky eyes and mouth ulcer	Moderate Condition, Suspect Measles	• Clean the mouth with chlorhex-idine • Conduct Rapid Diagnostic Test, and if positive, treat for malaria with: Artemether-Lumefantrine (AL) 20/120mg • 6-8 years: 15-24 kg - 2 Tablets given at 0 hour, 8hours later, Then, 12hourly for 3 days 9-12 years: 25-34 kg - 3Tablets given at 0 hour, 8hours later, Then, 12hourly for 3 days Artesunate-Amodiaquine (AA) • 7-12 years: 100 kg - 270 mg - 1 Tablet daily for 3 days Give Tablet Paracetamol 500mg 8hourly for 3 days Give Vitamin A capsules of 200,000 I.U. daily for 2 days. Tablet Co-trimoxazole 480 mg 12hourly for 5 days Clean eyes with cotton soaked in cooled boiled water. Chloramphenicol eye ointment 8hourly for 7 days Clean the mouth with Hydrogen Peroxide solution 1 in 5 parts of water for 5 days Dip a cotton swab into Gentian Violet solution, sparingly apply to the mouth affected area Review in two days. if no improvement, REFER to the appropriate health facility.
V. Fever with ear pain with or without discharge for less than14 days	Ear discharge; watery or pus or bloody. Warm to touch. Temp. 37.5°C or above. Tenderness.	Moderate condition Suspect ear infection	1.Give Tablet Paracetamol 500mg 8hourly for 3 days 2Clean ears with TCP-soaked cotton wool. 3.Cap. Ampicillin/Cloxacillin 250mg 6hourly for 5- 7days OR 4.Amoxicillin 250mg 8hourly for 5-7 days 5.Tablet Vitamin C 100mg 8hourly for 5 days 6.Review in 5 days, If there is no improvement, REFER immediately.

VI. Fever or history of fever in the past 24 hours.	Temp. above 37.2°C no other finding	Moderate condition Suspect uncomplicated malaria	1.Conduct rapid diagnostic test, if positive treat for uncomplicated malaria: a. Artemether-Lumefantrine (AL) 6-8 years: 15-24 kg - 2 Tablets given at 0 hour, 8hours later, Then, 12hourly for 3 days 9-12 years: 25-34 kg – 3 Tablets given at 0 hour, 8hours later, Then, 12hourly for 3 days OR b. Artesunate-Amodiaquine (AA) • 7-12 years: 100 kg - 270 mg - 1 tablet daily for 3 days 2.Give Tablet Paracetamol 500mg 8hourly for 3 days 3.Encourage intake of fluids, fruits, and vegetables 4.Review after 3 days 5.If fever persists, refer to the hospital

Health Education

Advise Parent / Caregiver on;

1.Environmental sanitation
2.Prevention of disease transmission
3.Use of insecticide treated nets/materials in malaria control.
4.Personal and food hygiene
5.Adequate nutrition
6.Safe water and disease prevention.
7.Adherence to treatment
8.Importance of taking home-based records to the clinic or hospital.

3.3 CONVULSION

Convulsion is jerky movements of the arms and legs and twitching of the muscles. It may be caused by injury and infection of the meninges, brain and spinal cord. Tetanus or lockjaw must not be mistaken for convulsion. It begins with tightness of mouth, jaw and muscle twitching.

HISTORY	EXAMINATION/ACTIONS
1.Has the child had fever? Any accident? Head injury? 2.Did the child take any medicine, or any substance accidentally? 3.Has the child had any previous convulsion? With or without fever? 4.Has the child been given any medication? If yes, what? Does any other family member suffer from convulsion? 5. Is there difficulty in breathing before the episode of Convulsion	1.General appearance: convulsing, conscious, drowsy, unconscious, smelling of cow's urine. 2.Nose: flaring of alae nasi 3.Chest: respiratory rate, lower chest indrawing 4.Head: swelling, sign of injury 5.Eyes: pupils for size and reaction to light 6.Neck: stiffness 7.Ears: pus, bulging drums or perforation of drums 8.Temperature and pulse: check and record 9.Weight: Weigh and chart.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. History of convulsion in the past 24hrs with or without history of head injury	Stiffness of neck, Signs of head injury, drowsy, vomiting	Severe condition Suspect Meningitis	I. Explain findings to mother 2 Give IM Paraldehyde 0.2-0.4ml/kg immediately (stat) 3 IV. Phenobarbitone 15-20mg/kg over 10-15minutes 2. REFER appropriately If bleeding, pack with Adrenalin
II. Convulsion with fever	Convulsing; With Temp. 37.5°C or above	Severe condition Suspect febrile convulsion	1.Remove from dangerous environment 2.Lie patient on side, clear airway 3.Give I.M Paraldehyde 0.2-0.4ml/kg immediately (stat) 4.Do rapid diagnostic test, if positive give pre -referral treatment for severe malaria, 5.Give Intramuscular (IM) Artesunate 2.4-3.0mg/kg body weight at the time of admission (stat), then at 12h and 24h, then once a day until patient can take oral medication. OR 6.IM Quinine 10mg/kg immediately (stat) on the anterior aspect of the thigh OR 7.Artesunate suppository 200mg rectally when available. 8.If is negative, give IM Amoxicillin 250mg immediately(stat) 9.Encourage intake of Glucose water or sugar in water if patient can take it orally. 10.Expose child REFER immediately to appropriate health facility.
III. One or more episodes of convulsion without fever	Convulsing; Temp 37.5°C	Moderate condition Suspect Epilepsy	I. Give IM Paraldehyde 5 ml stat 2. IV. Phenobarbitone 15-20mg/kg over 10-15minutes 3. REFER immediately to the appropriate health facility.

Health Education

Advise Parent / Caregiver on;

1.Causes and prevention of convulsion
2.Management of fever
3.Management of a person during convulsion episode,
4.Adherence to treatment
5.Adequate nutrition
6.Use of Long Lasting Insecticidal Nets (LLINs)
7.Importance of taking home-based records on child's next visit to the clinic or hospital.

3.4 COUGH

Cough is a sign of irritation of the throat and wind pipe which comes suddenly. In children, common causes of cough include pneumonia, asthma, mucus blocked nostrils, adenoidal, aspiration of foreign body or poison e.g. kerosene. Heart diseases may also be present with cough.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had the cough or difficulty breathing? 2.Is there wheezing? Have there been any previous episodes? Has the child been vomiting? 3.Has the child experienced convulsions? 4.Is there a fever? If so, how long has it been present? Is the child able to drink? 5.Has any medication been administered to the child? If so, what medication?	1.General Appearance: alert or abnormally sleepy or difficult to wake, severe malnutrition, respiratory distress. 2.Nose: flaring of alae nasi 3.Chest: a). respiratory rate (count for 1 full minute in a calm child) b.subcostal retraction lower chest-indrawing c.stridor d.wheeze 4.Eyes: redness, discharge Throat: soreness or redness Feet: swollen 5.Temperature, pulse: check and record Weight: Weigh and chart.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Cough, fever with one of the following: difficult to sleep and fast breathing.	Chest in drawing, warm to touch. Temp. 37.5°C or above. Respiratory rate of 40 or more per min. Decreased breath sound and crepitation. Clay colour sputum	Severe condition Suspect Pneumonia	1.Give IM. Amoxicillin 250mg immediately (stat) 2.Give Tab. Paracetamol 500 to 1000 mg 8hourly for 3 days 3.REFER appropriately.
II. Cough, fever with noisy and difficult breathing	•Chest in-drawing. •Harsh noise on inspiration or soft noise on expiration. Warm to touch. Temp. •3.7.5°C or above. •40 breath or more per min	Severe condition Suspect Asthma	1.Give IM Amoxicillin 250mg stat 2.IV Aminophylline 5ml given slowly over 10 -20 minutes immediately (stat) 3.Tablet Salbutamol 2mg 12hourly for 4 days 4. REFER appropriately.
III. Cough of 3 wks or more with or without weight loss. Blood in sputum	Loss of weight Cough with or without blood stain sputum	Severe condition Suspect Tuberculosis	1.Collect sputum for investigation 2.If 2 out of 3 specimens are positive. REFER for treatment 3.Direct Observed Treatment (DOT) see appendix 4.Follow up at home.
IV. Persistent or recurrent cough more than one month, skin rash, mouth ulcer, fever, weight loss, with or without diarrhea.	•Skin rash marked loss of weight. Oral thrush, warm to touch. Temp. 37.5°C or above, mouth ulcers	Severe condition Suspect HIV/AIDS	1.Conduct Respiratory Syncytial Virus (RSV) and Sputum test. 2.REFER appropriately.

V. Cough, fever with fast breathing	•Fast breathing •Temp. 37.5°C or above. •Reduced breath sound and crepitation. Clay color sputum	Suspect Pneumonia Moderate condition	1.Give Cap Amoxillin 250mg 8hourly for 5 days 2. Give Tablet Paracetamol 500mg 8hourly for 2 days 3.Conduct rapid diagnostic test (RDT); if positive treat for uncomplicated malaria 4.Home care.Warm tea and honey. 5.Review after 3 days. 6.If no improvement, REFER.
VI. Cough with no other complaint	No fast breathing	Mild condition	I. Parent to manage child at home with warm tea, lemon, honey, 2. Review after 2 days, if no improvement REFER.

Health Education

Advise Parent / Caregiver on;

Overcrowding and disease transmission
Personal hygiene and prevention of airborne diseases
Air pollution and respirations problem
Recommendations for good ventilation
Role of adequate nutrition
Adequate fluid intake in prevention of infections and promotion of quick recovery.
Use of Long Lasting Insecticidal Nets (LLIN)
Importance of taking home-based records to the clinic or hospital

3.5 DIARRHOEA

Diarrhea is the passage of watery stool more than 3 times a day. Diarrhea in children is one of the leading causes of morbidity and mortality. It kills children more easily than adults and therefore the replacement of fluids is the most important treatment. Causes of diarrhea include dysentery, typhoid, cholera, ingestion of poisons, and contaminated foods and HIV/AIDS. Persistent diarrhea is one that has lasted for two or more weeks.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been ill? 2.How many times do you usually pass stool each day? 3.Have you passed stool since coming to the clinic? 4.Are the stools watery? Is there blood or mucus? 5.What is your source of drinking water? 6.What type of toilet facilities do you use? 7.Are you nauseated or have you vomited? 8.Do you have a fever? Abdominal pain? Cough? 9.Are you sometimes constipated 10.Is there anyone else in the household who is sick? 11.Have you taken any medication or treatment for this? 12.Have you eaten plenty of fresh fruits (e. g. mangoes) or plenty of pepper?	1.General appearance alert, restless, irritable, thin, dehydrated, lethargic, unconscious 2.Eyes, normal, sunken, very sunken and dry. 3.Mouth and Tongue; moist dry, very dry 4.Chest, respiratory rate 5.Neck: nodes enlarged; throat inflamed 6.Ears: purulent discharge, tenderness behind ear 7.Abdomen: tenderness, distended 8.Skin: skin pinch goes back quickly, slowly or very 7 slowly 9.Weight weigh and chart 10.Temperature: check and record 11.Offer the child something to drink, and note if the child drinks normally, eagerly or poorly (or is not able to drink) SEE APPENDIX=r-r-for chart on assessment of dehydration

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Sudden onset of frequent watery stool	Weak with severe dehydration; skin pinch goes back very slowly; Mouth, 'Tongue dry; very sunken and dry eyes; Tears absent	Severe condition Suspect severe dehydration	1. Start IV infusion, Ringer's Lactate immediately100ml/kg body weight 2. If no Ringer's Lactate, give normal saline in divided doses as follows: 130ml/kg body weight in 30 min, then 70ml/kg body weight in 2 hours 3. If child is able to drink, give Low Osmolarity Oral Rehydration Salt (Lo-ORS) by mouth while the IV set is up on, re-assess every 1-2 hours 4. Give dispersible Zinc tablet 20mg daily for 10 days 5. REFER Immediately.
II. Frequent stool and vomiting (in area where there is Cholera	Irritable sunken eyes, Mouth and Tongue dry. Tears absent child is thirsty and drinks eagerly, Skin pinch goes back slowly. Some dehydration	Moderate/Severe Suspect Cholera	1. Explain condition to mother 2. Give IVF as above if severe 3. Give Oral Rehydration Salt (ORS); One sachet dissolved in 1litre of water, to be taken over 24 hours. 4. Capsule Amoxicillin 250mg 8hourly for 5 days 5. Metronidazole Tablet 200mg 12hourly for 5 days 6. Zinc sulphate Tablet 20mg daily for 5 days 7. Metoclopramide Tablet 5 mg immediately (stat) 8. If no improvement, REFER.

III. Frequent stool with visible blood. Some dehydration	Irritable, Mouth and tongue dry. Skin goes back slowly Visible blood in stool	Moderate/severe Suspect dysentery	1. Oral Rehydration Salt (ORS) (to be prepared in the facility) 2. Continue feeding 3. Tablet Co-trimoxazole 480mg 12hourly for 5 days 4. Review in 2 days 5. If there is still blood in stool after 2 days, give Tablet Metronidazole 200mg 8hourly for 5 days 6. Review in 2 days 7. If no blood, complete Metronidazole 8. If blood persists, REFER to appropriate health facility.
IV. Frequent stool with severe weight loss	Miserable looking wasted muscles Weight low for age	Severe condition Suspect severe malnutrition	1.Give mother Oral Rehydratiion Salt (ORS)to give slowly on the way to hospital 2. REFER immediately to appropriate health facility.
V. Frequent stool, no mucus or blood, no vomiting	Alert, eyes normal, mouth and tongue moist, child drinks normally, skin pinch goes back immediately	Moderate condition Suspect Mild dehydration	1. Low Osmolarity Oral Rehydration Salt (Lo-ORS) (demonstrate to mother how to prepare and administer). Continue feeding frequently. 2.Advise mother to give more fluids at home. 3. Mother to return if not better in 3 days or if any danger sign appears. 4. Zinc sulphate 20mg daily for 10 days
VI. Intermittent diarrhoea for more than one month with or without fever, skin rashes, mouth ulcers,	Emaciated, lethargic, oral thrush, skin rashes, some dehydration	Severe condition Suspect HIV/AIDS	1. Low Osmolarity Oral Rehydration Salt (Lo-ORS) (demonstrate to mother how to prepare and administer, continue feeding frequently). 2. REFER to appropriate health facility for further investigation.

Health Education

Advise Parent / Caregiver on;

1.Personal, food and environmental hygiene.
2.Proper Hand washing.
3.Adequate nutrition (conduct food demonstration).
4.Preparation of salt-sugar solution.
5.Water sanitation.
6.Importance of taking home-based records with her anytime she takes baby to the clinic or hospital.

3.6 VOMITING

Vomiting is the expulsion of the contents of the stomach through the mouth. Common causes of vomiting include gastroenteritis, typhoid, cholera, meningitis, ear infections, liver disease, and abdominal problems. A child who is vomiting should not be given large or heavy meals but rather small, frequent, and freshly prepared foods. Children who are vomiting should receive urgent care through fluid replacement.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been vomiting? How many times a day do you vomit? 2.What is the vomitus like? Projectile (think of obstruction), greenish? With blood? 3.Is there diarrhea? If not, when was the last stool? 4.Do you have abdominal pain? 5.Do you have any cough, fever or headache? 6.When was your last menstrual period? (if female) 7.What was your last meal? 8.What medication has the child taken?	General appearance: ill-looking jaundiced, dehydrated, distressed. 1.Mouth: tongue coated, clean. 2.Throat: redness, inflamed tonsils 3.Neck: stiffness. 4.Abdomen: tenderness, rigidity, visible peristalsis, distended. 5.Ear: redness, bulging or perforation of drum, discharge. 6.Hydration status: eyes, tongue, mouth, skin 7.Temperature and pulse: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Vomiting more than 3 times daily with any of the following: a) Blood (brownish, vomitus b) Neck stiffness Abdominal pain	• Lethargic and ill looking with bloody vomitus or some severe dehydration or abdominal rigidity or tenderness	Severe condition Suspect PUD	1. Give IM. Amoxicillin 250 mg stat immediately 2. REFER immediately. 3. Give Low Osmolarity Oral Rehydration Salt (Lo-ORS) on the way to referred facility 4. IM Promethazine 12.5 mg stat immediately
II.Vomiting more than 3 times daily with Abdominal pain	•Ill looking. Some dehydration and abdominal rigidity or tenderness	Moderate condition Moderate dehydration	1. Explain findings to mother 2. Give Lo-ORS 3. IM. Metoclopramide 5mg immediately (stat) 30minutes before meals. 4. Tablet Hyoscine 10mg 8hourly for 1 day 5. Counsel about giving small but frequent meals. 6. If not improving, REFER to the appropriate health facility.
II.Vomiting less than 3 times daily with no other sign	No dehydration Eyes normal Child happy	Mild condition Suspect Allergy	1. Explain findings to mother. 2. Counsel about giving small but frequent meals. 3. Give Low Osmolarity –Oral Rehydration Salt (Lo-ORS) en route to referred facility 4. IM Promethazine 12.5 mg immediately (stat) 5. If still vomiting, REFER to the appropriate health facility.

Health Education

Advise Parent / Caregiver on;

1.Personal and food hygiene
2.Proper hand washing
3.Environmental sanitation
4.Use of Long-lasting Insecticidal Nets (LLINs)
5.Causes of vomiting.
6.Adequate fluid intake and prevention of dehydration (excessive loss of body fluids).
7.Importance of taking home-based records to the clinic or hospital.

3.7 HEADACHE

Headache is a common complaint which may be a symptom of many diseases such as Malaria, sinus infection, high blood pressure, ear, nose, and throat problems, nervous tension, meningitis, brain abscess and typhoid fever. It can come on when there is constipation or inadequate sleep and rest.

HISTORY	EXAMINATION/ACTIONS
1. Where do you feel the headache most? a) In the front (thick of nose, eyes, malaria) b) In the back (think of high blood pressure, tension, kidney failure) 2. How long have you had it? 3. Do you have fever or chills? Convulsion? 4. Do you have nausea, vomiting or abdominal pain? (typhoid or malaria) 5. Is headache worse in the morning? (sinuses). What time of the day is headache worse? 6. Do you have constipation or diarrhea? 7. Are you on birth control pills? 8. Have you had any head injury? If so, when 9. Do you have any other problem?	1. General appearance: in pain, wasted 2. Temperature and pulse: check and record 3. Blood pressure: measure and record 4. Face: swelling, puffiness 5. Eye: redness 6. Nose: pus or mucus, press both sides below the eyes for tenderness 7. Ears: redness or dullness of drums, mastoid pain 8. Mouth: acetone or alcohol on breath 9. Neck: stiffness, nodes 10. Conduct RDT, if positive treat for uncomplicated malaria . 11. Chest: difficulty in breathing, creps, abnormal breath sounds. 12. Abdomen: tenderness, distended, rigidity, enlarged spleen, masses. 13. Limbs: weakness, paralysis 14. Urine: protein

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Headache with fever	Temp 37.5°C or above	Moderate condition Suspect malaria	1.Administer Tablet Paracetamol 250 -500 mg 8hourly for 3days 2.Encourage fluid intake 3. Conduct mRDT, if negative explore other causes of fever. 4.Consider use of antibiotics 5.Bed rest for patient and observe
II. Headache with any of these conditions. Toothache, Pus in the ear, eye pain. c) visual problems i swelling in the neck, d) Runny or stuffy nose.	•Palpable nodes, in the neck Stuffy nose Tenderness in the ear lobe Temperature above 37 °C	Severe condition Suspect. Sinusitis	1.Capsule Ampiclox 250 - 500mg 6hourly for 5days 2. REFER immediately to the appropriate health facility.
III. Headache, Disturbed sleep, restlessness	Raised blood pressure	Severe condition Suspect childhood hypertension	1.Tablet Paracetamol 500mg 8hourly for 3 days 2.Encourage bed rest 3.REFER immediately.
IV. Headache with stuffy nose,, catarrh, hoarse voice	Stuffy nose	Mild condition Suspect upper respiratory tract infection.	1. Tablet Paracetamol 500mg 8hourly for 3 days 2. Tablet Chlorpheniramine 2mg 12hourly for 3 days 3. Tablet Vitamin C 100 mg 8hourly for 7 days 4. Bed rest, warm tea and honey 5. Encourage fluid intake 6. Treat malaria when positive for RDT 7. Review in 3 days.

Health Education

Advise Parent / Caregiver on;

1. Healthy living (exercise, adequate rest, avoidance of stress)
2. Avoidance of animal fat in food in case of raised blood pressure.
3. Avoidance of overcrowding
4. Personal food hygiene
5. Use of Long-lasting Insecticidal Nets (LLINs)
6. Importance of taking home-based records to the clinic or hospital.

3.8 PALLOR

Pallor is paleness or loss of color from normal skin tone, which is a major sign of anemia, reduced number of red blood cells or reduced amount of hemoglobin in each red cell (below 10grams)/deciliter. A child can develop anemia because of not eating adequate nutritious foods, malaria infection, malnutrition, parasites (hook worm or whipworm), infections and sickle cell disease. Anemia can lead to death.

HISTORY	EXAMINATION/ACTIONS
1. Is there bleeding from any source? a) Nose, gums b) Coughing or vomiting c) In the stool or urine d) Black stools, injury 2. Does the child have a fever? For how long? 3. Has the child had painful swelling of the fingers, toes, the back of hands and feet? Or bone pains? 4. Does the child have a good appetite? What food does he/she eats? 5. Does the child reject some food? 6. Does the child go barefooted 7. Has the child been swimming or washing in the stream? 8. Does anybody in the family have similar problems? 9. Has any medication been given? 10. Is the mother using camphor?	1. General appearance: tall and thin, malnourished 2. Chest: respiratory rate, lower chest indrawing 3. Heart: rate, murmur 4. Head: bossing of the skull 5. Eyes: jaundice or conjunctiva} pallor 6. Abdomen: enlarged spleen, distension 7. Limbs: a) Swollen and tender joints, feet and/or digits, or back of hands and feet b) Nails for pallor 8. Hemoglobin: estimate and record 9. Temperature: check and record 10. Urine: color, blood, protein 11. Conduct RDT if positive, treat for malaria.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Pale child with swollen or puffy eyes. Swollen feet	Nose flaring, Lower chest in drawing, grunting, fingers clubbing.	Severe condition Suspect Heart failure	1.REFER urgently 2.Place in proper position(cardiac position) 3.Reassure parent/caregiver.
II. A pale child with generalized bone pain, joint pains.	• Swollen hands and feet • Severe Conjunctival and Palmar pallor • Hb lower than 10g/dl	Severe condition Suspect Sickle Cell crisis	1. Encourage fluid intake2. Give Paracetamol 500mg 8hourly for 3days 2. Give folic acid 5mg daily for 3days 3. Review in 3 days if no improvement, REFER.
III. Yellow eyes, weakness, breathlessness and body pains	• Jaundice of conjunctiva and palmar pallor • Hb estimation lower than 10gm/dl	Severe condition Suspect Sickle cell Anemia	1. Give ACT. Artemether-Lumefantrine (AL) 20/120mg • 6-8 years: 15-24 kg - 2 Tablets given at 0 hour, 8hours later, Then, 12hourly for 3 days • 9-12 years: 25-34 kg - 3Tablets given at 0 hour, 8hours later, Then, 12hourly for 3 days 2. Give Capsule Ibuprofen 200mg 8hourly for 3 days (After food) 3. Give Tab. Folic acid 5mg daily for 2 weeks 4. Encourage fluid intake 5. Review in 3 days, if no improvement REFER immediately to the appropriate health facility.

IV. Pale, not growing well with or without yellow eyes. Weakness	Some pallor, bossing of forehead, apparent long limb, thin hair, may be stunted	Severe condition Suspect Sickle Cell Disease	1. Give Tablet Folic acid 5mg daily for 2 weeks 2. Tablet Paracetamol 500mg 8hourly for 3 days 3. Review in 3 days, if no improvement REFER immediately to the appropriate health facility.
V. Weight loss, may or not pass worm	Some pallor- mild to moderate. Check Hb, PCV	Moderate Condition Suspect Sickle Cell Disease	1. Give Tablet Pyrantel Pamoate 3 tabs immediately (stat). 2. Tablet Ferrous Sulphate 300mg (60mg elemental iron)1 Tablet daily for 2 weeks 3. Tablet Folic acid 5mg tablet daily for 2 weeks 4. Nutrition counselling 5. Follow up after 2 weeks.

Health Education

Advise Parent / Caregiver on;

1.Prevention and care for sickle cell disease and protection of feet.
2.Prevention of malaria (Environmental control).
3.Use of Long-lasting Insecticidal Nets (LLINs).
4.Personal, environmental and food hygiene.
5.Avoid extremes of weather and conditions that could trigger crisis.
6.Importance of taking home-based records to the clinic or hospital.
7.Genetic counseling.

3.9 YELLOWISH EYES (JAUNDICE)

Jaundice is yellowish discoloration of the conjunctiva and skin due to an increase in bile pigmentation. It is commonly due to excessive breakdown of red blood cells (hemolysis) as in Sickle Cell crisis, severe malaria and liver problem e.g. infective hepatitis, obstruction of the bile duct, etc. It is these conditions rather than the Jaundice itself which demands treatment.

HISTORY	EXAMINATION/ACTIONS
1. Is the child unwell? (sepsis & GIT obstruction can cause jaundice) 2. Is there yellow discoloration of eyes and skin 3. When did you notice the yellowish discoloration of the eyes? 4. Is there dehydration or poor weight gain? (both exacerbate jaundice) 5. Is there family history of hemolytic disease (ABO/G6PD, spherocytosis, sickle cell) 6. Is there dark urine or pale stools (obstruction) 7. What medication have you given (e.g. herbs)?	1. General appearance: conscious, drowsy, unconscious 2. Chest: respiratory rate, lower chest indrawing 3. Head: swelling, sign of injury 4. Eyes: yellowish 5. Ears: pus, bulging drums or perforation of drums 6. Hands: palms and nails beds for pallor and yellowish coloration 7. Temperature and pulse: check and record 8. Weight: Weigh and record 9. Conduct Rapid Diagnostic Test (RDT)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Yellow eyes, General body weaknesses May or may not have blood in vomit	Yellow eyes May have swelling of the abdomen with rigidity Marked abdominal tenderness Pallor Hb lower than 10g/dl	Severe condition Suspect sickle cell with liver condition	1. REFER immediately
II. Yellow eyes, General body pain, dark yellow urine	Yellow eyes Haemoglobin less than 10gm /dl Dark yellow urine	Severe condition Suspect liver involvement	1.Glucose in water 2.If RDT is positive , give Pre-referral treatment for severe malaria REFER immediately to the appropriate health facility.
III. Yellow eyes, Fever, nausea and/or vomiting, abdominal pain	• Yellow eyes • Temperature 37.5°c above • Abdominal tenderness • Diarrhea	Moderate condition Suspect Hepatitis	1.Conduct mRDT, . If positive treat for uncomplicated with ACT Artemether-Lumefantrine (AL) 20/120mg 6-8 years: 15-24 kg - 2 Tablets given at 0 hour, 8hours later, a.Then, 12hourly for 3 days 9-12 years: 25-34 kg - 3Tablets given at 0 hour, 8hours later, b.Then, 12hourly for 2 days 2. Give Capsule Amoxicillin 250mg 8hourly for 5 days Review in 3 days REFER if no improvement

Health Education

Advise Parent / Caregiver on;

- Prevention and care of Sickle cell disease.
- Personal and food hygiene.
- Environmental sanitation.
- Use of Long-lasting Insecticidal Nets (LLINs).
- Proper nutrition.
- Importance of carrying home based records to the clinic.

3.10 NUTRITIONAL DISORDERS

Nutritional disorders are conditions caused by improper diet between what an individual eats and what he require to maintain health. This can result from eating too much or excess or an inadequate diet such as protein, fat and carbohydrate.

Children with nutritional disorders have higher risk of developing other diseases, and deaths, identifying children with nutritional disorders and treating them can help prevent many severe diseases and deaths. Stunting is common among this age group especially children from families with low income in rural communities and urban slums.

HISTORY	EXAMINATION/ACTIONS
1. How long has the child been losing weight? 2. Has the child been having diarrhea or vomiting? If yesfor how long? 3. Does the child have cough? 4. Does the child have fever? 5. Has the child been eating well? 6. What food is the child eating? (probe type of food for 24hour period) 7. Has the child been ill recently of Measles, Diarrhea, Prolonged cough etc? 8. Has the child recently passed worms? 9. Social History i. Does the mother work outside the home ii. Who takes care of the child? iii. How many other children are in the family? iv. How much do the mother and father earn? v. Does the child live with the parents? If not, whom does he live with and why? vi. Are the parents separated or disvorced?	1. General appearance: listless, irritable, alert, wasted. Oedematose, weak 2. Hair: color, texture, quantity 3. Skin: peeling, sore, rashes, dark and white patches on legs 4. Hydration status: fontanelle, eyes, mouth, skin 5. Eyes: pallor, cloudy, cornea or famy patches on white of the eye 6. Ears: discharge, red drums 7. Mouth: sores, thrush, cracked lips 8. Neck: enlarged lymph nodes 9. Chest: fast breathing, lower chest indrawing, wheeze 10. Heart: rate, rhythm and murmurs 11. Abdomen: masses, enlarged spleen 12. Legs: cedemia, muscle wasting 13. Temperature and pulse: check and record 14. Urine: test for protein and record 15. Weight: weigh and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Child has poor weight gain with any of the following: a) Severe wasting b) Oedema of hands and feet c) White spot or cloudy cornea d) Severe pallor	• Looks emaciated, edematous, weak, muscle wasting • White spot on cornea • Severe pallor	Severe condition Suspect severe malnutrition	1. Conduct Mid-Upper Arm Circumference (MUAC) 2. REFER immediately to the appropriate health facility.
II. Poor weight gain	• Not pale, active • Growth graph flat	Moderate condition Suspect Malnutrition	1. Give Tablet Pyrantel pamoate 3 tabs. Immediately (stat) 2. Tablet Folic acid 5mg daily for 2 weeks 3. Tablet Vitamin B- complex 1 tab 8hourly for 7days 4. Tablet Ferrous sulphate 200mg (60mg elemental iron) for 7 days 5. Give micronutrient powder if available (see annex for proper use) 6. Review weekly for one month and monthly subsequently 7. If gaining weight satisfactorily: a) Explain findings to mother b) Nutritional counselling 8. Arrange for: a) Home visit weekly b) Discuss with family 9. If no improvement, REFER appropriately.

III. Child continues to lose weight Failure to gain weight for more than one month, with or without loss of appetite	Weight loss mildly pale, active,	Moderate condition Suspect moderate acute malnutrition	1. Give multivitamin 2. Nutrition counselling 3. Give micro nutrient powder if available (see annex for proper use) 4. Ask of recent illness 5. Give Tablet Pyrantel Pamoate 3 tabs immediately (stat) 6. Review weekly for one month 7. If no improvement, REFER to the appropriate health facility.
IV. Child's weight is low for age on first child visit. (small for age) Child appears healthy	Alert and appears healthy	Mild condition	1. Explain findings to mother 2. Nutrition counselling 3. Give multivitamin 1 12hrly X 1 week 4. Give micronutrient powder if available (see annex for proper use) 5. Give Tablet Pyrantel pamoate 3 tabs immediately (stat)
V. Persistent loss of weight for 3 months with or without intermittent diarrhea, Sore mouth, skin rashes, may or may not have cough	. Child looks emaciated and miserable . Oral thrush . Skin rashes . Some dehydration . Enlarged lymph nodes.	Severe condition Suspect HIV/AIDS or TB	Treat opportunistic infections REFER for further investigation, HCTS

Health Education

Advise Parent / Caregiver on;

1.Proper Nutrition/Food Demonstration
2.Periodic deworming
3.Personal, food and Environment hygiene
4.Use of Long-lasting Insecticidal Nets (LLINs)
5.Regular clinic attendance and weight measurement
6.Importance of taking home-based records to the clinic or hospital

3.11 MEASLES-TYPE RASH

Measles is an acute viral infection characterized by the appearance of rashes in little groups made up of macules (flat spot) and papules (raised spots). It begins with high fever, cough, catarrh, and redness of eyes and mouth. The mouth may later become sore, rashes appear on the whole body after the fourth day of onset. The child may also get other serious illness like pneumonia, diarrhea, or malnutrition at the same time or soon after the child has had the disease. It is a serious illness because it can easily lead to complications especially in poorly nourished children.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child been ill? 2.Does the child have fever? 3.Does the child have cough? Is it a barking cough? 4.Is there runny nose? 5.What food is the child taking? Is the child refusing to feed? 6.Is there diarrhea? Vomiting? (If so, may be force -feeding) 7.Has any medicine or treatment been given? 8.Did the child receive measles vaccine? 9.Are there any other children in the family under 2 years who have not been immunized for measles? 10.Are there children in the neighborhood with the same complaint?	1.General appearance: alert, abnormally sleepy, or difficult to wake, unconscious, ill-looking or thin 2.Skin: rash, peeling, infected spots, hemorrhagic rashes 3.Chest: respiratory rate, lower chest indrawing wheeze, stridor 4.Eyes: red, purulent discharge or sensitive to light (eyes closed), redness 5.Ears: bulging discharging pus or fluid, perforation of eardrum, redness 6.Nose: runny, discharge 7.Mouth: a. sores, Koplik's spots, thrush b) throat-inflamed tonsils, purulent exudates 8.Hydration status: sunken eyes, dry mouth and skin 9.Weight: weight and chart 10.Weight: weigh and record 11.Conduct Rapid Diagnostic Test(RDT), if positive treat for uncomplicated malaria with Artemisinin-based Combination Therapies (ACTS)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Measles with white spot on cornea cloudy cornea poor vision	• Miserable child, ill looking eyes closed, measles rash+++, white spots on cornea, cloudy cornea. warm to touch Temp 37.5°c or above	Severe condition Suspect measles infection complicated by Xeropthalmia Vit. A Deficiency	1. Counsel on the importance of going to the hospital 2. Give Capsule Vit A 200,000IU 3. Calamine lotion when necessary (PRN) 4. Give Capsule Amoxicillin 250mg 8hourly for 7days 5. Instill Chloramphenicol eye ointment 6hourly 6. Review in 3 days, if no improvement REFER immediately.

II. Measles with tender swelling behind the ear with fever	Ill-looking, , Temp. above 37.50C, , tender swelling behind the ear	Severe condition Suspect mastoiditis	1. Give (ACT) Artemether-Lumefantrine (AL) 20/120mg • 6-8 years: 15-24 kg - 2 Tablets given at 0 hour, 8hours later, Then, 12hourly for 3 days • 9-12 years: 25-34 kg - 3Tablets given at 0 hour, 8hours later, Then, 12hourly for 3 days 2. Give Capsule Amoxicilline 250mg immediately (stat) 3. Tablet Vitamin C 100mg 8hourly for 7 days 4. Give Tablet Paracetamol 500mg immediately (stat) 5. REFER immediately to the appropriate health facility.
III. Measles with any of the following : Lower chest in drawing Convulsion Inability to drink Difficulty in sleeping, weakness	Very ill looking, irritable, Lower chest in-drawing, Abnormally sleepy Malnourished , Temp above 37.5°C	Severe condition , Suspect complicated Measles Suspect severe malaria	If mRDT is positive give pre-referral treatment for malaria IM Artesunate 2.4mg stat OR IM Quinine 10mg/kg stat OR IM Artemeter 3.2mg stat or Artesuanate suppository 3.REFER urgently.

IV. Measles with cough, runny nose and fever	• Warm to touch . Temp. 37.5°C and above. Eyes sore and sticky, Mouth sore- Koplik's spots. • Measles type rash	Severe condition Suspect Measles infection	1. Instill gentamicin eye drops 6hourly for 5 days. Teach mother how to apply 2. Gentle mouth care with a pinch of salt in water 3. Paint mouth with Gentian Violet 1 % , Dip a cotton swab into the gentian violet solution and sparingly apply it to the affected area in the mouth 8hourly. Avoid excessive application, as it can lead to staining of the mouth and surrounding tissue. 4. Daily bath with soap and water 5. Give Capsule Vitamin A supplement 200,000IU for 2 days 6. Give Tablet Paracetamol 500mg 8hourly for 2 days 7. Apply calamine lotion when necessary (PRN) 8. Nutritional and immunization counselling 9. Follow up next day and every other day until well 10. Look out for diarrhoea, difficult breathing and eye problems. If present, REFER immediately
v. Measles with cough, runny nose without fever	• Alert, runny nose, cough sore eyes, Koplik's spots, well nourished	Moderate condition Suspect Measles	1. Advise mother on the following 2. Importance of nutritious food (without force-feeding) 3. Gentle mouth care with a pinch of salt in water 4. Daily bath with soap and water 5. Give Capsule Vitamin A supplement 200,000IU for 2 days 6. Follow up next day and every other day until well 7. Manage other symptoms 8. Apply calamine lotion when necessary (PRN) 9. If no improvement, REFER to the appropriate health facility.

Health Education

Advise Parent / Caregiver on;

1.Proper Nutrition/Food Demonstration
2.Periodic deworming
3.Personal, food and Environment hygiene
4.Use of Long Lasting Insecticidal Nets (LLINs)
5.Regular clinic attendance and weight measurement
6.Importance of taking home-based records to the clinic or hospital

3.12 SKIN DISEASES

Skin diseases are conditions that affects the skin. It may be due to poor hygiene, allergy, malnutrition, poor housing condition as well as the environment. Some rashes can easily spread to other members of the family therefore, early referral and correct treatment is important for the welfare of the whole family. Some skin diseases affect the external skin (ringworm) or affect the skin internally (pyoderma). Others affect many parts of the body systems and are generally more important (measles, leprosy, diabetes, HIV/AIDS, jaundice). Others are scabies, eczema, tinea vesicular, vitiligo, athletes' foot, urticaria and contact.

HISTORY	EXAMINATION/ACTIONS
1.How long has the rash been there? 2.Does it itch? Is the child scratching? 3.Are there any other symptoms? Fever, cough, headaches, difficult breathing, sore throat? 4.Has the child eaten any unusual foods? 5.Does anyone else in the family have a similar rash or skin problem? 6.How often do you bathe the child? 7.Where and with whom does the child sleep? 8.What medicine (including injections or home treatment) has the child been given lately? 9.Has the child been passing excessive urine? 10.Has the child been having excessive thirst?	1.General appearance: thin, malnourished, distressed/ uncomfortable 2.Skin: a) location, distribution and size of lesions b)colour of rash or lesions c)lesions or rash: flat (macules), raised, pus or fluid –filled weeping, crusted, ulcerated, scratch marks or d)combination of (a) to (c) 3.Ears: red or discharging 4.Eyes: redness, discolouration of eyelids (dark -due to repeated scratching) 5.Mouth: redness, sores 6.Chest: respiratory rate, wheeze 7.Temperature: check and record 8. Urine; sugar, colour

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Painful boil-like blister particularly in the lower limbs, History of other members of the household with similar blister	• Non-tender, boil- like blister, • wWorm maybe seen	Moderate to severe condition Suspect Guinea Worm	1. Give Tablet Paracetamol 500mg 8hourly for 2 days 2. Clean and dress the wound with Cicatrin powder daily 3. Tablet Vitamin C 100-200mg 8hourly for 5 days 4. Gently wind worm on a stick 5. Ensure compliance with Health Education in conjunction with Village Head and health authority. 6. Give Capsule Amoxicillin 250 mg 6hourly for 5 days 7. Since eggs may survive, a second treatment may be necessary if client shows no clinical improvement after 1 week

II. Very itchy skin rash	• Small, raised red spots or pustular rash between fingers, wrists, buttocks and ankles	Moderate condition Suspect Scabies	1. Bathe daily with antiseptic soap for 1 week 2. Paint the entire body except the scalp, hair and face with Benzyl Benzoate lotion (25%) at night for 5 days 3. Additional weekly treatments are warranted only if live mites are present 4. All household contacts of clients should be promptly treated 5. Give Capsule Amoxicilline 250mg 6 hourly for 5 days 6. Decontamination of articles e.g. Advise client to wash clothing, beddings, towels in hot water, sundry and iron. 8. Review in 1 week.
III. History of painful lump or swelling on any part of the body.	Warm or tender mass	Mild to Moderate Suspect Boil	1. Give Capsule .Amoxicillin or Ampicillin/Cloxacillin 250 mg 6 hourly for 5 days. 2. Tablet Paracetamol 500mg 8 hourly for 3 days 3. Warm compress 3 times daily until ready to drain 4. Incise and drain abscess when fluctuant 5. Daily cleansing and dry dressing when back in the community
IV. Flat itchy, dry patches on scalp or trunk.	Scaly macule or light spots with raindrop pattern or active borders	Mild condition	1. Examine the rest of family for similar problems 2. Apply Whitefield's ointment for 4 to 6 weeks 3. if no improvement give Tablet Griseofulvin 125mg tab 8 hourly for 14 days 4. If no improvement Refer
V. Rashes, may or may not have fever, or look ill	Small blister-like rashes mostly on the trunk. May be warm to touch with runny nose. May or may not look ill	Mild to moderate Suspect Chicken pox	1. Give Tablet Paracetamol 500 to 1,000mg 6 hourly for 3 days 2. Give Capsule Vitamin A 200,000IU daily for 2 days 3. Apply calamine lotion 4. Encourage fluid intake especially fruit juice 5. Give Tablet Promethazine 12.5 mg stat then PRN 6. Review in 3 days, If no improve REFER to the appropriate health facility

VI. Severe itching with skin changes which disturbs sleep at night	Maybe warm to touch Apply calamine lotion with runny nose. May or may not look ill. Thickening dis-coloration of the skin	Mild condition	1.Investigate cause of rash 2.Tablet Chlorpheniramine(Piriton) 2 – 4 mg 12 hourly for 3 days 3.Apply calamine lotion. Review in 3 days
VII. Itching of the skin followed by raised red lesion which disappears after some time	Skin changes with severe itching especially at night with or without skin nodules (Noncholera nodules) Raised lesion of the skin, Scratch mark Small light patches not raised, not itchy on the face, back etc	Mild condition	1.Investigate cause of rash 2.Tablet Chlorpheniramine (Pirton) 2 – 4 mg 12 hourly for 3 days 3.Tablet Vitamin C 100 to 200 mg 8 hourly for 3 days 4.Review in 3 days, If no improvement Refer appropriately
VIII. Small light patches not raised, with or without itching	Small light patches not raised, not itchy, on the face, back etc.	Mild condition	1.Apply Miconazole ointment for 4-6 weeks 2.If itchy, give Tablet Chlorpheniramine 2 to 4mg 12 hourly for 3 days 3.Personal hygiene and care of clothing Review in 3 days
IX. History of very itchy scalp and pubic area	Lice present	Mild condition Suspect Hair/Pubic lice infestation	1.Examine the rest of the family for similar problem 2.Apply 100% Gamma Lorexane lotion or shampoo 3.Apply to scalp close to roots of hair. 4.Shave pubic hair 5.Wash and fine comb hair after 24 hours 6.Repeat treatment in one week

Health Education

Advise Parent / Caregiver on;

1.Personal hygiene; use of disinfectants
2.Discourage sharing of underwear and clothes
3.Importance of washing and ironing of clothes (emphasis on not spreading clothes on grass and other inappropriate places)
4.Proper Nutrition
5.Roles of water in skin infection (protection and treatment of water sources e.g ponds with insecticide - Abate)
6.Overcrowding and transmission of skin infections
7.Importance of taking home-based records to the clinic or hospital

3.13 OEDEMA

Oedema is the abnormal collection of fluid in the tissues resulting in the affected part of the body. It could be generalized or localized e.g. one foot or finger. Generalized oedema is most noticeable on the parts of the body which are lowest at the time e.g. legs when standing. Causes include congestive heart failure, severe anemia, kidney and liver diseases and kwashiorkor. Oedema may also be the result of reaction to toxins, drugs, and insect sting. Ascites, which is fluid in the abdominal cavity, may also occur.

HISTORY	EXAMINATION/ACTIONS
1. How long has the child had the swelling? 2. Has the child ever had this swelling before? 3. Is the swelling painful? 4. Was the child bitten by an insect? 5. Is there itching? 6. Is there cough? 7. Is there any problem with urination? 8. Does the child eat well? What foods? 9. Has any medication been given if yes, what? 10. Is there any other problem?	1. General appearance: pale, lethargic, puffy, distressed 2. Head: a) hair-color. Texture b) face-oedema, particularly around eyes 3. Chest: a) difficulty in breathing b) creeps or decreased breath sounds 4. Heart: murmur, rate 5. Eyes: pallor, jaundice 6. Abdomen: swelling, tenderness 7. Limbs: swelling, pitting oedema 8. Urine: color, cloudiness, protein, volume 9. Weight: weigh and record 10. Temperature, Pulse and Respiration: check and record 11. Hemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Swelling of limbs and/or face with yellowness of eyes	Jaundice, wheezing, stridor or fast breathing	Severe Condition Suspect Liver problem	Place in cardiac position REFER immediately
II. Swelling of legs, arms and/ or face	• Swollen feet (pitting) • and/or face • Protein or blood in urine	Moderate condition Suspect kidney problem	1. Tablet Furosemide 20mg daily for 3 days Review in 3 days, If no improvement, REFER
III. Swelling of limbs and/ or face with itching, with or without allergic reaction to toxins drugs, stings or some food	• Swollen legs, urticarial rash • itching	Moderate condition Suspect allergic reaction	1. If no respiratory distress, give IM. Hydrocortisone 50 - 100 mg stat immediately 2. Observe for 4-6 hours, 3. If no improvement, REFER
IV. Swelling of legs and/or face	Pallor, Swollen legs and face, breathlessness Fast heartbeat with or without fever Hb < 10g/dl	Moderate condition Suspect Anemic Heart Failure	1. REFER Immediately

Health Education

Advise Parent / Caregiver on;

1. Management of oedema (postural)
2. Relationship between infection, anemia and heart failure
3. Proper nutrition
4. Environmental sanitation
5. Importance of taking home-based records to the clinic or hospital

3.14 BURNS AND SCALDS

Burns are injuries to the body (skin) caused by fire or hot liquid. Burns are dangerous because it could lead to severe pains and shock resulting from loss of large amount of body fluids. The area of body surface affected is important in deciding the degree of a burn. The chart below shows approximately the proportion of total body surface area affected.

S/N	Rule of nines for burns (William, 2023)	Percentage of Body Surface
1.	Head and neck	9%
2.	Chest	9%
3.	Abdomen	9%
4.	Upper back	9 %
5.	Lower back	9%
6.	Right arm	9%
7.	Left arm	9%
8.	Genital area	1%
9.	Left leg	18%
10.	Right leg	18%
	Total	100%

HISTORY	EXAMINATION/ACTIONS
1. How did the child get burnt? And with what? 2. When did it happen? 3. Has any mediation been given or applied since then? 4. Has the child been urinating since the burn? 5. Has the child been immunized against tetanus?	1. General appearance; restless, distressed, dehydrated, unconscious 2. Temperature, Pulse and Respiration: check and record 3. Chest: respiratory distress, wheezing, stridor 4. Skin: a) size and area of burns, blisters, leathery areas with no sensitivity to pin-prick b) signs of infections-pus, redness, warmth

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Burns / scalds	Burns of 9% or more with any of the following involved: Face, hands, groin or joints, Lethargic, without sensation or hoarse voice, has not passed urine since burn and sign of shock (weak, thready pulse and cold extremities, low BP	Severe condition Suspect Burns with imminent shock	1. Give IM. Tetanus Toxoid (T.T.) 0.5ml, if not recently immunized 2. REFER immediately 3. If referral is difficult, give Ringers Lactate I.V. one third of the total amount (500ml) 8 hourly and Darrow's half strength with 5% dextrose 4. Tablet Ascorbic acid (Vitamin C) 200mg 8 hourly for x 7 days 5. Apply Vaseline gauze 6. Give IM. Amoxicillin 250mg 6 hourly for 2 days, then convert to oral 7. DO NOT GIVE ORAL ANALGESIC, IF BURNS ABOVE 10%

II. Burns / scalds	Less than 9%; no blisters, lethargic; no involvement of face, hands, groin or joints	Moderate condition	1. Vaseline gauze dressing 2. Tablet Phenobarbitone 30mg 12hourly for 3 days 3. Tablet Paracetamol 500mg 8hourly for 3 days 4. IM. Tetanus Toxoid (T.T.) 0.5ml, if not previously immunized in the last 18 months; Review daily until well 5. Tablet Ascorbic acid (Vitamin C) 100mg 8 hourly for 7days
III. Burns/ scalds	Less than 9% with blisters, no lethargy, involvement: face, hands, groin or joints	Moderate condition	1. Paint with Genettian violet; Vaseline gauze dressing 2. Tablet Phenobarbitone 30mg 12 hourly for 3days 3. Tablet Paracetamol 500mg 8 hourly for 3 days 4. Capsule Amoxicillin 250mg 6 hourly for 5 days 5. Tablet Ascorbic acid (Vitamin C) 100-200mg 8 hourly for 7days 6. IM. Tetanus Texoid (T.T.) 0.5ml stat immediately; if immunized in the last 18 months 7. Oral Rehydration Salt (ORS) Liberally 8. Review daily until well

Health Education

Advise Parent / Caregiver on;

1.Prevention of burns/ scalding
2.Personal hygiene
3.Proper Nutrition.
4.Prevention of burns.
5.Compliance with treatment.
6.Importance of taking home-based records to the clinic or hospital

3.15 WOUNDS, SORES, ULCERS, FRACTURES AND VARIOUS BITES/STINGS

Children can have wounds, sores, ulcers and various bites similar to adults. They are prone to accidents both in the home and outside the home particularly when they go to school.

HISTORY	EXAMINATION/ACTIONS
1. How did it happen? 2. When did it happen? 3. Was there any bleeding? If yes, how much? 4. What have you applied on the wound? 5. Can the child move the area? What immunization has the child taken?	1. General appearance: in pain 2. Signs of shock: cold, clammy, pale, 3. Wound: size, bleeding, dirt, or foreign body, pus, swelling, redness. 4. Temperature, Pulse and Respiration: check and record. 5. Haemoglobin: Check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Injury, Pain	Wound resulting from injury. Tenderness over wound or sore area Evidence of fracture unable to move the limb Lethargic Wound more than 4-6cm	Severe condition Suspect Fracture	1. If bleeding, stop by applying pressure 2. Immobilize limb (splint) 3. Give IM. TT 0.5ml, if not immunized 4. Give Cap. Amoxicillin 250mg 8hourly for 5 days 5. Tab. Ibuprofen 200mg 8hourly for 3 days 6. Tab. Vitamin C 100- 200 mg 8hourly for 7days 7. REFER appropriately
II. Wound as a result of injury	Laceration Bleeding Unable to move the limb Wound more than 4cm No evidence of fracture	Severe condition Suspect lacerated wound	1. Clean with available antiseptic lotion 2. Stop bleeding by pressure 3. Suture and dress wound 4. Give IM. Tetanus Toxoid (T.T) 0.5ml immediately, if not immunized, if wound is less than 24hours, apply sutures 5. Give Capsule Amoxicillin 250mg 6hourly for 5 days 6. Tablet Paracetamol 500mg 8hourly for 3days 7. Tablet Vitamin C 100-200mg 8hourly for 7days 8. Review the next day then at 1 week to remove suture 9. Ask to return in 6 weeks for 2nd dose of Tetanus Toxoid (T.T)

III. Wound as a result of injury	Laceration Less than 4cm with no sign of shock, No infection	Moderate condition Suspect lacerated wound	1. Clean with available antiseptic lotion 2. Stop bleeding by pressure 3. Give IM. Tetanus Toxoid (T.T.) 0.5ml, if not immunized, if wound is less than 24hours, sutures, 4. Give Capsule Amoxicillin 250mg 6hourly for 5 days 5. Tablet Paracetamol 500mg 8hourly for 3 days 6. Tablet Vitamin C 100 -200mg 8hourly for 7 days. 7. Review the next day then at 1 week to remove sutures 8. Ask to return in 6 weeks for 2nd dose of T.T
IV. Wound as a result of injury	Laceration Bleeding	Mild condition	1. Pull pressure on site for 10 minutes, if slowing, observe and maintain pressure for 1 2. If bleeding does not stop and wound is less than 24hrs, suture 3. If more than 24hrs, Clean with available antiseptic lotion 4. Stop bleeding by pressure 5. Give IM. Tetanus Toxoid (T.T.) 0.5ml, if not immunized 6. IM. Amoxicillin 250mg stat then capsule 250mg 6hourly for 5 days 7. Tablet Paracetamol 500mg 8hourly for 3days 8. Tablet Vitamin C 100-200mg 8hourly for 7 days. 9. Review the next day then at 1 week to remove sutures 9. Ask to return in 6 weeks for 2nd dose of T.T
V. Bleeding from circumcision	Bleeding	Moderate condition Suspect haemophilia	1. Put pressure on site for 10 mins 2. If slowing down, observe and maintain pressure for 1 - 2hrs 3. If bleeding does not stop, REFER immediately 4. Teach mother how to apply pressure, enroute referral facility

VI. Dog Bite	• Evidence of bite	Moderate condition	1. Wash wound with soap under running water 2. Advise parents to observe dog for 10 days and report any changes in the dog's behavior or death 3. Anti-Rabies injection according to manufacturers instructions, if not available REFER 4. Give IM. Tetanus Toxoid (T.T.) 0.5ml, if not previously immunized 5. Dress wound daily, 6. If dog is alive after 10 days. Stop anti-rabies injection 7. Review daily
VII. Snake bite	Wound at site of bite	Severe condition Suspect venomous snake poisoning	1. Apply crepe bandage above site, but do not use as tourniquet 2. Keep client, calm and clean the area thoroughly with copious soap water 3. Give anti-snake venom (read and follow manufacturers instruction) 4. If not available, REFER 5. Give IM. Hydrocortisone 50-100mg immediately 6. Give IM. Tetanus Toxoid (T.T) 0.5ml if not immunized 7. Give Capsule Amoxicillin 250mg 6hourly for 5 days 8. Tablet Paracetamol 500mg 8hourly for 3 days 9. Review in 5 days 10. If there is complications such as bleeding from orifices or Cardiovascular manifestation, REFER immediately
VIII. Scorpion sting	Evidence of sting (Painful Swelling, itching and may be red)	Moderate condition Scorpion sting	1. Tablet Paracetamol 500-1000mg 8hourly for 3 days 2. Lignocaine Sec stat directly on the site of sting 3. Cold compress 4. Review within 24hours
IX. Human bite	Wound at the site of the bite	Moderate condition	1. Clean wound well 2. If more than 4cm suture 3. Give Capsule Ampicillin/Cloxacillin 250mg 6hourly for 5 days 4. Dress daily 5. Review after 3 days

Health Education

Advise Parent / Caregiver on;

1.Protection from bites/stings.
2.Personal hygiene.
3.Importance of compliance with treatment.
4.Proper nutrition.
5.Environmental sanitation.
6.Importance of taking home-based records to the clinic or hospital.

3.16 HEAD DISORDERS

The head contains the brain, the organs for the sense of hearing, seeing, smelling and tasting. The bones of the head are very strong but sometimes a serious head injury may be caused by a fall or a blow to the head. Other head problems include infestation with lice, dandruffs and headaches. Injuries and infection to the head are of great danger because of the large blood vessels and nerves which is contains and the possibility of damage to the brain.

Some signs of serious injury are bleeding from the nose, mouth or ears, severe headache or unconsciousness. There also may be vomiting or the client may be confused. Any delay in referring a client with head injury for proper medical treatment may cause permanent injury or death.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Has he fallen or hit his head? When? 3.If he has had an injury, has he been drowsy? 4.Has he vomited? How many times? 5.Does he have a headache? 6.Has he convulsed? 7. Is he playing and behaving normally for his age as he usually does?	1. General appearance and behavior dull, abnormally, sleepy, or difficulty to wake, restless, delirious or unconscious 2. Head: a) circumference, shape b) external injuries, sores, or rashes c) swelling or depressions 3. N eek: stiffness 4. Eyes: hemorrhage, pupils' size (equal or unequal), reaction to light, vision, jaundice 5. Gait: steady or unsteady 6. Temperature, Pulse and Respiration: check and record 7. Hemoglobin: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. History of trauma to head, abnormally sleepy or difficult to wake ,vomiting	Pupil abnormal size or not reacting to light. Fracture, vomiting Bleeding from ear or nose	Severe Condition Suspect fracture of the base of the skull	1.Immobilize the patients head and neck 2. Refer immediately
II. Headache with vomiting, fever neck stiffness	• Warm to touch (temp . 37.50(or more) • Stiff neck	Severe condition	1. IM. Benzyl Penicillin 3cc (300,000 unit) stat 2. Refer immediately
III. Head with a small Swelling	• Injury or swelling abnormal behavior • No drowsiness	Moderate condition Suspect moderate head trauma	1. Explain findings to mother 2. Tablet Paracetamol 500mg 8hourly for 3 days 3. Apply cold compress 4. Treat local injury 5. Review after 24 hours 6. Advise mother to return child if worse, become drowsy, start vomiting show any strange behavior, If no improvement, REFER
IV. Bulging forehead	Bossing of forehead	Mild condition Suspect sickle Cell disease	1. Tablet Folic acid 5mg daily for 2 weeks 2. Tablet Paracetamol 500mg 8hourly for 3 days 3. Review in 5 days, if no improvement, refer

Health Education

Advise Parent / Caregiver on;

1.Prevention of burns/ scalding.
2.Personal hygiene.
3.Proper Nutrition.
4.Prevention of burns.
5.Compliance with treatment.
6.Importance of taking home-based records to the clinic or hospital.

3.17. NECK DISORDERS

Neck problem is the presence of enlarged nodes and lumps around the neck. There could also be stiffness of neck. These are usually caused by infection of ears, mouth or on the scalp, tuberculosis, leukemia (cancer of the blood), meningitis, other childhood malignancies, swollen neck glands or goiter.

HISTORY	EXAMINATION/ACTIONS
1. How long has the child had the problem? 3. Is there weight loss or loss of appetite? 4. Is there fever or sore throat? 5. Is there excessive sweating at night? 6. Is there headache? Earache? 7. Is there pain or vomiting? 8. Is the child restless of hyperactive? 9. Has the child convulsed?	1. General appearance: pale, thin, or wasted 2. Head: sore or abscess 3. Mouth: a) condition of teeth b) sores c) tonsils 4. Ears: discharge, red or bulging or perforated ear drum 5. Neck: stiffness, nodes, lumps, glands 6. Chest: stridor 7. Temperature, Pulse and Respiration: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Stiffness of neck with history of a fall and no Fever	Stiff neck	Severe condition	1. Support the neck 2. IM. Paracetamol 300mg immediately(stat) 3. Refer to appropriate health facility.
II. Neck pain with Fever, with or without vomiting, no history of neck injury	• Stiff neck Temp. 37.5°C or above, vomiting, Kerning's sign positive	Severe condition Suspect Meningitis	1. If Kerning's sings is positive, IM Benzyl Penicillin 3cc (300,000IU) immediately (stat) 2. REFER immediately to appropriate health facility
III. Neck pain with no Fever	Lump and Nodes present, no tenderness, deformities, swelling,	Moderate condition Suspect tumour	1. REFER immediately to the appropriate health facility
IV. Neck pain with Fever	Tender lumps or nodes Warm to touch (Temp 37.50.c or above) kernig sign positive	Moderate condition	1. Give Capsule Amoxicillin 250mg 6hourly for 5 days 2. TabletParacetamol500mg 8hourly for 3 days 3. Review in 3 days, if no improvement, REFER immediately to the appropriate health facility
V. Swelling of both sides of the neck.	Enlarged salivary gland behind the jaw under the chin. Testicles may or may not be involved (swollen and/or painful)	Mild condition Suspect mumps	1.Encourage fluids 2.Saline mouth wash, Explain findings to mother, if male, advise on rest 3.Tablet Vitamin C 100 – 200mg 8hourly for 7days 4.Review in 3 days, If no improvement, refer to the appropriate health facility

Health Education

Advise Parent / Caregiver on;

1.Prevention of neck injuries
2.Prevention of transmission to others
3.Complications of mumps in males
4.Oral hygiene
5.Importance of taking home-based records to the clinic or hospital

3.18 EYE DISORDERS

Eye problems, especially redness and discharge are commonly caused by injuries, chemicals e.g soap and infections. The health worker must however be careful not to treat all eye conditions as if they were caused by infection. There are other more serious conditions such as pain and problems with vision which is missed, may lead to blindness e.g. lack of Vitamin A. visual problems require correction. Cataract occurring in children may be a manifestation of Rubella infection.

HISTORY	EXAMINATION/ACTIONS
1. What is the matter with his eyes? 2. How long has he had the problem? 3. Has he had it before? 4. Did the problem come suddenly or gradually? 5. Is there a great deal of pain in the eye? 6. Does he rub eye or hold it? 7. Did anything enter his eye? 8. Did the mother or anyone else put something in his eye? What was it? 9. Has any medicine been put in his eye? If so, what medicine? 10. Does the child have runny nose and cough? (think of measles) 11. Does the child have severe prolonged bouts of coughing (think of whopping cough) 12. Does anyone else in the family have the problem? 13. Is he receiving treatment for any other disease? 14. Does he take palm oil or eat carrots, Fresh mangoes, pawpaw, tomatoes? 15. Do you think the child sees?	1. General appearance: in pain, ill, thin, wasted 2. Skin: rashes, scratch marks? 3. Eyes: a) Redness, where is the redness: (around the edges, in a) circle at a colored part, all over or bleeding into white part of eye?) b) tears c) swelling around the eye(s) d) pus or other discharge e) pupils: (a) normal or small in size, equal f) protruding g) squint h) Cloudiness or whiteness of pupil(s) i) Test of vision j)Check if the child can follow a bright object, or reach out to receive an object from you without assistance 4. Nose: discharge

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Injury to eye ball with sharp object	Injured white part of eye(s)	Severe condition	1. Apply eye pad 2. IM. Gentamicin 3mg/kg immediately (stat) 3. Instill Gentamicin eye drops 6hourly to affected eye 4. Refer immediately to see an Ophthalmologist in the hospital
II. Injury to eye ball as a result of burns, Hot water or chemicals	Painful watery eyes	Moderate to Severe condition	1. Irrigate eye(s) with saline water immediately 2. Instill Gentamicin eye drops 3.Reassure parents/ caregivers 4. Refer immediately to hospital
III. Complaint of abnormalities of the eyes	• Injury or swell• Drooping of eye lids (ptosis) • Eyes protruding (pushing out) • Eyes watery or discharging • Painful, Red eyes, with or without visual loss • Cataract (Growth of white small patches which do not react to light) poor vision (less than 6/9 visual acuity)ing abnormal behavior • No drowsiness	Moderate to severe condition	1. Explain condition to parents/ caregiver 2. REFER immediately to an Ophthalmologist in a hospital

V. Foreign body in the eye with pain	Watery eyes; Restlessness, May see foreign body, impaired vision, eyes may be red.	Moderate condition	1. Irrigate with Saline 2. Tablet Paracetamol 500mg 8hourly for 3days 3. Instill Gentamicin eye drops to affected eye 6hourly 4. Cover eye with pad; 5. Refer immediately to appropriate facility
VI. White bubbly material at corner of eye or dry eye(s)	White bubbly material seen at corner of eye(s)	Mild condition	1. Capsule Vitamin A 200,000IU for 2 days 2. Encourage consumption of colored (yellow /red) fruits, green leafy, vegetables and palm oil 3. Refer to hospital for proper assessment
VII. Sticky eye(s) or red eye(s); no pain, normal pupils and vision	Sticky yes; No pain; Normal vision	Mild condition	1. Saline bath to eye 3 times a day 2. Instill Chloramphenicol eye drops 8hourly to affected eye 3. Review in 3 days If no improvement, Refer to hospital

Health Education

1. Personal food and environmental Hygiene
2. Prevention of eye injuries
3. Eye care
4. Nutrition Counselling
5. Regular eye checkup
6. Importance of taking home-based records to the clinic or hospital

3.19 EAR, NOSE AND THROAT DISORDERS

Ear Disorders

Ear problems are disease affecting the ear and are commonly caused by middle ear infection; usually with pain in the ear, dullness or redness of the ear drum sometimes with perforation and drainage of pus or fluid. It is important to examine the bone behind the ear, because if this is tender to touch, it may mean that the child has mastoiditis. This is danger sign. The other major complication is deafness which can be tested by clapping your hands behind the child to see if he will turn and look.

Nose Disorders

The nose is the organ for breathing and for the sense of smell. Disease of injury in the throat or sinuses can affect the nose. This could bring about watery discharge, mucous blood or pus from the nose. Also, injury to the nose can affect the throat. The health worker should think of a foreign object if there is discharge from only one side of the nose.

Throat Disorders

A red throat may be caused by infection. It may be combined with nasal congestion and middle ear infection especially in young child. The tonsils, normally enlarge during childhood, but unless they show sign of difficult breathing or inflammation, no treatment is needed.

HISTORY	EXAMINATION/ACTIONS
2. Does the child have hearing problems? 3. Is there any discharge from the ear? 4. Has the child had any ear surgery? 5. Is there any family history of deafness? 6. Does the child have runny nose? 7. Is there any bleeding from the nose? If yes, 8. Has the child had it before? 9. Is the child able to swallow food or fluids? 10. Has the child received any injection or medication recently? If yes, what is the name? (antibiotics (gentamicin), diuretics, 11. cyst 12. Has the child been exposed to loud noises Exposure to noise (e.g. pneumatic drill or sound) 13. What treatment have you given for the child's condition?	1. General appearance: well, ill-looking, pale, irritable 2. Nose: watery discharge, mucus, blood, pus, any visible swelling, deformity, or foreign 3. Throat: pus or redness, enlargement of tonsils 4. Head: tenderness over bone behind the ear (mastoid) 5. Ear: a) condition of canal b) pus or blood, perforated drum c) foreign body d) red or bulging drum 6. Test for hearing: clap 7.Neck: stiffness Throat 8. Temperature, Pulse and Respiration: check and record 9.Conduct mRDT, if positive, treat for malaria

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Ear pain with history of foreign body with bleeding or Neck Stiffness	Bleeding from ear Stiff neck Dull ear drum Warm to touch (Temp > 37.50C)	Severe condition Suspect Ear problem	1. IM. Diclofenac sodium 30-50mg stat 2. Give first dose of IM Amoxicillin 250mg immediately (stat) 3. Refer Immediately to the appropriate health facility
II. Pain behind the Ear	Tender swelling behind the ear Dull ear drum Warm to touch (Temp > 37.50C)	Severe condition Suspect mastoiditis	1. IM. Amoxicillin 250mg immediately (stat) 2. Tablet Paracetamol 500mg 8hourly for 3 days 3. REFER appropriately to the health facility

III. Ear pain with discharge for less than 14 days with or without fever	Pus draining from the Ear Dull ear drum with or without perforation Warm to touch (Temp > 37.50C)	Moderate condition Suspect acute ear infection	1. Conduct mRDT, If positive treat with (AA) 2. Capsule Amoxicillin 250mg 8hourly for 5 days 3. Tablet Paracetamol 500mg 8hourly for 3 days 4. Tablet Vitamin C 100mg 8hourly for 5 days 5. Review in 5 days, If no improvement, REFER to the appropriate facility
IV. Ear pain with discharge for more than 14 days with or without fever	Pus is seen draining from the ear. Dull ear drum Warm to touch (Temp > 37.5°C)	Moderate condition Suspect Chronic Ear Infection	2. Capsule Amoxicillin 250mg 8hourly for 5days 1. Explain condition to mother 3. Tablet Paracetamol 500mg 8hourly for 3 days 4. Review in 3 days, If no improvement, REFER to the appropriate facility
V. Boil in ear	Boil seen	Moderate condition	1. Capsule Amoxicillin 250mg 8hourly for 5 days 2. Tablet Paracetamol 500mg 8hourly for 3 days 3. Review in 3 days, If no improvement, REFER to the appropriate facility
VI. Difficulty in Swallowing	Grey and dirty membrane on tonsil Warm to touch (Temp. > 37.5°C)	Severe condition Suspect tonsillitis	1. IM Amoxycillin 250mg immediately (stat.) continue with Oral Capsule Amoxicillin 250mg 8hourly for 5 days 2. REFER appropriately to the appropriate facility
VII. Pain in the throat difficulty with swallowing	Enlarged red tonsils with pussy exudate with or without tender lymph nodes in the neck	Moderate condition Suspect Tonsillitis	1. Give Capsule Amoxycillin 250mg 8hourly x 5 days 2. TabletVitamin C 100mg 8hourly for days 3. Tablet Paracetamol 500mg 8hourly for 3 days 4. Gargle with salt water 6hourly daily if able 5. Review in 3 days, If no improvement, REFER to the appropriately facility
VIII. Pain on either side of the nose and Headache	Pus from nose tenderness Smelly discharge	Moderate condition Suspect acute sinusities	1. Tablet Co-trimoxazole 480mg 12hourly for 5 daysOR Capsule Amoxicillin 250mg 8hourly for 5 days 2. Tablet Paracetamol 500mg 8hourly 3 days 3. Tablet Chlorpheniramine (Pirton) 1-2mg 12hourly for 3 days 4. Tablet Vitamin C 100-200mg 8hourly for 7 days 5. Review in 3 days, If no improvement refer to the appropriate facility

IX. Bleeding from nose	Evidence of bleeding	Moderate condition Suspect Epistaxis	1. Teach mother to pinch side of the nose intermittently for 15minutes 2. Apply Adrenaline with gauze onto the nose 3. Apply Cold Compress to nose for 30minutes and re - examine, If no improvement, REFER to the appropriate facility
X. Stuffy nose, with or without Fever; Sneezing with or without watery eyes, exposure to dust or pollen	Watery eye, Catarrh Warm to touch (Temp. > 37.50C)	Moderate condition	1. Conduct Rapid Diagnostic Test (RDT), if positive, treat for uncomplicated malaria 2. Tablet Chlorpheniramine (Piriton) 2mg 12hourly for3 days 3. Tablet Paracetamol 500mg 8hourly for 3 days 4. Tablet Vitamin C 100 -200mg 8hourly for7 days 5. Review in 3 days
XI. Pain in the throat, Fever, inability to swallow	Sore throat with or without redness Tonsils not inflamed and no pus Warm to touch (Temp. > 37.50C)	Moderate condition	1.Give Capsule Ampiclox 250mg 6hourly for 5 days 2. Tablet Paracetamol 500mg 8hourly for3 days 3. Tablet Vitamin C 100-200mg 8hourly for 7 days 4. Take fruits
XII. Boil in nose	Boil can be seen	Mild condition	1. Capsule Ampicillin 250mg 8hourly for 5 days 2. Tablet Paracetamol 500mg 8hourly 5 days 3. Tablet Vitamin C 100-200mg 8hourly for 5 days 4. Review in 3 days, if no improvement, REFER to the appropriate facility
XIII. Foreign object or Polyp in the nose	Foreign object seen	Moderate condition Foreign object	1. Try to remove foreign body with blunt forceps 2. If not removable, REFER to the appropriate facility

Health Education

Advise Parent / Caregiver on;

1. Prevention of infections
2. Care of the ear and nose
3. Management of throat problems (gargle and discourage uvulectomy)
4. Proper Nutrition
5. Personal hygiene
6. Prevention of exposure of air pollution (fumes, dust and pollen)
7. Discouragement of home removal of foreign body
8. General child care
9. Encourage intake of fruits
10. Importance of taking home-based records to the clinic or hospital

3.20 ORAL HEALTH PROBLEMS

Oral health is a state of being free from chronic mouth and facial pain, oral and throat cancer, oral sores, birth defects such as cleft lip and palate, periodontal (gum) disease, tooth decay and tooth loss, and other diseases and disorders that affect the oral cavity.

The most common complaint is pain, which may or may not be accompanied by swelling inside the mouth or on the face. The condition of the mouth is also a reflection of the client's general health, so it should be inspected as part of a physical examination. Early recognition of oral problems, with prompt attention or referral, will reduce morbidity and mortality arising from otherwise simple disease conditions.

HISTORY	EXAMINATION/ACTIONS
1.Where is the problem? 2.Is there pain from teeth or gum? 3.How long has it been noticed or present 4.Are you able to eat your regular diet? 5.Does the pain increase with or cold drink or food? 6.Does the pain keep you awake at night? 7.Is there any bleeding from gum or other parts of the mouth? 8.Do you notice foul or bad breath 9.What do you use for cleaning your teeth? 10.How do you clean your teeth? How many times a day? Do you notice any sore in the mouth?	1.General appearance: ill, facial swelling, malnourished emaciated 2.Mouth a). sore: swelling, discharging, bleeding, white patches b). tongue: sore, coating, redness, fixtures, smoothness, dryness d)lips: cracks 3.Teeth: carries discoloration, redness, mobility, missing, plaque Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Slow or fast growing swelling of the jaws with or without pain	Hard or Soft growth in the jaws or soft tissue	Severe condition Suspect Burkitt's lymphoma or other swellings	1. Need to relief pain, give Paracetamol 500- 100mg immediately stat 2 Refer to the appropriate facility.
II. Ulcers in the mouth. Oral thrush, weight loss, bleeding gums with or without Skin rash	Oral thrush, Bleeding gum	Suspect HIV/AIDS	Refer to the appropriate facility.
III. Injury to mouth and /or face	Tooth missing. Tooth broken Laceration, Mobile teeth, bleeding	Moderate to severe condition	1. Arrest bleeding by pressure 2. Tablet Paracetamol 500-1000mg immediately (stat) 3. Refer to the appropriate facility
IV. Mouth Ulcer with fever with or without general weakness.	Vesicle on tongue. Mouth Ulcers	Moderate condition: Suspect Oral Herpes	1. Ensure adequate fluid 2. Tablet Paracetamol 500mg 8hourly for 2 days 3. Capsule Ampicillin 250mg 6hourly for 5 days 4. Tablet Metronidazole 200mg 8hourly for 3 days 5. Tablet Vitamin C) 100-200mg 8hourly for 14 days 6. Review in 5 days If no improvement, refer to the appropriate facility

V. Cracks at the corner of the mouth	Tongue is smooth and red. Stinging sensation in the tongue Whitish cracks at corner of the mouth, tends to be bilateral.	Moderate condition: Suspect Angular Stomatitis, Avitaminosis	1. Tablet Vit. B Complex 8hourly for 2weeks 2. Tablet Vitamin 100-200mg 8hourly for 2weeks 3. Tablet Multivite 1 daily for 2weeks 4. Advise on diet 5. Oral hygiene instruction
VI. Pains in the mouth that keeps patient awake at night with or without jaw	Holes in tooth/teeth (Caries)	Moderate condition Suspect dental carries	1. Tablet Paracetamol 500-1000mg 8hourly for 2 days 2. Gargle salt water 3. Cap. Ampicillin 250mg 6hourly for 5 days 4. Tab. Metronidazole 200mg 8hourly for 3 days 5. Refer to the appropriate facility
VII. Severe pain in the mouth that, keeps patient awake at night with or without jaw swelling	• Holes in tooth/teeth (Caries)	Mild condition Suspect dental caries	1. Tablet Paracetamol 500-1000mg 8hourlyfor 2 days 2. Capsule Ampicillin 250mg 6hourly for5 days 3. Tablet Metronidazole 200mg 8hourly for 3days, Refer to the appropriate facility
VIII. Sharp pain in the tooth, Pain elicited by hot/cold foods	Holes in tooth/teeth	Mild condition	1. Explain condition 2. Refer to the appropriate facility
IX. Overlaping teeth	Teeth not well arranged	Mild condition	1. Refer to the appropriate facility

Health Education

1 Oral hygiene, emphasizing proper utilization of cleaning materials e.g brush, chewing stick etc.
2. Nutrition counselling
3. Discouragement of uvulectomy
4. Prevention of accident
5. Importance of taking home-based records to the clinic or hospital

3.21 CHEST/HEART DISORDERS

Very rapid .low or irregular heart rate is always a serious sign. Blueness of the lips or fingers, clubbing of the fingers with difficult breathing are sign that the heart is not able to perform its functions effectively. Congestive heart failure occurs when the heart becomes too weak to pump all the blood around and some of fluid flow back into the lungs and retained in the abdomen and legs .Any history of shortness of breath should make health worker think of a heart disease.

HISTORY	EXAMINATION/ACTIONS
1.How long have you had the problem ? 2.Do you have cough, chest pain or shortness of breadth? If pain where. Does deep breathing or bending make it worse? 3.Can you sleep flat or do you need several pillows at night? 4.Do you sometimes have to get up at night to breathe? 5.If cough is present, do you bring up sputum? Color? Any blood in sputum? 6.Do you eat well? 7.Have you had any night sweating or weight loss 8.Does the pain / discomfort start on exertion, made worse by exertion or even come at rest? 9.Are you on any medication? If yes, what?	1.General Appearance: ill, in pain, thin, pale, cyanosed 2.Neck: nodes 3.Chest: a)Respiration for rate, difficulty in breathing b)Crepitationess, rhonchi, wheezes, decreased breadth sounds c)Heart: rate, rhythm, murmur 4.Abdomen: tender liver, distension 5.Legs: Oedema 6.Temperature: pulse, check and record 7.Blood pressure: measure and record 8.Conduct mRDT

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Shortness of breath, easily tired, blue lips or finger nails	• Coughing, • Swelling of limbs, • Fast breathing, • More comfortable sitting up, • Blueness of lips and finger nails, Fatigue	Severe condition Suspect congestive} heart problem	1. Refer to health facility immediately in propped up position (cardiac position)
II. Easily tired, not playing with friends, squats	• Slow or fast heart rate, • Cough, • Blueness of lips and finger mails.	Severe condition Suspect congenital heart problem	1. Refer to the appropriate health facility immediately preferably in a propped up position
HI. Easily tired with any of the following: fever, or infection IV.Weakness, tires easily	• Fast heart rate • Normal Heart rate • 4 - 8years - 90/min • Warm to touch (Temp • 37.5°C or above) Heart murmur Hb <110/dl	Moderate condition Suspect Infective Heart Disease Moderate condition Suspect anemic heart failure	1. If Rapid Diagnostic Test (RDT) is positive, treat for uncomplicated malaria 2. Capsule Amoxicillin 250mg 8hourly for 5days 3. Tablet Paracetamol 500mg 8hourly for 3 days 4. Tablet Salbutamol 4mg 12hourly for 3 days 5. Review in 3 days, If no improvement, REFER to the appropriately

Health Education

Advise parent /Caregiver on:

1. Adequate rest
2. Proper care of person with chest, heart problem (keep warm and allow for adequate rest)
3. Adequate nutrition
4. Regular medical examination
5. Importance of taking home-based records to the clinic or hospital.

3.22 ABDOMINAL DISORDERS (INCLUDING HERNIAS)

The common causes of abdominal pain are gastro-enteritis, malaria, appendicitis, rupture of the intestine, typhoid, sickle cell crisis and obstruction, e.g. strangulated inguinal and umbilical hernias. Constipation, worm infestation may also cause abdominal pain. Abdominal pain may lead to hardness or rigidity of the abdominal wall.

An abdominal hernia is when a part of the gastro-intestinal tract protrudes through a gap or weakness in the abdominal wall. Hernia can be found on the umbilicus, groin and scrotum

HISTORY	EXAMINATION/ACTIONS
1. Is there vomiting? How many times? 2. Is there constipation? When was the last stool? 3. Is there any abdominal pain? 4. Has he/she had the problem before? 5. Has any medication been given?	1. General appearance: pallor, weakness. distress 2. Respiration. difficult or rapid 3. Abdomen: a) enlarged, distended b) masses, enlarged spleen visible intestinal movement c) presence of hernia tenderness 4. Hydration status: fontanelle, eyes, mouth, tongue, skin 5. Hemoglobin: estimate and record 6. Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Abdominal pain; Not passing stool	Child ill looking, crying and in pain, abdominal rigidity.	Severe condition Suspect Abdominal Obstruction	1. REFER immediately.
II. Abdominal pain with rigidity; vomiting but no diarrhoea	Child in pain and very uncomfortable, tender abdomen	Moderate condition Suspect Worm Infestation	1. Give Tablet Pyrantel pamoate 3 tabs immediately (stat) 2. Tablet Hyoscine 10 mg 8hourly for 2days 3. Advice on consumption of fruits and vegetables 4. Review in 2 days, If no improvement, REFER appropriately to the health facility
III. Abdominal pain with Constipation;	Well child; swelling of abdomen increased bowel sounds	Moderate condition Suspect Constipation	1. Review child's diet and fluids; 2. Advise fruits and vegetables 3. Give Liquid paraffin 5 - 10mls daily for 2 days, If no mprovement in 2 days, REFER to the appropriate health facility
IV. Abdominal pain with swelling on left side of abdomen with or without history of injury.	Ill looking, Pale, • Enlarged tender mass with painful swelling on the left upper part of the abdomen	Severe condition Suspect injury to spleen	1. REFER Immediately to the appropriate health facility
V. Abdominal pain with bloody stools; Strains in passing stool	Pale looking; blood in stools, Tenesmus on passing stool	Moderate to Severe condition Suspect; Dysentery, Amoebiasis or Schistosomiasis	1. Give Tablet Co-trimoxazole 480mg 12hourly for 5 days 2. If no improvement after 5 days; order for Stool test; 3. If amoebiasis, give Metronidazole 200mg 8hourly for 7 days 4. If Schistosomiasis, Give Praziquantel 600mg 6hourly for 2 doses

VI. Abdominal pain and paleness	Thin and pale looking (Pallor of conjunctiva and fingernails). No tenderness	Mild/Moderate condition Suspect Sickle Cell disorder	1. Check Hemoglobin, 2. If less than 10g/dl but more than 7g/dl 3. Give Tablet Folic Acid 5mg daily for 14days 4. Tablet Pyrantel pamoate 3 tabs immediately (stat) 5. Review in 1 week, 6. If Hemoglobin less than 7g/dl. REFER immediately to the appropriate health facility
VII. Abdominal pain with the history of passing worms. No Constipation	Appears well; Abdomen distended; no tenderness; No constipation	Mild condition; Suspect Worm Infestation	1. Give Tablet Pyrantel pamoate 3 tabs immediately(stat); 2. Review in 2 weeks 3. Advised fruits and vegetable 4. Self-pain persists- REFER immediately to the appropriate health facility
VIII. Swelling in Scrotum	• Child in Pain • Scrotum tender and warm • Swelling in one or both	Severe condition Suspect Epididymitis	1. Explain condition to mother 2. Tablet Paracetamol 500mg 8hourly for 5 days 3. REFER immediately to the appropriate health facility
IX. Lump or swelling in the groin/ the scrotum	Child in distress, Presence of tender swelling	Moderate Condition Suspect: - Inguinal, hernia - Scrotal hernia hydrocele	1. Explain findings to mother 2. REFER immediately to the health facility
X. Umbilical swelling (Hernia)	X. Umbilical swelling (Hernia)	Mild/Moderate Suspect Umbilical Hernia	1. If reducible, no treatment needed 2. If not reducible, REFER immediately to the health facility.

Health Education

Advise parent /Caregiver on:

1. Adequate nutrition, prevention of constipation and adequate fiber in diet
2. Food and personal hygiene
3. Adequate fluid intake.
4. Discourage using force to reduce hernia
5. Surgical intervention as appropriate.
6. Importance of taking home-based records the clinic or hospital.

3.23 ANAL DISORDERS

The Anus is the terminal end of the alimentary canal. Common problems affecting the anus include fissures, redness and prolapse. An anal fissure is painful linear sore at margin of anus. In rectal prolapse, the rectum turns inside out and protrudes out of the anus as a pink mass. It is common in young children who are poorly nourished having worms and diarrhea frequently

HISTORY	EXAMINATION/ACTIONS
1. How long have you' had the problem? 2. Is there diarrhea? How many times? 3. Is there constipation? When was the last stool? 4. Is there bright blood in stool? S.Is there abdominal pain? 6. Has the child had this problem before? 7. Has any medication been given? If yes what?	1. General appearance: pale, weak, in distress 2. Hydration status: eyes, mouth, tongue, skin 3. Anus: prolapse, redness, fissure 4. Hemoglobin: estimate and record. 5. Temperature: check and record 6. Weight: weigh and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Prolapse of rectum not reducible	Child distressed, Not dehydrated Rectal prolapse not reducible	Moderate condition Suspect prolapsed of the rectum	1. Do-sit bath 2.observe tor 2 hours, if not reduced, REFER immediately to the health facility
II. Prolapse of rectum reducible with or without bleeding	Child distressed Rectal prolapse reducible	Mild condition Suspect mild prolapsed of rectum	1.Apply warm saline, compress 2.Teach mother how to strap the buttocks 3.Capsule Amoxicillin 250mg 8hourly for 5days 4.Give Tablet Folic Acid 5mg daily for 2 wks 5.Tablet Ferrous sulphate 200mg(60mg elemental iron) 8hourly for 2 weeks 6.Give Tablet Zinc 20mg daily 7.Nutritional counseling 8.Review in 2 days, If no improvement, REFER immediately to the health facility
III. Fissures or sore on anus, Redness of anus	Anus red and sore. Child uncomfortable	Mild condition suspect fissures	1. Clean with soap and water 2. Dry thoroughly 3. Apply Zinc Oxide topically 4. Give Tablet Paracetamol 500mg 8hourly for 5 days 5. Dietary advice 6. Review in 5 days, If no improvement, REFER immediately to the health facility

Health Education

Advise parent /Caregiver on:

1. Personal, food and environmental hygiene
2. Early detection and treatment.
3. Adequate nutrition
4. Protection of genital area
5. Discouragement of sharing of sponge, towel, underwear and clothing
6. Bowel training (going to toilet at a regular time daily)

3.24 WORM INFESTATION

Worm infestation is the presence of large amount of worms in the gut which causes discomfort to the patient., Ascaris (roundworm) is the commonest worm .infestation. Heavy roundworm infestation produces abdominal pains, vomiting, constipation and sometimes, obstruction of the intestine. Hookworm infestation is common in rural areas where heavy infestation causes blood in stool, anemia, malnutrition. Severe Trichuris (whipworm) infestation causes bloody diarrhea, abdominal pain, anemia and prolapse of the rectum. Threadworm also lives in the intestine and causes' itching wound the anus. Tapeworm is a flat ribbon like Worm that can also be found in the intestine.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? When did it start? 2. Have any worms been passed? If yes, describe 3. Is there diarrhea? Blood in stool? 4. Is there vomiting? Is the child coughing? Any wheezing? 5. Is there anal itching? 6. Is there abdominal pain and rectal prolapse? 7. Is there any other problem? 8. Has any medication been given? If yes, What?	1. General appearance: thin, pale, 2.Eyes: pallor 3.Chest: wheezing, rhonchi 4.Abdomen: tenderness, distension, visible peristalsis 5. Anus: redness, prolapse 6. Temperature Pulse and Respiration: check and record 7.Haemoglobin: estimate and record 8.Weight: weigh and record 9.Stool examination: for ova of worm (if laboratory available1

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Passing worms from the rectum, itchy rectum, protruding rectum, blood in stool	Rectal prolapse	Moderate condition Suspect worm infestation with rectal prolapse	1.Tab. Pyrantel pamoate 3 tablets stat 2.Anusol suppository apply topically or insert into the rectum each night at bedtime 3.Give Tab. Vitamin C 100 – 200mg 8hourly for 5days 4.Review in 1 week, If no improvement REFER immediately to the health facility
II. Passing roundworms par rectum, diarrhoea, vomiting, rectal itching, abdominal pain.	• There may be pallor • May be underweight and have mild abdominal pain or distension	moderate condition Suspect worm infestation	1. Tablet Pyrantel pamoate 3 tablets stat 2. Lo ORS with Zinc DT 3. Review in 1 wk, If no improvement REFER immediately to health facility

Health Education

Advise parent /Caregiver on:

Advise Mother /Caregiver on:

1. Personal and food hygiene
2. Adequate nutrition
3. Environmental sanitation
4. Importance of taking home-based records to the clinic or hospital

3.25 URINARY AND GENITAL DISORDERS

Urinary genital problems are diseases or conditions affecting the male or female urinary tract or genitalia. Children may present with abnormality of the genitalia such as undescended testis, imperforate hymen, hydrocele and hypospadias and may be hermaphrodite. They may also present with painful urination with or without discharge redness or itching of the genitalia may be due to infection.

Sexually active older children are especially susceptible to sexually transmitted infections (gonorrhea, syphilis, HIV/AIDs and other nonspecific bacterial infection), patient may complain of discharge, pain and burning sensation on passing urine and blood in urine may occur as a result of schistosomiasis infection. Other causes of urinary problems are kidney diseases and diabetes which can cause frequent micturition.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? How long have you had it?' 2. Do you urinate very often? In the night or in the day? 3 Do you have pain or burning on micturition? 4. Do you drink more than usual (increased thirst)? 5. Do you have pain in the flank? 6 Is there blood in the urine? 7. Do you have night sweats? Or are you losing weight? 8. Do you have chills? 9. What area do you come from? (think of Schistosomiasis area) 10. Have you ever had this problem before? Any treatment? 11. Do you have discharge from the vagina(or if male, from the pennis)	1. General appearance: ill, thin, puffy. 2. Temperature, pulse: heck and record 3. Blood pressure: measure and record 4. Eyes: pallor. 5. Chest crepitations, decreased breath sounds. 6. Heart: rate, rhythm, murmur. 7. Abdomen: tenderness, swelling or masses. 8. Flanks: tenderness 9. Limbs: a) legs for swelling b) Nails for pallor 10. Genitals: sore, pus, or discharge 11. urine: a) color or cloudiness b) protein c) sugar d) Microscopy

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. No Urine for more than 12 hours	Ill, Restlessness, Anxious	Severe condition Suspect Kidney problem	REFER. immediately to the health facility
II. Excessive urination with history of ants crowding round urine. Excessive thirst	Sugar in Urine	Severe condition; Suspect diabetes	1. REFER immediately the health facility.
III. Urethral or Vaginal Discharge with painful micturition	Urethral/Vaginal discharge	Moderate condition Suspect Urinary Tract Infection	1. Risk assessment to know if the child has been sexually abused 2. Tablet Paracetamol 500mg 8hourly for 3 days 3. Tablet Ciprofloxacin 250mg 12hourly for days 4. Review in 2 days, If no improvement. REFER immediately to the health facility
IV. Painful urination with or without frequency, cloudy urine or fever	Fever; Protein in Urine; painful Micturition	Moderate condition Suspect nephrotic syndrome	1. Encourage fluid; 2. Tab. Paracetamol 250-500mg 8hourly for 3 days 3. Tab. Co-trimoxazole 480mg 12hourly for 5 days 4. Review in 2 days, If no improvement, REFER immediately to the health facility

V. Pain in the flank, fever	Pain on tapping flank, muscle spasm	Severe condition Suspect Kidney Stones	1.Tab. Paracetamol 250-500mg 8hourly for 3 days 2.Tab. Co-trimoxazole 480mg 12hourly for 5 days 3.Encourage fluids; 4.REFER immediately to the health facility.
VI. Redness or Swelling of Genitalia with or without painful urination, itching.	•Redness and swelling of Genitalia •Scratch marks Urethral discharge with or without pain	Moderate condition Suspect Urinary Tract Infection Suspect Fungal infection	1.Wash the area with soap and water 2.Apply Zinc Oxide and Mycostatin cream or paste to the are 3.Encourage fluids if male retract foreskin 4.Tablet Co-trimoxazole 480mg 12hourly for 4 days 5.Review in 7 days, If no improvement REFER immediately to the appropriately health facility
VII. Blood in last stream of Urine with or without pain, fever or discharge. History of swimming in stream/river	Blood in Urine Terminal hematuria	Moderate condition Suspect Schistosomiasis	-Encourage liberal fluid intake -Give Metriphonate 200mg (2 tab) every 2 weeks X 6 weeks or praziquante 1.2g stat(2 tablets) -If not improvement. REFER immediately to the appropriate health facility to further
VIII. Painful Urination; No discharge	No Discharge	Moderate condition Suspect urinary Tract Infection	-Capsule Amoxicillin 250mg 8hourly for 5 days -Tablet Paracetamol 500mg 8hourly for 3 days Encourage fluid and fruits intake
IX. dark yellow urine, no other complaint	Dark Urine with or without yellow eyes	Mild to Moderate condition Suspect dehydration	1.Explain finding to mother and re-assure her; REFER immediately to the appropriate health facility
X. abnormality of the Genitalia	• Un-descended testes, imperforate hymen. Hermaphrodite, Large clitoris, • Hydrocoele; • Hypospadias (penis in abnormal place or several penial openings).	Mild to Moderate condition	1.Explain finding to mother and re-assure her 2. Give fluids. 3. Review in 3 days, if no improvement, REFER immediately to the appropriate health facility
XI. Difficulty in passing urine	Penis blocked by tight foreskin not easily retracted, bladder full	Moderate condition	1. Catheterize. 2. Circumsise

Health Education

Advise parent /Caregiver on:

1. Personal hygiene.
2. Adequate fluid intake.
3. Proper parental care and supervision of the child.
4. Side effects of medications.
5. Drug compliance.
6. Environmental sanitation.
7. Importance of taking home-based records to the clinic or hospital.

3.26 ARM AND LEG DISORDERS

Deformities of the arms and legs, if not corrected early could lead to permanent disabilities. The deformities or swellings may be due to bruises, fractures or a dislocation. It is important to test for normal movement of the parts whenever there is a history of injury, swelling or deformity. Swelling of a joint may be due to sprain of the tendons surrounding the joint. Sickle cell disease may also be a cause of pain in the arms, legs and the hip bone. Tuberculosis can occur in the joints. Polio can seriously affect one or both legs or arms first causing weakness then wasting, or paralysis. Diabetes Mellitus may cause tingling sensation and numbness of the fingers and toes, foot ulcer, that fail to heal and black discoloration of toes and feet (Gangrene).

HISTORY	EXAMINATION/ACTIONS
1. When did it begin? 2. Was there an accident or injury? 3. If an injury, has the child been able to use that part since the injury? 4. Is there any pain or tingling or numbness? Is it recurrent? 5. Is there fever? 6. Is the child a Sickler? 7. Is the child passing excessive urine 8. Is the child having excessive thirst?	1. General appearance: ill looking, wasted or deformed 2. Limbs a) Swelling b) Bleeding c) Fracture d) Tenderness e) Warmth f) Loss of function g) Wasting 3. Eyes: Jaundice 4. Temperature: check and record 5. Hemoglobin: estimate and record 6. Urine: sugar and color

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Inability to move arm and/or leg; Pain in joints and limbs	Weakness of limbs with or without pain Paralysis with or without swelling at the joints Wasting of the limb	Severe condition Suspect Polio, Sickle Cell Disorder	1. Immobilize the patients arm or leg 2. Refer immediately to the appropriate health facility
II. Swelling of one or more joints without fever or tenderness	•. Pain Tenderness Swelling of the joint • No fever	Moderate condition Suspect Arthritis	1. Tablet Paracetamol 500mg 8hourly for 5 day's 2. Capsule Ampiclox 500mg 6hourly for 5 days 3. Review in 3 days, if no improvement, REFER immediately to the appropriate health facility
III. Deformity without swelling, tenderness or pain or bowing of leg or difficulty in movement.	Club foot, Bow leg, walking with difficulty	Mild condition	1. REFER immediately to the appropriate health facility.
IV. Tender lumps and nodes in groin	Painful lumps and nodes in the groin Source of infection identified	Moderate condition	1. Tablet Paracetamol 500mg 8hourly for 5 days 2. Capsule Amoxicillin 250mg 8hourly for 5 days Treat open wound if any 3. Review in 5 days, If nodes get bigger; REFER immediately to the appropriate health facility

V. Swelling on and off especially of fingers, toes, back of hands and feet	Swelling of fingers, toes, back of hands and feet. Hb below 10gms	Mild to moderate condition Suspect Sickle Cell Disease	1. Tablet Folic acid 5mg daily for 2 weeks 2. Tablet Paracetamol 500mg 8hourly for 3 davs 3. Review in 5 days, if no improvement, REFER immediately to the appropriate health facility

Health Education

Advise parent /Caregiver on:

1. Prevention of accidents
2. Genetic Counseling
3. Adequate diet (Calcium rich foods)
4. Proper posturing
5. Importance of taking home-based records to the clinic or hospital.

3.27 CHANGES IN ALERTNESS AND BEHAVIOUR

Changes in alertness and behavior mean that the client does not act normally sometimes. He/she may be drowsy or give a history of fainting or there may be obvious changes such as unconsciousness or delirium which are easy to identify. All cases of change in alertness or behavior should be taken seriously no matter how mild to prevent complications.

HISTORY	EXAMINATION/ACTIONS
1 How long has the child been like this? 2. Did anything happen to the child? Injury? 3. Did the child have a fit or convulsion, or is he clumsy? 4. Has any medication been given? 5. Has the child been ill otherwise? 6. Has he had fever, diarrhea, vomiting or ear pain? 7. Has he been drinking more recently? 8. Has he been urinating more frequently? 9.Was there any difficulty with the child's pregnancy or delivery? 10.Was the child jaundiced at birth or shortly after? 11. Has he been frothing at the mouth or passing urine when he faints?	1. General appearance:(especially for alertness and irritability signs of trauma) 2. Chest: a) respiratory rate, lower chest in drawing 3. Head: signs of injury 4. Neck: stiffs 5. Ears: pus, redness or perforation of the drum 6. Limbs: arms and legs for abnormal movements 7.Temperature and pulse: check and record 8. Hemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Unconscious with fever or history of fever difficult to arouse confusion more irritable than normal abnormal movements of arms and legs	Fever of 37.5C and above Irritable. No stiffness. Patient may be delirious difficult to arouse abnormal movements of arms and legs confused	Severe condition Suspect Severe Malaria	I, Conduct mRDT, if positive, give pre -referral treatment for severe malaria 2. REFER immediately to the appropriate health facility
II. Unconscious or more irritable than normal abnormal movements of arms and legs	Irritable. No stiffness. May be ; delirious difficult to arouse	Severe condition	REFER immediately to the appropriate health facility
III. Fainted suddenly but is arousable.	Child disoriented, with or without stiff neck	Moderate condition Suspect hypoglycemia	1. Give child sugar or glucose at once and let him rest for 30mins. 2. If no improvement, REFER appropriately. 3. If improved, advise on proper feeding. 4. Test Urine 5. Review next day.
IV. Fainted (fit) once or twice with or without fever or frothing from mouth or family history of epilepsy	Conscious, normal temperature No fits	Severe condition Suspect epilepsy	1. Explain condition to mother 2. Tab Phenobarbitone 30mg 12hourly 3. REFER immediately to the appropriate health facility
V. Unconscious or More irritable than normal Abnormal movement of arms and legs	Irritable, no stiffness, maybe; Delirious Difficult to arouse	Severe condition	1. Refer immediately to the appropriate health facility

Health Education

Advise parent /Caregiver on:

1. Prevention of injuries
2. Proper parental care and emotional support
3. Avoidance of self- medication
4. Compliance with prescribed drugs regimen
5. Healthy home environment
6. Need for therapy as appropriate
7. Importance of taking home-based records to the clinic or hospital

3.28 POISONING

Poisons are toxic substances which may enter the body through swallowing, breathing, absorption through skin or mucus membrane or infections. They can cause fits marked restlessness and pain. Liitle children are adventurous and may therefore be easily exposed to harmful substances.

Mothers should therefore be advised to keep kerosene, medicine, bleach, dye, alcohol and other harmful substances out of the reach of children. Kerosene, medicine and other harmful substances should be labelled and must not be stored in soft drink bottles.

HISTORY	EXAMINATION/ACTIONS
I. What and how much poison did the child take? 2. When was the poison taken? 3. Has any medication been given (modem or traditional)? 4. Has he vomited? Is there blood in the vomitus?	I. Try to examine poison container. 2. General appearance: consciousness, alertness, pallor, and cyanosis, 3. Eyes: size and reaction of pupils to light? 4. Chest: respiration rate, difficulty, lower chest, indrawing, deep, sighing. 5. Abdomen: tenderness, rigidity 6. Mouth: bums, bleeding, dribbling, color of membrane 7. Pulse: check rate, volume and record. 8. Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Ingestion of petrol or kerosine or corrosive substance e.g. alkali, caustic soda and acid.	Irritable, swelling and mouth burns	Moderate to severe condition Suspect substance poisoning	1. Do NOT induce vomiting 2. Give enema saponins 3. GiveIM. Amoxicillin 200mg stat, REFER immediately to the appropriate health facility
II. Ingestion of poison with convulsion	Irritable, Restless, may or may not be convulsing No mouth burns and swelling	Severe condition Suspect Poison leading to convulsion	1. IM Paraldehyde immediately (stat) if convulsing: 2. Gastric lavage/washout 3. NB: DO NOT INDUCE VOMITING IN A CONVULSING OR UNCONSCIOUS PATIENT
III. Ingestion of any harmful substance. E.g. Kerosene, acid or petrol	Irritable, Restless, No mouth burns or swelling May or may not be warm to touch	Moderate condition	1. Identify the poison 2. Milk drink or Mist magnesium trisilicate 5mls 8hourly for 3 days 3. REFER immediately to the appropriate health facility
IV. Aspirin poison with or without fever or other signs	Temp. normal or above 37.5°C Irritable Thirsty Restlessness	Moderate condition	1. Give milk drink or Mist magnesium trisilicate. 5 mls 8hourly for 5 days 2. Give plenty of fluids 3. Review if no improvement REFER immediately to the appropriate health facility

Health Education

Advise parent /Caregiver on:

1. Prevention of home accidents
2. Adequate nutrition and effect on healing
3. Avoid induced vomiting
4. Proper storage of chemical and harmful liquid
5. Prevention of bites and stings
6. Importance of taking home-base records to the clinic or hospital

3.29 SCHOOL HEALTH SERVICES

The child has certain rights, which are recognized national and international, treaties and conventions. These include the right to education and access to health and other social services. Six years is the age when the Nigeria child commences primary school education in public schools. The Nigerian child has the rights to quality health care which should be provided in the school health program including routine physical examination, screening tests e.g. for hearing and vision, preventive and therapeutic dental services, management of minor ailment, immunizations, health education, provision of school meals, health promoting physical exercises and referrals.

HISTORY	EXAMINATION/ACTIONS
1. Is there any problem? 2. What type of food do you eat? (ask about main meals and snacks) 3. How many brothers and sisters do you have? 4. Whom do you live with? (parents, grandparents, spouse, others, specify) 5. How many times do you eat in a day? Where do you get your food? 6. What type of work does your father do? Mother? 7. What is your source water? 8. What type of toilet facility do you have in your house or hostel? 9. Have you had any serious illness in the past?	1. General appearance: well nourished, ill, tired, clean, thin, congenital abnormality 2. weight and height: weigh, measure and record 3. Head: lice, ringworm 4. Eyes: discharge, pallor, jaundice, cataract, test of vision 5. Nose: discharge, blockage, test of hearing 6. Ears: redness, discharge, test of hearing 7. Mouth: odor, caries, missing teeth 8. Gum: healthy, bleeding 9. Throat: redness, inflamed tonsils 10. Chest: respiratory rate, wheezes, stridor, lower chest indrawing 11. Heart: rate, rhythm, murmur 12. Abdomen: scars, tenderness, rigidity, masses, palpable spleen and liver 13. Skin: dryness, sores, rashes, multiple scars or wounds (think child abuse) 14. Limb :deformity, paralysis, oedema 15. Pulse: blood pressure, check and record 16. Urine: Sugar, protein 17. Hemoglobin: estimate an record 18. Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Well, first visit	No abnormality	Healthy child	Refer for further laboratory investigations Blood group, Genotype, Chest X-ray, urine test, stool for ova,
II. Well visit (School Food Vendor)	No Physical Abnormality	Healthy individual	Refer vendor for the following test at least every six months Chest X-ray Sputum test for AF Blood for widal test Stool for ova and parasites If any abnormal finding, stop from providing food until treatment given and completed.

Health Education

Advise on;

1. Personal hygiene
2. Food hygiene
3. Environmental sanitation
4. Importance of taking home-based records to the clinic or hospital

3.30. SEXUALITY EDUCATION

Children, in particular older ones need to have correct information about their anatomy, physiology and developmental changes. They should be able to handle their psycho-sexual feelings in a healthy manner and therefore require counseling.

HISTORY	EXAMINATION/ACTIONS
1. Has she started menstruation if a girl? 2 At what age did she experience menarche? 3 Are you living with your parents? If No Who? 4 Have you been sexually molested by anybody before? 3Any other thing you may want to share with me\ 4When did you notice changes in your physical appearance 5Have you had sexual intercourse before 6 Do you have wet dreams	1. General appearance: healthy, ill looking 2. Check temperature, pulse , respiration and blood pressure 3. Eyes: pallor, jaundice 4. Heart: Pulse rate ,rhythm Axilla presence of hair Breast developed size Pelvic public hair , vulva for perforate/imperforate hymen Haemoglobin estimate and record 6 Weight and Height : measure and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Abdominal pain with menstruation; Foul smelling discharge per vagina; Fever	Lower abdominal tenderness, High temperature	Moderate condition Suspect infection	1.Reassure the client/patient 2. Counsel on personal hygiene (use of sanitary pad) 3. Capsule Amoxicillin 250mg 8hourly for 5 days 4. Conduct RDT, if positive, treat for uncomplicated malaria 5. Review in 5 days, if no improvement REFER immediately to the appropriate health facility
II. Heavy menstrual flow with clots, prolonged menstruation	May be pale Blood clots per vagina	Moderate condition Abnormal menstrual flow	1.Reassure the client/patient 2.Tablet Paracetamol 500mg 8hourly for 3 days
III. Lower abdominal pain associated with menstruation	Lower abdominal tenderness, no fever, No foul smelling discharge	Mild condition Suspect dysmenorrhea	1.Tablet Paracetamol 500 mg 8hourly for 3days Reassure and counsel
IV. Mucus or blood per vagina	Normal mucus, no smell Normal menstrual flow	Mild condition	1. Counsel and reassure patient.
V. Anxiety about new relationship with the opposite sex	Normal	Mild condition	1. Reassure, counsel on physiology and developmental changes and advise on the negative consequences of early sexual relationships.
VI No sexual abuse	Healthy child	Normal	Intensify health education.

Health Education

Advise Parent /Care giver on:

1. Personal hygiene.
2. Adequate nutrition.
3. Sex education.
4. Implication of teenage pregnancy.
5. Use of Long Lasting Insecticidal Nets (LLNs).
6. Importance of taking home-based records to the clinic.

SECTION FOUR

ADOLESCENT HEALTH (10 – 19 Years)

4.0 ADOLESCENT HEALTH

PREAMBLE

Adolescence is the period of physical, psychological and social changes from childhood to adulthood. It spans the ages of 10 to 19 years.

It is subdivided into two periods: early adolescence (between 10-14 years) and late adolescence (between 15-19 years). It should also be noted that the term "young people" covers the ages between 10-24 years. True adolescence, therefore, is a transition that may fall anytime between years and hence whenever health status or behaviour is being considered, 'youths" are quiet often include among "adolescents" leading to the addressing of both categories as "young people"

It is acknowledged globally that this group form an important part of any society and in Nigeria, young people aged 10-24 constitute at least 30% of the population. It is also widely known that adolescence is a period of learning, excitement and overall increased prevalence of high risk behaviors.

Rapid societal changes place adolescents in a situation where they are confronted with conflicting and at times contradictory social expectations. Transition to adulthood often involves period of stress, innovation, experimentation and disorganization.

Some of the problems with the consequences among adolescents include unwanted pregnancy, rape, unsafe abortion, and sexually transmitted infections including HIV/AIDS, drug and substance abuse and other behavioral changes.

To reduce these consequences, it is important to have adolescents-friendly clinic where they are provided amongst other things, information on their health needs, counseling especially during crisis situations and have access to health services.

This section addresses the issues that are peculiar to adolescents. Other conditions are discussed in the child/ adult section.

4.1. ADOLESCENTS CONCERNS ABOUT PUBERTY (FEMALE CONCERNS)

The period of development between the onset of puberty and adulthood (10 - 19 years), involves physical growth of an adolescent from a child into an adult. A common concern is how to distinguish between "normality" and "abnormality".

HISTORY	EXAMINATION/ACTIONS
1. How are you? 2. How may I help you? 3. When did you notice changes in your physical appearance (presence of breast, axillary and pubic hair)? 4. Are you comfortable with the size of your breast? Any family history of large/small breast? 5. Have you had sexual intercourse before? 6. Have you started menstruating? 7. When was your last menstrual period? For how many days? 8. Do you have abdominal pain with menstruation? 9. Family history (onset of puberty, chronic illness, stress) 10. What are your feeding habits like? 11. Is there anything else you would like to share with me?	1. General appearance: Healthy, ill, thin, obese, anxious 2. Check temperature, pulse, respiration & blood pressure and record 3. Eyes: Pallor, jaundice 4. Nose: Catarrh, nasal flaring 5. Axilla: Presence of hair 6. Breast: Developed, size 7. Abdomen: Tenderness, palpable uterine masses 8. Pelvis: Pubic hair, vulva for perforate/imperforate hymen, check flow if menstruating 9. Skin: Sweating, cold and clammy, oily, acne 10. Hemoglobin: Estimate and record 11. Weight and height: Measure and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Early onset of physical changes of sexual maturity about age 8 years	Presence of breast, axillary and pubic hair Menstruation	Mild moderate condition Suspect precious puberty	1.Allay fears 2. Counsel the client/parent on physiological changes of the adolescents
II. No menstruation in a child 16 years or older	Hymen may be imperforate or perforate	Moderate condition	Allay fears REFER immediately to the appropriate health facility
III. No menses for more than one month after onset, abdominal pain with or without vaginal bleeding, dizziness	Abdominal tenderness, with or without vaginal bleeding, pallor, fast and weak pulse, cold and clammy hands and legs, sweating	Severe condition Suspect ectopic pregnancy	1. REFER immediately to the appropriate health facility.
IV. No menses for more than one month after onset	Palpable uterus	Mild condition Suspect pregnancy	1. Conduct pregnancy test 2. Explain condition to client and parent 3. Encourage antenatal care 4. Give social and psychological support
V. Prolonged and heavy menstrual flow	Heavy flow with clots May or may not be pale	Moderate to severe condition Suspect heavy menstruation	1. Explain the condition to client and parent 2. Check: Hb; if below 10g/dl give Tab folic acid 5mg daily for 2 weeks
VI. Scanty Menstrual flow	Small quantity of blood on pad May or may not be pale	Mild condition Suspect scanty menses	1. Explain condition to client and parent 2. Check Hb, if below 10g/dl give, Tab folic acid 5mg daily for 2 wks 3. Give tab ferrous sulphate 200mg daily for 2 wks 4. If Hb is below 8g/dl REFER appropriately 5. Give dietary advice 6. Advise on adequate rest
VII. No menses for more than one month. has not had sexual intercourse	uterus not palpable	Mild condition	Counsel and allay fears

VIII. lower Abdominal pain with menstruation.	* May or may not have lower abdominal tenderness. * No mass in abdomen	Mild to moderate condition Suspect dysmenorrhea.	1. Explain condition to client and parent 2. Encourage regular exercise 3. If below 12 yrs, give Tab Hyoscine 10mg 12hourly for 3 days or Tab Ibuprofen 200mg 12hourly for 5days (After meal) Tab. Paracetamol 500mg 8hourly for 3 days 4. If 12yrs and above, give tab paracetamol 1000mg 8hourly for 3 days 5. Advise on personal hygiene/care
IX. Large breasts	* Anxious * Excessively large breast *Breasts appear healthy. * No abnormality discovered.	Mild condition Macromastia (large Breasts)	1.Explain condition to client that it's normal she will get used to a changing body. 2. Advise on correct size of brassier 3. If abnormality is detected, REFER immediately to the appropriate health facility
X. Small breasts	* Anxious * Small breast * No other abnormality detected	Mild condition	1.If pubertal signs of development are normal and have started menstruating, counsel to allay fears. 2.If other puberty characteristics are absent or poorly developed, REFER appropriately. 3.Give dietary advice.
XI Unequal breast size	Anxious No other abnormality detected	Mild condition	1.Reassure her if unequal breast size is during puberty (when the breasts are still developing). 2.advice if the breasts are of unequal size at the end of puberty. 3.refer to an endocrinologist (doctor specialising in conditions related to hormones).
XII Bedwetting	*Child is anxious & Depressed	Mild condition Nocturnal Enuresis is also called involuntary urination	1.Counsel client, parents, and/or guardian. 2.Advise on urinating before going to bed. 3.Support by waking up to urinate. Avoid fluid intake 3 hours before bedtime. Advise parent/guardian to avoid punishment for wet nights but rather commend for dry nights.

Health Education

Dietary advice/Nutritional counselling

1.Dietary advice/nutritional counseling.
2.Personal hygiene.
3.Solicit social & psychological support.
4.Sexuality education.
5.Importance of taking home-based records to the clinic or hospital.

4.2 ADOLESCENTS CONCERNS ABOUT PUBERTY (MALE CONCERNS)

The period of development between the onset of puberty and adulthood (10-19 years) involves the physical growth of an adolescent from a child into an adult. A common concern during this time is how to distinguish between 'normality' and 'abnormality'

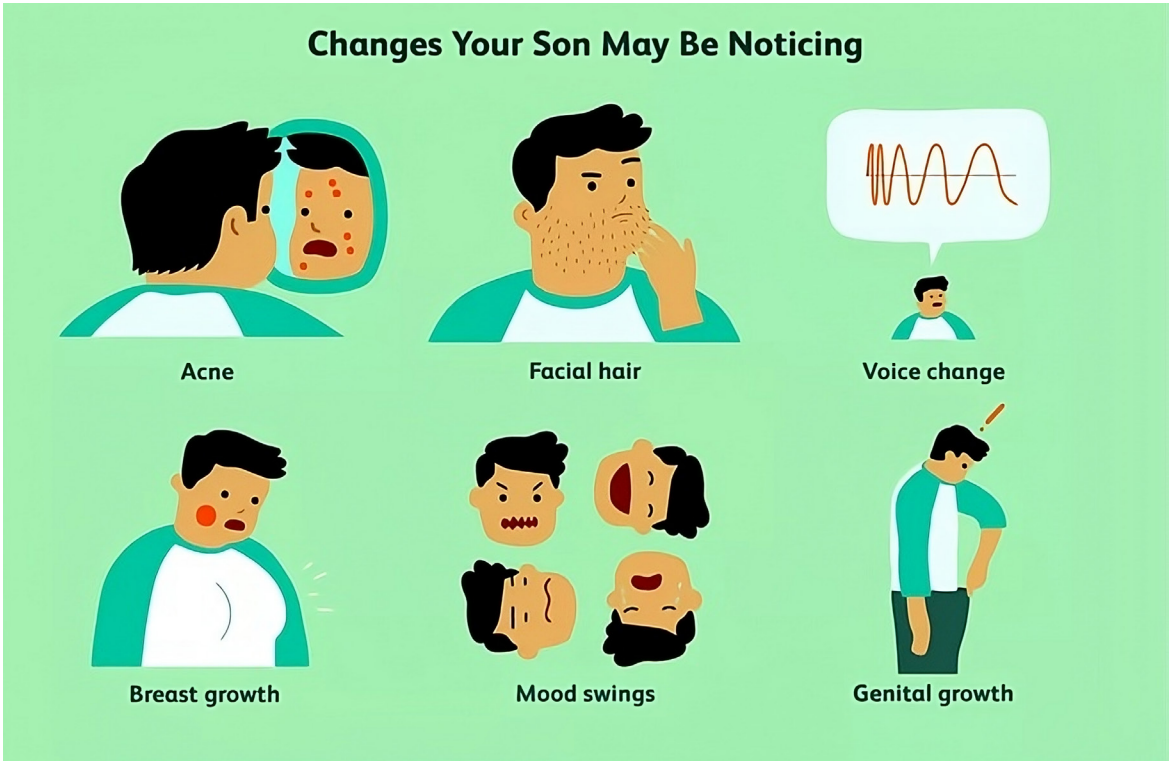
HISTORY	EXAMINATION/ACTIONS
1. How are you? 2. How may I help you? 3. When did you notice changes in your physical appearance (pubic hair, axillary hair, breaking of voice)? 4. Any swelling/pain in the scrotum? 5. Have you had sexual intercourse before? 6. Any urethral discharge? 7. Are you comfortable with the size of your penis? 8. Do you have wet dreams? 9. Family history (onset of puberty, chronic illness, stress) 10. What are your feeding habits like? 11. Any other thing you may want to share with me?	1. General appearance: Healthy, ill, thin, obese, anxious. 2. Temperature, pulse, respiration & blood pressure and record. 3. Eyes: Pallor, jaundice. 4. Axilla: Presence of hair. 5. Scrotum: Presence of testes, swelling, tenderness. 6. Penis: Size, deformity, presence of pubic hair, urethral discharge. 7. Skin: Oily and acne. 8. Weight and height: Measure and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Onset of puberty at about age 8 years	* Any of the following may be present: ~ Public hair ~ Large hair ~ Large penis ~ Broken voice	Mild condition Suspect precious puberty	1. Explain condition to client and parent. 2. Allay fears 3. Follow up.
II. Late onset of pubertal changes	• May or may not have family history • May or may not be small for age • No evidence or history of chronic illness or stress • No abnormality detected	Mild condition Suspect delayed puberty	1. Explain condition to client & parent 2. Allay fears 3. Follow up
III. One or both testes missing	Anxious. One or both testes missing in the scrotum Scrotal sac retracted Scrotal sac empty	Moderate condition Suspect undescended testis	1. Explain condition to client & parent 2. Allay fears 3. REFER immediately to the appropriate health facility
IV. Painless swollen scrotum	The presence of fluid in scrotum	*Moderate condition *Suspect hydrocele	1. Explain condition to client & parent 2. Allay fears 3. REFER appropriately
v. Scrotum swelling with pain with or without urethral discharge	* Client in distress Swollen tender scrotum Presence or absence of urethral discharge Temp. 37.5°c. or above	*Severe condition *Suspect epididymo-orchitis	1. Give Tab. Paracetamol 1000mg 8hourly for 3 days 2. Give caps Amoxicillin 500mg 8hourly for 5 days 3. REFER immediately to the appropriate health facility

VI. Scrotal pain with or without swelling	• Client in distress • Scrotal tenderness • Temperature may be Normal	Severe condition Suspect testicular torsion	1.REFERimmediately to the appropriate health facility
VII. Small penis	• Anxious • Penis may or may not be relatively small • No penile deformity • No other abnormality found	Mild condition	1. Reassure client and parent that size of penis does not affect sexual function
VIII. Bed wetting	Child is anxious and depressed No other abnormality detected	Mild condition Suspect Enuresis	1. Counsel clients, parents/guardian 2.Advice urinating before going to bed 3. Support fluid intake 3 hours before sleeping time 4. Support by waking up to urinate 5.Advise parent/guardian to avoid punishment for wet nights, rather commend for dry nights

Health Education

1.Dietary advice/nutrition counseling.
2.Personal hygiene.
3.Solicit social & psychological support.
4.Use LLIN (Long-Lasting Insecticidal Nets).
5.Sexuality education.
6.Importance of taking home-based records to the clinic or hospital.



4.3 CONTRACEPTIVE CHOICE FOR ADOLESCENTS

It is important to note that abstinence should be encouraged and emphasized among adolescents due to the spread of HIV infection and other STIs. Many are already sexually active or have had sexual intercourse at least once. Helping adolescents make responsible and healthy choices in terms of the various contraceptive methods that are appropriate in preventing unwanted pregnancy is therefore crucial. Myths and misconceptions should also be dispelled and clarified.

HISTORY	EXAMINATION/ACTIONS
1. Can you tell me what you know about contraceptives? 2. Have you used any contraceptive method before (if yes, which one did you use)? 3. Which clinic did you attend? 4. Why do you want to attend this clinic? (who referred you)? 5. Do you have regular sexual intercourse? (How many partners)? 6. Do you have any other problems?	1. General appearance: Healthy, ill, thin, obese. 2. Temperature, pulse, respiration & blood pressure and record. 3. Eyes: pallor, jaundice. 4. Heart: pulse rate, rhythm. 5. Abdomen: tenderness, palpable uterus, masses. 6. Weight and height: measure and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Request for contraceptive advice	*No abnormalities Detected	Normal condition	1. Discuss sexuality in detail. 2. Avoid situations that can provoke sexual arousal 3. Advise abstinence as the best choice. 4. Discuss various methods of contraception such as condoms (male and female), diaphragm, spermicidal, vaginal foaming tablet, pills, injectable. 5. Demonstrate the proper use of CONDOM (male and female) in preventing HIV/AIDS, STIs, and pregnancy.
II. HIV positive adolescent requesting for contraceptive	• HIV positive • Looks healthy	Severe condition	1. Give adequate counseling on condom (male and female) use plus second contraceptive device (i.e., dual protection). 2. Ensure confidentiality. use Medical Eligibility Criteria (MEC) Wheel For Contraceptive Use REFER to HIV counseling and testing (heart-to-heart) center.

Health Education

Advise Mother /Caregiver on:

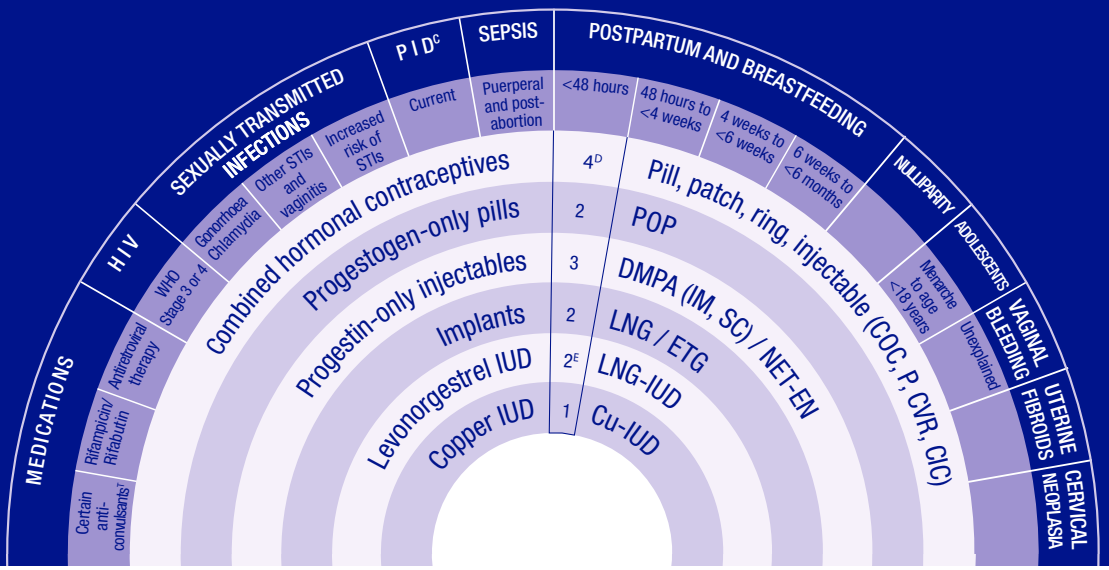
1. Counsel on choice of contraceptive (emphasizing abstinence).
2. Discuss all methods of contraception.
3. Discuss the side effects and complications of each of them.
4. Nutritional counseling.
5. Sexuality education.
6. Advise on personal hygiene.
7. Importance of taking home-based records to the clinic or hospital.

World Health Organization

WHO

MEDICAL ELIGIBILITY CRITERIA WHEEL FOR CONTRACEPTIVE USE

2015



4.4 SEXUAL DYSFUNCTIONS/MALFUNCTIONS

Sexual dysfunction/malfunction is difficulty experienced by an individual or a couple during any stage of normal sexual activity, including physical pleasure, desire preference, arousal or orgasm. There is a need to understand the human sexual response cycle to be able to help a young person with a problem of sexual dysfunction. It may be as a result of fear, guilt, anger, anxiety, relationship difficulties or medical problems and many of these can be solved through honest communication.

HISTORY	EXAMINATION/ACTIONS
1. What is your view about sexual intercourse? 2. Do you have a sexual partner? If yes, I. How often do you engage in sexual practice? II. Do you have erectile problems? III. Do you experience any pain during sexual intercourse? 3. Any history of sexual abuse, rape? 4. Drug/substance history 5. How long does it take you to ejaculate? 6. Do you have any other problem?	1. General appearance: anxious looking angry; fearful; healthy 2. Check and Record the vital signs (temperature, pulse, respiration & blood pressure) and record 3. Penis: turgid and erect, flaccid 4. Vagina: check for bleed in bruise, tear.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Prolonged painful erection may or may not be associated with sexual stimulation	Penis turgid and erect rigid	Severe condition. Suspect Priapism	1. Give Tab paracetamol 1000mg [2 tabs] stat 2. REFER immediately to the appropriate health facility
II. Painful intercourse may or may not have bleeding	Anxious looking	Moderate condition Suspect dyspareunia	1. Give tab paracetamol 1000mg [2 tabs] stat 2. REFER immediately to the appropriate health facility
III. Failure to have an erection while with partner	Anxious Healthy looking	Mild to Moderate condition Suspect psychological erectile dysfunction or infection	1. Explain condition to client. 2. Explain that client should exercise patience when about to have sex. 3. Counsel on abstinence and risk of STIs/HIV Infection 4. After Review if no improvement Refer to the appropriate health facility
IV. Quick ejaculations	Healthy looking May or may not look anxious	Mild condition Suspect premature ejaculation	1. Explain condition to client. 2. Encourage physical exercise 3. Advice use of condom to reduce sensitivity 4. Counsel on abstinence and HIV/AIDs and STIs

Health Education

1. Counsel on sexuality education
2. Advise regular exercises
3. Nutrition counselling
4. Advise on personal hygiene
5. Importance of taking home-based records to the clinic or hospital.
6. Sexuality education.

4.5 GENITO-URINARY TRACT DISORDERS

Genitourinary tract disorders are problems that develop along a patient's urinary or genital tract, or both, in some cases. They are very common, especially among sexually active people and those with certain risk factors like a history of genitourinary anomalies. Genitourinary problems can be caused by gonorrhea, syphilis, non-specific bacterial infection, and schistosomiasis. Patients may complain of discharge, pain and burning sensation on micturition, and blood in urine.

HISTORY	EXAMINATION/ACTIONS
1. How long have you experienced the problem? 2. Do you have pain, burning sensation when passing urine? 3. Do you pass urine normally? Is it more or less frequent? 4. What is the colour of your urine-is it reddish, cloudy, dark or yellowish? 5. Do you have urethral discharge? 6. Do you have pains on your flanks? 7. Is there swelling of your legs? 8. Do you have shortness of breath, nausea or vomiting?	1. General appearance: ill, puffy face, pale 2. Temperature, Pulse, respiration & blood pressure, check and record 3. Eyes: check for pallor and peri-orbital oedema 4. Chest: check breath sounds 5. Abdomen: distension, check for flank tenderness 6. External genitalia: any abnormality, redness, swelling, discharges 7. Limbs: check hands and feet for oedema 8. Urine: colour, odour, protein, cloudiness and blood

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Urinary disorder with any of the following condition: a) Shortness of breath b) Nausea and vomiting c) No urine for more than 24 hours	Chest not clear Brown or dark urine. Swelling of legs or Abdominal distension	Severe condition Suspect acute renal failure	REFER immediately to the appropriate health facility
II. Pain or burning sensation on micturation, painful flank, swelling of legs	• Protein in urine • Cloudy urine • Swelling of legs with or without facial swelling	Moderate to severe condition Suspect nephroitic syndrome	REFER immediately to the appropriate health facility
III. Pain or burning sensation on micturation,	• urine plain colour Dysuria (pain during urination)	Mild condition Suspect STIs	1.Encourage a person to drink more fluids as this dilutes urine, making it less painful to pass. 2.Resting and taking medications as directed can usually help relieve most symptoms. 3.Tab ibuprofen 400mg 12hourly for5days 4.Tab Nitrofurantoin 100mg 8hourly for 5days REFER appropriately

IV. Blood in Urine (terminal) may or may not have painful micturition	Blood in Urine	Moderate condition Suspect schistosomiasis	1. REFER for bilharzias test 2. If Schistosomiasis test is negative, REFER appropriately 3. If positive and staff trained, give praziquantel60mg/kg/ day in divided doses or metrifonate 4mg tab every 2 weeks for 6 weeks, If no improvement, REFER immediately to the appropriate health facility
IV. Cloudy or dark yellow urine without any other symptoms	Dark yellow urine No jaundice	Mild condition Suspect dehydration	1. Encourage adequate fluid intake 2. Review in 3 days If no improvement, REFER appropriately

Health Education

Advise Mother /Caregiver on:

1. Advise on personal hygiene
2. Advise on protection of genital area
3. Advise on drug compliance
4. Encourage adequate fluid intake
5. Counsel on side effects of medications
6. Importance of taking home-based records to the clinic or hospital

4.6 ABORTION

Abortion is the termination of a pregnancy before the age of viability. Abortion can occur spontaneously or may be induced. Unsafe abortion is a common phenomenon among adolescents and young adults in Nigeria and has led to many cases of infertility and maternal deaths. Usually, spontaneous abortions occur in the first 12 weeks of pregnancy.

HISTORY	EXAMINATION/ACTIONS
1. When was your last Menstrual Period (LMP)? 2. Is there bleeding? When did it start and for how long? Any injury? 3. Did you expel any blood clots/products of conception? 4. Is there abdominal pain? 5. Is there any vaginal discharge? Does it smell? 6. Do you have fever? 7. Is there any other problem?	1. General appearance: ill, weak, pale 2. Temperature, pulse, respiration & blood pressure: measure and record 3. Vagina: check for bleeding and discharge, blood clots/products of conception 4. Haemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. History of bleeding that is subsiding without abdominal pain, Developing embryo completely expelled	Bleeding subsiding Temperature: normal No abdominal tenderness	Moderate condition. Suspect complete abortion.	1. Give Paracetamol 1000mg (2tabs) 8hourly for 3 days 2. Counsel to minimize emotional and physical effects of abortion 3. Encourage abstinence 4. Advise client on the use of contraceptive 5. Advise on personal hygiene 6. Demonstrate how to wash vagina 7. If bleeding continues, after 3 days, REFER appropriately
II. Pain in the abdomen, Severe bleeding	*Abdominal tenderness *Bleeding from the Vagina	Severe condition Suspect incomplete abortion	1. Conduct Manual Vacuum Aspiration (MVA) if trained 2. If not trained, Refer immediately to the appropriate health facility
IV. Feverish, Foul smelling discharge from the vagina, Trauma.	Temp. 37.50C or above Abdominal tenderness with or without bleeding from the vagina. Foul smelling discharges from the vagina.	Severe condition Suspect septic abortion	Tab. Paracetamol 1000mg 8hourly for 3 days Tab. Metronidazole.400mg 8hourly for 5days. Tab. Ciproflaxacin 500mg 12hourly for 5days, Advise on personal hygiene. REFER to the appropriate health facility

Health Education

Advise Mother /Caregiver on:

1. Advise on personal hygiene
2. Sexuality education
3. Advise on abstinence
4. Counsel on the use of contraceptive
5. Advise on adequate fluid intake
6. Counsel on the danger of unsafe abortion
7. Importance of taking home-based record with anytime she goes to the clinic or hospital

4.7. NUTRITIONAL DISORDERS

Teenagers have high nutritional needs. At this stage, the bone develops to its maximum density and this (growth) depends on adequate nutrition. Adequate nutrition also provides protection against diseases. The nutritional status of an adolescent like other group is influenced by socio-economic factors.

The adolescents are very active physically and their nutritional requirements depend on such factors as physiological state, level of activities, stress, etc. Peer group influences also affect their eating habits especially girls. Therefore efforts should be made to ensure adequate nutrition for this group.

4.7.1 NUTRITION DISORDERS OF ADOLESCENT

HISTORY	EXAMINATION/ACTIONS
1.Have you been eating well? 2.What food are you eating? (Probe type of food for a 24 hour period)? 3.What is your favourite food? 4.Do you have appetite? 5.Are you comfortable with your weight? 6.Did you have Significant weight loss over several days,weeks or months? 7.Have you experience vomiting after eating? 8.Are you pregnant (if Female)?	1. General appearance: obesse, thin, pale, anxious or aggressive looking, restless, moody, unkempt, mode of dressing 2. Orientation: person, place, time 3. Eyes: redness, pallor of the conjunctiva, dilated pupils 4. Mouth: smell of alcohol and other substances 5. Speech: slurred, incoherent in nature 6. Skin: for dryness 7. Perception: hallucination, thought disorders 8. Pockets: drugs in the pocket 9. Temperature, Pulse, Respiration & Blood pressure: check and record 10. Weight and height: measure and record (calculate Body Mass Index – BMI and or MUAC) 11. Urine test for pregnancy

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Excessively fat with or without increased appetite	Obese (BMI: 30 kg/m2 and above) May or may not be depressed	severe condition Suspect Obesity	1.Encourage exercise 2.Nutritional counseling If no improvement, 3.REFER immediately to the appropriate health facility
II. Excessively fat	Overweight (BMI: 25 to 29..9 kg/m2)	Moderate condition Suspect Over weight	1. Encourage exercise 2. Nutritional counseling 3. Review in 1month. If no improvement, REFER appropriately
III. Excessively fat	Malnutrition ≥ 18.5 to < 25.0 Normal	Normal to mild	1. Encourage exercise 2. Nutritional counseling 3. Review in 1 week. If no improvement, REFER appropriately

I. Girl/boy losing weight II. Significant weight loss over several days, weeks or months III. Induces IV. vomiting after eating	Thin, BMI Nutritional status < 16.0 Severe malnutrition or MUAC< 185 mm < 190 mm (SAM) May look anxious Intense fear of gaining weight. Shortness of breath. Feeling overweight or "fat," even if you're underweight. Fear of certain foods or food groups. Experiencing thoughts of self-harm or suicide. May or may not be pale	severe condition Suspect Anorexia nervosa	1.Explain condition to client and parent 2.Nutritional counseling 3.Conduct food demonstration 4.Tabs Vitamin B complex 8hourly for 2 weeks 5.Tab. Folic acid 5mg daily for 2weeks 6.Try to elicit the reason for inducing vomiting and counsel appropriately. If no improvement, REFER to the appropriate health facility
Adolescent weight loss with any of the following	BMI Nutritional status ≥ 16.0 to < 17.0 Moderate malnutrition or MUAC ≥ 185 to < 220 mm Bloating and/or abdominal pain. Muscle weakness and loss of muscle mass. Dry skin, brittle nails and/or thinning hair. Poor wound healing and frequent illness. Bluish or purple coloring of the hands and feet.	Moderate condition Suspect Anorexia nervosa	1.Explain condition to client and parent 2.Nutritional counseling 3.Conduct food demonstration 4.Tab Vitamin B complex 1tab 8hourly for 2 weeks 5.Tab. Folic acid 5mg daily for 2weeks 6.Try to elicit the reason for weight loss and counsel appropriately If no improvement, REFER immediately to the appropriate health facility
Adolescent weight loss with any of the following	≥ 17.0 to < 18.5 Mild malnutrition MUAC ≥ 190 to < 230 Dizziness and/or fainting. Feeling tired. Slow heartbeat (bradycardia) or irregular heartbeat (arrhythmia). Low blood pressure (hypotension). Poor concentration and focus. Feeling cold all the time. Absent periods (amenorrhea) or irregular menstrual periods.	Mild condition Suspect Anorexia nervosa	1.Explain condition to client and parent 2.Nutritional counseling 3.Conduct food demonstration 4.Tab.Vitamin B complex 8hourly for 2 weeks 5.Tab. Folic acid 5mg daily for 2weeks 6.Try to elicit the reason for weight loss and counsel appropriately If no improvement, REFER immediately to the appropriate health facility

Health Education

1. Counsel on dangers of not eating balance diet.
3. Advise on personal hygiene.
4. Nutritional counseling.
5. Counsel on negative peer influence.
6. Importance of taking home-based records with anytime he/she goes to the clinic or hospital.

4.7.2 DRUG AND SUBSTANCE ABUSE

A substance could be a drug, toxin, which its abuse alter the mind, perceptions, feelings, moods and behavior of an adolescent.

The most common substance/drugs that are abuse include:

- Nicotine e.g. cigarettes,
- Caffeine e.g. kolanut and coffee
- Cannabis e.g. Indian hemp,
- Heroin, cocaine, amphetamines,
- Solvents e.g. herbs, solution, petrol, gum,
- Hallucinogens (LSD),
- Barbiturates and sedatives e.g. benzodiazepines (Valium, Lexotan)
- Medications drugs eg Panadol, sudrex, codeine, Pentazocine etc.

Note during history taking: Denial is a common element of substance abuse and addiction, the adolescent may not easily admit to substance use because it will lead to loss of benefits from parents, guardians and families.

HISTORY	EXAMINATION/ACTIONS
1.What substance do you take that may make you eat well, or sleep, or stay awake for long hours? Or strong and active to perform better in skills/studies/work? 2.What job are you doing? Student, physical work etc. 3.What was going on in your life that makes you start using/drinking substance/drugs? 4.What kind of substance/drugs are you using? 5.When you use (or drink), how do you feel? 6.When was your last use substance/drugs? 7.What were the circumstances surrounding your last substance/drugs use? 8.Do you ever feel guilty or ashamed after substance/drugs use? 9.Has anyone ever expressed concern about your drinking habits? 10.Have you ever stopped or tried to limit your substance/drugs use? 11.Have you ever tried to quit using substance/drugs? If so, how long ago was your last attempt? 12.Do you have any medical conditions that could be exacerbated by using substance/drugs? 13.Have you ever experienced withdrawal symptoms when you stopped using substance/drugs? 14.Do you have any other problem?	1. General appearance: pale, anxious or aggressive looking, restless, moody, unkempt, mode of dressing 2. Orientation: person, place, time 3. Eyes: redness, pallor of the conjunctiva, dilated pupils 4. Mouth: smell of substances etc 5. Speech: slurred, incoherent in nature 6. Skin: check for needle pricks 7. Perception: hallucination, thought disorders 8. Pockets: search for drugs in the pocket 9. Temperature, Pulse, Respiration & Blood pressure: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Chronic drugs/substance intoxication	Restless Aggressive, violence Anxiety depression Bloodshot eyes (miosis). Sudden weight loss severe loss of appetite Interrupted sleep patterns: insomnia (difficulty sleeping at night) & or Hypersomnia (sleeping for too long). Disorder affecting: liver, kidney and heart disorder or cancer, Seizures, stroke, and brain damage	Severe drugs or substance abuse Chronic drugs intoxication/addiction	1.Counsel the patient, close friends and family as Addiction is treatable. It's crucial to seek help as soon as possible. 2. REFER immediately to the appropriate health facility
Moderate drugs intoxication	Secretive behavior: The person withdraws from friends and family. Mental confusion Change in complexion: Repeated drug use can lead to acne, paleness and jaundice. There may also be scabs, scars, bruises and track marks on the body.	moderate drugs or substance abuse	1.Invite/counsel close friends and family 2.encourage Cessation of substance use 3.REFER immediately to the appropriate health facility
Acute drugs intoxication	Slurred speech Smell of substances Unexplained change in personality or attitude. Sudden mood swings, irritability, or angry outbursts. Poor personal hygiene: Bathing, brushing teeth and physical cleanliness may decline in anyone who is drug dependent Financial distress: (borrowing money without a reason, sell their belongings to get money for drugs). Ignoring responsibilities: avoid their daily responsibilities such as work, school or taking care of the home. Dental health disorder: dental carries Reckless behavior, isolation	mild drugs or substance abuse	1.Give activated charcoal 2.If not available, give milk drink, 3.Invite and counsel close friends and family on the dangers of substance abuse. 4.friend or loved one should help to observe the patient under round-the-clock 5.encourage Cessation of substance use as it help to minimize the physical harm caused by the abuse of substances. If no improvement REFER appropriately

4.7.3 ALCOHOLISM

Alcohol use disorder:Also called alcohol dependence or alcohol addiction: is a chronic condition characterised by uncontrolled drinking and preoccupation with alcohol, resulting to inability to control alcohol drinking due to physical and emotional dependence.

HISTORY	EXAMINATION/ACTIONS
1. How much alcohol do you drink? 2. How often do you drink? 3.What kind of alcohol do you drink? 4.When was your last drink? 5.What were the circumstances surrounding your last drink? 6. Do you ever feel guilty or ashamed after drinking? 7. Has anyone ever expressed concern about your drinking habits? 8. Have you ever stopped or tried to limit your drink? 9. Have you ever tried to quit using alcohol ? If so, how long ago was your last attempt? 10. Do you have any medical conditions that could be exacerbated by using alcohol? 11. Do you have any mental health conditions that could be exacerbated by using alcohol? 12. Have you ever experienced withdrawal symptoms when you stopped using alcohol?	General appearance: pale, anxious or aggressive looking, restless, moody, unkempt, mode of dressing 2. Orientation: person, place, time 3. Eyes: redness, pallor of the conjunctiva, dilated pupils 4. Mouth: smell of alcohol 5. Speech: slurred, incoherent in nature 6. Skin: check for needle pricks 7. Perception: hallucination, thought disorders 8. Pockets: alcohol in the pocket 9. Temperature, Pulse, Respiration & Blood pressure: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Chronic Alcoholic Dependence	A patient who is dependent on alcohol, may drink: a lot of the time first thing in the morning every evening every day and with any of the following: Slurred or incoherent speech Thought disorders Looking Restless Aggressive, Violence Visual hallucinations or seizures. shaking hands (tremor)	Severe drugs or substance abuse Chronic drugs intoxication/addiction	Counsel the patient, close friends and family as Addiction is treatable. It's crucial to seek help as soon as possible. REFER immediately to the appropriate health facility
Moderate Alcoholic Dependence	Moody, Hallucinations-evidence of perception disorder seizures (fits) anxiety feeling anxious and upset feeling irritable or angry difficulty sleeping (insomnia)	Moderate to severe Alcoholic Dependence	Counseling of patient and family Detoxification (also known as 'detox') can be a key stage of treatment. Detox involves stopping drinking completely REFER appropriately
Acute Alcoholic Dependence	Looking unkempt Altered orientation Physical injury physical withdrawal symptoms, such as shaking (tremor), sweating and sickness (nausea) You may also have more severe symptoms, such as feeling sick (nausea) sweating	Mild alcoholic dependence	Counsel the patient, close friends and family as Addiction is treatable. It's crucial to seek help as soon as possible. REFER appropriately

4.8 MENTAL HEALTH DISORDERS

Mental health problems are those illnesses with psychological or behavioral manifestations and/or impairment in functioning due to social, psychological, genetic, physical biological disturbance. Mental disorders are characterized by abnormalities in a person's emotions, thoughts, cognition, sensory perceptions, belief and behavior. Mental-problems usually manifest in parents in many ways.

The affected may become restless, afraid, suspicious and confused. They may think that people are after them. They may imagine/or complain of seeing strange people or things. They may become violent and attack other people or themselves without cause. The condition may have come on after a fever due to malaria, typhoid, or viral infection, it may also be as result of severe emotional disturbance or abnormal emotional experiences or excessive drinking of alcohol or drugs over a significant period. If nothing is done immediately there may be danger to life and property. Other symptoms include sleeplessness, hearing and seeing things others cannot see and suicidal tendencies.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long has he/she had the problem? 3. Has he/she had this or similar problem in the past? 4. Does the client use alcohol or drug such as Indian hemp? 5. How is he/she functioning as school/work? 6. What is his/her relationship with family members? 7. Is he/she in any love relationship with family members? 8. Is there a recently stressful event? 9. Does he/she sleep well? 10. Does he/she eat well? 11. Any abnormal behavior, speech or experience (e.g excessive spending)? 12. Has he/she caused or threatened injury to others or himself/herself? 13. Has he/she had recent encounter with police, public?	1. General appearance: unkempt, calm, restless, threatening or physically aggressive, ill, withdrawn, confused, tensed, frightened,fidgety, trembling/shaking, sweaty, cold clammy palms, acting strangely, tearful 2. Reaction to the health worker-cooperative, withdrawn, aggressive 3. Eyes: redness, staring, bulging, lid retraction, pallor 4. Mouth: smell of alcohol or other substances 5. Speech: audible, shouting, understandable, meaningful, relevant to the questions asked or to the situation, slurred, rambling, excessive, stammer, hesitant, expressing worry, any strange statements or beliefs expressed. 6. Orientation: responds to name, knows where he is, has idea of time 7. Perception: Seeing or hearing things other people cannot see or hear 8. Pocket: drugs of other substances in the pocket 9. Sign of tension-suspicious, hyper-alertness 10. Thought: expression of hopelessness, suicidal tendencies 11. Weight: weight and record 12. Temperature, pulse, respiration & blood pressure: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Stressful life events such as: a) Parental disharmony b) Examination failure c) Depressed mood	* Nervous * Jittery * Excessive worry * Sad * Tearful	Moderate condition Suspect depression	1. Counsel client/family member separately and jointly 2. If necessary, intervene at home/ school 3. Home visits for follow-up 4. If no improvement, REFER appropriately
II.Complaint of any of the following: a) Excitement, irritability b) Poor, sleep, talkative c) Sexual promiscuity d) Excessive spending e) Changes in appetites	Restless Talkative Irritable Withdrawn Weight loss	Moderate to severe condition Suspect social maladjustment	REFER immediately to the appropriate health facility
III. Behavioural disorders such as: Stealing b) Truancy c) Aggression to people d) Destruction of property	Evidence of Drug use/abuse Aggressive	Severe condition Suspect behavioural disorder	Counsel client/family member REFER immediately to the appropriate health facility

Health Education

Advise Mother/Caregiver on:

1. Counsel on drug compliance
2. Counsel on social behavior
3. Counsel on dangers of drug/substance abuse
4. Advise on personal hygiene
5. Importance of taking home-based records to the clinic or hospital

4.9. VIOLENCE ASSAULTIVE BEHAVIOURS AND SEXUAL ABUSE

Violent Behaviour

Violence is an outwardly directed destructive behavior either to persons (including self) or to properties. It might be behavior learned from the home (family) and can occur in mentally and non-mentally ill persons. It may arise from disturbance of thoughts, feelings or emotions and in adolescence, from a variety of reasons such as alcohol and other substance abuse and diseases affecting the brain. Assaultive Behaviour This refers to actions that involve threatening or attempting to cause physical harm to another person. This type of behaviour includes a range of aggressive acts that can create fear or actual injury in the targeted individual.

Sexual Violence and Abuse

This is any form of sexually related activity that takes place without consent being given and may range from just touching to rape. Sexual abuse goes beyond sexual intercourse alone and includes pornographic materials, writings, verbal comments or non-verbalization e.g. winking, licking lips etc.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long has it been noticed? 3. What caused the problem? 4. Any injury threatened or inflicted? 5. Has he/she had similar behavioral problems in the past? 6. Any recent encounter with police or public? 7. Is there fever, stresses/or recent life problem? 8. Is there any history of alcohol or other substance abuse? 9. Is there any exposure to pornographic materials? 10. Is there any history of fits or head injury?	1. General appearance: unkempt, restless, threatened, or physically aggressive, ill, depression, withdrawn 2. Level of consciousness: respond to sound, touch and light 3. Mouth: smell of alcohol or other substances 4. Speech: slurred, rambling, excessive 5. Orientation: responds to name, knows where he is, idea of time 6. Signs of tension: suspicious, hyper-alertness 7. Evidence of violence, assault or rape (e.g blood stains, semen and physical injuries) 8. Temperature, pulse, respiration & blood pressure: check and record.

I. Uncontrolled anger a) Aggressive to people b) Sexual disorders c) Alcohol intoxication and substance abuse. d) Emotional outbursts e.g. impulse control disorders.	Ill looking, Aggressive Sign of recent drug use Unkempt Level of consciousness: response to sound, touch, light Mouth: smell of alcohol or other substances Tense, suspicious, hyper-alert Speech: Slurred, rambling, excessive. Orientation: Identify person, time, place.	Severe condition Suspect psychosis	1. calm the patient: give IM diazepm 10mg stat (PEP) if available 2. Counsel the family/client 3. REFER immediately to the appropriate health facility 4. Follow-up visit
II. Sexual abuse (Rape)	Anxious Withdrawn Sad Evidence of rape e.g. blood stain, semen or physical injuries	Severe condition	1. Give Post-Exposure Prophylaxis (PEP) if available 2. Counsel the family/client 3. Emergency birth control. It is important to receive birth control and treatment for STDs within 72 hours of the assault for maximum effectiveness. Victims can get emergency contraception up to 5 days after rape 3. REFER immediately to the appropriate health facility 4. Follow-up visit

Health Education

Advise Mother /Caregiver on:

1. Advise patient not to harbor grievances nor harbour bitter revengeful thoughts
2. Advise against alcohol and other substance abuse
3. Advise patient not to get too angry as this may result in loss of self-control
4. Solicit religious/traditional leaders to help in counseling

4.10.1 Sexually Transmitted Infections (STIs) (FEMALE)

Sexually Transmitted Infections (STIs) are infections that are spread through sexual intercourse. Common STIs include gonorrhea, syphilis, herpes, Chlamydia, trichomoniasis, candidiasis, genital warts, HIV/AIDS, etc. STIs can be painful and uncomfortable. They can also have tragic consequences such as pelvic inflammatory diseases and infertility. Therefore, all cases of STIs should be adequately treated.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. Do you have a discharge? 3. How long have you had the discharge? Is it excessive? What is the colour? Is it smelling and itching 4. Do you soil your under wear? 5. Do you feel any burning sensation or pain on urinary? 6. Do you have lower abdominal pain? If yes, for how long? 7. When was your last menstrual period? 8. Do you have any sore or wound in the genitalia? Any bleeding from the vagina? 9. Is there any history of sexual intercourse, pregnancy, delivery or abortion 10. Is there fever 11. Are you on any medication? 12. Risk assessment (see appendices)	1. General appearance: normal uncomfortable, irritable, anxious 2. Temperature, pulse, respiration & blood pressure: check and record 3. Abdomen: tenderness, masses 4. Pelvic examination: vulva-soreness, redness (Speculum) Urethra-discharge Vagina-redness, discharge (colour, smell) Cervix-redness, discharge (smell, colour, sore)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Vaginal discharge itching with or without painful urination	* Profuse muco-purulent discharge from cervix * Risk assessment positive	Severe condition Suspect cervicitis	1. Give Tab Ciprofloxacin 500mg tab (single oral dose) 2. Give Cap Doxycyline 100mg 12hourly for 7 days 3. Review in 7 days, If no improvement, REFER immediately to the appropriate health facility
II) Vaginal discharge Profuse, thin, watery, Creamy, or thick whitish discharge with or without itching Painful Urination.	Discharge is Profuse, Thin, watery, creamy or thick whitish Risk Assessment negative	Severe condition. Suspect the following: Candidiasis Vaginitis Trichomoniasis	1. Nystatin Vaginal pessaries 100,000 units x 14 days OR 2. CLOTRIMAZOLE vaginal pessaries 100mg inserted at night for 6 days AND 3. Tab. Metronidazole 400mg 8hourly for 7 days orally 4. Review in 7 days If no improvement, REFER immediately to the appropriate health facility

III. Severe lower abdominal pain with vaginal discharge	*Vaginal discharge *Tenderness on moving cervix	*Severe condition * Suspect pelvic inflammatory diseases (PID)	1. Give Tab. Ciprofloxacin 500mg stat. (single oral dose) 2. Give Cap. Doxycyline 100mg orally 12hourly for 14 days 3. Tab. Metronidazole 400mg 8hourly for 14 days 4. Review in 4 days, If no improvement, REFER immediately to the appropriate health facility
IV) Lower abdominal pain with any of the following present: a) Late or missed period (menses) b) Recent delivery/abortion c) Vaginal bleeding d) Fever	Vaginal discharge Tenderness on moving cervix Temperature 37.5°C or above	Moderate to Severe condition. Suspect Pelvic Inflammatory Disease (PID)	1.Give Tab. Ciprofloxacin 500mg (single oral dose) 2.Give Cap. Doxycyline 100mg 12hourly for 14 days 3.Tab. Metronidazole 400mg 8hourly for 14 days 4.Review in 7 days, If no improvement, REFER immediately to the health facility
V) Vaginal discharge with or without genital sore/ulcer or painful blister a) sexual exposure with or without groin swelling	Group of painful blisters (often recurrent) Risk Assessment positive	Moderate condition. Suspect Genital Herpes, HIV/AIDS	1.Tab. Acyclovir 400mg 8hourly for 7 days (also in pregnancy) 2.Tab. Ibuprofen 400mg 8hourly for 3days (After food) 3. Keep lesion dry 4. HIV Testing (HT) 5.Avoid sex during treatment 6. 4Cs (Counsel, Compliance, 7.Condom and Contact treatment). 8. Review after 7 days, If no improvement, REFER appropriately

Health Education

1.Advise on the 4Cs (Counsel, Compliance, Condom and Contact treatment).
2.Advise on personal hygiene.
3.Importance of taking home-based records to the clinic or hospital.

4.10.2 Sexually Transmitted Infections (STIs) (MALE)

Sexually Transmitted Infections (STIs) are infections that are spread through sexual intercourse. Common STIs include gonorrhoea, syphilis, herpes, genital warts, HIV/AIDS etc.

STIs can be painful and uncomfortable. They can also have tragic consequences such as infertility. Therefore, all cases of STIs should be adequately treated.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. Do you have a discharge? 3. How long have you had the discharge? Is it mucoid, purulent, profuse or scanty? 4. Have you ever had urethral discharge before? If yes, how long ago? 5. Do you notice the discharge after a particular sexual intercourse? 6. Have you had any sexual contact after you noticed the discharge? If yes, with who? 7. Do you feel any burning sensation or pain on urination? 8. Is there fever? 9. Are you on any medication? 10. Risk assessment (see appendices)	1. General appearance: normal, uncomfortable, irritable, anxious 2. Temperature, pulse, respiration & blood pressure: check and record 3. Throat: redness, pus 4. Skin: rashes 5. External genitalia-discharge ('milk' the urethra) 6. Uncircumcised male-retract the foreskin, be sure the discharge is from the urethra 7. Check glands for redness, ulcer 8. Check scrotum for sores, redness, swelling, (just look, NOT PALPATE 9. Check anus for purulent discharge 10. Groin, swelling, tenderness.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I) Groin (inguinal swelling) with or without pain a) Patient not sexually active b) Under 15 years of age c) Skin rash	Inguinal swelling with or without tenderness Skin rashes	Severe condition Suspect Granuloma Inguinale	1. Give Tab. Erythromycin 500mg 6hourly for 7 days 2. Give Benzathine Penicillin G 2.4mu Intramuscularly (IM) in a single dose 3 Review in 4 days, If no improvement, REFER immediately to the appropriate health facility
II. Scrotal swelling with or without pain a) urethral discharge b). Sexually active	• urethral discharge	Severe condition suspect urethritis	1. Give Erythromycin 500mg tab orally8 hourly for 7 days. 2. Give doxycycline 100mg tab orally 6 hourly for 7 days. 3. Review i 4 days, if no improvement, REFER immediately to the appropriate health facility
III. Genital sores or ulcer a) history of sexual exposure b) Swelling in the groin c) Urethral discharge with or without painful micturition	• sore/ulcer present • Swelling in the groin • Risk assessment positive	Moderate to severe condition Suspect Genital Ulcer Disease (GUD)	1. Give erythromycin 500mg tab orally 6 hourly for 7 days 2. Give benzanthine penicillin G2 IM in a single dose Review in 4 days, if no improvement REFER to the appropriate health facility
IV. Patient with painful micturition, no discharge	• No penile discharge	Moderate condition	1. Give cotrimoxaole 960mg 12 hourly for 10 days 2. Review in 5 days. If no improvement refer immediately to the appropriate health facility

V. Genital sore/ulcer: a) History of sexual exposure b) With or without painful swelling in the genital	• Group of painful blisters (often recurrent) present • Risk assessment positive	Suspect urinary tract infection (UTI) Severe condition Suspect genital herpes and HIV/AIDS	1. Acetylsalicylic acid 600mg (2 tabs) 8 hourly for 3 days (with food) OR 2. Ibuprofen 400mg 8 hourly for 3 days 3. Refer for HTS 4. Avoid sex during treatment 5. Adhere to the 4Cs 6. Review in 7 days. If no improvement, refer appropriately 3. Tablet Paracetamol 500mg 8hourly for 3 days

Health Education

Advise Mother /Caregiver on:

1. Advise on the 4Cs, (Counsel, Compliance, Condom and Contact Treatment)
2. Advise on personal hygiene
3. Importance of taking home-based records to the clinic or hospital

4.10.3 HIV/AIDS (MANAGEMENT OF PATIENT)

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long have you had the problem? 3. Are you losing weight? 4. Have you had infections? 5. Have you had blood transfusion?Scarification? 6. Do you have casual sexual partner or sexual partner of same sex? 7. Have you had contact with relatives or partners with similar problem? 8. Have you been treated for HIV/ AIDS? 9. What other problems do you have? Any cough? Diarrhoea? 10. Repeat infections? Any fever?	1. General appearance: healthy, wasted, pale 2. Temperature, pulse, respiration & blood pressure: check and record 3. Weight: weigh and record 4. Hair: thin or scanty 5. Eyes: sunken, pale conjunctiva 6. Mouth: thrush, red and white patches 7. Neck: and Armpit: enlarged nodes 8. Skin: rashes, ulcers, fungal infection 9. Chest: crepitations, decreased breath sounds 10. Abdomen: palpable organ 11. Groin: enlarged nodes, ulcer 12. Chlamydia: ulcer, rashes, discharges 13. HIV screening after pretest counseling

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I) History of recurrent or persistent diarrhoea of more than 1 month duration a) Generalized itching, skin rash b) unexplained cough longer than 1 month c) Recurrent genital and mouth sores d) Prolonged fever for more than one 1 month e) Severe weight loss f) General weakness g) Ulcer that has refused to heal	Severe dehydration Temperature 37.50C or above May or may not have weight loss greater than10% of body weight Generalized dermatitis Oral candidiasis Genital ulcer Enlarged generalized lymph nodes HIV test positive	Severe condition	1. Symptomatic treatment 2. Give Paracetamol 1000mg 8hourly for 5 days 3. Give ORS solution for dehydration 4. Home-based care includes: a. Provision of counseling to patient and family members on compliance b. Provision of pastoral support/ care c. Prevention of opportunistic infections by encouraging patient to visit the H/F as soon as any complication/complaint arise 5. Management of opportunistic infections 6. Review in 3 days. if no improvement REFER immediately to the appropriate health facility
II)Chronic diarrhoea Chronic cough Mouth and genital ulcer Oral thrush Skin rash	Weight loss General skin rashes Enlarged lymph mode Mouth and genital ulcers Laboratory test positive	Moderate to Severe condition	REFER immediately to the appropriate health facility
III) Claim to be HIV positive patient No other complaint	Laboratory result may or may not be positive and have weight loss.	Mild condition.	1.Counseling 2.Refer for confirmation test.

Health Education

Advise Mother /Caregiver on:

1.Advise on positive lifestyle e.g avoidance of stress
1b. Advice on personnel hygiene
2.Advise on the 4Cs (Counsel, Compliance, Condom and Contact Treatment)
3.Advise on personal hygiene
3b. Advice on dietary intake (balance diet)
4.Importance of taking home-based records with anytime he/she goes to the clinic or hospital

4.11. INJURIES, FRACTURES AND BITES

Injuries occur in different part of the body tissues. They may be on the skin and/or deeper tissues which may be small or large and may include cutting of nerves and tendons. There could be wounds that have been for several days and have become infected or healing poorly; or Sores which are injured, infected or diseased part of the skin. Some may be Ulcers which are injuries in a bodily membrane.

A fracture is a break in the continuity of a bone. It is more serious if there is also a compound fracture, profuse bleeding resulting from wounds. There is an urgent need to stop bleeding to avoid the patient getting into shock. Early cleaning and treatment of wounds or sores is important to prevent infections including tetanus. Adolescent are prone to injuries, fractures, dislocations, wounds, bites from snakes, dogs, scorpions or humans.

Some injuries involve the head which comprises of the brain and the sense organs (for hearing, sight, smelling and tasting). Serious injury to the head may be caused by a fall or a blow to the head. Some signs of this injury are severe headache, unconsciousness, vomiting, and confusion, bleeding from the nose, mouth or ears. Delay in prompt care and/or referral of a patient with head injury may cause permanent disability or death.

HISTORY	EXAMINATION/ACTIONS
1. How did the injury occur? 2. Was there any bleeding? 3. Has it bled much? (Try to carefully assess the amount) 4. Has anything been put on the wound? 5. Have you been immunized with Tetanus Toxoid? Number of doses? 6. Did anyone see it happen to describe the incident? 7. What bit him/her? A Dog, A Snake, A person? Scorpion? Or any other thing? 8. Where is the pain? How bad is it? 9. Can you move/use limb? Is there limitation of movement? 10. Which movement causes greatest pain? 11. Was there loss of consciousness? Irritability? Fits? 12. Do you have headache, nausea, or vomiting? 13. Is there any other thing you like to tell me or any additional injuries?	1. General appearance: a. Conscious, confused, irritable, unconscious b. Pale, cold or clammy c. Increased sweating. d. Increased respiration and pulse rate e. Patient agitated or restless 2. Check the vital sign: Temperature, Pulse, Respiration, Blood pressure record 3. Head: Visible wound, swelling, bleeding a. Ear: Bleeding b. Eyes: Subconjuctival hemorrhage, pupils for size reaction to light c. Mouth: Teeth, fractures, bleeding d. Nose: Bleeding, fractures 4. Hemoglobin: Estimate and record 5. Chest: Ribs for fractures 6. Abdomen: For tenderness, rigidity 7. Urine: Monitor output WOUNDS: 8. Size and site 9. Bleeding. 10. Pus or dirt in wound. 11. Swelling, redness or warmth around wound. SPRAINS, dislocation AND FRACTURES 12. General appearance: a. Signs of shock – increased sweating, pallor, 13 restlessness, rapid pulse and rapid respiration. b. Degree of pain. c. At Limb or joint. d. Tenderness or swelling. e. Warmth or redness. f. Crepitus (crackling). g. Associated Open wound.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Wound on the limb resulting from injury	Evidence of fracture (unable to move the limb) BP below 110 / 70 mmHg or above Lethargic Wound with blood loss	Severe condition Suspect fracture	1. Immobilize the limb 2. Dress wound 3. Give Tetanus Toxoid 0.5cc IM stat 4. Caps Amoxicillin 500mg 8hourlyfor 5days 5. Tabs Metronidazole 400mg 8hourly for 5days 6. Tabs Ibuprofen 400mg 8hourly for 3days or Paracetamol 1000mg 8hourlyfor 3days 7. Tab Vitamin C 200mg 8hourly for 7 days 8. Review in 3 days. if no improvement, REFER immediately to the appropriate health facility
II Wound as a result of deep laceration	• Warm to touch (temp . 37.50°C (or more) • Stiff neck Bleeding from wound, Torn tendon Check hemoglobin	Moderate to Severe Condition Suspect laceration	1. Clean well with available diluted antiseptic solution 2. Control bleeding by pressure 3. Give Tetanus Toxoid 0.5ml IM stat 4. Perform stitches and dress the wound were possible 4. If hemoglobin is less than 10g/dl 5. Review in 3 days. if no improvement, REFER immediately to the appropriate health facility
III Wound from dog bite	Evidence of bite (wound from fangs seen at site of bite) With or without bleeding	Mild condition Suspect dog bite	1. Wash wound with soap or detergent/ diluted antiseptic solution 2. Advise patient to observe the dog for 10 days and report any change in the dog's behaviours or death of dog. 3. Give anti-rabies injections 1ml on days 0, 3, 7, 14 and 28 on the deltoid. Depending on the manufacturers instruction. 4. Tab Ibuprofen 400mg 8hourly for 3 days 5. Tetanus Toxoid 0.5ml I.M stat 6. Daily dressing of wound after cleaning well. 7. If dog is alive after 10 days, discontinue anti-rabies injection. 8. REVIEW daily.

IV Wound from human Bite	Wound at the site of bite	Moderate condition Suspect human bite	1. Tablet Folic acid 5mg daily for 2 weeks 2. Tablet Paracetamol 500mg 8hourly for 3 days1.Wash with diluted antiseptic solution or normal saline 3.Tetanus Toxoid 0.5ml I.M stat 4. Caps Amoxicillin 500mg 8hourly for 5days 5.Vitamin C 200mg 8hourly for 14 days 6. Dress wound daily 7. Review in 5 days 8. If no improvement, REFER appropriately
V History of snake bite	Wound from fangs seen at the site of the bite. Swelling of the affected part Bleeding gum	Severe Condition Suspect snake envenomation	1. Keep client calm, clean area thoroughly. 1.2. Give anti-snake venom e.g. 10mls stat Snake venom antiserum sensitivity test is done before by administering 0.1ml of the serum to the victim. (read and follow manufacturer's instructions) 3. IV Hydrocortisone 100mg stat 4.Tetanus Toxoid 5ml IM stat 5.Tab Paracetamol 1000mg 8hourly for 3days 6. Caps Amoxicillin 500mg 8hourly for 5days 7. Review in 3 days. If no improvement, REFER immediately to the appropriate health facility
VI . Scorpion/Bee/Wasp sting	*Spots with or without history or sign of leprosy *Ulcers, particularly on the hands and feet with or without deformity Hypo-pigmented spot with or	evere condition	1.Wash ulcers thoroughly with warm salt water 2. Clean and dress daily 3. If no improvement, REFER
VII. History of wound	* Greater than 4cm * No signs of shock Evidence of infection (septic wound)	Moderate condition Suspect wound	1. Clean wound with/diluted anti-septic solution 2.Tetanus Toxoid 0.5ml IM stat 3.Tab Ciprofloxacin 500mg 12hourly for 5days 4. Dress wound with Eusol 5.Vitamin C 200mg 8hourly for 14days 6. REVIEW in 5 days if no improve-ment, 7. REFER to the appropriate health facility

VIII. History of wound	*Wound less than 4 cm * No signs of shock *Without evidence of infection	Mild condition Suspect wound	1. Clean wound with diluted antiseptic solution 2. Stop bleeding by pressure 3. Daily dressing with Sofratulle or vaseline gauze 4. Tetanus Toxoid 0.5ml IM stat 5. If wound is a laceration and less than 6 hours, suture if needed 6. Cap Amoxicillin 500mg 8hourly for 5 days 7. Tab Paracetamol 1000mg 8hourly for 3 days 8. Tab Vitamin C 200mg 8hourly for 7 days 9. REVIEW in 3 days, if no improvement, refer immediately to the health facility
IX. Injury to the head	*Bleeding from ear or nose *May or may not be conscious * Stiff neck *May or may not be bleeding from ear or nose * May have convulsion *Wound seen *Abnormal behavior	Severe condition Suspect fracture of the skull	1. Asses rapidly using the ABC method 2. REFER immediately to the appropriate health facility
X. Injury to the head with transient loss of consciousness	*With or without wound seen * Pupils reacting normally to light *With or without tenderness	Moderate to severe condition	1. Asses rapidly using the ABC method 2. Tab Paracetamol 1000mg tds x 3days if conscious 3. Review in 3 days. if no improvement, REFER immediately immediately to the appropriate health facility
XI. Injury to the head without transient loss of consciousness	*Pupils reacting normally to light *No tenderness	Moderate to severe condition	1. Monitor vital signs hourly for 4 to 6 hours 2. If condition deteriorates, REFER appropriately
XII. Injury to the head, No other	* Conscious * No other signs of head injury	Mild condition	1. Check vital signs and advice to return immediately if vomiting or headache or drowsiness occurs. 2. if patient returns, REFER immediately to the appropriate health facility

Health Education

1. Advise on use of protective clothing to avoid bites.
2. Importance of compliance with treatment.
3. Give nutritional counseling.
4. Advise on environmental sanitation.
5. Advise on reducing risk of accident.
6. In cases of snake bite, if possible, take along killed snake to the clinic.
7. Advise on prevention of head injuries e.g. use of crash helmets, and seat belts.
8. Prevention of home and road accidents.
9. Importance of taking home-based records during visit to the hospital.

4.12. NECK DISORDERS/INJURIES

Neck problems can result from physical injury or infections (tuberculosis, meningitis, HIV/AIDS etc.) or it is common among adolescents, which include an injury (fall from high object, slippery, or domestic accident resulting to affect the structures in the neck This may manifest as enlarged nodes and/or lumps around the neck which may be associate with stiffness of the neck. Lumps can be located in the centre or front of the necks (for example goiter)

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. Do you have difficulty swallowing? Hoarseness? 3. Do you have pain and/or stiff neck? Neck swelling? 4. Do you have pain or headache? 5. Did you injure your neck? 6. Do you have fever? 7. Do you have weakness or numbness of the arms, hands or legs? 8. Is there weight loss or night sweats?	1. General appearance: Ill-looking, distressed. 2. Check the vital sign: Temperature, Pulse, Respiration, Blood Pressure and record 3. Estimate weight and record 4. Head: Abscesses, sores or pustules, lumps a. Ears: Signs of infection. b. Mouth: Abscesses or sores 5. Neck: Swelling, nodes, stiffness, tenderness 6. Arms and Legs: Weakness or decreased sensation.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Neck swelling with or without difficulty in breathing	*Palpable mass in the front of the Neck *May be in respiratory distress	Severe condition Suspect goiter	REFER immediately to the appropriate health facility
II. Pain and stiff neck without history of fever or injury	*Stiff Neck *No weakness of arms and legs *No evidence of injury	Moderate condition Suspect torticollis	1. Reassure client 2. Tabs Ibuprofen 400mg 8hourly for 3 days (After meal) 3. Methyl Salicylate Ointment 4. Review in 2 days, 5. If there is no improvement, REFER immediately to the appropriate health facility
III. Enlarged, painful nodes around the neck with or without fever	Palpable, tender nodes	Severe condition Suspect infection of the head and neck region	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tab Metronidazole 400mg 8hourly for 5 days 3. Tab Ibuprofen 400mg 8hourly for 3 days. 4. Review in 3days. 5. If no improvement, REFER immediately to the appropriate health facility

Health Education

1. Advise on prevention of neck injuries.
2. Advise on complications of mumps in males.
3. Advise on prevention of transmission to others.
4. Give oral hygiene instructions.
5. Importance of taking home-based records during to the clinic.

4.13 EAR, NOSE AND THROAT DISORDERS

The ear, nose and throat are linked and whatever affect one may affect the other. The ear is used for hearing and to maintain a balance, and any disease of the ear which is not urgently and adequately treated may not only lead to deafness, but also lead to inability to maintain balance.

Ear problems commonly caused by middle ear infection often accompanied by pain, dullness and redness of the ear drum with drainage of pus or fluid infections of the nose, throat or Eustachian tube may spread to the middle ear. It is important to examine the bone behind the ear if it is painful (tender to touch); it may mean that the infection has spread to the mastoid bone. Testing an adult for hearing, one is advised to sit behind the client and ask him to repeat words or numbers you whisper to him.

The nose is the organ for sense of smell, and for breathing. When the nose is affected by disease or injury, it discharges watery fluid (mucus, blood or pus.) Blocking of the nose may lead to breathing difficulties.

The throat is the junction of the ear, nose and mouth. It contains the tonsils, pharynx, etc. Diseases affecting the throat can easily spread to the chest and brain.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long have you had the problem? 3. Is there catarrh? 4. Is there pain? Boil? Any discharge: pus, blood or water? 5. Do you feel pain while swallowing? 6. Is there fever? Watery eyes? Headache? 7. Has it happened before? 8. Has there been any blow or injury to the head? 9. Do you have nose bleeding? Can you estimate the volume? When did it start? 10. Are you having bleeding from any other part of the body? 11. Is there pain from the teeth or gum? 12. Do you notice bad or foul breath? 13. Is there diarrhea or vomiting? 14. Do you work in a noisy environment or with noisy machines? 15. Have you put anything into your ears, nose and throat? Did anything enter your ears, nose or throat? 16. Is there any trauma to the ear, nose or throat?	1. General appearance: Ill-looking, in distress. 2. Check the vital sign: Temperature, Pulse, Respiration and Blood Pressure and record 3. Ear: (a) Pus or blood discharge; (b) Red or bulging drum; (c) Perforated drum, foreign body (d) Cerumen (Wax); (e) Bone behind the ear (mastoid) for tenderness. 4. Eye: Redness, watery discharge 5. Nose: Mucus or catarrh 6. Neck: Stiffness or enlarged lymph nodes. 7. Throat: (a) Redness (b) Enlarged tonsils, pus. (c) Hoarseness or loss of voice. 8. Chest: Crepitations, wheezes, decreased breath sounds 9. Test for hearing and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Ear problem with vomiting *Tenderness behind the ear	*Temperature 37.5C or above *Tenderness behind the Ear Restlessness	Severe condition Suspect Mastoiditis	1. REFER immediately to the appropriate health facility
II. Boil in external ear	*Soft, tender swelling with well-defined margins	Moderate to severe condition Suspect boil	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Ibuprofen 400mg 8hourly for 3 days; OR 3. Tab Paracetamol 1000mg 8hourly for 3 days 4. Review in 3 days, If no improvement REFER immediately to the appropriate health facility

III. Pain in the ear with or without pus or water from the ear or vomiting	*Temperature 37.5 or above Red, dull bulging ear drum or *Perforation of ear drum Pus or blood discharge	Severe condition Suspect Acute Supurative Otitis Media with perforation	1. REFER immediately to the appropriate health facility
IV. Pain in the throat, Difficulty with swallowing	• Bossing of forehead • Enlarged and red tonsils	Moderate condition: Suspect Tonsillitis	1. Cap Amoxicillin 500mg 8hourly for 5days 2. Tab Metronidazole 400mg 8hourly for5 days 3. Tab Ibuprofen 400mg 8hourly for 3days 4. Warm saline gargle 5. Review in 3 days, If no improvement REFER immediately to the appropriate health facility
V. Pus from the nose, headache, Pain on either side	*Tenderness of the nose	Moderate to severe condition Suspect abscess	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Ibuprofen 400mg 8hourly for 3days 3. Tabs Vitamin C 200mg 8hourly for 7 days 4. Review after 3 days, If no improvement REFER to the appropriate health facility
VI. Bleeding from the nose	*Evidence of bleeding from the nose	Moderate condition Suspect Epistaxis	1. Sit patient in upright position 2. Teach patient to pinch side of the nose for 5-10 minutes 3. Re-examine, if bleeding does not stop, apply cold compress to nose for 10 minutes. 4. If no improvement, apply Adrenaline nasal pack. 5. If no improvement, REFER appropriately
VII. Stuffy nose Catarrh with or without fever Sneezing with or without water	*Temperature 37.5°C or above *Watery eyes Catarrh	Mild condition Suspect Sinusitis	1. Tabs Chlorpheniramine maleate 4mg 12hourly for 3days. 2. Tabs Paracetamol 1000mg 8hourly 3 days 3. Tabs Ascorbic acid 200mg 8hourly for 7 days 4. Tabs Cotrimoxazole 960mg 12hourly for 5days 5. Advise to avoid or reduce exposure to dust or pollen 6. Review in 3 days, if no improvement, REFER to the appropriate health facility
VIII. Pain in the throat Fever Inability to swallow Sore throat	*Throat inflamed * Tonsils not red and no pus	Mild condition Suspect pharyngitis	1. Tabs Ibuprofen 400mg 8hourly for 3days 2. Caps amoxicillin 500mg 8hourly for 5days 3. Tabs Metronidazole 400mg 8hourly for5 days 4. Tabs Vitamin C 200mg 8hourly for 7 days 5. Encourage fluids intake 6. Review in 3 days, If no improvement, REFER appropriately

Health Education

1. Advise on personal and environmental hygiene.
2. Advise on care of the ear, nose and throat.
3. Discourage traditional practices such as uvulectomy and tonsillectomy
4. Give nutritional counseling.
5. Advise on avoiding exposure to air pollution (fumes, dust and pollen).
6. Discourage home removal of foreign body.
7. Importance of taking home-based records during visit to clinic.

SECTION FIVE

ADULT HEALTH

(18 – 59 Years)

5.0 ADULT HEALTH

This refers to an adult often recognized as somebody who is matured, well developed, a fully grown and is 18 to 59 years and above. Section 5.1 is for adults who present with no complaints. The CHEW should be able to screen for health disorders in the well adults. Note that, other health conditions affecting adolescents (13 years and above) and some conditions affecting elderly are also treated in this section

5.1. ADULT FIRST VISIT

HISTORY	EXAMINATION/ACTIONS
1.How are you feeling? 2.If ill, what are your symptoms? 3.Are you taking any medication or treatment? If so, please describe? 4.Have you lost weight recently? 5.Have you had any operation(s)? If yes, what and when? 6.How are you able to cope with daily life at home? 7.Do you smoke, take snuff or alcohol? OBSTETRIC 1.Have you been pregnant before? If yes, how many times? 2.What was the nature of your delivery- CS, SVD or forceps delivery? 3.Have you had abortion before? If yes, how many times? 4.Are you pregnant now? If so, are you receiving prenatal care? What is your LMP? 5.How many children do you have? 6.How many are alive? IMMUNIZATION 7.Have you ever had TD immunization? How many doses? SOCIAL 1.How many wives do you have or how many wives does your husband have? 2.Do they all live with you? 3.Are you employed? What work do you do?	1.General appearance: healthy, ill-looking, thin or depressed 2.Vital Signs; Temperature, Pulse, Respiration and Blood Pressure: check and record. 3.Eyes: pallor, jaundice, hyperemia , discharge. 4.Ear and Nose: examine for discharge, redness, swelling, foreign bodies or excess wax 5.Mouth: conditions of the lips, mucous membranes, gums, teeth, uvula and throat 6.Neck: , any stiffness, swelling or enlarged thyroid, 7.Breasts: lumps, flat or inverted nipples. 8.Chest: a) difficult or rapid breathing; b) crepitation, rhonchi, wheezing, rales or any decreased breath sounds. 9.Heart: rate, rhythm, 10.Abdomen: scars, tenderness, masses, hernias 11.Limbs: Deformities, Varicosities, Oedema. 12.Weight and height: Measure and record, determine BMI and WHR 13.Urine: Colour, Odour, Protein, Sugar 14.Stool: Consistency, Parasites and Ova 15.Haemoglobin/PCV: estimate and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.First visit/Regular Check up	• Well, Healthy looking. • No abnormalities detected-well, Healthy looking. • No abnormalities detected	Normal condition	1.Encourage checkup twice yearly 2.Regular exercises; 3.Advise on : a.maintaining good posture; b.yearly PAP smear if female; c.regular breast examination, d.regular prostate examination in male 4.Educate on a.STI ,HIV/AIDS prevention; b.Child spacing; c.Nutrition 5.If female, within child bearing age, advise to take 5 doses of Tetanus Diptheria (TD) 6.Refer for urine and stool testing
II.First Visit	Abnormalities detected	Abnormal condition	REFER to appropriate section.

Health Education

1.Healthy lifestyle.
2.Personal hygiene.
3.Regular clinic attendance.
4.Environmental sanitation.
5.Nutrition.
6.Importance of taking home-based records to the clinic or hospital

5.2 FEVER

Fever is hotness of the body. It is body temperature of 37.2°C and above. Fever is a sign of infection or loss of body fluid. Some common causes of fever in adults are malaria, tuberculosis, respiratory tract infection, gastrointestinal tract infection and other infections. All patients with history of fever should be treated and encouraged to drink plenty of water, other fluids and fruits, and should be advised to wear light clothing. High fever (> 38.0°C) should be treated as emergencies

HISTORY	EXAMINATION/ACTIONS
1.How long have you had the fever? 2.Is there any ear pain, cough, catarrh or headache? If yes, for how long? 3.Is there any diarrhoea, nausea, vomiting or abdominal pain? If yes, for how long? 4.Do you sleep inside long lasting Insecticidal (mosquito) bednet? (LLIN)? 5.Are there any pain(s); muscle, joints, back, arms and legs? 6.Do you have pains or burning sensation on micturition? If yes, for how long? 7.Is anyone else sick in the family? 8.Are you taking any medication or treatment? If yes, describe 9.Have you been losing weight recently? 10.Any recent travel? If yes, to where? 11.Do you bleed from any part of your body? If yes, indicate	1.General appearance: ill-looking, weak, depressed, dejected 2.Vital Signs; Temperature, Pulse, respiration, BP - check and record 3.Skin: rashes, ulcers, discolorations, macular lesions 4.Eyes: Jaundice, pallor, redness or discharge. 5.Nose: catarrh, bleeding, polyps. 6.Mouth: Blisters, ulcers, redness or swelling of the gum, tonsils or uvula throat, 7.Ear: redness or dullness of drum, tenderness behind the ear, discharge or difficulty hearing 8.Neck: stiffness,. 9.Chest: a) crepitation, wheezes, decreased breath sounds; b) rapid or laboured respiration. 10.Breasts: tenderness, swelling. 11.Abdomen: tenderness, enlarged spleen or masses 12.Genitalia: discharge, sores, rashes, swelling, 13.Elicit Kernig's sign mRDT: Test and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. History of trauma to head, abnormally sleepy or difficult to wake, vomiting difficult to wake, vomiting	Temperature 37.2°C and above Yellowish conjunctiva, Tiredness	Severe Condition Suspect Hepatitis	1.Advise on: a.Bed rest b.Glucose drink and High carbohydrate diet No fatty food 2.Tabs Multivitamin one tab 8 hourly for 2 weeks 3.REFER immediately to the appropriately to the appropriate health facility
II. Fever, General body Pains, Headache and Stiff neck	Temperature 37.2°C and above Tenderness of Abdomen Stiffness of neck Positive Kernig's sign	Severe Condition Suspect Meningitis	I.IV Benzyl Penicillin 1,000,000units (600mg) 6 hourly for the first 48 hours Or 48 hours IV Ceftriaxone 2g 12 hrly for Or IV Chloramphenicol 1g 6 hourly for 48 hours 2. Review in 24 hours. If no improvement, REFER IMMEDIATELY to the appropriate health facility

III. Fever, known leprosy patient	joint pain on pressure Reddish or Painful Skin patches Inability to close the eyes Muscle weakness	Severe condition Suspect complications of leprosy	1.Give Acetylsalicylic Acid 600mg (2 tabs) stat, 2.If client has history of Ulcer,give a.Tabs Paracetamol 1g 8 hourly for 3 days. b.Treat according to TB / leprosy 3. Management if trained and have necessary drugs 4.If not trained, REFER immediately to MDT Clinic.
IV. Fever longer than 2 weeks. Abdominal pain with or without diarrhea Vomiting and headache	Temperature 37.2°C and above Tiredness Tenderness in the abdomen Rapid Diagnosis Test (RDT) for malaria negative	Moderate to severe Condition Suspect Typhoid fever	1.Low Osmolarity ORS. 2.Acetylsalicylic-Acid 600mg 8 hourly for 3 days, if client has history of Ulcer, give Tabs Paracetamol 1g 8 hourly for 3 days. 3.Caps Amoxicillin 500mg 8 hourly for 5 days. 4.Review in 3days If no improvement, REFER immediately to the appropriate health facility
V. Fever, Cough, Difficulty in Breathing, Headache	Temperature 37.2°C and above Difficult or fast breathing	Moderate to severe condition Suspect Pneumonia	1.Tabs Paracetamol 1g 8 hourly for 3days. 2.Caps Amoxicillin 500mg 8 hourly for 5 days. 3. Review 3 days, If no improvement, REFER immediately to the appropriate health facility
VI. Prolonged Fever for more than 1 month, with recurrent or persistent Diarrhoea, with weight loss, with or without Skin rashes	Dehydration Temperature 37.5°C or above Marked weight loss Skin rashes	Severe condition Suspect HIV/AIDs	1.Give Tabs Paracetamol 1g 8 hourly for 3days 2.Low Osmolarity ORS solution 3.Counsel on HIV and perfom test, if positive,REFER toART site, if negative REFER appropriately
VII. Fever with facial swelling, sore throat and bleeding from nose, eyes, ears, gums	Temperature 37.2°C and above Headache Joint and muscle aches Sore throat Weakness Abdominal pain Loss of appetite Vomiting or coughing up of blood Bloody diarrhoea	Severe Condition Suspect Hemorrhagic Fever	1. REFER immediately to the appropriate health facility NOTE: No specific treatment for the disease is yet available Efforts to help those who are infected include giving either oral rehydration salts/ORS or intravenous fluids
VIII. Fever: > 38°C Moderate to severe dehydration Sub- optimal level of consciousness Cough, respiratory distress, with or without history of exposure to COVID- 19	• Acutely ill- looking, toxic, • Dysnea as evidenced by labored breathing, fast breathing, respiratory rate > 18 cpm etc. • Moderate to severe dehydrated Sp02(peripheral capillary oxygen saturation) at room temperature < 92% Fever usually high grade > 38°C Generalized body weakness	Severe Condition Suspect COVID- 19	2. REFER immediately to the appropriate health facility 1. ENSURE ALL UNIVERSAL PRECAUTIONS ARE IN PLACE

IX. Fever less than 1 week, no other complaints	Temperature 37.2°C or above mRDT positive	Mild condition Malaria fever	1. Drug Age Wt.(kg) Dose Tab Arthemeter /Lumefantrin (AL) >14yr >35kg 80mg/480mg 1tab at 0hour, 8hours later, then, 12 hourly for 2days OR Tab Artesunate/Amodiaquine (AA) >14 >40 100mg/300mg 12hourly for 3 days Tabs Paracetamol 1g 8 hourly for 3 days. 2. Review in 3 days. If there is no improvement, REFER immediately to the appropriate health facility

Health Education

1.In case of known penicillin or cephalosporin allergy, give chloramphenicol inection.
2.Advise on environmental sanitation and prevention of disease transmission.
3.Encourage use of insecticide treated net/materials for malaria control.
4.Advise on Personal and food hygiene.
5.Give nutrition counseling.
6.Advise on safe water and disease prevention.
7.Importance of taking home-based records to the clinic or hospital.

5.3 COUGH

Cough is a rapid expulsion of air from the lungs in order to clear the lungs and airways of fluids, mucus, food particles or other materials. Cough is a sign of irritation of the throat and wind pipe which comes suddenly. Causes of cough are: common cold, inhalation of dust, fumes and pollens, heart and lung diseases, cigarette smoking, etc. Cough sometimes helps clear the lung of accumulated mucus . It is therefore not good to give cough suppressants unnecessarily when a patient has cough.

HISTORY	EXAMINATION/ACTIONS
1. How long have you been coughing? How often? 2. Do you have difficulty in breathing? 3. Do you wake up at night with shortness of breath? 4. Do you bring up sputum? If yes,what colour is it? (Yellow, Green,White or Blood stained) 5. Is there fever or nasal discharge or Sore throat? 6. Is there any loss of appetite? 7.Have you been losing weight or sweating at night recently? 8. Does anyone else in your family have cough? 9. Have you taken any medication? If yes, what type? 10. Is your feet swollen? 11. Does the cough come while frying, sweeping, or when exposed to fumes, dust, cigarette smoke, snuff, pollen or cold?	1. General appearance: ill-looking, thin, tired, 2.Vital signs:Temperature, Pulse, Respiratory rate, Blood pressure: check and record. 3. Nose: Nasal discharge , flaring of alae nasi, swelling. 4.Throat: reddish or purulent tonsils. 5. Chest: a) Difficult or rapid breathing. b) Tenderness, crepitation, rhonchi or decreased breath sounds. unkempt, depressed c) In-drawingHeart: beats and rhythms 6.Abdomen: Enlarged liver, spleen, tenderness 7.Ankles: Oedema 8. Sputum: if available, colour, quantity and appearance and AFB stain up to 3 times

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Cough with any of the following: a.Swollen feet b.Shortness of breath c.Blood in sputum d.Severe Chest pain e. Easy fatigability	• Pedal Oedema • Cough	Severe condition Suspect heart condition	1.REFER immediately to the appropriate health facility
II. Cough with wheezing, breathlessness	• Cough with mild wheezing with: • normal temperature (36.20C -37.2°C) • respiration 40 cycles/min • no signs of heart failure	Severe condition Suspect asthma	1. Give Broncho dilator: Inj IM Triamcinolone Acetonide 10 – 40mg stat slowly 2. Give Steroids : Prednisolone 5mg 12hourly for 5 days, OR 3. IM Hydrocortisone 100mg stat, OR 4. Dexamethasone 1mg 12hourly for 5 days 5. Review in 30-60 minutes 6. If there is improvement, discharge on oral broncho dilators: Salbutamol 4mg 12hourly for 7 days. 7. If no improvement, REFER to Chest Clinic

III. TB patient on treatment presenting with cough, blurred vision and dizziness	Jaundice, Skin rashes, Cough Severe abdominal pains Chest pain	Severe condition Suspect reaction to TB drugs	1. Stop all anti TB medications 2. REFER immediately to DOT clinic or hospital for better management
IV. Bulging forehead	Cough with yellowish green sputum	Moderate condition Suspect Pneumonia	1. Tabs. Paracetamol 1g 8 hourly for 3days. 2. Caps. Amoxicillin 500mg 8hourly for 5 days. 3. Review 3 days, If no improvement, REFER immediately to the appropriate health facility
V. Cough of more than 2weeks duration with or without chest pain a) Weight loss b) Shortness of breath c) Blood stained sputum	Cough Sputum may or may not have blood Weight loss Breathlessness AFB positive or negative	Moderate condition Suspect Pulmonary tuberculosis	1. If at least 2 out of 3 sputum tests are positive, treat for Pulmonary TB: 2. If trained on TB Management and have necessary drugs treat according to treatment regimen in the Annexure 3. If not trained, REFER to DOT clinic for better management 4. If negative, refer to DOT Clinic for further investigation.
VI. Cough with numbness or tingling sensation	TB patient on anti TB medications	Mild condition Suspectside effect of anti-TB medication	Give Tab: Vit. B Complex. 1 tab 8 hourly for 3 days REFER to DOT clinic
Persistent cough for more than one month, Weight loss, Itching,, Generalized skin rash, Diarrhoea for more than one month, Fever	Temperature 37.5°C or above Cough Weight loss Diarrhoea Positive or negative result for AFB stain	Severe condition Suspect HIV/AIDS	1. Tabs. Paracetamol 1000mg 8 hourly for 3 days 2. Tab. Chlorpheniramine (Piriton) 4mg at night for 3days 3. Low Osmolarity ORS 4. REFER For HCT Counsel and test for HIV 5. If positive, REFER to ART site

Health Education

1. Overcrowding.
2. Personal hygiene Regular clinic attendance.
3. Adequate ventilation.
4. Nutrition.
5. How to dentify and avoid trigger factors.
6. Importance of taking home-based records during visit to clinic or hospital.

5.4 DIFFICULTY IN BREATHING

Difficulty in breathing means the client has to make extra effort to breathe. Rapid breathing in an adult means breathing more than 25 times per minute. It is important to observe the client's breathing carefully as part of your general observation. Most common causes of difficult breathing are pneumonia, asthma, tuberculosis, bronchitis and heart failure. Wheezing in an adult can be caused by asthma or congestive heart failure.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been sick? 2.Do you have breathing difficulty with wheezing or swollen neck? 3.Did it come suddenly or slowly? At night? While walking? Lying? Sweeping? Fumes? Dust? Cigarette smoking? Snuffing? Pollen? Cold? 4.Can you sleep flat at night? 5.Do you have fever or sweating at night? 6.Is there cough? 7.Do you bring up sputum? If yes, what colour?(Yellow, Green, White or with Blood) 8.Any weight loss recently? 9.Is there pain? Is it worse with deep breathing? 10.Is there dizziness or tingling of the fingers? 11.Does your skin itch? 12.Are you on any medication? If so, what type?	1.General appearance: ill-looking, thin, depressed, unkempt, tired. 2.Vital signs: Temperature, Pulse, Respiratory rate, Blood pressure : check and record. 3.Nose: nasal discharge, polyp, swelling,, flaring of alae nasi. 4.Throat: reddish or purulent tonsils. 5.Chest: 6.a) Difficult or rapid breathing. 7.b) Tenderness, crepitation, rhonchi or decreased breath sounds. 8.C) In drawing 9.Heart: beats and rhythms 10.Abdomen: enlarged liveror spleen, tenderness 11.Ankles: Oedema 12.Sputum: if available, check colour, quantity and appearance 13.Back: straight or abnormally bent. 14.Haemoglobin/ PCV: estimate and record 15. Stool microscopy, culture and sensitivity (if services available)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Shortness of breath, Difficulty in lying flat on the back	Shortness of breath with or without the following: Enlarged liver. Low haemoglobin of 7gm/dl or PCV of 21% Rapid pulse Goitre.	Severe condition Suspect heart condition	REFER immediately to the appropriate health facility
II. Breathing difficulty with wheezing and itchy skin rash	Wheals or hives	Moderate to severe condition Suspect Allergy	1.Tabs Chlorpheniramine maleate 4mg 12hourly for 3 days 2. Review in 3 days, If no improvement, REFER immediately to the appropriate health facility
III. Difficulty in breathing with wheezing	Difficult breathing Wheezing	Moderate condition Suspect Asthma	1. Inj.Aminophylline 250mg stat slowly over 15 minutes 2. Review in 30 minutes, if improved, 3.Tabs Salbutamol 4mg 8hourly for 5 days. 4. Encourage fluid intake. 5. Review after 24 hours. If no improvement, REFER appropriate

Health Education

1.Avoid pollutants and smoking
2.Avoid overcrowded spaces.
3.Proper ventilation and adequate fluid intake.
4.Taking cold substance.
5.Steam Inhalation for congested nose or chest
6.Nutritional content.
7.Importance of taking home-based records during visit to clinic or hospital.

5.5. DIARRHOEA

Diarrhoea is the passage of frequent, loose or watery stools more than 3 times within 24 hours or the passage of stool more frequent than is normal for the individual. It is important to observe for mucus and blood in stool and to note the colour. Causes of diarrhoea include dysentery, typhoid fever, cholera, ingestion of poisons and contaminated foods, HIV/AIDS, anxiety, etc. Diarrhoea can lead to loss of body fluids and electrolytes. Persistent diarrhoea is one that has lasted for two or more weeks. The major complication of Diarrhoea is dehydration.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been having diarrhoea? 2.How many times do you pass stool each day? Today? 3.Have you passed any since coming to the clinic? 4.Are the stools watery? Is there blood or mucus? 5.What is your source of drinking water? 6.What type of toilet facility do you have/use? 7.Are you nauseated or have you vomited? 8.Do you have fever? Headache? Abdominal pain? Cough? 9.Are you sometimes constipated? 10.Is there anyone else in the household who is sick? 11.Have you eaten plenty of fresh fruits (e.g. mango, orange) or plenty of pepper? 12.Has the diarrhoea reoccurred or persisted for more than one month? 13.Any weight loss? Skin rashes? Oral thrush? 14.Are you on any medication? Local herbs? Have you taken any medication or treatment for this?	1.General appearance: ill-looking, thin, weak, dehydrated 2.Temperature, Pulse, Respiratory rate, Blood Pressure: check and record 3.Eyes: Sunken 4.Mouth: Dry lips, tongue 5.Skin: turgor 6.Abdomen: tenderness, masses, distension. 7.Weight: Measure and record 8.mRDT(If fever present)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Recurrent or persistent diarrhoea for more than one month, with any of the following: a) weight loss, b)skin rashes, c) oral thrush, d) genital ulcers	Marked weight loss Oral thrush Lethargy Genital ulcers present.	Severe condition Suspect HIV/AIDS	1. Give Low Osmolarity ORS 2. Tabs Paracetamol 1g 8 hourly for 2days 3. REFER for HTS
II. Profuse diarrhoea with abdominal mass	Abdominal tenderness and palpable abdominal mass	Severe condition Suspect Abdominal tumor	REFER immediately to the appropriate health facility
III. Acute profuse diarrhea with vomiting, Weakness	Severe dehydration Sunken eyes Dry tongue Rapid pulse Low urine output	Severe condition Suspect Gastroenteritis	1.Give IV fluid (dextrose Saline) 2.Give IV Metronidazole 500mg 8 hourly for 24hours hours 3. REFER immediately
IV Persistent diarrhoea for 14 days with weight loss. Bulging forehead	Evidence of weight loss Dehydration Ova or Cysts seen or not	Moderate to Severe condition Suspect persistent diarrhoea	1.2. Give Low Osmolarity ORS in sips Tabs Metronidazole 400mg 8 hourly for 5days 3. If no improvement REFER immediately to appropriate health facility

V. Diarrhoea with headache, cough and fever	Temperature 37.5°C or above Cough Mild dehydration RDT positive	Moderate condition Suspect Malaria	1. Give Low Osmolarity ORS Drug Age Wt.(kg) Dose Artemisinin combination >14yr >35kg 4 tab twice daily x3days or Day 1st 2nd 3rd Artesunate Amodiaquine>13yr >40 4Tab (AA) 2.Tabs Paracetamol 1g 8 hourly for 3 days 3.Review in 3 days, If no improvement, REFER appropriately
VI. Diarrhoea with or without mucus or blood in stool	No dehydration	Mild to moderate condition Suspect Dysentery	1. Give Low osmolality ORS sips 2.Tabs Cotrimoxazole 960mg 12 hourly for 3 days 3.Tabs Metronidazole 400mg 8 hourly for 5 days If no improvement, REFER appropriately
VII. Frequent stool more than 5 times a day	Weakness Tiredness Some dehydration	Mild condition Suspect Mild Diarrhoea	1.2. Give Low Osmolality ORS for 24 hours Tabs Metronidazole 400mg 8 hourly for 5 days. If no improvement, REFER appropriately

Health Education

1. Advise on personal, food, water and environmental hygiene.
2. Give nutritional counseling.
3. Advise on preparation and use of Salt-Sugar-Solution (SSS) (if Low Osmolarity ORS is not available).
4. Encourage adherence to treatment.
5. Importance of taking home-based records during visit to clinic or hospital

5.6 VOMITING

Vomiting is the throwing up of the contents of the stomach. It may be due to illness, poisoning or other conditions, for example, malaria, early pregnancy, appendicitis, eating contaminated food or water as in cholera, gastro-enteritis etc. It is a serious problem because prolonged vomiting is a major cause of body fluids and electrolyte loss.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been vomiting? 2.How many times a day do you vomit? 3.How is the vomiting like? Projectile (think of obstruction), 4.Is there diarrhoea? When was the last stool? 5.Do you have any abdominal pain? Swelling? 6.Do you have any cough, fever or headache? Stiff neck? 7.When was your last menstrual period? (if female) 8.What was your last meal? 9.What is your source of drinking water? Are you on any medication? If yes, what type?	1.General appearance: ill-looking, Jaundice, dehydrated, distressed. 2.Vital Signs: Temperature, Pulse, Respiration, Blood Pressure: Check and record 3.Mouth: Dry/coated tongue greenish? With Blood? 4.Eyes: Sunken, Pallor 5.Throat: redness 6.Neck: stiffness. 7.Abdomen: tenderness, rigidity, visible peristalsis. 8.Ear: redness, bulging or perforation of drum, discharge. 9.mRDT (IF fever present): Test and record 10.Pregnancy test(if necessary)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Vomiting with any of the following: a) Abdominal swelling b) Constipation c) Stiff neck d) Severe Headache	Palpable abdominal mass Visible intestinal movement May have abdominal rigidity May have stiff neck Lethargy.	Severe condition	REFER immediately to the appropriate health facility
II.Vomiting with moderate abdominal pain	Abdominal tenderness	Severe condition Suspect Acute Abdomen	REFER immediately to the appropriate health facility
III.Vomiting one or two times a day with fever	Temperature 37.5°C and above No abdominal tenderness Sore throat mRDT positive	Moderate condition Suspect Malaria	1.Low Osmolarity ORS 2.Tabs Paracetamol 1g 8hourly for 3 days 3.Caps Amoxicillin 500mg 8hourly for 5days Drug Age Wt.(kg) Dose Artemisinin base combination therapy (ACT) 80/480mg>14y >35kg 1 tab at 0 hour, after 8 hours, then, 12hourly for 3days. OR Day Artesunate Amodiaquine>13yr >40 4Tab(AA) 12hourly for 3days 4. Review in 3 days, If no improvement, REFER to the appropriate health facility

IV. Vomiting with frequent loose stool with or without headache	Sunken eye Dry mouth Skin pinch goes back quickly	Moderate condition Suspect Gastro enteritis	1. Low Osmolarity ORS 2. IM. Metoclopramide 10mg stat Then 3. Tab Metoclopramide 10mg 12hourly for 2 days 4. Tab Metronidazole 400mg 8hourly for 5 days 5. Review the next day. If no improvement, REFER
V. Vomiting with Last Menstrual Period over a month	Mild dehydration Positive Rapid Pregnancy test	Mild condition Suspect pregnancy	1. Low osmolarity ORS 2. Do Urine or Serum PT If Positive 3. Eat dry biscuits in the morning. 4. Do not get out of bed in a hurry 5. Take small meals (not oily food). 6. Refer for ante-natal care. If pregnancy test is negative, REFER appropriately

Health Education

1. Avoid causes of vomiting.
2. Advise on personal, food and environmental hygiene.
3. Encourage adequate fluid intake and prevention of dehydration (excessive loss of body fluid).
4. Importance of taking home-based records during visit to clinic or hospital.

5.7 BODY PAIN

General body pain may be due to injuries or infectious diseases. Pain may be in any part of the body. Pain weakens the client and prevents him/her from getting normal sleep.
Strenuous work and vigorous exercise may lead to muscle strain and may cause general body pain. However, there is need to relieve pain and give comfort to the client.

HISTORY	EXAMINATION/ACTIONS
1.Where do you feel the pain? 2.How long have you had the pain and what is it like? Any inqury? 3.Does the pain worsen with movements or activities? 4.What type of work do you do? 5.Have you had fever or chills? 6.Have you had catarrh or stuffy nose?, or cough? 7.Have you been losing weight? 8. Does the pain radiate? Have you been bleeding from any where? Any black stool?	1.General appearance: ill-looking, wasting, unkempt,pale. 2.Vital Signs: Temperature, Pulse, Respiration, Blood Pressure - check and record. 3.Eyes: pallor, Jaundice. 4.Chest: a) crepitation, wheezes, decreased breath sounds; b) Rapid or laboured respiration. 5.Abdomen: tenderness, masses, rigidity, enlarged spleen. Back: a) tenderness in spine on thumping; b) deformity of the spine, masses; c) tenderness in the flank on thumping; d) tenderness where client says the pain is e) tenderness on standing, moving legs or bending. 6. Limbs: swelling, pallor of fingernails, weakness, numbness or paralysis on either side. 7.Weight and height: measure and record. 8. Urine: check for protein and glucose

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Severe back pain with generalized stiffness with difficulty in mouth opening	Stiffness and difficulty in mouth opening with or without spasm	Severe condition Suspect tetanus	1.IV Diazepam 10mg stat, 2.Give IV Phenobarbitone 30mg stat 3.Look for injury or old wound. 4. Clean and dress the wound 5. REFER immediately to the appropriate health facility
II. Bone pain, fever, Yellowish discolouration of the eyes	• Temperature 37.5°C and above • Yellowish conjunctiva • Generalized tenderness mRDT positive	Severe condition Suspect sickle cell diseases in crises	1.Tab Ibuprofen 400mg 8 hourly for 3days 2.Arthemether/Lumenantrine 80/40mg 12 hourly for 3 days 3.Artesunate/Amodiaquine 100/300mg 12 hourly for 3days 4.Tab. Folic Acid 5mg daily for 5 days 5.REFER immediately to the appropriate health facility

III. Pain radiating down the leg, Numbness, Weakness of one or both legs	Numbness and Abdominal tenderness	Severe condition Suspect Inter vertebral disc problem	1.Tabs Diclofenac Potassium 100mg 12 hourly for 3 days (After food) or 2.Tabs Paracetamol 1000mg 8hourly for 3days 3. IM Vitamin B-Complex 3ml daily for 3 days 4. REFER immediately to the appropriate health facility
IV. Pain in the flanks, Frequency of Micturition, Cloudy urine	Tenderness on palpation of the flank, x Cloudy urine	Moderate to severe condition Suspect Urinary Tract Infection (UTI)	1.Tabs Ibuprofen 400mg 8 hourly for 3 days (After food) 2. Caps. Amoxicillin 500mg 8hourly for 5days 3. Encourage fluids intake 4. Review in 3 days if no improvement, REFER
V. Pain in the joint or spine	Difficulty in walking	Mild condition Suspect Osteoarthritis	1.Tabs Diclofenac Potassium 100mg 12 hourly for 3 days (After food) 2.Topical application of Methylsalicylic acid cream/ ointment prn (when necessary) 3.Review in 5 days 4. Encourage moderate exercise If no improvement, REFER appropriate.
VI. Back pains with menstrual period	Lower abdominal tenderness	Mild to moderate condition Suspect Dysmenorrhoea	1.Tabs Ibuprofen 400mg 8 hourly for 3days (After food) 2. Review after menstruation. If no improvement, REFER appropriately

Health Education

1.Advise on adequate rest and exercise.
2.Advise on regular clinic attendance and compliance with drugs.
3.Advice to abstain from alcohol and tobacco.
4.Give nutritional counseling.
5.Advise on proper ventilation.
6.Importance of taking home-based records during visits to clinic or hospital.

5.8 CHEST PAIN

Chest pain can appear in different forms, ranging from a sharp stabbing pain, pain that radiates to the back, shoulder or neck, to a dull ache. It could sometimes present as a crushing or burning feeling. The causes of chest pain are diverse including heart and lung issues, heavy lifting, weightlifting, trauma to the chest, swallowing a large piece of food, etc. Chest pain could sometimes be an indication of a life-threatening situation and requires urgent medical intervention.

HISTORY	EXAMINATION/ACTIONS
1. Where do you feel the pain? 2. How long have you had the pain? And what is it like? Any injury? 3. Does the pain worsen with movement or activities? 4. What type of work do you do? 5. Do you have fever or chills? 6. Do you have catarrh or stuffy nose? Or cough? 7. Have you been losing weight recently? 8. Does the pain radiate? Have you been bleeding from anywhere? Any black stool?	1. General appearance: ill-looking, wasting, unkempt, pale. 2. Vital signs: Temperature, Pulse, Respiration, Blood Pressure: check and record. 3. Eyes: pallor, jaundice. 4. Chest: a) crepitation, wheezes, decreased breath sounds. b) Rapid or laboured respiration. 5. Abdomen: tenderness, masses, rigidity, enlarged spleen. 6. Back: a) tenderness in spine on thumping. b) deformity of the spine, masses. c) tenderness in the flank on thumping. d) tenderness where client says the pain is e). tenderness on standing, moving legs or bending. 7. Limbs: swelling, pallor of fingernails, weakness, numbness or paralysis on either side. 8. Weight and height: measure and record. 9. Urine: check for protein

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Chest pain with any of the following: Nausea Difficult breathing Vomiting Pain radiates to neck, arm, shoulder and back	Pulse < 50/min Or > 100/min BP > 180/110mmHg	Severe condition	REFER immediately to the appropriate health facility.
II. Chest pain	BP <90/60mmHg Temperature 37.2°C	Severe condition	1. IV Normal saline 500ml, stop if breathing becomes worse 2. IV/IM Ceftriaxone 1g stat If no improvement, REFER appropriately..
III. Chest pain becomes worse on lying down or deep breathing	Severe pain	Severe condition	REFER immediately to the appropriate health facility.
IV. Severe chest pain and known sickle cell disease	In pains	Severe condition	Low- Osmolarity ORS or IV Normal Saline 500ml 6hourly REFER if no improvement appropriately.

V. Central chest pain, radiates to the neck, jaw and arm that is : recurrent attack (joint pain) resulting from physical activity but relieved by rest :	Central chest pain	Severe condition (suspect angina)	REFER immediately to the appropriate health facility.
VI. Chest pain, Joint pain with cough		Moderate condition	Refer to section on cough
VII. Epigastric pain: while hungry, eating, or lying down	Burning, aching, or cramping pain in the upper abdomen.	Moderate condition Suspect Ulcer	1.Advise to avoid smoking, Alcohol, caffeine, Asprin, NSAIDS(Ibuprofen,) 2.Tabs Omeprazole 20mg daily for 4 weeks 3.Tab Mist Mag 2tabs 8hourly for 5 days If no improvement, REFER appropriately.

Health Education

1. Adequate rest and exercise.
2. Regular clinic attendance and compliance with drugs.
3. Abstinence from alcohol and tobacco.
4. Nutrition proper ventilation.
5. Importance of taking home-based records during visits to clinic or hospital.

5.9 HEADACHE

Headache is a common complaint among adults and may be a symptom of many diseases such as: Malaria, sinus infection, high blood pressure, ear, nose and throat problems, nervous tension, meningitis, brain abscess and typhoid fever. It can occur when there is constipation, inadequate sleep or rest.

HISTORY	EXAMINATION/ACTIONS
1.How long have you had it? How severe is it? 2.Do you have fever or chills? Convulsions? 3.Do you have nausea, vomiting or abdominal pain? (Typhoid or Malaria) 4.Is headache worse in the morning? (sinuses) What time of the day is it worse? 5.Do you have constipation or diarrhoea? 6.Are you on birth control pills? 7.Have you had any head injury? If so when? 8.Any other problem? (family, work)	1.General appearance: In distress, ill-looking, 2.Temperature, Pulse, Respiration, Blood Pressure: check and record. 3.Face: swelling, puffiness 4.Eyes: tenderness. 5.Nose: pus or mucus. Press both sides below eyes for tenderness. 6.Ears: redness or dullness of drums, mastoid pain. 7.Mouth: acetone or alcohol on breath. 8.Neck: stiffness, nodes. 9.Limbs: weakness, paralysis. 10.Urine: protein. 11. mRDT(If fever present)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Headache with puffiness, weakness on one side of the body.	Paralysis of one side of the face Slurred speech.	Severe condition Suspect Stroke	REFER immediately to the appropriate health facility.
II.Headache with any of these conditions: a) toothache, b) pus in the ear and/or pain, c) visual problems d) runny or stuffy nose e) Eye pain f) Swelling in the neck	Palpable nodes in the neck Stuffy nose Tenderness in the ear lobe May or may not have pus in the ear Diastolic BP equal to or less than 90mmHg Systolic BP equal to or less than 140mmHg	Moderate to severe condition Suspect Severe Ear Infection (Mastoiditis)	1. IV Ceftriaxone 1g daily 2. Tab Ciprofloxacin 500mg 12hourly 3. Tabs Ibuprofen 400mg 8 hourly for 3 days 4. Tab.Vit C 200mg 8 hourly for 5 days 5. Advise adequate rest 6. Review in 3 days, REFER to the appropriate health facility.
III. Headache Disturbed sleep, Restlessness	Severe paina Diastolic BP equal to or more than 90mmHg. Systolic BP equal to or more than 140mmHg BP 140/90mmHg and above	Moderate condition Suspect Hypertension	1.Tabs Diazepam 5mg 12hourly for 3 days 2. Tabs Paracetamol 1g 8hourly for 3 days 3. Advice the rest 4. REFER after 3 days if no improvement to the appropriate health facility.
IV. Pain with stuffy nose, headache, catarrh, hoarse voice	Stuffy nose	Moderate condition Suspect Sinusitis	1.Tabs Ibuprofen 400mg 8hourly for 3 days 2. Tabs Chlorpheniramine 4mg 12hourly for 3 days 3. Tabs Vit. C 200mg 8hourly for 5 days 4. Encourage bed rest, warm tea, honey and fluid intake 5. Advise on avoidance of cold 6. Review in 3 days If no improvement, REFER to the appropriate health facility.

V. Headache with Fever, aches, pain with or without pallor	Temperature 37.5°C and above, Pallor mRDT positive	Moderate condition Suspect malaria	1. Tabs Paracetamol 1000mg 8hourly for 3 days 2. Artemisinin / lumerfantrine combination >14yr >35kg 4 tab 12hourly for 3days OR Artesunate Amodiaquin >13yr >40 100/300mg 12hourly for 3days 3. Tab. Folic acid 5mg daily for 1 week 4. Review in 3days, if no improvement, REFER to the appropriate health facility.

Health Education

1. Advise on adequate rest and exercise.
2. Encourage regular clinic attendance and compliance with drugs.
3. Advise on abstinence from alcohol and tobacco.
4. Advice to cut off caffeine and kola nut.
5. Give nutritional counseling.
6. Advise on proper ventilation.
7. Importance of taking home-based records during visits to clinic or hospital.

5.10 PALLOR

Pallor is a major sign of anaemia which indicates low haemoglobin level below 10g/dl. The common causes of anaemia are bleeding from any part of the body, lack of iron or folic acid in food, hookworm infestation, sickle cell, and unspaced pregnancies, especially in women of child-bearing age. The place to look for pallor is the conjunctiva, the nail bed, the palm, the lips etc. Certain drugs like aspirin or phenacetin make anaemia worse, so we should not use them without good reason and never given to sickle cell and ulcer patients. Paracetamol should be used for such clients.

HISTORY	EXAMINATION/ACTIONS
1.Do you feel ill? 2.Is there any bleeding from the Nose or Mouth (gums), vaginal (if female) or other parts of the body? Do you feel ill? 3.Is there any bleeding from the Nose or Mouth (gums), vaginal (if female) or other parts of the body? 4.Are you coughing or vomiting blood? 5.Is there blood in urine or stool? Stool black or dark brown or urine black? 6.Do you have fever? 7.Do you have shortness of breath or do you wake at night short of breath? 8.How many pillows do you need to sleep on? 9.Is there abdominal pain or have you passed any worms recently? 10.Do you have numbness or tingling of the feet? Sore tongue? 11.What medication are you taking? (think of Aspirin or Phenacetin -containing medicines) 12.Are you pregnant? (if female) 13.Do you have joint pains? 14.Are you coughing or vomiting blood? 15.Is there blood in urine or stool? Stool black or dark brown or urine black? 16. Do you have fever?	1.General appearance: ill-looking, thin, malnourished. 2.Vital signs: Temperature, Pulse, Respiration, Blood Pressure: check and record 3.Eyes: pallor, jaundice. 4.Tongue: redness, swelling, smoothness. 5.Lips & nails: pallor. 6.Neck: nodes 7.Chest a) Breathing difficulty b) crepitation, wheezes, decreased breath sounds 8.Abdomen: spleen, liver masses. 9.Limbs: swollen hands or feet. 10.Heart: rate, rhythm 11.Haemoglobin: estimate and record. 12.Stool: hookworm ova. 13. mRDT: Test and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Paleness due to acute or profuse bleeding	Bleeding due to visible injury. HB < 10g/dl	Severe condition	1.Apply pressure to arrest bleeding. 2.Start patient on IV normal saline 3.REFER to the appropriate health facility.
II. Tiredness, Yellow eyes, Pain in the body and bone	Conjunctival pallor and lips, and mucus membrane Tenderness in the joints	Moderate condition Suspect sickle cell anaemia	REFER immediately to the appropriate health facility.
III. Paleness with fever	Palmar pallor	Moderate condition	REFER to section on fever
IV. Paleness, tiredness with abdominal pain	Temperature 37.5°C and above mRDT result positive or negative ova of worms seen	Moderate condition	REFER to section on worms

Health Education

1.Nutrition.
2.Prevention and care for sickle cell disease.
3.Personal, environmental and food hygiene use of LLIN.
4.Importance of taking home-based records during visits to clinic or hospital.

5.11 WEIGHT LOSS

Weight loss is a situation where an adult weighs less than normal standard body weight for age (see BMI chart in the Annexure). The common causes of weight loss are diarrhoea, vomiting, parasitic infection, chronic diseases, tuberculosis, HIV/AIDS and poor diet. Goitre (Thyrotoxicosis), Tuberculosis, cancer, and diabetes can also cause weight loss.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been losing weight? 2.How is your appetite? 3.Do you have nausea or vomiting? 4.Do you have diarrhoea or abdominal pain? Persistent or reoccurring? 5.Have you been having fever? 6.Do you have cough? How long? Is there sputum? What colour s it? 7.Do you have chest pain? 8.Are you having skin rash? Oral thrush? 9.Do you feel thirsty often? 10.Frequent urinating?	1.General appearance: ill-looking, wasted. 2.Temperature, Pulse. Respiration, Blood Pressure: check and record. 3.Eyes: pallor. 4.Mouth: condition of teeth and tongue Chest a) Difficulty in breathing; b) Crepitation or rhonchi, decreased breath sounds. 5.Abdomen: tenderness, masses, rigidity. 6.Weight and height measure and record. 7. Haemoglobin: estimate and record. 8. Urine: protein and sugar. 9. Conduct HCT 10. Sputum for AFB x 3

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Weight loss, Persistent or recurrent diarrhoea, Generalized skin rashes, Oral thrush, Prolonged fever	Coughing Wasted Skin rash Oral thrush Temperature 37.5'C or above HCT positive or negative	Severe condition Suspect HIV/AIDS	If HT result is positive, REFER to the appropriate health facility.
II.Weight loss, Palpitations	Rapid pulse above 100 beats per minute Evidence of muscle wasting with or without pallor. Swelling in the neck Excessive sweating Tremor BP 140/90 mmHg or above	Moderate to severe condition thyrotoxicosis	REFER immediately to the appropriate health facility.
III.Weight loss, Frequent urination, Excessive drinking of water	Urinalysis – sugar in urine Waste	Severe condition Suspect Diabetes mellitus	REFER immediately to the appropriate health facility.
IV.Weight loss, Prolonged cough (more than two weeks), chest pain	Coughing Wasted Blood in sputum Sputum Test for Acid- Fast Bacilli positive or negative	Severe condition Suspect Tuberculosis	1. If 2 out of 3 sputum specimens are positive, commence treatment for TB. 2. HTS, If positive, refer to the to the appropriate health facility. 3. If not trained, REFER to MDT Centre

Health Education

1.Advise on personal, food and environmental hygiene.
2.Give nutritional counselling.
3.Encourage regular clinic attendance and weight measurement.
4.Advise on compliance with prescribed medications.
5.Advise on importance of taking home-based records during visits to clinic or hospital.

5.12 JAUNDICE

Jaundice is the yellowish discolouration of eyes and skin due to increase in bile pigment in the body. Jaundice results from the build-up of bilirubin- a reddish pigment resulting from heme metabolism in the body. It may or may not be as a result of liver problem. The build-up of this yellowish pigment may be due to abnormalities in its formation, transportation, metabolism and excretion. At times, the level may not be clinically recognizable until the level is so high. Jaundice occurs in liver problems, sickle cell disease, haemolysis, amoebiasis and tumour of the pancreas.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long have you noticed the condition? 3. Has it happened before? If yes, when? 4. Is there any fever or pain anywhere in your body? If yes, where? 5. Is there any change in the colour of your body? 6. What is the colour of your urine? 7. Is there any diarrhea? If yes, what is the colour of the stool? 8. Is there nausea and/ or vomiting 9. Is there any itching of skin? 10. Are you taking any medication? If yes, what?	1. General appearance: irritable, in distress, may be in coma 2. Eyes: jaundice 3. Mouth: dry tongue coated 4. chest: a) difficult or rapid respiration b) crepitation, wheezes, decreased breath sounds 5. Abdomen: tenderness rigidity, masses, spleen liver 6. Skin: rash, yellow, petechial haemorrhage

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Yellowness of the eyes, general body weakness, may or may not have vomited	Yellow conjunctiva May or may not have swelling of the abdomen with rigidity Major may not have marked abdominal tenderness pallor, Hb less than 10gm/di	Severe condition Suspect hepatic Disorder	REFER Immediately to the appropriate health facility.
II. Yellowness of eyes, general body pain, dark yellow urine	Yellow conjunctiva Hb less than 10gm/di Dark yellow urine Pallor	Severe condition Suspect Hepatic disorder	REFER Immediately to the appropriate health facility.
III. Fever, nausea and/or vomiting, abdominal pain. Diarrhoea with blood in stool	Yellow eyes Temperature 37.5 °C or above Abdominal pain Diarrhoea	Moderate condition Suspect Amoebic Liver Disease	REFER Immediately to the appropriate health facility.

Health Education

Advise Mother/Caregiver on:

1. Advise on adequate fluid intake
2. Advise on personal food and environmental hygiene
3. Give nutritional counseling
4. Advise on periodic de-worming
5. Importance to taking home-based records during visits to clinic or hospital

5.13 SKIN DISORDERS

Skin disease is the presence of itchy or non-itchy, raised or non-raised lesions on the body. It may be due to poor hygiene, allergy, malnutrition, infection. Some skin diseases can easily spread to other members of the family, therefore early referral and correct treatment is important for the welfare of the whole family. Some common causes of skin diseases are: ringworm, measles, leprosy, jaundice, HIV/AIDS, diabetes, scabies and eczema. Others include tinea vescular, vitiligo, athlete's foot, urticaria, contact dermatitis.

HISTORY	EXAMINATION/ACTIONS
1.How long has the rash been there? 2.Does anyone else in the family have a rash? Describe? 3.Does it itch? 4.Are there any other symptoms? Fever, Headache, Sore throat? 5.Any change in the skin condition: Colour? Patches? Visible veins? Swelling of the legs? Swelling with or without Pus? Blisters? 6.What type of accommodation do you have? 7.How many people share the room? 8.What do you sleep on? 9.How often do you bath? 10.What medication, (including injections or home treatment) have you received lately? 11.What cream/lotion are you using for your skin?	1.General appearance: thin, malnourished, distressed/ uncomfortable 2.Skin: a)location, distribution and size of lesions b)colour of rash or lesions c)lesions or rash: flat (macules), raised, pus or fluid-filled, weeping, crusted, ulcerated, scratch marks or d)combination of(a)to(c) 3.Ears: red or discharging 4.Eyes: redness, discolouration of eyelids (dark - due to repeated scratching) 5.Mouth: redness, discolouration of eyelids (dark-due to repeated scratching) 6.Chest: respiratory rate, wheeze 7.Vital signs: Temperature: check and record 8.Urine: sugar, colour.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Skin patches or lesions with light colouration with or without deformities, Fever, General malaise and or joint pains, May or may not be confirmed	* Ill patient * warm and tender joints * Skin patches with loss of sensation * Temperature above 37.2° C or above * There may be weakness of hand or feet or foot drop * Inability to close the eyes	Severe condition Suspect complications of Leprosy	1. If not trained to manage, REFER to DOT centre 2. If trained on leprosy management and equipped with drugs, manage according to treatment regimen in Appendix
II. Skin patches with light colouration with or without weakness of hands and feet	Skin patches with light colouration Loss of sensation in the patch (cotton wool test), with or without weakness of foot or foot drop	Moderate - Severe condition Suspect Leprosy	1.If not trained to manage, REFER to hospital. 2. If trained to manage, commence treatment according to treatment regimen in appendix 13
III. Severe itching with skin changes which disturbs sleep at night	* Scratch marks * Thickening discolouration of the skin * Skin changes with severe itching which disturbs skin and eye * With or without loss of vision * Skin nodules	Moderate-severe condition Suspect Onchocerciasis	1. REFER immediately for further investigation 2. Give Ivemenctin 4 tabs stat 3. Enrol in LGA Onchocerciasis control programme for 5 years.

IV. Visible veins on legs	Engorged veins	Mild condition Suspect Varicose veins	1. Elastic bandage when standing 2. Elevate legs when siting 3. Review in 5 days If no improvement, REFER to the appropriate health facility.
V. Itching of the skin followed by raised red lesions which disappears after some time	* Raised lesion of skin * Scratch marks	Mild condition Suspect Allergy	1. Give IM Promethazine 25mg stat 2. Tab. Chlorapheniramine 4mg 12hourly for 2 days 3. Review in 3 days if no improvement refer to the appropriate health facility.
VI. Itching of the scalp or public area		Mild condition Suspect lice infestation	1. Shave the hair 2. Apply chlorhexidine gel to scalp and other areas 3. Leave for 24 hours 4. wash and comb If no improvement, REFER to appropriate health facility.
VII. Itching rashes between the fingers, wrists and buttocks		Mild condition Suspect scabies	1. Wash with antiseptic soap and water 2. Apply Benzyl benzoate emulsion locally daily for 2 days (avoid face) 3. If pus is present give Amoxicillin 500mg 8hourly for 5 days 4. Give Chlorpheniramine 4mg 12hourly for 2 days 5. Review in 1 week 6. If no improvement, REFER
VIII. Itching with rough, raised, round rashes	Rough, raised, round rashes	Mild condition Suspect skin rashes	1. Wash with antiseptic soap and water 2. Shave hair of affected part 3. Examine other members of the family 4. Apply miconazole ointment daily for 4-6 weeks 5. Tabs Griseofulvin 500mg 12hourly for 3 weeks 6. Review after 6 weeks 7. If no improvement, REFER
IX Small light patches, Not raised not itchy on the back face, etc.	Small light patches , not raised	Mild condition Suspect fungal infection	1. Miconazole ointment for 4-6 weeks OR whitfied ointment OR clotrimazole cream

X. Painful swelling with pus	Tender purulent swelling	Mild condition Suspect Abscess	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Ibuprofen 400mg 8hourly for 3 days (After meal) 3. Drain and dress daily 4. Vit. C 200mg 8hourly for 7 days 5. Review in 3 days if no improvement, REFER to appropriate health facility.
XI. Putular rash with fever on face and trunk, loss of appetite	Pustular rashes Temperature > 37.5 °C	Mild condition Suspect Chicken pox	1. Give Chlopheniramine 4mg 12hourly for 3 days 2. Tabs Paracetamol 1000mg 8hourly for 3 days 3. Vit. C 200mg 8hourly for 7 days 4. Calamine lotion 5. Review in 3 days REFER to appropriate health facility.
XII. Itching of affected part with blisters, fever and headache, swelling of punctured blister site	* Swelling of affected part * Temperature>37.2 °C	Moderate to severe condition Suspect herpes virus infection or HIV/AIDS	1. Tabs Paracetamol 100mg 8hourly for 3 days 2. Tabs Chlopheniramine 4mg 12hourly for 3 days 3. Counsel for HTS

Health Education

1. Personal hygiene.
2. Sharing of underwear and clothes.
3. Importance of washing and ironing of clothes.
4. Nutrition.
5. The avoidance of overcrowding.
6. Adequate care of bedding.
7. Importance of taking home-based records during visits to clinic or hospital.

5.14 OEDEMA

Oedema is an abnormal collection of fluid in the tissues resulting in swelling of the affected part of the body. It could be generalized or localized e.g one foot or finger. Generalized oedema is most noticeable on the parts of the body which are lowest at that time e.g legs when standing. Causes include congestive heart failure, severe anaemia, kidney diseases, liver disease and kwashiorkor. Oedema may also be the result of a reaction to toxin, drugs and insect sting. Ascites, which is fluid in the abdominal cavity, may accompany oedema.

HISTORY	EXAMINATION/ACTIONS
1.How long have you had this complaint? Is this the first time? 2.Have you had a good appetite? Any nausea or vomiting? 3.Have you been urinating more frequently at night? Any pain on micturition? 4.Any bleeding from nose? *Vomiting or coughing blood? * Blood in the urine or stool? Black stool? 5.Any cough? Or breathlessness at night? 6.Do you sleep with extra pillow? 7.Have you ever been treated for jaundice?	1. General appearance: Ill-looking 2. Temperature, pulse respiration, blood pressure: check and record 3. Face: Puffy 4. Eyes:pallor jaundice 5. Chest: crepitation, wheezes, decreased breath sounds 6. Heart: rate, rhythm, 7. Abdomen: large tender liver, enlarged spleen 8. Limbs: a) legs i) for oedema ii) does swelling pit on pressure (if not pitting, think of elephantiasis) b) hands for oedema 10. Urine: check for protein and sugar

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Swollen legs with large abdomen, Persistent cough, Tiredness	Swollen legs Distended abdomen Enlarged ,tender liver	Severe condition Suspect heart failure	REFER immediately to appropriate health facility.
II. Oedema with Pallor, Cough, Breathlessness	* Crepitations in the chest * Pedal oedema * Paleness of the conjunctiva and mucous membrane * Blueness of lips * Haemoglobin below 10gm	Severe condition Suspect anaemic heart failure	REFER immediately to appropriate health facility.
III. Swelling of the leg	Swelling of the leg Not pitting on pressure	Mild condition Suspect elephantiasis	1.Test for filariasis 2.Tab; Ivermection 12mg (4tabs) once 3. Daily washing and drying of affected limb 4. Limbs should be elevated when sleeping 5. Review in 5 days, if no improvement refer to appropriate health facility.

Health Education

1.Reassure patient
2.Give nutrition counseling
3.Explain the relationship between infection, anaemia and heart failure
4.Encourage environmental sanitation
5.Importance of taking home based records during visit to the clinic or hospital

5.15 BURNS

This is injury to the skin that may be caused by boiling water, hot liquid, chemicals or fire. Serious burns can cause shock because large amount of body fluid is lost. The size of the area of body surface affected is important in deciding the severity of burns.

HISTORY	EXAMINATION/ACTIONS
1. How did you get burnt and with what? 2. When and where did it happen? 3. Has any medication been given or applied since then? 4. Have you been immunized against Tetanus? When? Number of doses?	1. General appearance: restless, distressed, dehydrated 2. Skin: (a) size and area of burns, blisters, leathery (b) signs of infection-pus, redness, warmth 3. Temperature, blood pressure: check and record 4. Pulse: check rate and volume and record 5. Chest: respiratory rate, wheezing

Part	Percentage of Body Surface(%)
Head and neck (front and back)	9
Chest (front and back)	18
Abdomen (front and back)	18
Uooer limb (both)	18
Lower limb (anterior and posterior) - each 18	36
Genetalia	1
Total	100

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Large area Burns or Scalds	* Large area of burns or scald * Burn 9% or more * Any of the following involved * Face, hands, groin or joints lethargy * Loss of sensation * Hoarse voice since burned	Severe condition Suspect severe burns	1. Give IM paracetamol 600mg stat 2. IV normal saline and maintain line 3. RFER immediately to appropriate health facility.
II. Chemical, electrical burns	large or small area of burns or scald caused by either chemicals or electricity	Severe condition Suspect severe burns	1. Wash profusely with water 2. REFER immediately to appropriate health facility.
III. Small area of burns or scalds	Burns less than 9% No blisters No lethargy Face and Hands not involved	Mild condition	1. Clean burns with cool, clean or sterile solution 2. Do not break blister 3. Tabs Paracetamol 1000mg 8hourly for 3 days 4. Apply clean dressing

Health Education

1.Advise on personal and environmental hygiene.
2.Advise on prevention of burns.
3.Give nutritional counseling.
4.Encourage adequate fluid intake.
5.Advise on proper storage of chemicals.
6. Importance of taking home-based records during visit to the hospital

5.16 WOUNDS, SORES, ULCERS, FRACTURES & BITES

Wounds are injuries to the different body tissues. They are breaks in the skin and deeper tissues which may be small or large and which may include cutting of nerves and tendons. There could also be wounds that have been for several days and have become infected or healing poorly. Sores are injured, infected or diseased part of the skin. Ulcer s a break or discontinuity in a bodily membrane that impedes the organ of which that membrane is apart from continuing its normal function. The client could also have bites from snakes, dogs, scorpions or humans.

A fracture is a break in the continuity of a bone, it is more serious if there is also a cut in the skin and other soft tissues (compound fracture). If there is profuse bleeding to avoid the patient going into shock. Early cleaning and treatment of wounds is important to prevent infection and tetanus.

HISTORY	EXAMINATION/ACTIONS
1.How did the injury occur? 2.Was there any bleeding? 3.Has it bled much? (Try to carefully assess the amount) 4.Has anything been put on the wound? 5.Have you been immunized with Tetanus Toxoid? Number of doses? 6.What bit him/her? A Dog, A Snake, A person? Scorpion? Or any other thing? 7.Where is the pain? How bad is it? 8.Can you move/use limb? Is there limitation of movement? 9.Which movement causes greatest pain?	1. General appearance: a. Pale, cold or clammy b. Increased sweating c. Increased respiration and pulse rate d. Patient agitated or restless 2.WOUNDS: a. Size and site b. Bleeding c. Pus or dirt in wound d. Swelling, redness or warmth around wound. 3.SPRAINSANDFRACTURES a. General appearance: b. Signs of shock-increased sweating, pallor, restlessness, rapid pulse and rapid respiration. c. Degree of pain d. Limb of joint e. Tenderness or swelling f. Warmth or redness g. Creptus (crackling) h. open wounds

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1.Wounds on the limb resulting from injury	Evidence of fracture (unable to move the limb). BP below 110/70mmHg or above. Lethargic Wound with blood loss.	Severe condition	1. Immobilize the limb 2. Dress wound 3. Give TT 0.5ml stat 4. Caps Amoxicillin 500mg 8hourly for 5days 5. Tabs metronidazole 400mg 8hourly for 3 days 6. Tabs Ibuprofen 400mg 8hourly for 3 days OR Tab Paracetamol 1000mg 8hourly for 3 days 7. Tab vitamin C 200mg 8hourly for 7 days 8. Review in 3 days, if no improvement REFER to appropriate health facility.

II. Wound as a result of laceration.	Bleeding from wound Torn tendon. Haemoglobin > 10gm/dl	Moderate to severe condition	1. Clean well with available diluted antiseptic solution 2. Stop bleeding by pressure 3. Give tetanus toxoid 0.5ml stat 4. If haemoglobin is less than 10gm/dl 5. Review in 3 days, if no improvement REFER immediately to appropriate health facility.
III. Wound from dog bite	Evidence of bite. Anxious Wound, swelling and redness of site. Tenderness at the site of bite with or without bleeding	Moderate to severe condition	1. Wash wound with soap or detergent/diluted antiseptic solution 2. Advise patient to observe the dog for 10 days and report any change in the dog's behaviours or death of dog 3. Give anti-rabies injection 1ml on days 0,3, 7, 14 and 28 on the deltoid. 4. Tab Ibuprofen 400mg 8hourly for 3 days 5. Tetanus toxoid 0.5ml I.M stat 6. Daily dressing of wound after cleaning well 7. If dog is alive after 10 days, discontinue antirabies injection 8. If no improvement, REFER to appropriate health facility.
IV. Wound from human bite	Wound at the site of bite.	Moderate condition	1. Wash with diluted antiseptic solution or normal saline. 2. IM Tetanus toxoid 0.5ml stat. 3. Caps. Amoxicillin 500mg 8 hourly for 5 days. 4. Tab. Ascorbic acid (Vit. C) 200mg 8 hourly for 14 days. 5. Dress wound daily. 6. Review in 5 days. If no improvement, REFER to appropriate health facility.
V. History of snake bite	* Wound at the site of the bite * Swelling of the affected part * Bleeding gum	Severe condition Suspect snake bite	1. Keep client calm, clean area thoroughly 2. Give anti-snake venom e.g ECHITAB 10mls stat after test dose (read and follow manufacturer's instructions) 3. IV Hydrocortissone 100mg stat 4. Tab Paracetamol 1000mg 8hourly for 3 days 5. Caps Amoxicillin 500mg 8hourly for 5 days 6. Review in 3 days, if no improvement REFER immediately to appropriate health facility.

Health Education

Advise on:

1. Use of protective clothing to avoid bites.
2. Importance of compliance with treatment.
3. Nutritional counseling.
4. Environmental sanitation.
5. Reducing risk of accident
6. In cases of snake bite, if possible, take along killed snake to the clinic.
7. Importance of taking home-based records during visit to the hospital.

5.17 HEAD INJURIES

The head comprises of the brain and the sense organs for hearing, seeing, smelling and tasting. A serious head injury may be caused by a fall or a blow to the head. Some signs of serious head injury are severe headache, unconsciousness, bleeding from the nose, mouth or ears. There also may be vomiting or the patient may be confused. Any delay in referring a patient with head injury for proper medical treatment may cause permanent disability or death.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.What was the cause of the problem? 3.When did it happen? 4.Was there any loss of consciousness? Irritability? Fits? 5.Can you remember what happened? 6.Did anyone see it happen to describe the accident? 7.Do you have headache, nausea, vomiting? 8.Are there any other complaints or additional injuries?	1. General appearance: 2.Vital signs: Temperature, pulse, respiration, blood pressure: check and record 3. Head: visible wounds, swelling, bleeding 4. Ears: Bleeding 5. Eyes: subconjuctival haemorrhage, pupils for size, reaction to light 6. Mouth: teeth, fractures, bleeding 7. Nose: bleeding, fractures 8. Chest: ribs for fractures 9. Abdomen: tenderness, rigidity 10. Limbs: fracture or injury 11. Urine: monitor output 12. Haemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Injury to the head, Bleeding from ear or nose	* may or may not be conscious * Stiff neck * May or may not be bleeding from ear or nose * May have convulsion	Severe condition Suspect fracture of the skull	1.Ensure airway is clear 2. REFER immediately to appropriate health facility.
II. Injury to the head with transient loss of consciousness	• With or without wound seen • Pupils reacting normally to light • With or without tenderness	Moderate to severe condition	1. Tab Paracetamol 1000mg tds x 3 days 2. REFER immediately to appropriate health facility.
III. Injury to the head without transient loss	• No wound seen • Pupils reacting normally to light	Moderate condition	1. Monitor vital signs hourly for 4 to 6 hours 2. If condition deteriorates, refer to appropriate health facility.
IV. Injury to the head, No other complaints	Conscious No other signs of head injury	Mild condition	1. Check vital signs and advice to return immediately if vomiting, headache or drowsiness occurs 2. REFER if no return, to appropriate health facility.

Health Education

Advise on:

1.Prevention of head injuries e.g use of crash helmets, seat belts, placing children at the back seat and use child lock while driving
2.Prevention of home and road accidents
3.The importance of taking home-based records during visit to the clinic.

5.18 NECK DISORDERS/INJURIES

Neck problems can be as a result of infections or injuries to the structures in the neck or from secondary causes, this may manifest as enlarged nodes and/or lumps around the neck. There could also be stiffness of the neck. Examples of such infections are tuberculosis, meningitis, HIV/AIDS etc lumps can be located in the centre or front of the necks (for example goiter). There can also be an injury to the neck following a fall.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Do you have difficulty swallowing? Hoarseness? 3.Do you have pain and/or stiff neck? Neck swelling? 4.Do you have pain or headache? 5.Did you injure your neck? 6.Do you have fever? 7.Do you have weakness or numbness of the arms, hands or legs? 8. Is there weight loss or night sweats?	1.General appearance: ill-looking, distressed. 2.Temperature, Pulse, Respiration, Blood Pressure: Check and record 3.Head: abscesses, sores or pustules, lumps 4.Ears: signs of infection. 5.Mouth: abscesses or sores a.Neck: swelling, nodes, stiffness, tenderness 6.Arms and Legs: weakness or decreased sensation.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Injury to the head, bleeding from ear or nose	* may or may not be conscious * Stiff neck * May or may not be bleeding from ear or nose * May have convulsion	Severe condition Suspect fracture of the skull	1. Ensure airway is clear 2. REFER immediately to appropriate health facility.
II. Injury to the head with transient loss of consciousness	• With or without wound seen • Pupils reacting normally to light • With or without tenderness	Moderate to severe condition	1.Tab Paracetamol 1000mg 8hourly for 3 days 2. REFER immediately to appropriate health facility.
III. Injury to the head without transient loss	• No wound seen • Pupils reacting normally to light	Moderate condition	1. Monitor vital signs hourly for 4 to 6 hours 2. If condition deteriorates, refer immediately to appropriate health facility.
IV. Injury to the head, no other complaints	• Conscious • No other signs of head injury	Mild condition	1. Check vital signs and advice to return immediately if vomiting, headache or drowsiness occurs
I. Neck swelling With or without difficulty in breathing	Palpable mass in the front of the neck May be in respiratory distress	Severe condition Suspect goiter	REFER immediately to appropriate health facility.
II. Pain and stiff neck without history of fever or injury	• Stiff neck • No weakness of arms and legs • no evidence of injury	Moderate condition Suspect torticollis	1. Reassure client 2.Tabs Ibuprofen 400mg 8hourly for 3 days (After food) 3. Methyl Salicylate Ointment 4. Review in 2 days

III. Enlarged, painful nodes around the neck with or without fever	Palpable, tender nodes	Suspect infection of the head and neck region	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Metronidazole 400mg 8hourly for 5 days 3. Tab Ibuprofen 400mg 8hourly for 3 days (After food) 4. Review in 5 days 5. If no improvement, REFER to appropriate health facility.

Health Education

Advise on

1.Prevention of neck injuries.
2.Complication of mumps in males.
3.Prevention of transmission to others.
4.Oral hygiene instructions.
5.Importance of taking home-based records during to the clinic.

5.19 EYE DISORDERS

Eye disorders in adults can be a result of infection, foreign body, congenital malformation not treated in childhood, deficiencies of vitamins and micro-nutrients, complication of diabetes mellitus, hypertension, thyrotoxicosis etc. Eye disorders are often associated with defect of vision especially with aging. Burns from hot water and chemical can also affect the eye.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem with the eyes? 2. How long have you had the problem? Have you had it before? 3. Did the problem come suddenly or gradually? 4. Is there pain in the eye? How severe? 5. Does your eye itch? 6. Did anything get into your eye? 7. Have you (or anyone) put anything into the eye? What? 8. Has any medicine been put in the eye? What? 9. Do you have high blood pressure? 10. Do you have diabetes or any other disease? 11. Do you smoke or drink alcohol? 12. Is there any family history of blindness? 13. If so, is anyone else in the family receiving eye treatment? What type?	1. General appearance: in distress, ill-looking 2. Skin: rashes 3. Eyes: a.) Redness b) watery b.) Squint c.) Cloudiness or whiteness of pupil(s) swelling around the eye d.) Eyelid: sticky, inversion, or eversion, pus or other discharges e.) Eyelashes: irregular, missing or touching the surface of the eyeballs f.) Pupils: normal or small in size, equal, irregular in shape, reactive to light. g.) Eyeballs protruding (pushing out) h.) conduct visual acuity test of both eyes and record 4. Nose: catarrh 5. Blood pressure: measure and record 6. Urine: sugar

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Burns eye with chemical or hot water	*Watery, painful discharge from the eyes	Severe condition	1. Wash eye immediately with cold water 2. Tab. Paracetamol 1g 8hourly for 3days 3. REFER immediately to appropriate health facility.
II. Redness of eyes, pain in the eye, worsening Vision/difficulty in seeing with one or both eyes, foreign body in the eye, swelling of eye shadowy vision	* Red eyes *Watery discharge from the eye *Pain in the eye *Squint	Moderate to severe condition	REFER IMMEDIATELY to appropriate health facility.
III. Discharge of pus from the eyes Gritty sensation in the eye	* Purulent discharge * Sticky eyes * Red conjuctiva	Moderate condition (conjunctivitis)	1. Clean eyes with sterile water 2. Chloramphenicol eye drop 1 to 2 drops in each eye 6hourly for 5 days 3. Review in 5 days 4. If no improvement, REFER to appropriate health facility.
IV. Pustule or small abscess on eyes lid	* Swelling on the eye lids	Mild condition Suspect Stye	1. Reassure client/patient 2. Tab Ibruprofen 400mg 8hourly for 3 days (After food) 3. REVIEW in 5 days, If no improvement, REFER to appropriate health facility.

Health Education

Advise on:

1. Personal and environmental hygiene.
2. Regular eye check-up.
3. Prevention of eye injuries.
4. Nutrition.
5. Importance of taking home-based records with anytime he/she goes to the clinic or hospital.

5.20 EAR, NOSE AND THROAT DISORDERS

The ear, nose and throat are linked with each other and whatever affect one may affect the other. The ear not only helps us to hear but also helps us maintain our balance and any disease of the ear which is not urgently and adequately treated may not only lead to deafness, but may also lead to inability to maintain balance.
Ear problems are commonly caused by middle ear infection often accompanied by pain, dullness and redness of the ear drum with drainage of pus or fluid infections of the nose, throat or Eustachian tube may spread to the middle ear. It is important to examine the bone behind the ear if it is painful (tender to touch); it may mean that the infection has spread to the mastoid bone. To test an adult for hearing, sit behind the client then ask him to repeat words or numbers that you whisper to him.
The nose is the organ for breathing and for sense of smell. When the nose is affected by disease or injury, it discharges watery fluid(mucus, blood or pus) blocking of the nose may lead to breathing difficulties.
The throat is the junction of the ear, nose and mouth, it contains the tonsils, pharynx etc. Diseases affecting the throat can easily spread to the chest and brain.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long have you had the problem? 3. Is there catarrh? 4. Is there any pain? Boil? Any discharge: pus, blood or water? 5. Do you feel pain while swallowing? 6. Is there fever? Watery eyes? Headache? 7. Has it happened before? 8. Has there been any blow or injury to the head? 9. Do you have nose bleeding? Can you estimate the volume? When did it start? 10. Are you having bleeding from any other part of the body? 11. Is there pain from the teeth or gum? 12. Do you notice bad or foul breath? 13. Is there diarrhea or vomiting? 14. Do you work in a noisy environment or with noisy Machines? 15. Have you put anything into your ears, nose and throat? Did anything enter your ears, nose or throat? 16. Is there any trauma to the ear, nose or throat?	1. General appearance: in distress, ill-looking, in distress 2. Temperature, pulse respiration, blood pressure: check and record 3. Ear: a) Pus or blood discharge b) Red or bulging drum c) Perforated drum, foreign body d) Cerumen (Wax) e) Bone behind the ear (mastoid) for tenderness 4. Eye: redness, watery discharge 5. Nose: Mucus or catarrh 6. Neck: stiffness or enlarged lymph nodes 7. Throat: a) Redness b) Enlarged tonsils, pus c) Hoarseness or loss of voice 8. Chest: crepitation, wheezes, decreased breath sounds 9. Test for hearing and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Ear problem with vomiting Tenderness behind the ear	Temperature 37.5C or above Tenderness behind the Ear Restlessness	Severe condition Suspect Mastoiditis	REFER immediately.
II. Boil in external ear	Soft, tender swelling with well-defined margins	Moderate to severe condition: suspect boil	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Ibuprofen 400mg 8hourly for 3 days (After food); OR 3. Tab Paracetamol 1g 8hourly for 3 days 4. Review in 3 days If no improvement REFER to appropriate health facility.

III. Pain in the ear with or without pus or water from the ear or vomiting	Temperature 37.5°C or above Red, dull bulging ear drum or Perforation of ear drum Pus or blood discharge	Severe condition: Suspect Acute Suppurative Otitis Media with perforation	REFER immediately to appropriate health facility.
IV. Pain in the throat, Difficulty with swallowing	Enlarged and red tonsils	Moderate condition Suspect Abscess	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Metronidazole 400mg 8hourly for 5 days 3. Tab Ibruprofen 400mg 8hourly for 3 days (After food) 4. Warm saline gargle 5. Review in 3 days, If no improvement REFER to appropriate health facility.
V. Pus from the nose, Headache, Pain on either side of the nose	Temperature 37.5°C or above Pus draining from the nose Tenderness of the nose	Moderate to severe condition Suspect abscess	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Metronidazole 400mg 8hourly for 5 days 3. Tab Vitamin C 200mg 8hourly for 7 days 4. Review in 3 days, If no improvement REFER to appropriate health facility.
VI. Bleeding from the nose	Evidence of bleeding from the nose	Moderate condition: Epistaxis	1. Sit patient in upright position 2. Teach patient to pinch side of the nose for 5-10 minutes 3. Re-examine, if bleeding does not stop, apply cold compress to nose for 10 minutes. 4. If no improvement, apply Adrenaline nasal pack. If no improvement, REFER to appropriate health facility.
VII. Stuffy nose Catarrh with or without fever Sneezing with or without watery eyes	Temperature 37.5°C or above Watery eyes Catarrh	Mild condition Suspect Sinusitis	1. Tabs Clorpheniramine maleate 4mg 12hourly for 3 days 2. Tabs Paracetamol 1000mg 8hourly3 days 3. Tabs Vitamin C 200mg 8hourly for 7 days 4. Tabs Co-trimoxazole 960mg 12hourly for 5 days 5. Advise to avoid or reduce exposure to dust or pollen 6. Review in 3 days
VIII. Pain in the throat, fever inability to swallow, sore throat	Throat inflamed Tonsils not red and no pus	Mild condition Suspect Pharyngitis	1. Tabs Ibruprofen 400mg 8hourly for 3 days 2. Caps Amoxicillin 500mg 8hourly for 5 days 3. Tabs Metronidazole 400mg 8hourly 5 days 4. Tabs Vitamin C 200mg 8hourly for 7 days 5. Encourage fluids 6. Review in 3 days 7. If no improvement, REFER to appropriate health facility.

Health Education

Advise on

1.Personal and environmental hygiene.
2.Care of the ear, nose and throat.
3.Discouraging traditional practices such as uvullectomy and tonsillectomy.
4.Nutrition.
5.Avoiding exposure to air pollution (fumes, dust and pollen)
6.Discouraging home removal of foreign body
7.Importance of taking home-based records during visit to clinic

5.21 ORAL HEALTH DISORDERS

Oral health is a state of being free from chronic mouth and facial pain, oral and throat cancer, oral sores, birth defects such as cleft lip and palate, periodontal (gum) disease, tooth decay and tooth loss, and other diseases and disorders that affect the oral cavity.

The most common complaint is pain which may or may not be accompanied with swelling inside the mouth or on the face.

The condition of the mouth is also a reflection of the client's general health, so it should be inspected as part of physical examination. Early recognition of oral problems with prompt attention or referral will reduce morbidity and mortality arising from otherwise simple disease condition.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. Where is the problem? 3. Is there pain from tongue? Palate, mucus membrane, teeth or gum? 4. How long has it been noticed or present? 5. Are you able to eat your regular diet? 6. Does the pain increase with hot or cold drink or food? 7. Does the pain start on its own? 8. Does it keep you awake at night? 9. Is there swelling anywhere? 10. Is there any bleeding from gum or other part of the mouth? 11. Do you notice bad or foul breath? 12. What do you use for cleaning your teeth? 13. How do you clean your teeth? How many times a day? 14. Do you notice any sore in the mouth?	1. General appearance: in distress, ill-looking, facial swelling, malnourished 2. Temperature,: check and record 3. Mouth: a) Gum: sore, swelling, discharging sinus, bleeding, white patches b) Tongue: sore, coated, redness, fissures, smoothness, dryness c) Lips: crack 4. Teeth: caries, discolouration, tenderness, mobility, missing, plaque

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Severe tooth ache, disturbs patients sleep, aggravated By chewing, with or without headache	*Holes in tooth or teeth	Moderate condition Suspect irreversible pulpitis	1.Caps Amoxicillin 500 mg 8hourly for 5days 2.Tabs Metronidazole 400mg 8hourly for 5 days 3.Tabs Ibuprofen 400mg 8hourly for 3days (After food) or Tabs paracetamol 1g 8hourly for 3days 4. Review in 3 days, if no improvement REFER to the dental clinic.
II. Severe tooth pain, Disturbs patient's sleep, Aggravated by chewing, With or without headache, External swelling of the cheek	Hole on the offending tooth/ teeth Swelling of the jaw in relation to the offending tooth Tooth may be mobile	Severe condition Suspect Dental Abscess	1.Caps Amoxicillin 500mg 8hourly for 5days 2.Tabs Metronidazole 400mg 8hourly for 5days 3.Tabs Ibuprofen 400mg 8hourly for 3days or Tabs Paracetamol 1000mg 8hourly for 3 days 4. Review in 3 days, if no improvement REFER to the dental clinic
III. Sharp, short pain triggered by chewing, hot/cold drinks, Stops as soon as stimulus is removed.	Hole on the offending tooth/ teeth Wear of the offending tooth	Moderate condition Suspect reversible pulpitis Dentine hypersensitivity	1. Caps Amoxicillin 500mg 8hourly for 5 days 2.Tabs Metronidazole 400mg 8hourly for 5 days 3.Tabs Ibuprofen 400mg 8hourly for 3 days (After food) OR Tabs paracetamol 1000mg 8hourly for 3 days 4. Review in 3 days, if no improvement REFER to the dental clinic

III. Sharp, short pain triggered by chewing, hot/cold drinks, Stops as soon as stimulus is removed.	Hole on the offending tooth/ teeth Wear of the offending tooth	Moderate condition Suspect reversible pulpitis Dentine hypersensitivity	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Metronidazole 400mg 8hourly for 5 days 3. Tabs Ibuprofen 400mg 8hourly for 3 days (After food) OR Tabs paracetamol 1000mg 8hourly for 3 days 4. Review in 3 days, if no improvement REFER to the dental clinic
IV. Hole on the tooth, With or without pain	Hole on tooth/teeth	Dental caries Moderate condition	REFER to the dental clinic
V. Gum bleeding while brushing or biting into fruits e.g apple with or without gum bledding	* Gum may be red and slightly swollen	Moderate condition Suspect chronic Gingivitis (Gum Disease)	1. Oral hygiene instructions 2. REFER to dental clinic for scaling and polishing

Health Education

1.Advise on

1.Brush twice daily, in the morning and last thing before going to bed at night.
2.Use fluoride-containing toothpaste.
3.Change your toothbrush as soon as the bristles begin to flare/scatter.
4.Visit your dentist at least twice a year for routine cleaning (scaling and polishing).
5.Avoid use of tooth pick.
6.Avoid intake of sugary foods (refined sugar) in-between meals. Eat mouth cleansing fruits instead.
7.Cultivate the habit of rising your mouth with water after meals.
8.Discourage uvulectomy.
9.Emphasize the importance of bringing home-based records during visit to the clinic/hospital.

5.22 CARDIOVASCULAR DISORDERS

Cardiovascular problems are increasingly becoming major health concerns in adults. These are problems that affect mainly the heart and blood vessels. Congestive heart failure occurs when the heart becomes too weak to pump all the blood around and some of the fluids flows back into the abdomen and legs. Any history of shortness of breath especially while lying down should make the health worker think of cardiovascular disorders.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? How long have you had it? 2. Do you have cough, chest pain? Difficulty in breathing? If pain, where? 3. Does deep breathing and bending make it worse? 4. Can you sleep flat or do you need more than one pillow to sleep at night? 5. Do you sometimes get up at night to breath? 6. If cough is present, do you bring up sputum? Colour/ any blood in the sputum? 7. Do you take tobacco and/ or other substances, do you drink alcohol? 8. Do you eat well? 9. Do you have drenching night sweats 10. Have you lost weight? 11. Does the pain/ discomfort come or worsen with physical activity or when you are resting? 12. Are you on any medication? If yes, what?	1. General appearance: ill-looking, in distress, wasted, pale, cyanosed 2. Vital signs: Temperature: pulse, respiration, blood pressure: check and record 3. Neck: nodes 4. Chest: (a) respiration for rate, difficulty in breathing, wheezes (b) Crepitation, rhonchi, decreased breath sounds 5. Heart: rate, rhythm, murmur 6. Abdomen: distension, tender liver 7. Legs: Oedema.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Severe chest pain, Pain may radiate to other parts of the body. body, Unable to sleep	Painful distress	Severe condition Suspect Angina Pectoris	REFER immediately to appropriate health facility..
II. Tightness of the chest, Discomfort, Chest pain increase on exertion and not relieved by rest, Difficulty in breathing	* Rapid heart rate less than 60 beats per min. Increases with movement Difficulty in breathing	Severe condition Suspect Myocardial Infarction	1. Give Oxygen (if available) 2. Give tabs Acetyl salicylic Acid 150mg immediately (stat) 3. REFER Immediately to appropriate health facility.
III. Fast heartbeat ,feeling anxious	*Rapid heart rate 90-130 beats/min *Anxious	Severe condition Suspect cardiac Arrythmia	REFER immediately to appropriate health facility.

Health Education

1.Adequate rest.
2.Moderate exercise.
3.Regular clinic attendance and compliance to drugs.
4.Nutrition.
5.Importance of home-based records during visit to the clinic/hospital.

5.23 ABNORMAL BLOOD PRESSURE

Blood pressure refers to the force with which the heart pumps blood into body circulation. Blood pressure usually increases with age as the blood vessels become less elastic. Women tend to have lower blood pressure than men of the same age. The normal systolic pressure for young adult is 100-120 mmHg. High blood pressure is referred to as hypertension. Hypertension is a very common cardiovascular disease affecting a lot of people. It is thought to result from the narrowing of the blood vessels. The narrowing causes the heart to beat harder in order to pump the blood. Hypotension refers to low blood pressure and can lead to shock and organ failure.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? How long have you had it? 2. Do you get headache, dizziness and swollen legs? 3. Do you use to hear of feel your heartbeat? 4. Are they related to any physical exercise? 5. Do you have difficulty in breathing? Unable to sleep at night? 6. Are you unable to sleep at night? 7. Have you ever had high blood pressure or low blood pressure? 8. Are you taking any medication? 9. Are you on any birth control pills? (if female) 10. When was your last menstrual period? (if female) 11. Do you have family history of high blood pressure? 12. Do you have any existing illness?	1. General Appearance: Stable, ill-looking, pale, flushed, anxious, cyanosed, icteric, dehydrated 2. 2. Temperature, pulse, weight, blood pressure: check and record (pay special attention to both systolic and diastolic pressure) 3. Neck: thyroid engorged veins 4. Chest:(a) respiration for rate (b) Crepitation or decreased breath sounds (c) Heart sounds, rate and murmur 5. Signs of shock: (a) pale, cold or clammy extremities (b) restless (c) Increased perspiration 6. Legs: Swellings 7. Urine: protein, sugar, colour, deposit, odour, specific gravity.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Headache	Lethargic Systolic Blood Pressure less than 90mmHg Cold extremities Feeble pulse Rapid respiration Anxious looking.	Severe condition Suspect Hypotension	1. Give IV normal Saline 1 litre stat fast 2. Give IV hydrocortisone 100mg stat 3. Elevate foot of the bed. 4. Adequate ventilation 5. Keep warm 6. REFER immediately to appropriate health facility.
II. Headache, difficulty in breathing, unable to sleep at night, shortness of breath on walking, weakness or inability to walk. Cough swelling of feet.	*. May or may not look ill * Shortness of breath * Swollen feet * Pitting Oedema	Severe condition Suspect Congestive Cardiac failure	1. Give IV Frusemide 80mg stat 2. REFER immediately to the nearest general or tertiary hospital
III. headache and/or dizziness	* Blood pressure greater than 160/ 100 mmHg, no other symptoms	Severe condition Suspect severe hypertension	1. Sit client up 2. Give Step medication from the treatment protocol 3. REFER Immediately within one hour to the nearest general and tertiary hospital.

Management of Simple and Uncomplicated Hypertension	Additional points
This excludes cases of severe hypertension (BP ≥ 180/110mmHg), hypertension in pregnancy, hypertension with evidence of end-organ damage -kidney failure, heart disease, and stroke. All cases above should be referred immediately. 1.If initial BP ≥ 140/90 mmHg, start Amlodipine 5mg. 2.After 1 month, measure BP again. If still high, start Amlodipine 5mg + Losartan 50mg. 3.After 1 month, measure BP again. If still high, start Amlodipine 10mg + Losartan 100mg. 4.After 1 month, measure BP again. If still high, start Amlodipine 10mg + Losartan 100mg + Hydrochlorothiazide 25mg. 5.After 1 month, measure BP again. If BP is still high, refer for specialist hypertension management.	1.If the initial BP is ≥ 160/100 mmHg but less than 180/110 mmHg, start at step 2. 2.If initial BP is ≥ 180/110 mmHg, give step 3 drugs and refer to the emergency unit of the nearest general/tertiary hospital within an hour. 3.Telmisartan 40mg and 80mg, if available, is preferable to Losartan 50mg and 100mg. 4.Hydrochlorothiazide 25mg, if unavailable, may be substituted with Amiloride 2.5mg/HCTZ 25 mg. Losartan and hydrochlorothiazide should not be given to pregnant women and women of child-bearing age that are not on effective contraception.

Medication	Starting dose	Intensification dose	Side effects
Amlodipine	5mg	10mg	Headaches (7%) and palpitations (1-3%) – usually self-limiting, observe on treatment. Lower limb/pedal edema (1-10%), especially at 10mg dosage without ARBs, e.g., losartan – Observe on treatment. Consider alternative medicine if it is excessive or persistent after using a diuretic. Muscle cramps – usually self-limiting; observe on treatment. Male sexual/erectile disorder – explore for other causes and reassure.
Losartan	50mg	100mg	Cough occasionally Hypotension with dizziness (10%) - Monitor blood pressure and consider dose reduction if systolic BP is < 100mmHg or the person is frail. Hyperkalemia (4-10%) - Monitor potassium and creatinine. Stop the drug when potassium is > 4.5mmol/L and or eGFR <30ml/min.
Telmisartan	40mg	80mg	
Hydrochlorothiazide	25mg	100mg	Frequent urination – might have to ask the patient to take the medicine in the morning rather than at night. Metabolic derangement (hyperlipidemia, hyperuricemia, etc.) – monitor while on treatment. Hypokalemia – monitor while on treatment via serum urea and creatinine.
Amiloride/Hydrochlorothiazide	2.5mg/25mg	80mg	

Laboratory monitoring		
Baseline and every 12 months		
Fasting blood sugar	Lipid profile	ECG
Urine protein		Fundoscopy
Potassium and sodium – if on Hydrochlorothiazide/diuretic therapy		
Serum urea and creatine – if on losartan/telmisartan/ARB		

Health Education

1.Explain the condition and diagnosis to the patient.
2.Educate the patient that hypertension is a life-long condition that can be controlled with medications and lifestyle changes.
3.Emphasize that without treatment for hypertension, there is an increased risk of stroke, heart attack, kidney failure, vision problems, and death.
4.Ask the patient for a treatment supporter and elucidate on the importance.
5.Counsel the patient on stopping tobacco and harmful use of alcohol.
6.Encourage the patient to increase physical activity to at least 30 minutes daily and get adequate rest.
7.Counsel the patient to lose weight if overweight, using the BMI and waist circumference as a guide.Aim for BMI within 18 – 25kg/m2; or waist circumference <94cm in men, and <80cm in women.
8.Give dietary counseling (a heart-healthy diet low in salt, trans-fat, and added sugar)
a.Eat 5 servings of fruits and vegetables per day.
b.Eat nuts, legumes, whole grains, and foods rich in potassium.
c.Eat fish at least twice per week.
d.Use healthy oils like sunflower, flax seed, soybean, peanut, and olive.
e. Limit red meat to once or twice per week.
f. Use less salt – restrict to less than 5g daily (1 teaspoon)
g. Limit consumption of ultra-processed, canned, and 'fast' foods.
h. Avoid donuts, cookies, sweets, fizzy drinks, and juice with added sugar.
9.Counsel on the importance of adherence to medications, clinic, and laboratory appointments.
10.Educate the patient on self-management practices at home.

5.24 ABDOMINAL DISORDERS

Abdominal disorders affects the structures surrounding or within the abdominal cavity. They can manifest as pain, swelling or masses on the abdominal wall. The common causes of abdominal pain are gastro-enteritis, malaria, appendicitis, rupture of the intestine, typhoid, sickle cell crisis and obstruction e.g. hernia. constipation, dysentery, amoebiasis, peptic ulcer diseases (PUDs), worm infestation etc. may also constitute abdominal disorders.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? When did it begin? How severe is it? 2.Where is it? 3.Do you have or have you had nausea or vomiting? 4.Have you had any constipation or diarrhoea? If diarrhoea, how many times do you pass stool? If constipation, when was your last stool? 5.Any blood in vomiting or in stool? Is the stool black? 6.Does food make the problem better or worse? Or no difference? 7.Is there abdominal pain? Where? Is it continuous or intermittent? 8.Is there anything that relieves it or makes it worse? 9.Have you passed any worm in the stool? 10.Do you have fever or cough? Painful and/or frequent urination? 11.Are there any urinary compliant? 12.Have you taken any medication? If yes, what are they? **ADDITIONAL** **IF FEMALE** 1.Are you having any problem with menstruation? Such as dark, heavy, scanty, painful menses etc. 2.When was your last menstrual period (LMP) 3.Are you using any contraceptive? 4.Is there any abnormal vaginal discharge? Is there any history of trauma?	1. General appearance: Stable, ill-looking, pale, in distress, dehydrated, icteric, cyanosed. 2.Temperature, pulse, respiration, blood pressure, weight: check and record 3. Eyes: Pale or jaundice 4.Mouth: dry tongue, coated tonsils 5. Chest: a) difficult or rapid breathing b) Abnormal or decreased breath sounds c) Crepitations or wheezes 6.Abdomen: a) Spleen, liver, other masses b) Swelling, distension, ascites, scaphoid c)Tenderness or board-like rigidity, rebound tenderness d) Bowel sounds, absent, normal, increase, decreased 7.Flanks: tenderness over spine or lower ribs 8.Anus: prolapsed, fissures, redness, 9. Hermia: umbilical, inguinal, scrotal, etc. 10. Haemoglobin: Estimate and record 11. Stool: Sent to lab for analysis and microscopy

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Abdominal pain with rigidity, vomiting blood, Swelling of abdomen	Signs of shock such as low BP less than110/70 mmHg Weak pulse Rebound tenderness Rigidity of abdomen	Severe condition Suspect Acute Abdomen	1. IV normal saline 1 litre stat 2. REFER immediately to the appropriate health facility
II.Abdominal pain with or without vomiting,Vaginal bleeding or discharge, Last period more than one month	Abdominal tenderness with or without signs of shock Vaginal bleeding	Severe condition Suspect Ruptured Ectopic Pregnancy	1. IV normal saline 1 litre stat 2. REFER immediately to the appropriate health facility
III.Abdominal pain with swelling on the left side	* Enlarged spleen * warm to touch * Temperature 37.5°C or above * Palpable mass * May or may not be pale	Severe condition Suspect Rupture Spleen	1. IV normal saline 1 litre stat 2. REFER immediately to the appropriate health facility

IV. Abdominal pain with jaundice with or without diarrhea, vomiting	*Slight jaundice *Upper abdominal Tenderness	Moderate to severe condition Suspect Hepatitis	1. Bed rest, high carbohydrate diet a) Glucose fluids orally b) Cap. Amoxicillin 500mg 8hourly for 5 days c) Review in 3 days, If no improvement, REFER to the appropriate health facility
V. Abdominal pain, Painful and frequent urination, May or may not have fever	Tenderness in the flank Cloudy urine May or may not be warm to touch Temp 37.5°C or above	Moderate condition Suspect Pyelonephritis	1. IV Gentamicin 80mg 8hourly x48hrs continue with 2. Caps Amoxicillin 500 mg 8hourly for 5days 3. Tab. Paracetamol 1000mg 8hourly for 3days 4. If no improvement in 3days, REFER appropriately
VI. Abdominal pain before or after food, Heart burn, Black stool VII. Non-painful swelling in the abdomen	Tenderness in the upper abdomen May or may not have black stool	Moderate condition Suspect Bleeding Peptic Ulcer Diseases	1. Suspension Mist Mag Trisilicate 10ml 8hourly for days OR Tabs. Mist Mag trisilicate 2 Tabs 8hourly for 5 days 2. Caps. Amoxicillin 500mg 8hourly for 5 days 3. Tab. Metronidazole 400mg 8hourly for 5 days 4. Tab. Omeprazole 20mg dly for 10 days 5. Review in 1 week, If no improvement, REFER appropriately
VII. Non-painful swelling in the abdomen	* Reducible * non tender * Swelling in the groin or * umbilical area	Mild to moderate condition Suspect Hernia	1. Explain condition to client 2. REFER appropriately
VIII. Abdominal pain with Constipation with or without blood and or mucus in stool	*Abdominal discomfort with or without distension	Mild condition Suspect Dyspepsia	1. If abdomen not distended advise to take fruits and vegetables 2. Encourage fluid intake 3. Suspension Mist Mag trisilicate 10ml 8hourly for 5 days OR Tabs. Mist Mag trisilicate 2 tabs 8hourly for 5 days 4. Review in 3 days, If no improvement, REFER appropriately

Bleeding during passage of stool, Irritation around the anus	*Cut in the anus *Redness with or without visible blood	Mild condition Suspect Anal Fissure	1. Zinc oxide ointment locally 2. Sitz bath with warm saline water 3. Advise high vegetable diet and more fluid intake 4. Review in 3 days, if no improvement, REFER appropriately.

Health Education

Advise on:

1.Personal and environmental hygiene.
2.Nutrition (emphasis on avoiding tobacco and alcohol consumption).
3.Use of portable water.
4.Regular moderate exercise.
5.Rug compliance
6.Take home-based records during visit to clinic or hospital.

5.25 WORM INFESTATION

Worm infestation is the presence of large amount of worms in the gut, which causes discomfort to the patient. Ascaris (roundworm) is the commonest worm infestation. Heavy roundworm infestation produces abdominal pains vomiting, constipation and sometimes obstructions of the intestine. Hookworm infestation is common in rural areas where heavy infestation causes blood in stool, anaemia and malnutrition. Severe Trichuris (whipworm) infestation causes bloody diarrhea, abdominal pain, anaemia and prolapse of the rectum. Threadworm also lives in the intestine and causes itching around the anus. Tapeworms is a flat ribbon-like worm that can also be found in the intestine.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. Have any worms been passed? If yes, describe the appearance? How many? 3. Do you eat well? 4. Has there been abdominal pain? Vomiting? Constipation? 5. Is abdomen tender to touch? 6. Is there any diarrhoea? Blood in the stool? 7. Is there rectal itching? 8. Is there any other illness? 9. Has any medication been used?	1. General appearance: Distressed, pale 2. Vital signs: Temperature, pulse, respiration, blood pressure: check and record 3. Eyes: pallor 4. Chest: wheezing, rhonchi 5. Abdomen: tenderness 6. Anus: redness, rashes 7. Haemoglobin: estimate and record 8. Weight and height: measure and record 9. Stool examination: if laboratory is available

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Abdominal pain	* Abdominal tenderness with signs of obstruction * Ova or Oyst seen	Severe condition Suspect intestinal obstruction	1. IV normal Saline 1 litre 2. REFER immediately to the appropriate health facility
II. Abdominal pain	* Abdominal tenderness without signs of obstruction * Ova or oyst seen	Moderate condition Suspect worm infestation	1. Tablet Albendazole 600mg stat 2. if no improvement, REFER appropriate
III. Passing worms per rectum	* Underweight with or without mild abdominal pain or distention * Presence of ova or oyst from laboratory result	Mild to moderate condition Suspect worm infestation	1. Tabs. Albendazole 600mg stat OR 2. Tabs. Pyrantel pamote 4-6 tablet immediately(stat) 3. If tapeworm give Tab. Niclosamide 200mg immediately (stat)

Health Education

Advise on:

1. Personal food and environmental sanitation (proper disposal of excreta).
2. Periodic deworming.
3. Nutrition (wash fruit and vegetable before eating).
4. Proper care of feet (use of footwears).
5. Importance of taking home-based records during visit to the clinic or hospital

5.26 GENITO-URINARY DISORDERS

Genito-urinary disorders are common and can be due to metabolic/degenerative changes, infections, abnormal growth etc. examples of metabolic/ degenerative changes include: diabetes, kidney stones, vaginal/ urine prolapse etc. examples of abnormal growths include: enlarged prostrate, cervical cancer etc. example of common infections include gonorrhea, syphilis, non-specific bacterial infection, schistosomiasis etc. Patient may complain of discharge, pain and burning sensation on passing urine (micturition), blood in urine, frequent micturition, urinary retention etc.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? How long have you had it? 2. How old are you? 3. Do you urinate often? How often night and day? 4. Do you have swelling in the scrotal sac with or without pain? 5. Did you have any trauma? Or Mumps? 6. Do you have abnormal pain or burning sensation on micturition? 7. Do you have weight loss? 8. Do you drink more than usual? (explain) 9. Do you have pain in the flank? 10. What is the colour of your urine? Is there blood in the urine? 11. Do you have Ulcer in the mouth? Oral thrust? Bleeding gum? Skin rash? Night sweats? Weight loss? 12. Do you have fever? Nausea? Vomiting? 13. Do you have history of wading in stream during childhood? 14. Are you on any birth control method? (if female) 15. Have you ever had this problem before? Any treatment? 16. When was your last menstrual period? (if female) 17. Do you have discharge or itching from the vaginal (female) or penis rmale)?	1. General appearance: ill-looking,wasting, puffy 2. Vital signs: Temperature, pulse, respiration, blood pressure check and record 3. Eyes: pallor 4. Chest: crepitations, decreased breath sounds 5. Abdomen: tenderness, swelling or masses 6. Flanks: tenderness 7. Limbs: (a) legs for swelling (b) palms, nail bed, for pallor 8. Genitals: sores, pus or discharge 9. Urine: (a) colour or cloudiness (b) protein (c) Sugar

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Swelling in one or both scrotal sacs with pain	*Swollen Scrotum *Tender	Severe condition Suspect strangulated Scrotal Hernia	1. Tab Ibruprofen 400mg 8hourly for 3 days 2. REFER immediately to the appropriate health facility
II. Blood in urine with history of trauma to the external genitalia	*Blood urine (initial)	Severe condition	1. REFER immediately to the appropriate health facility
III. Swollen scrotum not painful	*Swollen scrotum, may or may not be Reducible	Moderate to severe condition Suspect uncomplicated scrotal Henia	1. Reassure client 2. REFER immediately to the appropriate health facility
IV. Swollen painful scrotum with or without trauma	*Swollen tender scrotum	Severe condition Suspect Testicular Torsion	1. Apply cold compress 2. Untwist the testicular cord (left or right until comfortable) 3. REFER immediately to the appropriate health facility

V. Swollen painful scrotum with or without discharge	*Swollen tender scrotum *with or without discharge	Moderate to severe condition Suspect epididymo-orchitis	1. Tabs. Ciprofloxacin 500mg 12hourly for 7 days 2. Elevate the scrotum 3. Apply cold compress 4. bed rest 5. If no improvement after 3 days, REFER to the appropriate health facility
VI. Urinary problem with any of the following: a) Shortness of breath b) Nausea and/or vomiting c) No urine for more than 24hrs	Chest may or may not be clear *Brown or dark urine *Swollen legs or Abdomen	Severe condition Suspect Acute Renal Failure	REFER immediately to the appropriate health facility
VII. Pain or burning sensation on micturition, may have lower abdominal pain	*Protein in urine *Cloudy or clear urine	Moderate to severe condition Suspect urinary tract infection	1. Tabs. Ciprofloxacin 500mg 12hourly for 7days 2. Tabs. Paracetamol 1000mg 8hourly for 3 days 3. Encourage fluids 4. If no improvement in 3 days, REFER to the appropriate health facility hospital
VIII. Blood in urine, painful micturition, losing weight weak	*Blood in urine (total) *Pallor	Severe condition Suspect bladder tumour	1. Conduct test 2. If negative, REFER immediately to the appropriate health facility
IX Blood in urine	*Blood in urine (terminal)	Moderate condition Suspect schistosomiasis	3. If positive, give praziquantel 240mg of 4 tabs stat 4. Review in 2 weeks If no improvement, REFERto the health facility
X. Dark yellow urine without any other symptom	*Dark yellow urine *No jaundice	Mild condition Suspect dehydration	1.Encourage high fluid intake 2.Review in 3 days If no improvement, REFER to the appropriate health facility

Health Education

Advise on:

1.Personal hygiene.
1.Protection of genital area.
2.Drug compliance.
3.Adequate fluid intake.
4.Side effects of medications.
5.Importance of taking home-based records during visit to the clinic or hospital.

5.27 UPPER AND LOWER LIMBS DISORDERS

The limbs may be affected by athritis of various kinds, trauma and localized or systemic infections which may manifest as dislocation, fracture, swelling, pain and deformity to adults. Infection in the bone and sickle cell crises can also cause pain in the limbs.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? When did it start? 2. Has it happened before? 3. Was there any accident or injury? 4. If there is injury, have you been able to move the part since? 5. Is there any pain? Tingling? Numbness? Swelling? 6. Do you have joint pains or are they stiff? 7. Do you have back pain? Do you have fever?	1. General appearance: ill-looking wasted, deformed 2. Vital signs: Temperature: pulse, respiration, blood pressure: check and record 3. Eyes: jaundice 4. Spine: localized tenderness 5. Limbs: a) deformity b) tenderness and swelling c) bleeding d) fracture e) not able to move part f) wasting g) varicosity h) sign of paralysis 6. Haemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Painful swelling on the upper and/or lower limbs as a result of an accident (home or road traffic)	Painful swelling of the upper and/or lower limbs with deformity	Severe condition Suspect fracture	1. Splint 2. Tabs Paracetamol 1000mg stat 3. IM Diclofenac sodium 75mg stat 4. REFER immediately to the appropriate health facility
II. Painful swelling on the upper and/or lower limbs as a result of an accident (home or road traffic)	Painful swelling of the upper and/or lower limbs without deformity	Moderate condition Suspect soft tissue injury	1. Splint 2. Tabs Diclofenac sodium 50mg 12hourly for 5 days (After food) 3. Tabs Chymoral 2 tabs 12hourly for 5 days 3. Review in 3days, 4. If no improvement, REFER appropriately
III. Pain in the bone or Muscle with localized redness, pus accumulation and tenderness.	* Local redness * Swelling and tenderness of bones	Mild condition Suspect infection	1. Tabs Ciprofloxacin 500mg 12hourly for 5 days 2. Tabs Ibuprofen 400mg 8hourly for 3 days 3. Review in 3 days, If no improvement, REFER appropriately
IV. Pains and swellings in the joint with difficulty in moving the affected part.	Tenderness in the joint Stiffness in the joint	Mild condition Suspect Arthritis/sprain.	1. Splint. 2. Tabs Ibuprofen 400mg 8hourly for 3days (After food) 3. Cold compress/poultice alternate with warm compress. 4. Review in 3 days, If no improvement, REFER appropriately

Health Education

Advise on:

1. Prevention of accidents.
3.Adequate diet (calcium rich foods).
4.Proper posturing.
5.Importance of taking home-based records during visit to the clinic.

5.28 MENTAL HEALTH DISORDERS

Mental health disorders are those conditions with psychological or behavioral manifestations and/or impairment in functioning due to social, psychological, genetic, physical or biological disturbance. Mental disorders are characterized by abnormalities in a person's emotions, thoughts, cognition, sensory perceptions beliefs and behavioral. Mental problems usually manifest in many ways.

The affected patients may become restless, afraid, suspicious and confused. They may think that people are after them. They may imagine/or complain of seeing strange people or things. They may become violent and attack other people or themselves without cause. The condition may have come on a fever due to malaria, typhoid or viral infection. It may also be as result of severe emotional disturbance or abnormal emotional experiences or excessive drinking of alcohol or drugs over a significant period. If nothing is done immediately, there may be danger to life and property. Other symptoms include sleeplessness, hearing and seeing things others cannot see and suicidal tenderness.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long has the client had the problem? 3. Has the client had this or similar problem in the past? 4. Did he recently have febrile illness? If yes describe 5. Any head injury? Accident? Severe stress? E.g loss of job, divorce, arm robbery attack, loss of relation etc. 6. Does the client use alcohol or drug such as Indian Hemp? 7. Any headache? Jerking of limb? Clinching of teeth? Biting of tongue? 8. How does the client function at work? With relations at home 9. Has the client had prolonged fasting and praying or Ritual dancing? 10. Is there change in the way the client speaks? 11. Does any other person in the family have the same problem? 12. What makes you think that the client's behavior is abnormal?	1. General appearance: unkempt, clam, restless, threatening or physically aggressive, ill, withdrawn, confused, tensed, frightened, fidgety, trembling/shaking, sweaty, cold, clammy palms, acting strangely, tearful, hallucinating. 2. Vital signs: Temperature, pulse, respiration, blood pressure, check and record 3. Reaction: to the health worker-cooperative, withdrawn, aggressive 4. Eyes: redness, starting, bulging, lid retraction, pallor 5. Mouth: smell of alcohol or other substances 6. Speech: audible, shouting, understandable, meaningful, relevant to the questions asked or to the situation, slurred, rambling, excessive, stammering, hesitant, expressing worry, any strange statements or beliefs expressed. 7. Orientation: responds to name, knows where he is, has idea of time 8. Pocket: drugs or other substances in the pocket 9. Signs of tension: suspicious, hyper-alertness 10. Thoughts: expression of hopelessness, suicidal tendencies 11. Weight: weigh and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. low energy, fatigue, sleep or appetite problems, persistent, sad or anxious mood, irritability, low interest or pleasure in activities that used to be interesting	Weak, unkempt, withdrawn, sad, irritable, smell of alcohol/ tobacco or other substances.	Severe condition Suspect depression	1. Counsel on sleeping habit, drug usage and participation in community activities. 2. Group psychotherapy 3. REFER to Mental health Clinic
II. Incoherent or irrelevant speech, Delusions (force, firmly held beliefs), Hallucination (hearing voices or seeing things that are not real), Neglecting usual Responsibilities	Unkempt, self-neglect, aggressive, hallucinating	Severe condition Suspect psychosis	1. IM Chlorpromazine 100mg stat 2. REFER immediately to the appropriately health facility

III. History of frothing in the mouth after signs or premo-nitions, loss of consciousness with any of the following: a) Jerking of limbs b) Clenching of teeth c) Biting of tongue d) Incontinence (urine or faces)	* May or may not be Jerking * Frothing * Biting of tongue * Unconscious or drowsy	Moderate condition Suspect epilepsy	1. Prevent Client from injury to self 2. Keep airway clear 3. Check if client is on medication 4. If unable to take drugs, give 5. IV. Diazepam 10-20mg immediately(stat) (if CONVULSING) 6. Tabs Phenobarbitone 30mg stat 7. REFER immediately to the appropriate health facility
IV. Violent behaviour	* Injection marks * Drugs/substance in pockets * Agitated *Violent behavior * May or may not be drunk * Perceive smell of alcohol and/or other substances	Moderate condition Suspect alcohol or drug abuse	1. Restrain and observe until effects have worn off 2. IV Diazepam 10-20mg stat 3. Seek police and Community/ religious Leaders assistance if necessary 4. Review in 2 weeks 5. Counsel client and family members.
V. Confusion, restlessness, dizziness or convulsing, known or unknown diabetic	* Restless or confused * may be unconscious or convulsing * Ketone smell (sweet breath) * Check urine, blood sugar *(greater than 11. lmmoooooo llitre)	Severe condition Suspect hyperglycemia	1. IV normal saline 1 litre stat 2. REFER immediately to the appropriately health facility
VI. Confused, Restlessness, Drowsy or unconscious, Known or unknown Diabetic	* May be unconscious * Sweating profusely * Anxiety * Restlessness * No sugar in urine * Check blood sugar	Severe condition Suspect hypoglycemia	1. Give IV 50% Dextrose stat 2. Give sugar in water until gets to the clinic 3. Re-assure client 4. REFER immediately to the appropriately health facility

Health Education

1.Counsel client and family members.
2.Give nutrition counseling.
3.Discourage abuse and use of alcohol, drugs and other substances.
4.Advise on psychotherapy.
5.Encourage participation in community and religious activities.
6.Encourage to visit friends and relatives.
7.Importance of taking home-based records during visit to the clinic.

5.29. POISONING

A poison is a toxic substance which may enter the body through swallowing, breathing, absorption through skin or mucus membrane or infections. It can cause fits, marked restlessness, abdominal pain, diarrhea and vomiting. Patient may feel dizzy, weak, lose consciousness and it may result in death. Therefore, it is advisable to keep, kerosene, medicine, bleach, dye, alcohol and other harmful substances properly. The type of poison consumed should be communicated to the health worker. The principles guiding treatment are the following: Ability to identify ingested poison, use of activated charcoal and providing support to vital functions (ABCD)

a)Airways - relief obstruction
b)Breathing - check the person's breathing, maintain an open airway and normal body temperature
c)Circulation - Treat hypo-tension
d)Dextrose - Avoid severe hypoglycemia by giving glucose or neutralizing the effect of poison by ingestion of milk or magnesium tables.

HISTORY	EXAMINATION/ACTIONS
1.What did you ingest? 2.When was the poison taken? 3.Have you taken any medication? If yes, what? 4. Did you vomit? If yes, any blood in the vomitus?	1. Examine poison container (if available) 2. General appearance: conscious, drowsy, sleepy, confused or unconscious 3.Vital signs: Temperature, pulse, respiration, blood pressure: check and record 4. Eyes: size and reaction of pupils to light 5. Chest: lower chest in drawing, fast breathing 6. Abdomen: tenderness, rigidity 7. Mouth:burns, bleeding, dribbling, colour of membrane

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Ingestion of overdose of drugs e.g Aspirin with or without fever	* Temp less than 37.50C or above * Irritable * No other sign	Moderate condition	1. Induce vomiting by giving salty water or use spatula 2. Milk drink, if not available, 3. Susp. Magnesium trisilicate 10ml 8hourly for 2days 4. Review 2nd day If no improvement, Refer
II. Ingestion of Kerosene or petrol or corrosive substance	* Irritable * Conscious	Moderate condition	1. Do not induce vomiting 2.Milk drink, if not available 3. Susp. Magnesium trusilicate10ml 8hourly for 2days Procaine penicillin 600,000iu daily for 7days Review on 2nd day, If no improvement, Refer appropriately
III. Iron poisoning	*Irritable, restless	Severe condition	1.Refer immediately to the appropriate health facility

Health Education

Advise on:

1. Proper storage and labeling of chemicals and harmful substances
2. Use of bleaching creams
3. Advise to Keeping chemicals and harmful substances out of reach of children
4. Importance of bringing home-based records during visit to clinic

5.30 MANAGEMENT OF HIV/AIDS PATIENTS

HIV/AIDS is of public health concern, it is responsible for increased number of deaths. It affects not only the patient but relations and friends. In recent times, with the introduction of Anti Retro Viral Drugs (ARVs) the condition can be successfully managed while lives of the infected can be prolonged. The health worker therefore, should be able to provide care and support for people living with HIV/AIDS and those affected by it.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long have you had the problem? 3. Any weight loss? General weakness? 4. Have you had infections? 5. Have you had blood transfusion? Scarification e.g tribal mark? Circumcision? Tattoo. 6. Are you married? 7. Do you have a sex partner? How many sex partners do you have? 8. Have you had contact with relatives or partner with similar problems? 9. Have you been treated for HIV/ AIDs before? 10. Do you have genital and/or mouth sores that have refused to go? 11. Do you have cough? Diarrhea? Repeated infections? Fever? For how long? Did it persist or reoccur after treatment? 12. Do you have itchy skin or skin rash? 13. What other problems do you have?	1. General appearance: healthy, thin, pale 2. Weight: weigh and record 3. Hair: thin, scanty 4. Eyes: sunken, pale conjunctiva 5. Mouth: thrush, red and white patches 6. Neck and Armpit: enlarge nodes 7. Skin: rash, ulcer, fungal infection, kaposis around the mouth 8. Chest: crepitations, dimished breath sound, respiration rate 9. Abdomen: palpable organ 10. Groin: enlarged nodes, ulcer 11. Genitalia: ulcers, rashes, discharge 12. Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. History of recurrent or persistent diarrhoea of more than 1 month duration Generalized itchy skin rashes unexplained cough longer than 1 month Recurrent genital and mouth sores Prolonged fever for more than one 1 month Severe weight loss General weakness Ulcer that has refused to heal	* Severe dehydration * Temp 37.2°C or above * May or may not have weight loss greater than 10% of body weight * Generalized dermatitis * Oral candidiasis * Genital ulcer * Generalized enlarged lymph nodes	Severe condition Suspect HIV/AIDS	1. Observe universal precautions with patient's body fluids, see appendix 2. Ensure CONFIDENTIALITY 3. Confirm contact address 4. Follow up client 5. SYMPTOMATICTREATMENT 6. For fever, diarrhea, cough, sore mouth and throat, skin problems pains, anxiety and depression etc a) Low Osmolarity ORS b) Tab Piriton 4mg at night for 3 days c) Apply Nystatin oral gel to mouth 7. Conduct HTS (HIV testing services)
II. Chronic Diarrhoea, Chronic cough, Mouth and genital ulcer, Oral thrush, Skin rash	* Marked weight loss * General skin rashes * Enlarged lymph nodes * Mouth and genital ulcers		1. Give Symptomatic Treatment 2. REFER immediately to the appropriate health facility.
III. Claim to be HIV positive, No other complaint	* Laboratory result may or may not be positive * weight loss may not be present		1.2. Give symptomatic treatment REFER immediately to the appropriate health facility.

Health Education

1.Counseling on HIV/AIDS prevention.
2.Counseling on nutrition.
3.Counseling on adherence to treatment.
4.Contact tracing and treatment.
5.Condom use.
6.Compliance with drugs.
7.Importance of taking home-based records during visit to the clinic or hospital.

5.31 SEXUALLY TRANSMITTED INFECTIONS (STIs) (FEMALE) REPRODUCTIVE TRACT INFECTIONS (RTIs) FEMALE

Reproductive Tract Infections are caused by organisms normally present in the reproductive tract or introduced from the outside during sexual contact, medical procedures or as a result of an imbalance in vaginal bacterial flora. The common organisms responsible are Neisseria gonorrhea, chlamydia trachomatis and Trichomonas vaginalis, others are Human Immunodeficiency Virus (HIV),cytomegalovirus, streptococci and Hepatitis B. The rationale for syndromic management is that it is effective, prevents complications or mother to child transmission, encourages treatment of partner, and promotes safe sex habits and use of cheap, locally available and effective drugs. The common syndromes are discharges, genital ulcer and lower abdominal pain.

HISTORY	EXAMINATION/ACTIONS
1. How long have you had the discharge? 2. Is the discharge purulent, mucoid, abundant or scanty? 3. Have you ever had a urethral discharge before? How long ago? 4. Did you notice the discharge after a particular sexual contact? 5. Have you had any sexual contacts after you noticed the discharge? 6. Do you feel any burning or pain on urination? 7. What treatment have you taken?	1. External Genitalia: discharge ('milk'the urethral) Uncircumcised male: retract the foreskin, 2. be sure discharge is from urethra, check glands for redness, ulcer. 3. Check anus for purulent discharge, ulcer. 4. Throat: Redness, pus

SYNDROME	STI/RTI	ORGANISMS	TYPE
1. Discharge	* Bacterial vaginosis * Yeast infection * Gonorrhea *Chlamydia * Trichomoniasis	* Gardnerella vaginalis *Anaerobe * Candida albicans * Neisseria gonorrhoea * Chlamydia trachomatis * Trichomoniasis	Bacterial Fungal Bacterial Protozoan Bacterial
II. Genital ulcer disease (GUD)	* Syphilis * Chancroid * Herpes rash * Granuloma inguinale (donovanosis) * Lymphogranuloma venereum (LGV) * Moluscum contagiosum	* Treponema palladium *Hemophilus ducreyi *2 Herpes simplex virus (HSV-2) *Inguinale granuolomatis *Chlamydia trachomatis *M. contagiosum	Bacterial Bacterial Viral Bacterial Bacterial
III. Lower abdominal pain (LAP)	* Chlamydia * Gonorrhoea * Scabies(crab) * Body lice * Genital warts	* Chlamydia trachomatis * Neisseria gonorrhoea * Sarcoptes scabies * Phthirus pubis Human papilloma virus	Bacterial Bacterial Tick Tick Viral
Adopted from WHO (2005)			

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Vaginal discharge	* Discharge is profuse * Muco-purulent discharge from Cervix * Risk Assessment positive (unmarried, under 21 and Sexually active, more than 1 partner in the last 12 months, new partner in past 3 months)	Severe condition Suspect cervicitis Vaginitis, urethritis	1. Tablet Ciprofloxacin 500mg immediately (stat) 2. Cap. Doxycycline 100mg 12hourly for 7 days 3. Vaginal pessaries 100,000 units inserted every night for 14 days 4. Tabs. Metronidazole 2gm orally stat then 400mg 8hourly 5 days 5. Review after 7 days, If no improvement, REFER to the appropriate health facility Refer to appendix on syndromic management HTS, if positive, REFER for ART and follow-up
II. Vaginal discharge Profuse, thin, watery or Creamy, thick whitish with or without, itching Painful Urination	* Discharge is Profuse Thin, watery or Creamy or thick whitish with or without itching * Risk assessment negative	Severe condition Suspect the following: Candidiasis Vaginals Trichomoniasis	1. Nystatin vaginal pessaries 100,000 units inserted every night for 14 days 2. Tabs Metronidazole 2gm stat 3. Review in 7 days, if no improvement 4. Tabs Ciprofloxacin 500mg orally immediately (stat) 5. Caps Doxycline 100mg 12hourly for 3 days 6. Tabs Paracetamol 1000mg 8hourly for 3 days Review in 7 days, if no improvement REFER to the appropriate health facility hospital 8. Refer to appendix on syndromic management
III. Severe lower abdominal pain with Vaginal Discharge	* Pain with Vaginal Discharge * Pain on moving Cervic	Severe condition Suspect cervicitis	1. Tabs. Paracetamol 1000mg 8hourly for 3 days 2. Tab Ciprofloxacin 500mg stat 3. REFER immediately to the hospital 4. If referral is difficult, stat Caps. Doxycline 100mg 12hourly for 7 days 5. Refer to appendix on syndromic management

IV. Lower abdominal pain with any of the following: History of late or missed period (menses) Recent delivery/abortion Vaginal bleeding Fever	Vaginal discharge Pain on moving cervix Temp. 37.50C or above	Moderate to severe condition Suspect Pelvic Inflammatory Disease (PID)	1. Tab Ciprofloxacin 500mg orally stat 2. Caps. Doxycycline 100mg 12hourly for 7 days 3. Tab Metronidazole 400mg 12hourly for 14 days 4. 4Cs (counseling, contact treatment, compliance with drug, condom use) 5. Review in 3 days 6. If patient improves, continue treatment until completion 7. bring male partner for treatment 8. HTS , if no improvement, REFER to the hospital

Health Education

Advise Mother/Caregiver on:

1. Counseling.
2. Contact treatment.
3. Compliance with drug.
4. Condom use.
5. Avoid sex intercourse during treatment and relapse.
6. Importance of taking home-based records to the clinic or hospital.

5.32 SEXUALLY TRANSMITTED INFECTIONS [STIs (MALE) REPRODUCTIVE TRACT INFECTIONS [RTIs (MALE)]

Sexually Transmitted Infections (STIs) are infections that are spread through sexual intercourse. Common STIs include gonorrhea, syphilis, herpes, genital warts, HIV/AIDs etc. STIs can be painful and uncomfortable. The common syndromes are discharge, genital ulcer and lower abdominal pain. They can also have tragic consequences such as infertility. Therefore, all cases of STIs should be adequately treated.

HISTORY	EXAMINATION/ACTIONS
1.How long have you had the discharge? 2.Is the discharge purulent, mucoid, abundant, or scanty? 3.Have you ever had a urethral discharge before? How long ago? 4.Did you notice the discharge after a particular sexual contact? 5.Have you had any sexual contacts after you noticed the discharge? 6.Do you feel any burning or pain on urination? 7.What treatment have you taken?	1. External Genitalia: discharge (;milk' the urethral) 2. Uncircumcised male: retract the foreskin, be Sure discharge is from urethra. Check glands for redness, ulcer. Check anus for purulent discharge, ulcer. 3. Throat: redness, pus

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Groin (inguinal) swelling, Patient not sexually active No ulcer No rashes	Inguinal swelling	Severe condition	1. REFER immediately to the appropriate health facility
II. Swelling in the groin, Genital sore or Ulcer, History of sexual exposure	* Sore/Ulcer present * Swelling in the groin * Risk assessment positive	Moderate to severe condition Suspect genital ulcer disease (GUD)	1. IM Benzathine Penicillin G. 2.4mega unit stat 2. Tab Erythromycin 500mg 6 hourly for 7 days 3. Tab. Paracetamol 1000mg 8hourly for 3 days 4. 4Cs-counseling, compliance, contact reatment and condom use. 5. Review in 7 days 6. HTS , if no improvement, REFER to the appropriate health facility
III. Patient with Scrotal swelling and painful Urethral discharge, Patient sexually active	Examine the genitals (just look (DO NOT PALPATE)	Severe condition	1.Tab Ciprofloxacin 500mg immediately (stat) 2. Cap. Doxycycline 100mg 12hourly for 7 days 3. 4Cs-Counselling, compliance, contact treatment and condom use. 4.Advise to return in 7 days; if no improvement 5. REFER immediately to the appropriate health facility

IV. Urethral discharge and or painful urination, Genital Ulcer, Sexual exposure	Penile discharge Ulcer present Risk Assessment positive	Moderate condition Suspect genital ulcer disease (GUD)	1. Tab Ciprofloxacin 500mg stat 2. Cap Doxycycline100mg 12hourly for 7 days 3. Tab Ibruprofen 400mg 8hourly for 3 days (After food) OR 4. IM Benzathine Penicillin G 2.4 Mega units stat 5. Tab Erythromycin 500mg 6 hourly for 7 days 6. Tab Metronidazole 2g immediately (stat) 7. 4Cs- counseling compliance contact treatment and condom use 8. Review in 7 days 9. If no improvement, REFERto the appropriate health facility
V. Patient with painful urination no discharge	No penile discharge	Moderate condition	REFER for Urine Microscopy Treat based on result.
VI. Genital sore/Ulcer, Group of painful blisters (often recurrent), History of sexual exposure, With or without painful swelling in the genital area	Group of painful blisters Risk Assessment Positive	Severe condition Suspect genital herpes HIV/AIDs	1. Tab. Ibuprofen 400mg 8hourly for 3 days keep lesion dry 2. 4Cs- counseling compliance contact treatment and condom use 3. HTS 4. Review in 7 days 5. If there is no improvement, REFER to the appropriate health facility

Health Education

1. Counselling.
2. Contact treatment.
3. Compliance with drug.
4. Condom use.
5. Avoid sexual intercourse during treatment and relapse.
6. Importance of taking home-based records to the clinic or hospital.

SECTION SIX
MATERNAL HEALTH

6.0 MATERNAL HEALTH

6.1 PRENATAL (ANTENATAL) CARE

Prenatal period is the time between conception and childbirth (pregnancy). It lasts for about 280 days or 40 weeks. It is associated with physiological changes in the woman's body which enable foetal survival as well as preparation for labour. The physiological changes are not confirmed to the reproductive organs alone, but every tissue and organ reacts to the stimulus of pregnancy. The metabolic, chemical, and endocrine balance of the body is also altered. Many of these changes are regarded as signs and symptoms of pregnancy and can sometimes manifest as minor discomforts. However, when these conditions become persistent and severe, they become major disorders that will require prompt medical attention.

Prenatal or antenatal care (ANC) is the advice, supervision and intervention given to pregnant women in order to obtain and maintain a state of good health throughout the prenatal period. Prenatal care provides a platform for the health worker to discover any medical, obstetric, and mental problem the woman may have and determine if she can be able to deliver her baby safely. ANC provides opportunity to screen pregnant mothers for signs and symptoms of mental health conditions such as; anxiety/depression, involvement in substance use/abuse and signs of domestic violence, for appropriate management and referral.

The 2016 WHO ANC model aims to provide pregnant women with respectful, individualized, person centered care at every contact and to ensure that each contact delivers effective, integrated clinical practices (tests and interventions), provides relevant and timely information, and offers psychological and emotional support by practitioners with good clinical and interpersonal skills working in a well- functioning health system. The WHO recommends a minimum of eight contacts; five contacts in the third trimester, two contacts in the second trimester and one contact in the first trimester. There should be emphasis on quality of care and great focus should also be maintained on birth planning and emergency preparedness.

HISTORY	EXAMINATION/ACTIONS
At all visits; 1.How are you feeling? 2.If ill, what are your symptoms? 3.Do you have problems sleeping? Do you sleep well? 4.Do you get easily irritated, anxious or depressed? 5.How is your appetite? 6.Have you lost weight? 7.Any bleeding? 8.What arrangement have you made for delivery? 9.Any other problem (s)? **Social History** 1.Any history of Alcohol Use? 2.Any history of Substance Use/Abuse? 3.Any history of Domestic violence? 4.Any history of Mental Problem? **At First Contact Only** **Medical History** 1.Do you have any particular medical condition, e.g hypertension, diabetes, sickle cell, epilepsy, asthma/allergies, kidney disease etc. 2.Have you had any Surgery (C/S, Hernioraphy, Appendectomy etc)? 3.Have you ever had T.d immunization? How many does? (ask for card)	1.General appearance: well or ill looking, facial expression 2.Skin: pallor, rashes, sores, oedema. 3.Check skin for signs of domestic violence (like scars and broken teeth etc) 4.Eyes: redness, jaundice, pallor, discharges 5.Nose: discharge, polyps 6.Mouth: condition of mucous membrane, tongue, gum & teeth 7.Chest: for respiratory rate & sign of respiratory distress 8.Breast: tenderness or lumps, nature of nipples whether normal, flat or inverted, scar 9.Height: (first visit only) Check and record 10.Weight: weigh & record (initial weight will serve as baseline for comparison with subsequent readings) 11.Blood pressure: measure and record (initial reading will serve as baseline for comparison with subsequent findings) 12.Temperature: Check and Record 13.Pulse Rate: Count and record **Obstetric Examination** Abdomen: Inspect for scar, shape and size Fundal height: measure and record Palpation: foetal parts, lie, presentation, position, masses Auscultation: foetal heart sound **Pelvic examination:** (after the 35th weeks) Inspection: check external genitalia Speculum should be correctly inserted for inspection of cervix for colour, smoothness, any lacerations, discharge, or bleeding. Pelvic measurement: diagonal conjugate, height of symphysis, prominent spines

Obstetric History 1.How many pregnancies have you had? (livebirth stillbirth and miscarriage) 2.How many of your children are alive? 3.How many died? Age at the time of death and cause of death? 4.How many miscarriages have you had? 5.Any special maternal complications in previous pregnancies? E.g pre-eclampsia/eclampsia, bleeding, assisted delivery etc. 6.LMP: Obtain the date of LMP and calculate the EDD by adding 7 days to the first day of the LMP and subtract 3 months from the month of LMP i.e. LMP + 7 days -3months = EDD	**Laboratory Investigation** **First Visit Only** HTS: for positive client to be enrol for ART VDRL/RPR (Rapid Plasma Reagin): for syphilis Blood grouping and rhesus determination Any appropriate investigation as indicated by patients' condition e.g HBs Ag, Widal test, mRDTetc HBs Ag for Hepatitis At All Visit: Perform the following for patient Urinalysis Haemoglobin estimation or PCV count Ultrasound Scan Refer clients for USS at least once before 24 weeks

ANC CONTACTS

First Trimester

- Contact 1: up to 12 Weeks

Second Trimester

- Contact 2: 20 Weeks
- Contact 3: 26 Weeks

Third Trimester

- Contact 4: 30 Weeks
- Contact 5: 34 Weeks
- Contact 6: 36 Weeks
- Contact 7: 38 Weeks
- Contact 8: 40 Weeks

WELL VISITS

- 1ST Contact (up to 12 Weeks) or as soon as pregnancy is confirmed
- Establish a friendly atmosphere.
- Obtain comprehensive history.
- Perform head-to-toe examination.
- Request relevant laboratory investigations.
- Carryout Ultrasound Scanning.
- Provide Routine Drugs: Tabs Folic Acid 5mg daily, Ferrous Sulphate 200mg 8 hourly, Multivitamin daily and Vitamin B Complex 8 hourly for 4 Weeks.
- Provide the first dose of T.d. immunization.
- Provide LLITN (if available).
- Educate and counsel the client on any topic relevant to her condition.
- Initiate discussion on birth and emergency planning.
- Give an appointment for the next visit.

2ND and 3RD CONTACT (20 - 26 WEEKS)

- Maintain a friendly atmosphere
- Take history and perform physical examination.
- Abdominal examination and fetal heart sound.
- Take action on the results of laboratory investigations.
- Check for anaemia, measure B.P., and check urine for sugar and glucose.
- Provide a second dose of T.d. immunization.
- Provide first dose of S.P. for IPT of malaria in pregnancy at 13 Weeks and repeat after 4 Weeks.
- Provide Routine Drugs (Folic acid and ferrous sulphate).
- Continue the discussion on birth and emergency planning.
- Give the next appointment.

4TH -7TH CONTACT (30 - 38 WEEKS)

- Maintain a friendly atmosphere.
- Take history and perform physical examination.
- Re-check Hb and treat for anaemia if present.
- Check urine for sugar and glucose.
- Measure B.P.
- Provide routine drugs (Folic acid and ferrous sulphate).
- Provide a second dose of S.P. for IPT of malaria
- Check the uterine size and palpate the fetal position to exclude multiple pregnancies and abnormal presentation.
- Discuss lactation and contraception.

8TH CONTACT (40 WEEKS)

- Maintain a friendly atmosphere.
- Take history and perform physical examination.
- Check the uterine size and palpate fetal lie, presentation, engagement, and Fetal Heart Sound.
- Conduct Pelvic examination
- Check Hb to exclude anaemia,
- Measure B.P.
- Provide Routine Drugs (Folic acid and ferrous sulphate).
- Determine decision on Postpartum Contraception.
- Educate and counsel the client on signs/symptoms of labour.
- Review the birth plan with the client.
- Review the Birth Plan with the client.
- If not delivered by the end of 4 weeks, advise the client to report back and take a decision on delivery.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Difficulty breathing, persistent cough, yellow eyes, pallor, abdominal pain	* Might be in pain * Rapid breathing above 30/min *Yellowness of eyes Pallor, HB < 7g/dl, * HBsAg+	Severe condition Suspect Severe Anaemia, Hepatitis B, Severe Chest Infection	1. Reassure Patient 2. Refer immediately to the appropriate health facility
II. Bleeding from the vagina before 20 weeks of gestation with or without lower abdominal pain	II. Bleeding from the vagina before 20 weeks of gestation with or without lower abdominal pain	Severe to moderate condition (suspect inevitable abortion if Os is dilated and threatened abortion when closed)	1. Assess vaginal bleeding for passage of clots or products of conception. 2. If inevitable abortion without shock and pregnancy is below 12 weeks, perform MVA and give Post abortion care (If trained) 3. If pregnancy is over 12 weeks and/ or patient is in shock, set intravenous infusion of Normal Saline and refer 4. If bleeding is not much and cervical Os is closed advise on Complete Bed Rest 5. Tab Folic acid 5mg daily for 4 weeks 6. Tab Ferrous sulphate 200mg 8 hourly for 4 weeks 7. Review in 3 days.
III. Bleeding from the Vagina after 20 weeks of gestation (exclude provocation by sex, foreign body or trauma)with or without abdominal pain	Vaginal Bleeding Low BP, high pulse rate, cold clammy skin	Severe condition Antepartum Haemorrhage (think of abruption placenta or placenta previa)	1. Do Not Do Vaginal Examination (V/E) 2. Set IV line Normal Saline 3. Refer immediately to the appropriate health facility
IV. Unconscious, convulsion/ fits (severe muscle spasm), No previous history of epilepsy	Gestational age >20 Weeks BP > 140/90mmHg Noisy breathing Protein in urine 1+ or above With or without Unconsciousness Facial contraction and mouth foaming	Severe condition Suspect Ante Partum Eclampsia	1. Position pt. in left lateral position, clear and maintain airways 2. Give Oxygen if available 3. Set up IV infusion of 500ml Dextrose-Water with cannula/needle 4. Give 50% MgSo4 loading dose (IV 4gm in 12mls of distilled water over a period of 10-15minute, then IM 5gm + 1ml of 1% Xylocaine into each buttocks) 5. Tabs Nifedipine 10 mg under tongue (sublingual) or Hydralazine 5-10 mg IV slowly over 10-15 minutes 6. If BP is_> 160/110mmHg give Tabs Nifedipine 20mg under tongue (sublingual) stat. 7. If woman is in advanced stage of labour deliver the baby, if not, Refer immediately to the appropriate health facility

V. Headache, visual disturbance, Epigastric pain, vomiting/ Nausea	* Gestational age>20weeks * Hyperreflexia * BP > 160/110mmHg * Protein 3 + in urine	Severe condition Suspect severe pre-eclampsia	1. Set up IV infusion of 500ml Dextrose-Water 2. Give 50% MgSo4 loading dose (IV 4gm in 12mls of distilled water over a period of 10-15minute, then IM 5gm + 1ml of 1% Xylocaine into each buttocks) 3. Tabs Nifedipine 20mg sublingual stat (if conscious) or Hydralazine 5-10 mg IV slowly over 10-15 minutes if unconscious Refer immediately to the appropriate health facility
VI. Pregnant woman with any two of the following present: Weight increase of 1 kg or more in a week Urine with protein of 1+ or more Pitting oedema	* Raised BP, protein of 1+ or more in urine * Pitting oedema	Severe condition Suspect toxaemia in pregnancy	1. Set up IV infusion of 500ml Dextrose-Water 2. Give 50%MgSo4 loading dose (IV 4gm in 12mls of distilled water over a period of 10-15minute, then IM 5gm + 1ml of 1% Xylocaine into each buttocks) 3. If BP exceeds 160/100, give Tabs Nifedipine 20mg under tongue (sublingual) stat (if conscious) or Hydralazine 5-10 mg IV slowly over 10-15 minutes if unconscious 4. Reduce Salt intake 5. Refer immediately to the appropriate health facility
VII. Pregnant Woman with any of the following present: substance use/abuse (alcohol, cough syrup, etc) Scars suspicious of domestic violence Prenatal anxiety/depression	Disorientation Incoherent speech Physical Scars/bruises Looks anxious or scared Looks distracted with no interest to her surroundings	Moderate/Severe condition Suspect Common Perinatal Mental Disorders (CPMD)	1. Council on implications of substance use/abuse on the fetus 2. Refer for Appropriate Care 3. Review in 3days
VIII. Headache, Puffiness of the face or, generalized body oedema IX. Pregnant Woman with any of the following present: Headache or Puffiness of face Polydipsia Polyuria Polyphagia (2nd or 3rd trimester)	* Gestational age > 20 weeks * BP > 140/90mmHg * Protein 2 + in urine * BP < 140/90mmHg * Protein – No trace * RBS - > 11.0mmol/l Or FBS > 7.0mmol/l BMI > 25kg/m	Moderate to Severe to Condition Suspect mild pre-eclampsia Severe to Moderate Condition Suspect Gestational Diabetes	1. Advice on the danger signs of imminent eclampsia 2. Tabs Nifedipine 20mgt 1. Advise on lifestyle modification; – Eat healthy diets (low carbohydrate with high fibre) – Adequate exercise etc – Reassure and refer immediately to the appropriate health facility
X. Constant Fatigue, Family History of Diabetes Prior history of Macrosomic baby (> 4kg at birth) Known diabetic Unconsciousness	* BP 140/90mmHg OR less * Protein – Present or not present * RBS - > 11.0mmol/l Or FBS > 7.0mmol/l BMI > 25kg/m	Severe to Moderate Condition Suspect Gestational Diabetes	1. Advise on lifestyle modification; – Eat healthy diets (low carbohydrate with high fibre) – Adequate exercise etc – Reassure and refer

XI Headache pedal oedema	Protein 1+ in urine, with or without pitting oedema BP not more than 140/90mmHg	Severe condition Suspect mild pre-eclampsia	1. Advice on adequate bed rest 2. Tabs Nifedipine 20mg daily for 5 days 3. Restrict salt intake 4. Review in 3 days If improving, continue on bed rest 5. If no improvement, Refer immediately to the appropriate health facility
XII. Headache, Pedal oedema, epigastric pain Known hypertensive	* No Protein in urine * BP=/> 140/90mmHg	Mild-Moderate Condition Suspect Essential Hypertension	1. Advice on adequate bed rest 2. Tabs Nifedipine 20mg daily for 5 days 3. Restrict salt intake 4. Review in 3 days, If improving, continue on bed rest If no improvement, Refer to the appropriate health facility
XIII Weakness, Dizziness, Breathlessness at rest, palpitation	* Pallor- paleness of the conjunctiva, tongue or palms of the hands etc. * Haemoglobin below 7g/dl or PCV of less than 30%	Severe condition Suspect severe anaemia in pregnancy	1. Reassure Patient 2. Refer immediately to hospital 3. Advice on hospital delivery
XIV. Leaking of watery fluid from the vagina, Not in labour	Wet pad or cloths No sign of labour, Abnormal colour and odour of liquor FHS = 160b/m and above	Severe to moderate condition (PROM)	1. Apply clean sanitary pad IM Gentamicin 80mg (immediately) stat 2. Caps Amoxicillin 500mg (immediately) stat 3. Refer immediately to the appropriate health facility
XV. Cough, Fever, sneezing, difficulty in breathing	* Cough chest not clear * Temp 37.50C or above * BP 140/90mmHg and above * Hb between > 7g/dl * mRDT = +ve	Moderate condition Malaria in Pregnancy Hypertension in Pregnancy	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Paracetamol 1gm 8hourly for 3 days 3. Tabs Artemether/Lumefantrine AL 80/480mg 1 tab at 0 hour, 8 hours later, then, 12 hourly for 3days if pregnancy is within 2nd and 3rd trimester 4. Tabs Nifedipine 20mg daily for 5 days 5. Continue routine haematinic 6. Encourage more fluid intake. 7. Nutritional counselling 8. Review in 5 days, If no improvement Refer appropriately
XVI. Pain or burning sensation on passing urine, increased frequency of micturition after 12 weeks gestation, lower abdominal pain	* Urine test protein 1+ or more With or without high temperature	Mild condition Suspect UTI	1. Encourage more fluids intake 2. Tabs Erythromycin 500mg 6hourly for 7 days 3. Tabs Metronidazole 400mg 8houly for 7days 4. Tabs Vitamin C 200mg 8hourly for 7 days 5. Tabs PCM 1g 8hourly for 3 days Review in 3 days, if no improvement Refer appropriately

XVII Vaginal discharge, painful sores/rashes. With or without Lower abdominal pain	* Excessive vaginal discharge * Abnormal colour or cloudy * Abnormal odour	Mild condition Suspect STIs	1. Treat according to National guideline in the appendix 2. Avoid drugs that are contraindicated to Pregnancy 3. Education on the need to treat partner and co-wives 4. Review in 3 days. If no improvement Refer appropriately
XVIII. Pain in the abdomen or back	* No abdominal tenderness With or without uterine contraction * Intermittent abdominal pain after 28 weeks gestation * Pain often associated with blood stained mucus discharge (show) * Watery vaginal discharge or a sudden gush of water * Cervical effacement i.e., the progressive shortening and thinning of the cervix during labour; and cervical dilatation i.e., the increase in diameter of the cervical opening measured in centimetres.	Moderate condition Suspect labour	1. Admit in labour ward 2. Monitor vital signs 3. PCV or HB estimation 4. Urinalysis (sugar, protein & ketones) 5. Allay fears of woman 6. Open Labour Care Guide if/ when cervix is 5cm or more dilated (see Appendix)
XIX. Nausea, Vomiting in the morning may or may not have seen period (menses)	* Anxious * No other findings * PT = +ve	Mild to moderate condition Suspect morning sickness	1. Advise on small frequent meals especially dry foods (energy foods) 2. Give Low Osmolality ORS 3. Advise to get up slowly from bed in the morning 4. Review in 3 days, If vomiting persists (Hyperemesis) 5. Refer If there is history of fainting, review Hb, BP, Urine for sugar 6. Ask for black stool or other bleeding 7. Nutritional counseling 8. Review weekly
XX. Heartburn	No other findings	Mild condition (suspect reflux)	1. Advice to avoid spicy and too oily food 2. Ask to take cold milk during attack 3. Tabs Mist Magnesium Trisliciate 2 8hourly for 5 days or when necessary (PRN) if condition is severe 4. Advise to stay chest-up after meals 5. Avoid fizzy drinks 6. Eat few hours before bed If symptoms persist, Refer appropriately
XXI. Inability to defecate for some days, abdominal discomfort	No other findings	Mild condition Suspect Constipation	1. Advice to take plenty of water, fruit, leafy vegetables, and fiber rich foods

XII. Visible blood vessel and leg cramps	Visible veins on the lower extremities	Mild condition Suspect Varicose Vein	1. Keep leg elevated while sitting or lying down 2. Advise on calcium rich foods (sea foods, soy beans) if in 3rd trimester
XIII. Pregnant Woman with sleeping difficulty	No other findings	Mild condition	1. Advice on sleeping less during the day. (take a nap only) 2. Avoid going to bed hungry and do not eat late. 3. Take enough fluids to aid digestion 4. Ensure the sleeping place is comfortable. 5. Avoid tight clothes, allow fresh air, take a shower, and relax before going to sleep.
XXIV. Pregnant (well), woman for registration or routine visit	Pregnant well No other finding Urine normal	Normal condition	Refer to the page on well visit (ANC) depending on the number of visit.

Health Education

1.Adequate rest (at least 8 hrs in the night and 2 hrs during the day time).
2.Nutritional counseling.
3.Personal, food and environmental hygiene.
4.Mental Health Counseling.
5.Drug compliance and avoidance of unprescribed drugs and herbal drugs.
6.Advice on maintaining good posture.
7.Discourage from lying flat on the back in late pregnancy (it reduces blood supply to the placenta).
8.Importance of hospital delivery and emergency preparedness.
9.Educate on the danger signs in pregnancy eg. Severe headache, vaginal bleeding, severe abdominal pain, convulsion etc.
10.Importance of taking home based records to the clinic or hospital at all time.
11.Care of the newborn and exclusive breast-feeding.
12.Postpartum family planning.

6.2 MATERNAL HEALTH (LABOUR AND DELIVERY)

Normal labour is characterized by the onset of strong, painful, and regular uterine contractions and retraction, resulting in the delivery of a live baby without injury to the mother or baby. Labour progresses in two phases: latent and active. The latent phase is marked by cervical effacement (0-4cm dilated), while in the active phase, cervical dilatation occurs much faster. The active phase of labour occurs when the cervix is dilated between 5 and 10cm.

Labour is divided into four stages:

First Stage: This stage begins with the onset of strong, painful, and regular uterine contractions, leading to the dilation of the cervix and the appearance of the show. It should not last for more than 12 hours.

Second Stage: It starts from full dilatation of the cervix to the expulsion of the baby and should not last for more than 1 hour.

Third Stage: This stage begins after the delivery of the baby and continues until the complete expulsion of the placenta and membranes, as well as the control of bleeding, which should not last for more than 30 minutes.

Fourth Stage: It encompasses the first hour after delivery, during which both the mother and baby are allowed to rest and are closely observed. This period is critical, as complications can occur.

HISTORY	EXAMINATION/ACTIONS
1.How many months pregnant are you? 2.How are you feeling? Do you have any concerns? 3.Do you have problems sleeping? Do you sleep well? 4.Do you get easily irritated, anxious or depressed? 5.Are you having strong contractions? When did it start? How often? Any now? 6.Is there any show (thick blood stained mucus)? 7.Have the membranes ruptured? When? Are you draining now? 8.Do you feel the baby's movement? 9.Have you had any abdominal operations before? (unbooked) 10.Did you attend the ANC Clinic? What tests were done, what information/instructions and treatments were given to you?	General appearance: in pains, distressed, anxious Temperature: check and record Pulse: check and record Respiration: check and record Blood pressure: measure and records Eyes: pallor, jaundice Mouth: pallor, dry tongue Heart: rate, rhythm, murmur Abdomen: (a) inspect- shape, scars, distention, masses, whether it moves with respiration. (b) palpate- presenting part, foetal parts, position, foetal lie (c) Contraction- frequency, duration, rhythm (at least 2 in 10 minutes each lasting for 40 — 45 secs) Foetal Heart sound: heard, rate and rhythm Limbs: oedema Vulva: protruding or ruptured membranes, liquor draining, bleeding Vagina: (a) Condition of vagina (b) Cord in vagina (c) Cervical dilation (c_ presenting part, foetal parts, position, cord prolapse Urine: protein,sugar, acetone, katone, etc Haemoglobin: estimate and record Use the Partograph (Labour Care Guide) to monitor the progress of labour once the woman is in the active phase. (sample of partograph)

MANAGEMENT OF NORMAL LABOUR

First Stage of Labour:

- Explain the condition to the client.
- Explain all procedures and seek permission for examination.
- Ensure and respect the privacy of the woman.
- Discuss findings with the client and keep her informed about the progress of labour.
- Encourage active movement alternating with rest between contractions and frequent emptying of the bladder.
- Encourage the woman to bathe or wash herself and her genitals at the onset of labour.
- Ensure infection prevention.
- Avoid vaginal examination during contractions.
- Do not shave the perineal area.
- Do not administer an enema during labour.
- Start a Labour Care Guide when labour is in the active phase.
- Allow the client to choose a more comfortable position.

Second Stage of Labour:

- Continue to monitor maternal and fetal vital signs.
- Swab the vulva with a mild antiseptic lotion.
- Discourage bearing down until the head is at the perineum (teach the woman to breathe through the mouth to prevent pushing before the head has distended to the perineum).
- Ensure controlled delivery of the head by taking the following precautions:
- Keep a hand on the head as it advances with contractions.
- Cover the anus with a perineal pad using your other hand.
- Observe the perineum for signs of tear.
- Ask the mother to breathe deeply and steadily with her mouth open and not to push during the delivery of the head.
- Feel gently around the neck for the presence of the cord; if the cord is loosely felt, deliver the baby through the loop, but when it is tight, clamp and cut it, then unwind it from the neck.
- Allow for natural restitution.
- Deliver the anterior shoulder by applying gentle pressure downward.
- Lift the baby upward toward the abdomen to deliver the posterior shoulder and the rest of the body.
- Place the baby on the mother's abdomen.
- Note the time of delivery.
- Tie/Clamp The Cord in 2 Places and Cut the Center After About 2-3 Minutes.
- Provide immediate newborn care (Refer to Newborn Section).
- Avoid the following:
- Rectal examination.
- Lithotomy position with or without stirrups.
- Fundal pressure during labour.

- Verbal abuse.

Third Stage of Labour (Use Active Management of Third Stage of Labour - AMSTL):

- Palpate the abdomen to rule out the presence of another baby.
- Administer a uterotonic IM oxytocin 20mg on the thigh within one minute after birth or IM or IV Carbetocin 100ug stat OR Tabs Misoprostol 600ug (3 tabs) under the tongue (sublingual).

Perform Controlled Cord Traction (CCT) as follows:

- Clamp the cord close to the perineum.
- Use the other hand to apply counter traction on the uterus, just above the pubic bone.
- Use a forward and downward motion to deliver the placenta.
- Maintain steady but gentle pressure.
- Cup the placenta in your hands and deliver it gently once it is visible at the perineum to ensure that the membranes are intact.
- Examine the placenta and membranes to ensure completeness.
- Clean the mother and check the perineum for laceration or bruising.

Fourth Stage of Labour:

- Massage the uterus every 15 minutes for the next 2 hours to ensure that it is well contracted.
- Continue to check vital signs.
- Ensure breastfeeding is initiated to promote bonding.
- Monitor the amount of bleeding.
- Observe for any danger signs in both mother and newborn.
- Teach the mother about perineal care.

Best Practices:

- Close monitoring during the first 6 hours.
- Monitor BP, Pulse, Vaginal bleeding, and uterine hardness; every 15 minutes for 2 hours, every 30 minutes for 1 hour, and every hour for the last 3 hours.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Labour pains	* Mal-presentation *Transverse lie * Shoulder presentation * Brow or face presentation * Breech presentation	Abnormal Condition	1. Explain the condition to client and relation 2. Set up intravenous infusion of Dextrose-Saline and Refer immediately to the appropriate health facility
II. Labour pains for more than 12 hours	* Non descent of presenting parts *Vomiting * Rapid pulse * Dehydration	Abnormal condition Prolonged labour Maternal exhaustion	1. Examine for cervical dilatation 2. Set IV line Dextrose saline 3. IM Paracetamol 600mg stat 4. Elevate the woman's leg 5. Refer immediately to the appropriate health facility, 6. Assign a skilled CHEW to accompany her if possible
III. Labour pains with cord seen outside the vagina	Cord still pulsating *Transverse lie	Abnormal condition Cord prolapse	1. Set up intravenous infusion of Dextrose-Saline 2. Give Oxygen to mother (if available) 3. Push Cord up in the vagina and pack introitus with moist gauze 4. Refer immediately to the appropriate health facility
IV. Labour pains with cord seen outside the vagina	* Cord may or may not be pulsating * Cephalic presentation * Cervix fully dilated	Abnormal condition Cord prolapse	1.If cord is pulsating: a.Do generous Episiotomy b.Push cord back up beyond the head or apply two clamps on cord and cut in-between c.Deliver the baby d.Repair episiotomy 2. If cord is not pulsating: a.Deliver the baby normally
V. Labour pains, pregnancy not up to term	* Gestational age less than 37 weeks * Labour in active phase	Abnormal condition Preterm labour	1.Explain the condition to the pt and relation 2. IV Hydrocortisone 100mg stat 3. Set up intravenous infusion of Dextrose-Saline and Refer appropriately 4. Conduct delivery only when there is no time to get to the health facility 5. Ensure baby is warm and provide immediate newborn care
Labour pains, no foetal movement felt for some time	* Cord may or may not prolapsed * Smelly or unclear liquor * FHR <120 beat or >160 beat per minutes	Abnormal condition Foetal distress	1. Explain condition to pt and relation 2. Administer Oxygen to mother 3. Set up intravenous infusion of Dextrose-Saline and Refer 4. Transport pt in left lateral position 5. Conduct delivery Only when it becomes imminent and no time to get to the health facility

VII. Labour pain, with fits or convulsion	* May or may not be Conscious * BP> 140/90mmHg * In urine	Abnormal condition Intrapartum eclampsia	1.Set up IV infusion, 500ml Dextrose-Water with cannula/ needle 2.IM 50% MgSo4 (magnesium sulphate) loading dose 5gm + 1ml of 1% Xylocaine into each buttocks 3.Tabs Nifedipine 20 mg sublingual stat immediately (if conscious) or Hydralazine 5-10 mg IV slowly over 10-15 minutes if unconscious 4. Refer immediately to the appropriate health facility
VIII. Bleeding after expulsion of placenta	* Bleeding in excess of 500mls or any sign of shock (rapid pulse, low BP, dizziness, Or unconsciousness) Uterus not firm (above umbilicus)	Abnormal condition Primary Postpartum Haemorrhage (PPH)	1.Shout for help 2.Uterine massage 3.Encourage woman to empty her bladder or catheterize if necessary 4. IM oxytocin 10iu stat (immediately)or Tabs Misoprostol 800 mcg sublingual (under the tongue) or rectal 5.Set IV Normal saline with 20iu oxytocin at 60 drops per minutes 6.If bleeding continues apply NASG (if available) 7. Resuscitate if in shock 8. Refer immediately to the appropriate health facility with the NASG in place
IX. Bleeding after 24 hours Fever	* Bleeding * Tempt: above 37.2°C	Abnormal Condition Suspect Puerperal Sepsis	1. IV Ceftriaxone 1gm Stat, then Tabs Cefixime 400mg 12hourly for 5 days 2. Tabs Metronidazole 400mg 8 hourly for 5 days 3. Tabs Vit C 200mg 8 hourly for 7 days 4. Tabs PCM 1gm 8 hourly for 3 days 5. Tabs Misoprostol 600mcg under tongue (sublingual) stat 6. Advice on Personal Hygiene 7. If no improvement Refer immediately to the appropriate health facility

X. Perineal tear with or without bleeding	* Cervical laceration present * Degree of perineal or vaginal laceration (1st, 2nd,or 3rd degree tear)	Abnormal condition	1. Clean the area, apply pressure with a perineal pad 2. Repair cervical tear 3. Repair 1st and 2nd degree perineal tear 4. If tear is extending to the anus (3rd degree), clean and apply perineal pad and the woman to put her legs together (Do Not Attempt to Repair) 5. Refer immediately to the appropriate health facility
XI. Retained placenta or placenta fragment	* Placenta products are visible through the * vaginal orifice May or may not be bleeding	Abnormal condition	1. If retained placenta and not bleeding, attempt delivery of the placenta manually. 2. If bleeding, manage as in PPH above 3. Encourage client to pass urine or catheterize 4. Leave uterus alone for 10 minutes 5. Attempt CCT, 6. if not successful, add 20 IU Oxytocin to infusion and Refer immediately to the appropriate health facility

Health Education

1.Adequate rest.
2.Vulva hygiene.
3.Nutritional counseling.
4.Exclusive breast feeding.
5.Routine immunization.
6.Importance of postnatal visit.
7.Importance of taking home-based records during visits to clinic or hospital.
8.Care of newborn.
9.Counsel on postpartum contraceptives.
10.Appropriate Mental health counseling.

6.3 POSTNATAL DISORDERS

The postnatal period is 6-8 weeks after delivery. This is the time that lactation is well established and the return of the uterus and other reproductive organs to the pre-gravid state. At least 3 postnatal visits for checkup should be encouraged to determine how the mother and baby are faring. This is also a special opportunity to screen postnatal mothers for signs of mental health conditions such as; puerperal psychosis, involvement in substance use/abuse and signs of domestic violence, for appropriate management and referral.

HISTORY	EXAMINATION/ACTIONS
1.How are you feeling? Is there any problem? 2.How many days or weeks since the delivery? 3.Is the baby being exclusively breast fed? Any feeding problems? 4.Do you observe any changes in his/her skin or umbilicus? 5.How is your appetite? 6.Is there any pain or discomfort? 7.Do you sleep well? Is there a sleeping challenge? 8.Do you get easily irritated, anxious or depressed? 9.How long do you want to rest before having your next baby? 10.What is the nature of lochia?	General appearance: tired or pale check skin for signs of domestic violence. Temperature: Check and record Blood pressure: measure and record Breast: -tenderness, redness, lumps, engorgement Nipples: soreness, cracks, Abdomen: size and firmness of uterus, Legs - any local inflammation Pelvis: (internal examination not done before six weeks postpartum Condition of perineum Cervix Os Uterus involution Discharge, Bleeding Vulva: Nature of lochia, condition of repaired episiotomy/ tear (if any) Weight: weigh and record Haemoglobin: estimate and record Urine: protein, sugar and record Examine the child: Refer to appropriate section)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Severe headache convulsion/fits at home or on the way to clinic delivery few days ago	* BP 140/90mmHg or above * Protein in urine	Severe condition suspect postpartum eclampsia)	1. Resuscitate if patient is unconscious 2. Set up intravenous infusion of 500ml of Dextrose- Water 50% MgSo4 loading dose (IV 4gm in 12mls of distilled water over a period of 10-15minute, then IM 5gm + 1ml of 1% Xylocaine into each buttocks) 3. Tabs Nifedipine 20mg stat under tongue (sublingual) 4. Refer immediately to the appropriate health facility

II. Continuous vaginal bleeding	* Bleeding not profuse (Frank bleeding) * Primary PPH within 24 hours or Secondary PPH after 24 hours	Moderate condition	1.Check for cervical tear or erosion 2.Apply clean pad to perineum 3.IM oxytocin 20iu stat (immediately) or tabs misoprostol 600mcg stat sublingual (immediately under the tongue) 4.Set intravenous infusion of Dextrose-Saline 5. Refer immediately to the appropriate health facility if bleeding is profuse
III. Intermittent vaginal bleeding, fever, vaginal discharge with or without smell	Discharge with or without smell Temp. 37.5C or above BP 90/50 or above Spotting	Moderate - severe condition	1. IV Ceftriazone 1gm Stat 2. Caps Amoxicillin 500mg 8hourly for 5 days 3. Tabs Metronidazole 400mg 8hourly for 5 days 4. Tabs Ferrous sulphate 200mg 8hourly for 5 Weeks 5. Tabs Paracetamol 1gm 8hourly for 3 days 6. Review daily If no improvement after 3 days, Refer immediately to the appropriate health facility
IV. Vaginal discharge, not smelly. no fever	* Discharge present * Not smelly * Temp, less than 37.5C	Mild condition suspect RTI	1. Tabs Erythromycin 500mg 6hourly for 7 days 2. Tabs Metronidazole 400mg 8hourly for 14 days 3. Tabs Vit C 200mg 8hourly for 7 days 4. Use 4cs 5. Review in 3 days If improvement continue treatment 6. Counsel to bring male partner for treatment 7. If no improvement, Refer immediately to the appropriate health facility
V. Excessive tiredness or fatigue with one or more of the following: Dyspnea on exertion Palpitation Facial or pedal oedema Pallor	B.P>140/90mmHg or <90/50mmHg	Severe to moderate condition Suspect heart conditions or Severe Anaemia	1.Reassure Patient 2. Refer to hospital
VI. Breast Engorgement	* Breast full and painful * Temperature > 37,5 °C	Mild condition	1.Teach proper positioning of the baby on breast and ensure good attachment 2. Encourage exclusive breastfeeding 3.Advice on the use of firm Brassier 4. 5.Warm compress Tabs Paracetamol 1000mg 8hourly for 3 days

VII. Sore or cracked nipples	* Pain on nipples while breast-feeding	Mild condition	1. Observe the mother breastfeed 2.3. Give appropriate health education Tabs Paracetamol 1000mg 8hourly for 3 days 4. If sores are extensive, Tabs Erythromycin 500mg 6hourly for 5 days Tabs Vit C 200mg 8hourly for 7 days 5. Teach how to express milk and feed with cup and spoon 6. Review in 3 days, If no improvement Refer appropriately
VIII. Breast abscess/inflam-mation	*Tenderness * Pus formation	Mild condition	1. Warm compress 2. Tab Paracetamol 1000g 8hourly for 3 days 3. Tabs Erythromycin 500mg 6hourly for 5 days 4. Tabs Vit C 200mg 8hourly for 7 days 5. If fully ripe do an I & D 6. Dress daily 7. Encourage more fluid 8. If no improvement Refer appropriately
IX. Vaginal wall comes down with coughing	*Visible vaginal protrusion	Mild condition Suspect Vaginal prolapse	1. Explain cause to woman 2. Teach postnatal pelvic floor exercise, and advice against lifting heavy objects 3. check in 1 month, if no improvement, Refer appro-priately.
X. Uterus still large or tender after 8th day and up to 2l days	* Tenderness * Uterine size still above the umbilicus	Mild condition Suspect Sepsis	1. Advice restricted activity 2. Caps Amoxicillin 500mg 8hourly for 7 days 3. Tabs Paracetamol 1000g 8hourly for 3 days 4. Tabs Metronidazole 400mg 8hourly for 7 days 5. Tabs Vit C 200mg 8hourly for 7 days 6. Review in 3 days, if improved, continue treatment, If no improvement Refer appropriately.
XI. Headache, Fever with or without nausea and vomiting, Joint pains, Cough	* BP < 140/90mmHg * Temp > 37.5°C * Hb>7g/dl * Lochia not smelly or profuse * mRDT =+ve	Mild condition Malaria	1. Tab Paracetamol 1000g 8hourly for 3 days 2. Tab Artemether-lumefantrine 80/480mg 1 tab at 0 hour, 8 hours later, then 12 hourly for 3 days Or 3. Tabs Artesunate+ Amodiaquine 100 mg AS/270 mg AQ base daily for 3 days 4. Encourage to take plenty of fluid Review in 5 days, If no improvement Refer appro-priately

XII. Sleeping difficulty	No other findings	Mild condition	1. Advice on good nap and being active during the day 2. Avoid going to bed hungry and do not eat late. 3. Find time to rest between baby care to avoid stress. 4. Ensure the baby is well fed, dry napkin, no heat and comfortable for sound sleep, to allow the mother to sleep well. 5. Take enough fluids to aid digestion and breastmilk formation 6. Ensure the sleeping place is comfortable. 7. Avoid tight clothes, allow fresh air, take a shower and relax before going to sleep.
XIII. Sleeping difficulty with any of the present: serious substance use/abuse (alcohol, cough syrup, etc) Scars of confirmed domestic violence Puerperal Psychosis	Suspect Common Perinatal Mental Disorders (CPMD)	Moderate/Severe condition	1. Advice on avoidance of sub-stance use/abuse 2. Refer for Appropriate Care 3. Review in 3days
XIV Fever after child birth	Temperature 38°C or more mRDT = -ve	Moderate condition	1. Lower temperature with fan or tepid sponging or antipyretic 2. Adequate hydration by mouth or IV 3. Encourage bed rest 4. Monitor vital signs and urine output 5. If blood transfusion is needed refer 6. Administer therapeutic antibiotics

Health Education

1. Prompt return for postnatal visit.
2. Adequate nutrition and plenty of fluids.
3. Immunization (both).
4. Importance of Exclusive breastfeeding for six months, & the introduction of appropriate complementary foods at 6 months.
5. Personal hygiene (wash breast with water only before feeding the baby)/pelvic exercise.
6. Educate on proper position and good attachment to the breast.
7. Importance of using firm brassier.
8. Danger signs in mother and the newborn.
9. Counsel on and refer for cervical cancer screening.
10. Counsel and give services for contraception.
11. Importance of taking home-based records during visit to the clinic or hospital.
12. Appropriate mental health counselling.

6.4 FAMILY PLANNING

Family Planning is the process of helping clients to have the number of children they want and when they want them or avoid being pregnant. Some women have difficulty taking pills properly and also sometimes have problems caused by the pills. Some of these problems are mild to severe. The health worker should be able to follow up and take necessary action. This is also a special opportunity to screen mothers for signs of mental health conditions such as; anxiety/depression, involvement in substance use/abuse and signs of domestic violence, for appropriate management and referral.

HISTORY	EXAMINATION/ACTIONS
1.Have you used any contraceptive method before? 2.What method were you using? Which Clinic did you attend? Why do you want to attend this particular clinic? 3.Who referred you to this Clinic? 4.How many children have you had? How many are alive? How many dead? When did you have your last baby? Did you have a normal vaginal delivery or were you assisted (caesarean section, forceps delivery or vacuum extraction). 5.Are you breastfeeding? 6.Are you menstruating now? When did you start? If not, when was the last menstrual period? 7.Do you normally have; a.Heavy periods with cramping pain or clots? b.Scanty period? 8.How often do you bleed? How long do you bleed? 9.Have you had any operations? When? Where? 10.Do you have any pains in the lower part of your abdomen with fever and discharge? Do you have severe headaches? 11.Are you on any special drugs or receiving treatment anywhere else? 12.Do you have any medical condition such as diabetic, sickle cell, disease, cancer, kidney issues etc? 13.Do you sleep well? Is there a sleeping challenge? 14.Do you get easily irritated, anxious or depressed? Social History 1.Any history of Alcohol Use? 2.Any history of Substance Use/Abuse? 3.Any history of Domestic violence? 4.Any history of Mental Problem?	General appearance: anxious thin, obese, ill, Skin: scars (signs of domestic violence) Weight: weigh and record Blood pressure: measure and record Hair: hair loss especially in temporal region Face: swelling Eyes: pallor or jaundice Neck: goitre, lymph nodes Chest: shape, respiratory difficulty or rapid creps or wheeze Heart: rate, rhythm, murmur Breast: lump, dimpling of the skin, abnormal appearance of the nipples, retraction of the nipples. Abdomen: scars, tenderness, masses Legs: varicose veins, oedema. Thigh: excoriation, ringworm lesions, warts. Pelvis: Bimanual palpation: position and size of uterus, tenderness Speculum: condition of the cervix, cervical polyps, lesions, vaginal infection, menstrual bleeding or signs of pregnancy. Haemoglobin: estimate and record Urine; protein, sugar

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. First visit for Child Spacing	Well (may or may not be pregnant)	Normal condition	1. If Pregnant, advise to book for ANC and come for child spacing after delivery 2. If not Pregnant, counsel on various family planning methods using the counselling tools 3. Use MEC wheel for any client with a known medical condition to determine the appropriate method
II. Intrauterine Device (IUD) thread not felt, suspected pregnancy, severe lower abdominal pain	* Intrauterine Device (IUD) thread not seen * Abdominal tenderness * No bleeding * PT = +ve	Severe condition Suspect pregnancy	1. Tab Paracetamol 1000mg immediately (stat) 2. Conduct a speculum examination, if tip of IUD stem is visible, use forceps to slowly draw it out 3. If not visible, Refer immediately to the appropriate health facility

III. Intrauterine Device (IUD) in place with sign and symptoms of pregnancy. With or without bleeding	* (IUD) in place * Pregnancy * Bleeding	Moderate to Severe condition	1. Tabs Multivitamin 8 hourly for 2 weeks 2. Tabs Ferrous sulphate 200mg 8 hourly for 2weeks 3. Tabs Folic acid 5mg daily for 2weeks 4. Refer immediately appropriate health facility
IV. Heavy menstrual bleeding for 10days or more, Intra (IUD in place with or without cramping	Bleeding with IUD in place	Moderate condition	1. Caps Multivatamin 1capsule 8 hourly for 14 days 2. Tab Ferrous sulphate 1 tablet 8 hourly for 14 days 3. Tab Folic acid 5mg daily for 14 days 4..Explain findings to client 5. Administer Tranexamic acid 1500mg 8 hourly for 3 days Then 1000mg once daily for 2 days beginning when heavy flow stat 6. Tab Ibuprofen 400mg 12 hourly after meal for 5days, If no improvement Refer immediately appropriate health facility
V. IUD thread not felt	IUD thread seen	Mild condition	1. Explain finding to client 2. Ask client to re-examine herself and feel the identified string or replace the IUD with another without trimming its string
VI. Back pain during or between menstrual period with IUD in place	Pain IUD in place	Mild condition	1. Tab Paracetamol 1000mg 8 hourly for 3days 2. Review during the next menstrual period
VII. Requested for IUD insertion Menstruating with any of the following: Foul smelling discharge Vaginal itching Heavy bleeding last 7 days or more with severe cramping History of D & C, or caesarian section in the last 6 months	* Signs of infection in the uterus, * Pallor * Heavy bleeding with or without fever (>37.5°C)	Moderate Condition Suspect RTIs	1. Do not insert IUD 2. Advise on use of alternative method based on Medical Eligibility Criteria (MEC) 3. Caps Doxycycline 100mg 12 hourly for 14days 4. Tabs Metronidazole 400mg 8 hourly for 14days 5. Tabs Ciprofloxacin 500mg immediately (stat) 6. Review in 7days If there is no improvement, Refer appropriately
VIII. Bleeding or spotting with any of the following after sexual intercourse, lower abdominal pain, fever, IUD in place	* warm to touch * Temp 37.5°C or above * Bleeding or spotting * mRDT = +ve or –ve	Severe condition Suspect PID Malaria	1. If mRDT =+ve, treat for malaria as appropriate 2. Caps Doxycycline 100mg 12 hourly for 14days 3. Tabs Metronidazole 400mg 8 hourly for 14days 4. Tabs Ciprofloxacin 500mg immediately (stat) 5. Review in 7days If there is no improvement, Refer to the appropriate health facility

IX. Lower abdominal pains during or after insertion IUD Increase amount of bleeding	* Fainting during or after insertion, * pains, * apid pulse or respiration, fatigue	Severe Condition Suspect Perforation	1. Stop insertion 2. If IUD is already inserted, remove 3. Place client in a horizontal position and observe for an hour 4. Monitor vital signs 5. Check for intra-abdominal bleeding 6. If no sign of bleeding, observe for 1hour and counsel on abstinence from sex for two weeks 7. Help her choose another method If vital signs are getting down, refer to a higher facility.
X. Irregular, heavy or prolonged bleeding continuous after several months of normal bleeding of long after IUD was inserted	Vaginal examination shows no signs of malignancy Abnormal discharge from the vagina PT = +ve or -ve	Severe Condition Suspect PID	If PT is +ve send for USS to rule out pregnancy If PT is –ve do not remove IUD. Treat for PID as in action VIII above
XI. Pain, redness, swelling or pus at the site of sub-dermal implant insertion	*Tenderness * Redness *Visible pus Tip of the implant may or may not be visible	Moderate to Severe Condition	1. If not infected and the tip of the implant is visible, remove and replace a new implant on the other hand on the client's request 2. If there are signs of infection, do not remove the implant 3. Clean the infected area with antiseptic 4. Make an incision to drain the pus (if abscess has formed) 5. Caps Amoxicillin 500mg 8hourly for 7days 6. Tabs Metronidazole 400mg 8hourly for 7 days 7. Tabs Vitamin C 200mg 8hourly for 7 days 8. Tabs Ibuprofen 400mg 12hourly for 5 days (After food) 9. If no improvement Refer appropriately
XII. Irregular or heavy vaginal bleeding after initiation of implant	* No underlying medical condition * Implant in place	Mild to Moderate Condition	1. Reassure the client that bleeding changes are common in women using implant, but they gradually subside with time 2. If the client finds the bleeding unacceptable, offer one cycle of low dose combined oral contra-ceptive (based on MEC)
XIII. Severe lower abdominal pain on implant Abnormal vagina bleeding or no monthly bleeding	* Signs of pregnancy *Abdominal tenderness * Fainting	Severe Condition Suspect Ectopic Pregnancy or Ovarian Cyst	1. Leave implant in place 2. Reassure client 3. Refer for pelvic USS 4. If ovarian cyst or ectopic pregnancy, Refer immediately to the appropriate health facility

XIV. Headache, with or without aura (blurred vision , temporary loss of vision or seeing flashy light) on implant	* BP > or < 149/90mmHg * PT = -ve * No underlying medical condition	Mild to Moderate condition	1. If headache without any other symptoms 2. Tabs PCM 1g 8 hourly for 3 days Or 3. Tabs Ibuprofen 400mg 12 hourly for 5 days (After food) 4. If headache with other symptoms after implant insertion 5. Reassure client 6. Remove the implant 7. Advise other non-hormonal method 8. Review after 3 days, If no improvement Refer appropriately
XV. Backache with signs of pregnancy on oral contraceptive	Uterus not enlarged Abdominal tenderness	Severe condition	1. Tab Paracetamol 1000mg stat(immediately) 2. Take client off oral contraceptive 3. Refer immediately immediately to the appropriate health facility
XVI. Menstruation with any of the following: Visible blood vessel History of jaundice or blood clots Breast abnormalities Breastfeeding less than 6 months	* Varicose veins seen on legs * Jaundice * Blood clots * Lump in breast * inverted nipples * cracked nipples * BP 140/90 or above	Severe condition	1. Advise on elevation of feet 2. Reduction of period of standing 3. Proper position and attachment of baby to breast 4. If on hormonal contraceptives. Stop use. 5. Use a non-hormonal method as a backup. 6. Refer immediately to the appropriate health facility
XVII. On oral Contraceptive with mild, moderate or sudden severe headache	* No sign of pregnancy * Normal BP	Moderate condition	1. If headache is sudden and severe, take client off oral contraceptive immediately 2. Use back up method 3. Tab Paracetamol 1000mg stat 4. Review in 2 days If no improvement, Refer appropriately
XVIII. On oral contraceptive with any of the following; Scanty menstrual bleeding Brownish discharge or spotting No signs of pregnancy	* No sign of pregnancy * Blood spot seen	Moderate condition	1. Counsel client 2. Reassure 3. Review during next menstrual period, if no improvement, take off oral contraceptive 4. Counsel on barrier methods (condom and diaphragm)
XIX. On oral Contraceptive with heavy bleeding 7 -10 days	* Bleeding * BP normal	Moderate to severe condition	1. Tabs Multivitamin 8 hourly for 2weeks 2. Tabs Ferrous sulphate 200mg 8 hourly for 2weeks 3. Tabs Folic acid 5mg daily for 2weeks 4. Explain findings to client 5. Administer Tranexamic acid 1500mg 8 hourly for 3 days Then 1000mg once daily for 2 days beginning when heavy flow stat 6. Tabs Ibuprofen 400mg 12hourly after meal for 5days, If no improvement Refer immediately to the appropriate health facility

XX. On oral Contraceptive with heavy bleeding 7 -10 day	* Bleeding * Systolic BP lower than 120mmHg and diastolic lower than 70mmHg	Severe condition Suspect hypo tension	Set up intravenous infusion of Normal Saline and Refer immediately to the hospital
XXI. On oral contraceptives, Persistent irregular bleeding bleeds after intercourse	No abnormality	Mild to Moderate Condition	1. Change pills 2. If heavy bleeding persists 3. Tabs Mefanamic Acid 100mg 12hourly for 5 days 4. If bleeding persists or become heavy review contraceptive method 5. Refer appropriately
XXII. On oral contraceptive, with or without complaint	BP 140/90mmHg or above	Mild-Moderate to severe Condition	Stop pills Change to barrier method If BP is not more than 140/90mmHg, Review in 2 weeks If BP is still raised or higher, Refer immediately to the appropriate health facility
XXIII. On oral contraceptive and bleeds after intercourse	No abnormalities	Mild condition	1. Reassure client 2. Advise on proper position during intercourse 3. Refer for approrpiate care. review in 2 weeks
XXIV. Delayed Pregnancy with Regular menstrual period stopped depo provera injection 15 months ago and anxious to get pregnant	No obvious abnormalities or infection	Mild condition	1. Explain to client that condition could be the side effect of Depo Provera 2. Instruct her to have intercourse during fertile period 3. Advice to remain lying down 1-2 hours after intercourse 4. Invite partner for counselling and investigation 5. Advise to keep a record of menstrual period 6. Review in 3 months
XXV. Irregular menstrual bleeding, spotting or scanty bleeding injectable contraceptive	* No sign of infection * PT = -ve	Mild Condition	1. Re-assure 2. Explain condition to client 3. If still worried discuss about other methods
XXVI. Weight gain. On Injectable contraceptives	* PT = -ve * No underlying medical conditions	Mild Condition	1. Re-assure 2. Explain condition to client 3. Advice on lifestyle modification (eat healthy and exercise)
XXVII. Headaches, hot flashes, mood changes and vaginal irritation	* No underlying medical condition * No signs of STIs	Mild to Moderate Condition	1. Reassure 2. Tabs Ibuprofen 400mg 12hourly for 3 days (After food) 3. Review in 3 days. If no improvement, discuss on other methods

XXVIII. Menstrual irregularity	* No evidence of other abnormalities * No infection	Mild condition	1. Advise to have her keep record of menstrual period for 3 months 2. Review in 3 months If irregularity continues, Refer appropriately
XXIX. On hormonal contraceptives, No menstruation for the past 3 months	* No symptoms * No signs of Pregnancy	Mild condition	1. Re-assure 2. Explain condition to client 3. If still worried do a pregnancy test
XXX. On oral contraceptives, Nausea and or vomiting	* Pregnancy test * Signs of pregnancy	Mild condition	1. Review contraceptive method If test is positive , counsel and provide Antenatal care
XXXI. On oral contraceptives. Nausea and or vomiting	* No signs of pregnancy * Pregnancy test	Mild condition	1. If pregnancy test negative, change to lower dose pills (if available) 2. If nausea persists, REFER appropriately
XXXII. On oral contraceptives, Excessive weight gain after starting pills, Increased appetite, Weight gain cyclical	Increased weight	Mild condition	1. Encourage diet control 2. Advise regular physical exercise 3. Change to lower dose pill 4. If weight gain persists, change to other method
XXXIII. On oral contraceptives. Depression or mood changes after starting pills, No associated social problems	* Depressed Unhappy	Moderate condition	1.2 Stop pills Change to barrier non hormonal method 3. If depression persists, refer appropriately
XXXIV. On oral contraceptives, Breast tenderness or fullness after starting pills, No signs of pregnancy	* Tender full breast, no lump * Pregnancy test negative	Mild condition	1. Advise client to wear firm, supportive brassiere 2. Give Tab Paracetamol 1000mg tds for 3 days 3. If discomfort persists, change to lower dose combined contraceptive, progestin only pills or barrier method
XXXV. On oral contraceptives, Acne (Pimples) following use of pills	No other abnormality	Mild Condition	1. Change to lower dose oestrogen pills 2. If acne persists or become more severe, stop pill change to barrier method and refer to the hospital
XXXVI. Sleeping difficulty	No other findings	Mild condition	Refer to Postnatal Disorders
XXXVII. Sleeping difficulty with any of the following Noticeable anxiety serious substance use/abuse (alcohol, cough syrup, etc) Scars of confirmed domestic violence	pect Anxiety/Psychosis/Depression	Moderate/Severe condition	Refer to Postnatal Disorders

Danger Signs

1 Severe Abdominal Pain
2 Severe Chest Pain
3 Severe Headache
4 Eye Problems (Blurred Vision, loss of vision, flashes of light)
5 Severe Leg Problems (swelling, pain in calf or thigh).
6 Restlessness and visible signs of anxiety
7 Severe back pain
8 Heavy/prolonged bleeding

Health Education

1 Healthy timing and spacing of pregnancy (HTSP).
2 Personal, food and environmental hygiene.
3 Nutritional counselling.
4 Compliance to drugs.
5 Adhere to treatment.
6 Appropriate Mental Health Counselling.
7 Importance of taking home-based records to the clinic or hospital.

6.5 INFERTILITY

Infertility is the inability of a couple to conceive after one year of unprotected regular intercourse. Male infertility may be caused by any of the following; deficiencies in sperm production, blockage of the sperm delivery system, injury to the testicle, disorders of hormone production, poor descent of one or both testes. Sometimes women also have difficulty getting pregnant, or after having one or more children. This may be due to infections, poor knowledge of the fertile period, or irregular menstrual period, mental health conditions such as anxiety, depression involvement in substance use/abuse and domestic violence can also affect a couple's ability to become pregnant. However, the Community health practitioner should be able to advise appropriately and take necessary action. This is also a special opportunity to screen mothers for signs of mental health conditions such as; anxiety/depression, involvement in substance use/abuse and signs of domestic violence, for appropriate management and referral.

HISTORY	EXAMINATION/ACTIONS
1. Menstrual History: When was her last menstrual period (LMP) 2. Gynaecological History: Any signs of infection, vaginal discharge, lower abdominal pain, pain or bleeding on intercourse? How was the infection treated? Any operation D & C. Caesarian section. Abortion? Where and when was it treated? Any problems after it bleeding, discharge, fever, pain. 3. Obstetric History: Has she ever been pregnant? With this man? Any problems after the pregnancy - bleeding, discharge fever, lower abdominal pain? What happened to the pregnancy? How long ago was it? 4. Contraception History: What method of birth control has she used? When did she stop? Any problem with the method - bleeding, discharge, fever, lower abdominal pain, no menstrual period? 5. Sexual History: How often do they have intercourse? Does her husband have other wives? Does her husband have children? Does her husband live with her? How long have they been trying? Any sexual problem – pain on intercourse? Sexual history is important in her husband. 6. Medical History: How old is she? How does she feel now? Has she come to the clinic or seen the doctor in the last 6 months? For Husband; Has the husband had mumps before? Any history of injury to the genitals? Any undescended testicles? 7. Social History Any history of alcohol use?	General appearance: tired or pale check skin for signs of domestic violence. General Inspection: evidence of sexual maturity (breast development, womanly shape, female hair distribution), obstacles to sexual intercourse (obesity, physical abnormalities) or general poor physical health or small vaginal opening. Abdomen: scars, masses. Speculum examination: position of the cervix, signs of infection, erosions, lacerations. Bimanual examination: The presence of a normal uterus, the size of the uterus, masses, tenderness of the uterus or adnexa. Blood pressure: measure and record. Temperature: - check and record Hemoglobin: estimate and record. Urine: protein, sugar.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Inability to impregnate a woman	Undescended or very small testis	Severe condition	Refer immediately to the appropriate health facility
II. Unable to get pregnant, May or may not have commenced menstruation	* Sexual immaturity * Too young	Mild condition	1. Explain that fertility might come with age (if she is under 16 years) 2. Refer to appropriate if the women is 18 years or more and has never menstruated

III. Unable to get pregnant, May or may not be menstruating, above 18 years	*Young woman No abnormality	Mild condition Suspect infertility	1. If she has commenced menstruation, instruct her to have intercourse during her fertile period 2. Advise on sexual position 3. Have her remain lying down for 1-2 hours after intercourse 4. Invite husband for counselling and investigation 5. Review in 3 months, if no menses Refer to the appropriate health facility
IV. Unable to get pregnant	* Obese * Anxious * Emaciated	Mild condition Suspect infertility	1. Nutrition counselling 2. Have her to keep dates of menstrual period 3. Advise on fertile period 4. Review in 3 months, If no improvement, Refer to the appropriate health facility
V. Inability to impregnate a woman	*Adult male * No abnormality detected	Mild condition	1. Counsel client on best time for intercourse. 2. Frequency of intercourse. 3. Advice to take wife to the clinic. 4. Review in 3 months. If no improvement, Refer to the appropriate health facility
VI. Couple unable to conceive for over a year of regular sex with the following; Painful urination Vaginal/Urethral discharge	Vaginal/Urethral discharge	Moderate to Severe Condition Suspect Infertility secondary to STIs	1. Reassure client 2. Treat according to Syndromic Management of STIs 3. If no improvement, Refer immediately to the appropriate health facility

Danger Signs

1Severe abdominal pain.
2Severe chest pain.
3Severe headache.
4Restlessness and visible signs of anxiety.
5Eye problems (Blurred Vision, loss of vision, flashes of light.
6Severe leg problems (Swelling, pain in calf or thigh).

Health Education

1. Maintain a record of menstruation.
2. Advocate against early marriage.
3. Provide nutritional counseling.
4. Stress the importance of maintaining home-based records during visits to the clinic or hospital.
5. Educate on the fertile period and the importance of intercourse within that timeframe.
6. Advise against the use of herbal medication.
7. Offer appropriate mental health counseling.

6.6 PRE-MENOPAUSE

Pre-menopause is a natural process of transitioning to the menopausal stage when the woman's period stops as a result of advancement in age marking the end of reproductive years. It is marked with changes in the menstrual cycle, along with other physical and emotional symptoms. The symptoms are caused by changes in the hormonal levels in the body. The pre-menopausal period can last between 2-10 years before menopause set in.

HISTORY	EXAMINATION/ACTIONS
1. Personal History: Age? When did symptoms start? 2. Menstrual History: When was your last menstrual period (LMP), how often do you menstruate 3. Gynaecological History: Any signs of infection, vaginal discharge, lower abdominal pain, pain or bleeding on intercourse 4. Obstetric History: Have you ever been pregnant? Any problems after the pregnancy - bleeding, discharge, fever, lower abdominal pain 5. Contraception History: What method of birth control have you used? For how long? When did you stop? Any problem with the method - bleeding, discharge, fever, lower abdominal pain, no menstrual period? 6. Sexual History: How often do you have intercourse? Any sexual problem – pain on intercourse, decrease libid 7. Medical History: How regular is your period How do you feel now? Do you have any known medical condition e.g. Diabetes, Hypertension, Liver or Kidney Problem? 8. Social History: Any history of alcohol use? a) Any history of substance use/abuse? b) Any history of domestic violence?	1. General appearance: anxious thin, obese, ill, 2. Skin: scars (signs of domestic violence) 3. Weight: weigh and record 4. Blood pressure: measure and record 5. Hair: hair loss especially in temporal region 6. Face: swelling 7. Eyes: pallor or jaundice 8. Neck: goitre, lymph nodes 9. Chest: shape, respiratory difficulty or rapid creps or wheeze 10. Heart: rate, rhythm, murmur 11. Breast: lump, dimpling of the skin, abnormal appearance of the nipples, retraction of the nipples. 12. Abdomen: scars, tenderness, masses 13. Legs: varicose veins, oedema. 14. Thigh: excoriation, ringworm lesions, warts. 15. Pelvis: a. Bimanual palpation: position and size of uterus, tenderness b. Speculum: condition of the cervix, cervical polyps, lesions, vaginal infection, menstrual bleeding or signs of pregnancy. 16. Haemoglobin: estimate and record 17. Urine; protein, sugar

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. A woman over 40 years with Sleep problems with two or more of the following; - Poor sleep - Hot flashes - Weight gain - Headache - Hair loss - Body itching - Anxiety - Mood changes	* No underlying medical condition * No signs of Infection * No skin lesions	Mild to Moderate Condition Suspect Premenopausal Syndrome	1. Reassure 2. Nutritional counselling 3. Encourage simple exercise
II. A woman over 40 years with any of the following; - Low libido - Painful intercourse - Not pregnant or breast feeding	No signs of Infection PT negative Pelvic examination –Normal	Mild to Moderate Condition Suspect Premenopausal Syndrome	1. Reassure 2. Nutritional counselling 3. Encourage simple exercise 4. Advice on use lubricants like K-Y Jelly for Painful Intercourse

III. A woman over 40 years who develop Irregular menstrual periods with heavy bleeding	No signs of Infection PT negative Pelvic examination -Normal	Mild to Moderate Condition Suspect Premenopausal Syndrome	1. Reassure 2.Nutritional counselling 3.Tabs Ibuprofen 400mg 12hourly for 3 days (After food) 4. Review in 3 days, if no improvement refer to the appropriate health facility.

Health Education

1.Have adequate rest.
2.Identify and avoid triggers of hot flashes.
3.Nutritional counseling.
4.Take a lot of fluid.
5.Avoid using substance in the vagina and maintain vulval hygiene.
6.Appropriate Mental Health Counselling.

SECTION SEVEN
ELDERLY HEALTH (60 YEARS & ABOVE)

7.0 GERIATRIC HEALTH

The World Health Organization (WHO) describes the elderly as persons aged 60 years and above. Thus, with an increasing number of elderly persons (65 years and above) in the population, the PHC worker will be confronted with a growing number of elderly patients. This group of patients is vulnerable and at risk of health, nutritional, and social problems. They have spent most of their lives contributing to nation-building, either in the public or private sector when they were younger, so they deserve the best care from the country they served. There are differences in the signs and symptoms of diseases presented in the elderly compared to younger people. Therefore, diagnosis in the elderly may require a more precise history and laboratory tests. Due to changes associated with the ageing process, such as difficulty in hearing, speech problems, memory loss, and so on, it may be necessary for a family member to be involved in their care. An understanding of the needs of the elderly is essential for PHC workers.

7.1. MUSCULO - SKELETAL SYSTEM DISORDERS

The musculoskeletal system is the collective term for the interconnected system of muscles, bones, joints, ligaments, tendons, and other connective tissues in the body. It provides structural support, allows movement, protects internal organs, and plays a vital role in maintaining postures. The muscles, controlled by the nervous system to enable various bodily functions, such as walking, lifting, and other forms of physical activity. This system is essential for overall body function and mobility.

As adults age, they begin to lose minerals and living materials from their bones. The bones become weaker and even more brittle. This physiological process makes the elderly suffer from more broken bones. Osteoarthritis is a disease that occurs with ageing and in this condition normal joint movement becomes painful and difficult.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. When did it begin? 3. Has it happened before? 4. Was there any accident or injury? 5. If injury happened, have you been able to move the affected part since? 6. Is there any pain, swelling and numbness? 7. Do you have joint pain or stiffness? 8. Do you have back pain? 9. Do you have fever? 10. When did the muscular pain start? Was it sudden or gradual? 11. Was there any accident or injury? 12. Have you been able to move the affected part since the injury happened? 13. Can you describe the specific location(s) of the pain? Is it in the joints, back or leg? 14. Does it radiate to other areas? 15. Is there any tenderness, swelling, stiffness and numbness? 16. How would you describe the pain? Is it sharp, dull, aching, throbbing, or stabbing? 17. Have you experienced similar muscular pain episodes in the past? If so, how were they managed? 18. Do you have any preexisting medical conditions such as arthritis, diabetes, or cardiovascular disease that may contribute to muscular pain? 19. Are you currently taking any medications, including over-the-counter or herbs	1. General appearance: Look out for distress 2. Temperature: Check and record 3. Limbs: Check for deformity, swelling on the affected part, localized severe tenderness/pain, inability to move part. 4. General appearance: Look out for distress 5. Temperature: Check and record 6. Blood pressure: Check and record 7. Check for deformity, swelling on the affected part, localized severe tenderness/pain, inability to move part 8. Evaluate the affected area for swelling, tenderness, or deformities. 9. Access range of motion and muscle strength 10. Check for signs of inflammation and other abnormalities

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Severe pain on limb on falling even on a flat surface	• General appearance: ill-looking and/or in pain • Temperature: 37.5°C or above • Limbs: deformity or swelling on the affected part or localized tenderness, inability to move part of the limbs.	Severe condition Suspect a fracture or osteoporosis	1.Tab. Paracetamol 1g (2 tablets) stat or Non-Steroidal Anti-inflammatory e.g Tab. Ibuprofen 400mg stat (After food). 2. Methyl salicylate ointment apply twice daily topically on the affected area. 3. Apply splint and firm bandage on affected limb. 4. Review in 7 days to the appropriate health facility.
II. Bone pains: Stiffness and cramps of extremities, Pains in the joints aggravated by use and relieved by rest	• General appearance Ill-looking and in pain • Joints: swelling and tenderness movement of the joint	Moderate to severe condition Suspect osteoarthritis	1.Tab. Paracetamol 1g 8hourly 3days or Non-Steroidal Anti-inflammatory Tab. Ibuprofen 400mg 8hourly 3days (after food). 2. Methyl salicylate ointment. Apply twice daily topically. 3.Tab.Vitamin C (Ascorbic Acid) 200mg 8hourly for 7 days 4.Advise on weight reduction (if necessary) 5. Keep joints warm through moderate exercise, such as walking and usage of walking sticks for support 6. Review in 5 days, if no improvement refer to the appropriate health facility.
III.General body pains No fever No headache	• General appearance not in any pain • Limbs: No swelling, no localized tenderness	Mild condition Suspect fatigue	1.Tab. Paracetamol 1g 8hourly for 3days. 2. Methyl salicylate ointment apply twice daily topically on the affected area. 3.Tab. Multivitamins 1 daily for 2 weeks. 4.Adequate rest moderate exercise, adequate food and fluid intake. 5. Small frequent meals, fruits and vegetables 6. Improve water intake

Health Education

1.Explain common causes of muscular pain in aging individuals, such as arthritis, muscle strains, and changes in joint flexibility.
2.Emphasize the importance of maintaining good posture and incorporating gentle exercises to improve muscle strength and flexibility.
3.Educate on the appropriate use of over-the-counter pain relievers and potential side effects. Stress the importance of consulting healthcare providers before starting any new medications.
4.Encourage warm-up exercises before physical activities and proper cool-down routines to prevent muscle strains.
5.Highlight the role of hydration and a balanced diet in supporting overall muscle health.
6.Discuss the impact of excess weight on joints and muscles, emphasizing the benefits of maintaining a healthy weight.
7.Regular, low-impact exercises like walking, swimming, or taichi to enhance muscle strength and joint flexibility.
8.Teach the importance of pacing activities to avoid overexertion, especially for those with chronic conditions.
9.Provide tips on maintaining proper ergonomics at home and work to prevent muscle strain.
10.Emphasize the role of adequate rest and quality sleep in promoting muscle recovery.
11.Encourage seniors to consult healthcare professionals for personalized advice and treatment plans tailored to their specific needs.
12.Discuss non-pharmacological pain management techniques such as heat or cold therapy, massage, and relaxation exercises

7.2. CARDIOVASCULAR SYSTEM DISORDERS

Cardiovascular diseases are conditions that affect the heart and blood vessels. When the heart and blood vessels are healthy, blood flows smoothly throughout the body. However, sometimes the circulatory system fails to work properly, and blood may not flow correctly through the arteries and veins, leading to serious health problems.

Some people are born with cardiovascular problems, but most cardiovascular diseases take years to develop. This is why there is an increased number of cardiovascular diseases in the elderly. Conditions such as high blood pressure, arteriosclerosis, atherosclerosis, heart attack, heart failure, and stroke are very common in this age group. Death from cardiovascular diseases can often be prevented if the disease is detected early and referred immediately.

CARDIOVASCULAR SYSTEM DISORDERS is a general term used to describe a disease of the heart and blood vessels. It occurs when there is a blockage or interruption of the heart muscles that supplies blood to the heart due to a buildup of fatty substances inside the artery that leads to hardening or narrowing of the artery known as atherosclerosis Heart disease is a major cause of disability limiting the activity and eroding the quality of life of millions of older people. People 65 years and above are much more likely to suffer a heart attack, stroke coronary heart disease and heart failure due to high blood pressure.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long have you had it? 3.Do you have cough, chest pain, chest tightness or shortness of breath? 4.If cough is present, do you bring out sputum? What is the color? 5.Can you sleep flat or do you need several pillows to sleep? 6.Do you sometimes have to get up at night struggling to breathe? 7.Have you ever lost consciousness? 8.Any sudden fall? 9.Do you have difficulty in moving any part of the body? 10.Do have any history of high blood pressure or diabetes? 11.Are you on any medication? 12.Do you suffer from or have a family history hypertension or diabetes? 13.Are there factors that trigger or alleviate these symptoms? 14.Do you experience symptoms like dizziness, nausea or swelling? 15.Do you smoke or drink alcohol? 16.Have you suffered from any cardiovascular disease before? 17. Do you experience trouble sleeping at night?	1.General Appearance: look out for respiratory distress. 2.Temperature: check and record 3.CVS; a.Pulse: check for rates rhythm, volume b.Blood pressure: check and record 4. Heart sound: check for heart sounds, murmur. 5. Chest: check for respiratory rate, signs of dyspnea. 6.Abdomen: check for distension, tenderness, enlarged liver. 7. Central Nervous System (CNS): check for level of consciousness and paralysis. 8.Lower limbs: check for pedal swelling/oedema.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1a. History of loss of consciousness. b.History of sudden fall. 3.Sudden inability to move any side of the body. c. Loss of speech. d. Slight deviation of the mouth or face	1.General appearance: ill-looking or uncomfortable 2.CVS: Pulse: fast, irregular. Blood Pressure: high (above 140/90mm/hg) CNS: partial or total loss of consciousness, partial or complete paralysis (one side). Loss of speech (if paralysis is on the right side); Memory may or may not be affected.	Severe condition Suspect stroke	1. Give first aid management for unconsciousness by maintaining free airways. 2. Treat any sustained injury. 3. Monitor blood pressure intermittently. 4. Then refer immediately to the appropriate health facility..

1a.Headaches b.Oedema c. Breathlessness at rest or on mild exertion e,g. after walking. d.Awareness of tachycardia e.Inability to lie down on the bed Coughing Easy tiredness	1.General appearance: Ill-looking or uncomfortable 2. CVS:\ Pulse: fast, irregular BP: above 140/90mm/hg 3.Chest: difficult or fast breathing 4.CNS: changes in level of consciousness. Lower limbs: pitting oedema	Severe condition Suspect hypertension with heart failure	Refer immediately
V1.a.Headaches b.Oedema c. Breathlessness at rest or on mild exertion e,g. after walking. d.Awareness of tachycardia e.Inability to lie down on the bed f.Coughing Eas ily tiredness	1.General appearance: Ill-looking or uncomfortable 2. CVS:\ Pulse: fast, irregular BP: above 140/90mm/hg 3.Chest: difficult or fast breathing 4.CNS: changes in level of consciousness. Lower limbs: pitting oedema	Severe condition Suspect hypertension with heart failure	Refer immediately
IIIa) Difficulty in falling asleep b) Waking up after very few hours of sleep c) Associated Social problems d) History of high Blood Pressure	General appearance: pale anxious and not happy CVS: BP 140/90mmHg	Mild Condition Suspect hypertension	Drugs that can be used in management of cardiovascular system problem in PHC settings; 1.Antihypertensives: Used to control high blood pressure. Examples include ACE inhibitors (e.g., lisinopril), beta-blockers (e.g. metoprolol), and diuretics (e.g. hydrochlorothiazide). 2.Antiplatelet Agents: Help prevent blood clots. Aspirin and clopidogrel are examples. 3. Statins: Used to lower cholesterol levels and reduce the risk of cardiovascular events. Atorvastatin and simvastatin are commonly prescribed. 4. Beta-blockers: Besides hypertension, they may be used for conditions like angina and heart failure. Examples include metoprolol and carvedilol. 5.Angiotensin II Receptor Blockers (ARBs): Similar to ACE inhibitors, they help manage blood pressure and may be prescribed for heart failure. Losartan is an example. 6. Calcium Channel Blockers: Used for hypertension and certain heart conditions. Amlodipine and diltiazem are examples. 7.Antiarrhythmics: Prescribed for irregular heart rhythms. Amiodarone and flecainide are examples. 8.Counseling on sleeping habits, drug usage, food intake and participation in community work 9.Occasional Tranquilizer e.g. Tab. Diazepam 5mg at night for5 days Tab. Ferrous gluconate 200mg 8hourly for 2wks. Tab. Multivite 1 daily for 2wks 10. Review after 5 days, if no improvement, refer.

Health Education

1.Provide simplified information about the cardiovascular system with emphasis on the importance of heart health and blood circulation.
2.Explain common risk factors such as high blood pressure, high cholesterol, diabetes, and lifestyle choices
3.Discuss how these factors contribute to cardiovascular issues.
4.Educate on the importance of adhering to prescribed medications, understanding dosage schedules, and potential side effects.
5.Encourage regular check-ups to monitor medication effectiveness.
6.Emphasize a heart-healthy diet with reduced sodium, saturated fats, and cholesterol.
7.Encourage the consumption of fruits, vegetables, whole grains, and lean proteins.
8.Provide practical tips for meal planning.
9.Encourage low-impact activities like walking or swimming
10.Stress the importance of regular physical activity for cardiovascular health.
11.Discuss stress-reducing techniques such as mindfulness, deep breathing exercises, or gentle yoga. Chronic stress can exacerbate cardiovascular issues.
12.Emphasize the detrimental effects of smoking on cardiovascular health.
13.Educate and support for quitting smoking.
14.Provide guidance on achieving and maintaining a healthy weight through a balanced diet and regular exercise.
15.Teach individuals to monitor vital signs regularly, such as blood pressure and heart rate. Encourage keeping a health journal to track changes and report them to healthcare providers.
16.Emphasize the importance of a supportive social network.
Encourage family involvement in understanding and promoting cardiovascular health

7.3 GENITO- URINARY DISORDERS

Genitourinary problems are common in the elderly and can be due to metabolic, ageing changes, infections, abnormal growths, etc. Examples include: diabetes, vaginal/uterine prolapsed, enlarged prostate, cancers, schistosomiasis etc. Patient may complain of discharge, pain and. burning sensation on passing urine (micturition), blood in urine, frequent micturition, and urinary retention (difficulty in passing urine), etc.
Genitourinary problems refer to conditions that affect the genital and urinary organs or the structures associated with them. These problems can involve various parts of the urinary and reproductive systems, including the kidneys, bladder, ureters, urethra, prostate (in males), testes, ovaries, and other related structures. Some common genitourinary problems include urinary tract infections (UTIs), kidney stones, incontinence, erectile dysfunction, prostate issues (such as prostatitis or benign prostatic hyperplasia), sexually transmitted infections (STIs), cystitis, urinary retention, and reproductive system disorders (like infertility or menstrual disorders in women).

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long have you had it? 3.Do you urinate very often? How often? Night and day? 4.Do you have difficulty in passing urine or pain/burning sensation when passing urine)? 5.Do you have feeling of incomplete passage of urine? 6.Are you able to control urine? 7.Do you drink more than usual? 8.Do you have pain in the flank? 9.Is there blood in your urine? 10.Do you have fever? 11.Do you have a history of wadding in streams in childhood (schistosomiasis)? 12.Have you had this problem before? 13.Have you had any form of treatment? If yes, which? 14.Do you have any vaginal discharge or itching (if male, any discharge from the penis?) 15.Any history of weight loss?	1.General appearance: look out for discomfort 2.Temperature: check and record 3. Pulse: check and record 4.Respiration: check and record 5. Blood pressure: measure and record 6. Eyes: check for pallor 7.Abdomen: check for tenderness, swelling or masses 8. Flanks: check for tenderness 9. Limbs: check a. Legs for swelling b. Palms and nails for pallor c. varicose veins 10. Genitals: check for sores, pus or discharge 11. Urine: check for a. Color b. Protein c. Sugar

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Sudden inability to pass urine	• General appearance: in distress, ill-looking • CVS: - Pulse: fast or normal - Blood pressure: above 140/90 mmHg Abdomen: swelling/ distension, lower abdominal tenderness Flanks: tenderness	Severe condition Suspect acute renal failure	REFER immediately.
II. Severe lower abdominal pain Painful urination with or without urethral discharge. There may or may not be Fever.	• General appearance: in painful distress • Temp.: 37.5°C and Above Abdomen: Lower abdominal tenderness • Genitals: normal or urethral discharge • Urine: clear or cloudy (presence of protein)	Severe condition Suspect urinary tract infection (UTI)	1.Tab. Ciprofloxacin 500mg 12hourly for 10 days 2.Tab. Nitrofurantoin 50-100 mg 6hourly with food for 5-7 days. 3.Tab. Trimethoprim/sulfamethoxazole (TMP-SMX): 960mg 12hourly mg for 3 days to 14 days. 4. If no improvement, refer appropriately

III. Blood in the Urine	General appearance: normal or ill-looking and pale Abdomen: hard lower abdominal mass schistosomiasis which may or may not be tender discomfort Temp.: 37.5°C and above Urine: blood in urine or terminal haematuria.	Severe condition Suspect bladder cancer or schistosomiasis	1. For bladder cancer, refer. 2. For schistosomiasis, give Tab. Praziquantel 1200mg(1.2g) stat then 600mg daily for5 days
VI. (Inability to control) Urine incontinence	• General appearance: smell of urine with wet pants and/ or clothing	Moderate condition Suspect UTI or trichomoniasis)	1.If severe, refer immediately 2. Tab. Ciprofloxacin 500mg 12hourly for 10 days Or IM. Ceftriaxone 1g daily for 5days 3. Tab. Metronidazole 400mg 8hourly for 5days 4. Counsel patient and relatives and explain that the changes are due to ageing 5. Encourage to avoid constipation 6.Personal hygiene 7. Reduce quantity of fluid intake after 6pm. 8. Encourage to wear adult diaper if necessary
V. Painful urination	General appearance normal or in pain Abdomen no tenderness Urine: normal or cloudy	Moderate condition Suspect UTI or trichomoniasis)	1. Tab. Ciprofloxacin 500mg 12hourly for 10 days Or IM. ceftriaxone 1g daily for 5days 2. Tab. Metronidazole 400mg 8hourly for 5days 3. Ask to return in 1 week If suspect trichomoniasis, give Tab. Metronidazole 2gm orally in a single dose. If no improvement, refer to the appropriate health facility.
VI. Difficulty in passing urine in the male; has to wait a while before urine begins to come out, may or may not have pain and .blood in urine	• General appearance: normal or anxious • Abdomen: lower abdominal tenderness • Urine: presence or absence of blood	Moderate to severe condition Suspect enlarged prostate	1. Tabs Paracetamol 1g 8hourly for 5 days 2. Tabs Co-trimoxazole 960mg 12 hourly for 5days 3.Encourage fluid intake 4. REFER and follow up

VII. Passing a lot of urine. Getting up more than 4 times at night to pass large quantities of urine but without pain. Feels very thirsty and drinks a lot of water/fluid, loss of weight	General appearance: ill- looking. weak or normal Urine: presence of sugar.	Severe condition Suspect Diabetes mellitus	1. Explain the condition fully to the patient and how he is responsible for about 75% of his own treatment 2. Given dietary advice and menu guideline 3. Give diet sheet and map out food menu 4. Do a fasting or random Blood Sugar 5. Refer – and arrange follow up 6. Check for fasting blood sugar(FBS) and Random blood sugar to confirm diagnosis 7. Give Tab. Metformin 500mg 12hourly for 5days 8. Give Tab. Glibenclamide 5mg daily for 5days 9.Monitor sugar level regularly If symptoms persists refer immediately to the appropriate health facility.

Health Education

1.Advise on compliance with advice, drugs and diet.
2.Encourage regular medical check up.
3.Encourage client to take care of feet.
4.Encourage client to avoid and report any form of injury that results from loss of sensation.
5.Encourage personal hygiene stressing the importance of good hygiene practices for genital and urinary health.
6.Emphasize on the importance of taking home-based records anytime he/she goes to the clinic or hospital.
7.Educate client about preventive measures, such as practicing safe sex, staying hydrated, and maintaining a healthy lifestyle.
8.Explain warning signs or symptoms that might indicate a worsening condition and when to seek immediate medical attention.

7.4 GASTROINTESTINAL DISORDERS

Gastrointestinal problems associated with the elderly can be due to infections, ulceration, malignant changes which may result in conditions such as, constipation, cholera, diarrhea, gastritis, peptic ulcer, cancer of the colon, stomach or of the oesophagus.

Gastrointestinal (GI) problems refer to a wide range of conditions that affect the digestive system, which includes the oesophagus, stomach, small and large intestines, liver, gallbladder, and pancreas. These issues can involve various symptoms and affect the normal functioning of the digestive tract, for example, Gastroesophageal Reflux Disease (GERD), Peptic Ulcers, Irritable Bowel Syndrome (IBS), Inflammatory Bowel Disease (IBD), Gastroenteritis, Gallstones etc.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. For how long? 3. Do have nausea or vomiting? If vomiting, how many times? 4. Do you have constipation? When was your last stool? 5. Do you have diarrhoea? If yes, how many times do you pass stool in a day? 6. Any blood in your vomit or stool? What colour is your stool? 7. Does food make the problem better or worse or no difference? 8. Is there abdominal pain? Where is it located? Is it continuous or intermittent? 9. Have you passed any worms in stool? 10. Do you have fever? 11. Have you taken any medications? What are they?	1. General appearance: look out for discomfort. 2. Eyes: check for pallor and jaundice. 3. Temperature: check and record. 4. Blood pressure: measure and record 5. Abdomen: check for a. Swelling b. Tenderness 6. Hernia: check for umbilical, inguinal and scrotal hernia 7. Anus: check for prolapse, fissure and redness 8. Haemoglobin: estimate and record 9. Abdominal Examination: a. Inspection: Observe the abdomen for any distension, scars, or visible abnormalities. b. Auscultation: Listen for bowel sounds in all quadrants using a stethoscope. c. Palpation: Gently palpate the abdomen for tenderness, masses, organ enlargement, or areas of discomfort. Assess for guarding or rebound tenderness. d. Liver and Spleen: Assess for enlargement or tenderness of these organs. e. Percussion: Perform abdominal percussion to assess for tympany or dullness.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Inability to pass stool, abdominal fullness and discomfort, flatulence.	• General appearance: restlessness. • Abdomen: distended, hard, generalized tenderness.	Moderate condition Suspect constipation	1. Encourage plenty fluids intake. 2. Eat fruits and vegetables 3. Give Liquid Paraffin 10ml as necessary for one week. 4. De-worm at least once or (twice) yearly (Albendazole 200mg stat) 5. If no Improvement, refer appropriately

II. Epigastric pain worsened by eating, Epigastric pain worsened by hunger/ disturbs sleep and relieved by eating, with or without blood In stool or black coloured stools	• General appearance: distressed Eyes: pale • Abdomen; epigastric tenderness • Stool: blood stained or black coloured	Moderate to severe condition Suspect Peptic Ulcer Disease (PUD)	1. Encourage eating small meals regularly (every three hours). 2. Drink milk for temporary relief 3. Give Susp. Mist. Magnesium Trisilicate 10ml 8hourly for one week 4. Tab. Metronidazole 400mg 8hourly for 5 days 5. Give Cap. Amoxicillin 500mg -1000gm 8hourly for5days 4. Give Tab Folic Acid 15mg daily for two weeks 5. If no improvement after 3 days, refer. 6. Tab. Omeprazole: 20-40 mg once daily before a meal. 7. Tab. Pantoprazole20-40 mg daily 8. Ranitidine 75-150 mg twice daily.
III. Frequent passing of stool (diarrhea) with or without blood and abdominal cramp/ pain	General appearance: weak and restless Eye: pale and sunken Abdomen: pain and cramps. Stool: Passing of loose/ watery stool for more than 4 times.	Moderate to severe Condition Suspect Diarrhea/ Gastroenteritis	1. Cap Loperamide 4mg after the first loose stool, then 2 mg after each subsequent loose stool, up to a maximum of 16 mg per day. 2. Tab. Metronidazole 400mg 8hourly for 5 days 3. Tab. Hyoscine butyl bromide10-20mg 12hourly for 3 days 4. In severe cases, give IM. Ceftriaxone 1gm stat and continue the above treatment 5. Encourage fluid intake to replenish lost electrolyte and water 6. Encourage adequate diet 7. Give Oral rehydration solution (ORS) regularly 8. If symptoms persist, refer to the appropriate health facility.

Health Education

1.Emphasize the importance of staying hydrated by drinking plenty of fluids, such as water, oral rehydration solutions, to prevent dehydration.

2.Provide guidance on a bland diet that is easy to digest, including foods like bananas, rice, applesauce, and toast (BRAT diet). Avoid spicy, fatty, and dairy-rich foods.

3.Explain the proper use of antidiarrheal medications and when to seek medical advice before using them, especially in cases of infectious diarrhea.

4.Stress the importance of thorough hand washing after using the bathroom to prevent the spread of infections that can cause diarrhea.

5.Help the individual identify potential triggers for their diarrhea, such as specific foods or stress, and explore ways to manage or avoid these triggers.

6.Educate on signs of dehydration, persistent diarrhea, or symptoms such as fever, blood in stool, or severe abdominal pain, which may require prompt medical attention.

7.Advise against consuming substances that can exacerbate diarrhea, such as caffeine and alcohol.

8.Discuss preventive measures, such as practicing good food hygiene, avoiding contaminated water, and being cautious when traveling to areas with a higher risk of gastrointestinal infections.

9.Encourage the individual to keep track of symptoms, diet, and fluid intake to provide useful information during medical consultations.

7.5 SPECIFIC PROBLEMS OF THE ELDERLY

The elderly from time to time have some problems as a result of ageing, such as difficulty in hearing and seeing. They may also have cough as a result of heart or respiratory problems. There are also problems of irregular meals because of their lack of strength to prepare their meals. This makes some of them malnourished. Diabetes is common among the elderly.

HISTORY	EXAMINATION/ACTIONS
1.Confirm the patient's name? 2.Where do you live? 3.What is the problem? 4.How long have you had the problem? 5.Is there pain in the eye/ear? 6.Did you (or anyone) put anything into the eye/ ear? If yes, what? 7.Do you have any chronic disease such as high blood pressure or diabetes? 8.Is there any family history of blindness, deafness, high blood pressure or diabetes? 9.Do you observe ants around your urine? 10.Do you eat food regularly? Any change in appetite? 11.Do you have fever? 12.Means of transportation? 13.Are you on any medication?	1.General appearance: look out for distress. 2. Temperature: check and record. 3. Blood pressure: measure and record. 4. Eyes: check for; a. Pallor. b. White patch in both pupils (cataract). c. Visual acuity using Snellen's chart. 5. Ear: (otoscopy) check for wax. 6. Chest: check for difficulty in breathing. 7. Haemoglobin estimation: check and record. 8. Mouth - Signs of cancer - Fungal infection - Bleeding gum - Dentures - Swollen tongue - Broken Teeth - Bad Breadth 9. Neck: enlarged thyroid 10. Skin: wrinkle, dry skin. 11. Evaluate Hearing: whisper two to three random words, face out of patient's view.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Difficultly in reading and seeing near objects	• General appearance: normal • Eyes: cannot see near objects	Moderate condition Suspect presbyopia (difficulty in seeing near objects)	1.Counsel on process of ageing 2.Give Tab Multivite 8hourly for 2weeks 3.Give Tabs Yeast 8hourly for 2weeks 4. Encourage adequate in-take of foods like green and yellow/leafy vegetables, fruits, fish, and dairy products. 5.Advise to read with good light and in larger prints. 6.REFER to the appropriate health facility.
II. Difficultly in seeing far objects	• General appearance: Normal • Eye: Normal or white patch in the black of the eye; decreased vision with Snellen'schart:	Moderate condition(suspect cataract)	1. Encourage routine screening Refer
III. Bumping into objects	• General appearance: Normal • Eye: Normal	Severe condition	Refer Immediately

IV. Difficulty in hearing	• General appearance: normal • Ear: hearing with some wax in the ear.	Moderate to severe condition Suspect partial to total ear blockage.	1.Apply a drop of warm olive oil to ears for about 3-4 days to soften wax and clean with cotton bud. 2.Review medication for potential toxicity . If no improvement Refer to the appropriate health facility.

Health Education

1.Encourage use of LLINs.
2.Encourage consumption of fluid, fruits and vegetables.
3.Discourage use of tobacco and avoid exposure to second hand tobacco smoke.
4.Discourage exposure to smoke from household cooking stove, lamp, fire wood etc.
5.Advise on the importance of taking home-based records anytime patient visits the clinic or hospital.
6.Educate on risk of nicotine and pulmonary injury.

7.6 MENTAL DISORDERS

Mental health problems may not be easily detected in the elderly. Symptoms that may indicate a mental health disorder in younger patients can have different causes in older adults. Hopelessness, sadness, and crying episodes may indicate depression, while irritability can be an affective symptom of depression. General anxiety is the most common mental disorder encountered by the elderly (Stefannacci, 2022).

The risk of developing mental health disorders increases with age. The WHO estimates that mental health problems affect over 20% of the elderly population. Despite this high prevalence, mental health problems are under-identified among the elderly. The most common mental health disorders in this age group are dementia and depression. Others include anxiety disorders, substance use problems, and self-harm. This group is also vulnerable to neglect and maltreatment by family members, which can contribute to the development or worsening of mental health challenges. Loneliness and social isolation are key risk factors for mental health problems, and one in six elderly individuals experiences abuse, often by their own carers (Stev, 2022).

HISTORY	EXAMINATION/ACTIONS
1.Confirm patient's name 2.What is the problem? 3. For how long has this problem been there? 4.Has it happened before? If yes, when? 5.Is there persistent tiredness? 6.Is there persistent sadness or anxiety? 7.Is there loss of interest or pleasure in activities that used to be interesting and enjoyable? 8.Is there loss of appetite, nausea? 9.Is there headache and/or fever? 10.Is there difficulty in carrying out usual work or daily task? 11.Are there any problems with sleeping or waking up? 12.Is there frequent complaint of multiple symptoms with no obvious physical cause? 13.Is there decline or problems with remembering things (memory loss) and orientation (loss of awareness of time, place and person)? 14.Is there loss of emotional control (easily upset, irritable or crying)? 15.Is there any family history of similar problems? 16.Do you usually drink to calm your nerves? 17.Does having drink help when you feel lonely?	1. General appearance: look out for state of personal hygiene, irritability, smell of alcohol or tobacco, signs of injury. 2. Temperature: check and record. Rule out hypothermia 3. Blood pressure: measure and record. 4. Eyes: check for pallor. 5. CNS: check for a. Ability to remember (memory) b. Awareness of time, place and person (orientation) 6. Neck; Flexibility 7. Chest and Back ; lungs irregularity 8. Breast (both male and female) 9. Pelvis

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Difficulty in falling asleep or waking up after very few hours of sleep, weakness, fatigue,sadness, anxiety and irritability. Low interest in activities that used to be interesting, difficulty in carrying out usual activities. Poor appetite.	• General appearance: unkempt/untidy, ill -looking, anxious and unhappy, weak, irritable, smell of alcohol or tobacco •Temperature: normal or high • Eye: pale • Blood pressure: 140/90mmhg or above	Mild, moderate condition to severe Suspect depression	1.Counsel on sleeping habits, drug usage, food intake and participation in community work 2. REFER to the appropriate health facility.
II. Complaints by relatives of decline or problem with memory (severe forgetfulness), loss of orientation (awareness of time, place and person), loss of emotional control (easily upset, irritable or tearful), loss of interest in the environment. Difficulties in carrying out usual vvork), vvandering about	• General appearance: unkempt appearing uninterested or irritable • CNS: cannot remember important events and dates , not oriented in time, place and person,	Severe condition Suspect dementia	1. Counsel patients and relatives. 2. REFER to the appropriate health facility.

III. Increased consumption of alcohol and use of drugs such as pain killers, sleeping tablets, coffee, tobacco leaves, cigarettes and snuff. Poor sleep, tiredness, headaches. Poor appetite, nausea, vomiting, indigestion	• General appearance:unkempt, untidy, weak, agitated, smell of alcohol or tobacco, unsteady gait, may have some drugs in the pocket of their shirt and or trousers, purse or bag, skin infection, injuries, etc. • CNS: Nervous fidgeting, slurred speech.	Moderate to severe condition Suspect alcohol or drug use disorders	1.Counsel patient and his family friends 2.Encourage participation in social activities within the family and the community. 3. Refer 4. Follow up on return from referral 5. Patient and family should be asked to bring all medication including O.T.P drugs periodically.
a)Not happy b)Loneliness c)Isolated Dejected	• Not cheerful • Weeps during the interview • Look dirty with sores on the body • Poor Oral Hygiene with dirty nails, dirty clothes.	Moderate condition	1. Counsel family of patient 2. Encourage him to participate in religious and community activities. 3. Encourage family involvement in the therapy 4. Encourage ownership and use of smart phone 5. Refer to the appropriate health facility.

Health Education

1.Counsel family members to give both physical and emotional support to the elderly.
2.Encourage good nutrition.
3.Encourage the elderly to participate in religious and community activities and encourage visiting friends and associates.
4.Encourage family members to show love, affection and provide for the general needs of the elderly.
5.Discourage use of alcohol, tobacco and unprescribed drugs.

7.7 ORAL DISORDERS

Oral health is a state of being free from chronic mouth and facial pain, oral and throat cancer, oral sores, birth defects such as cleft lip and palate, periodontal (gum) disease, tooth decay and tooth loss, and other diseases and disorders that affect the oral cavity. The most common complaint is pain, which may or may not be accompanied by swelling inside the mouth or on the face. The condition of the mouth also reflects the client's general health, so it should be inspected as part of a physical examination.

Oral health problems are common in the elderly. About 68% of the elderly have gum diseases and tooth loss, and untreated dental issues are found in 96% of the elderly. Early recognition of oral problems with prompt attention or referral will reduce morbidity and mortality arising from otherwise simple conditions. Healthy teeth and gums make it easy to eat well.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. Where is the problem? 3. Is there pain in the tooth or gum? 4. When was it noticed? 5. Are you able to eat your regular diet? 6. Does the pain increase with hot or cold drink or food? 7. Does the pain keep you awake at night? 8. Is there bleeding from gum or any other part of the mouth? 9. What do you use for cleaning your mouth? 10. How do you clean your teeth? How many times a day? 11. Do you notice any sore in the mouth?	1. General appearance: ill-looking, malnourished, facial swelling. 2 Mouth a. Gum: look out for sores, swellings, discharging sinuses, bleeding and white patches b. Tongue, look out for sore, coating, redness, dryness. c. Lips: look out for cracks. 3. Teeth: look out for cavities, discoloration, shaky or missing teeth, and plaque 4. Temperature: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Severe pain in the mouth; with or without facial swelling a) Pain disturbs patient's sleep and aggravated by chewing b) May or may not be associated with headache	• General appearance, ill looking and in pain, malnourished swollen face. • Mouth: Bad oral hygiene, foul smell, blood stain/ bleeding teeth, hole, discoloration.	Severe condition Suspect dental abscess	1. Tab. Paracetamol 1g 8hourly for 3 days with food, 2. Tab. Vit B complex 1 tablet 8hourly for 2 weeks 3. Tab. Vitamin C 100mg –200mg 8hourly for 2 weeks 4. Use warm saline wash as mouth rinse 5. Refer
II. Continuous dull pain in the mouth, Bleeding in the mouth	• General appearance: normal Mouth: • Teeth: shaking tooth or tender/painful denture • Gum: sores and/or bleeding	Moderate condition Suspect chronic periodontitis	1. Tab. Paracetamol 1g 8hourly for 3 days 2. Tab. Vitamin B complex 1 tablet 8hourly for 2 weeks 3. Tab. Vitamin c 100-200 mg8hourly for 2 weeks 4. Use warm saline wash as mouth rinse, if no improvement Refer
III. Injury to the mouth, tongue and/or face, In-ability to chew properly	• General appearance: traumatized face/mouth • Mouth: Jaw: broken Lip: laceration or bruises Teeth: broken/missing/ shaky tooth/teeth Tongue: laceration	Moderate to severe condition Suspect fracture of the teeth/jaw bone	1. Tabs Ibuprofen 400mg 8hourly for 5 days (after meals). 2. Tabs Vitamin C 100mg 8hourly for 2 weeks 3. Use warm saline water as mouth rinse REFER to a dental clinic.

Health Education

1. Explain process of ageing.
2. Brush twice daily (morning and last thing before going to bed at night).
3. Encourage regular intake of food rich in calcium.
4. Avoid intake of sugary foods (refined sugar) in-between meals.
5. Encourage to rinse mouth with water after meals.
6. Encourage good personal and oral hygiene.
7. Report any tooth/teeth pains as soon as possible.
8. Advise on regular dental checkup at least twice a year.
9. Discourage the use of tooth pick.

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APPENDICES

Measurement of Blood Pressure	Additional Points
Accurate diagnosis and treatment of blood pressure is hinged on accurate blood pressure measurement. This, in turn, is dependent on the use of accurate and validated BP measurement devices and the right patient technique. Proper patient preparation for blood pressure measurement includes: The patient should avoid tea/coffee, exercise, and tobacco use for at least 30 min before measurement. The patient should have rested for at least 5 minutes before measurement. Ensure s/he has emptied his/her bladder. When taking blood pressure, the patient should be seated, with the arm supported at heart level, back supported, and both feet resting on the ground. The patient should not speak while blood pressure is being measured. The ideal cuff bladder length and width are ≥80 percent and ≥40 percent of the patient's arm circumference, respectively. Ensure that the appropriate cuff size is selected for the individual who will receive the BP measurement. Arm circumference (cm) — Recommended cuff size (width x length in cm) 22-26 — 12x22 (small) 27-34 — 16x30 (medium) 35-44 — 16x36 (large) 45-52 — 45 53 (extra-large) Validated and automated/digital BP monitors are preferred for use to measure patients' blood pressure. Where this is not possible, a calibrated manual sphygmomanometer with a stethoscope can be used. **Diagnosis of hypertension** Hypertension is diagnosed if the systolic blood pressure is ≥ 140 mmHg and/or diastolic blood pressure is ≥ 90 mmHg on two different days within a week but not exceeding a month. However, if the blood pressure reading is very high (≥ 160 mmHg systolic or 100 mmHg diastolic), a second reading should be taken on the same day to establish the diagnosis at least 4-6 hours apart from the first reading. At each visit, two blood pressure measurements are taken, with at least 3-5 mins rest between both readings and the average is calculated and recorded as the BP reading for the patient.	Procedure for BP measurement: 1. Explain the procedure to the patient. 2. If the patient is wearing a thick-sleeved garment, ask the patient to uncover the arm by either removing the top garment or rolling up the sleeve. 3.If the cuff is tied over the clothing, ensure that it is tied snugly and evenly. 4. The patient's arm should be supported on the table and relaxed. 5. Make sure the arm cuff is properly deflated (air is out) before placing it around the patient's upper arm. 6. Wrap the cuff comfortably or snugly above the elbow, about 2 cm (or 2 finger breadths) from the elbow and secure the Velcro tape. 6. Keep the cuff at the heart level during measurement. 7. Taking a blood pressure measurement using a digital blood pressure device 8. Turn the monitor on from the on/off button. 9. Ask the individual to stay still, breathing normally, without talking or laughing. 10. Press the START/STOP button to begin the measurement. 11. The BP cuff will automatically start to inflate. This might be uncomfortable for some. Reassure the patient that the discomfort is temporary. The cuff will slowly deflate to take the measurement. When the reading is complete, the systolic (SBP) and diastolic blood pressure (DBP) readings and pulse rate will appear on the screen. Note the BP readings exactly as seen in the device in the patient record. Please do not round off. If the monitor/machine does not record the reading, re-position the cuff and try again after 1-2 minutes and repeat the above-mentioned steps. Turn the machine off and remove the cuff. Using a manual instrument when a digital, automated device is unavailable The cuff has a bulb at one end of the tube that you will squeeze to inflate the cuff. Apply the cuff on the individual's arm and place your stethoscope where you will be able to hear the sounds of blood flowing. The place is where you can feel the pulse. You can find the place by placing your index and middle fingers of one hand in the crease of a relaxed elbow. Put your stethoscope on and hold the head of the stethoscope in place. Screw the valve tight and inflate the cuff to 200 mmHg. Keep holding the stethoscope at the right spot. Slowly let the air out of the cuff while listening for the sounds of blood flowing. Keep a rate of deflation 2-3 mmHg per second (equals one line on the dial every second).

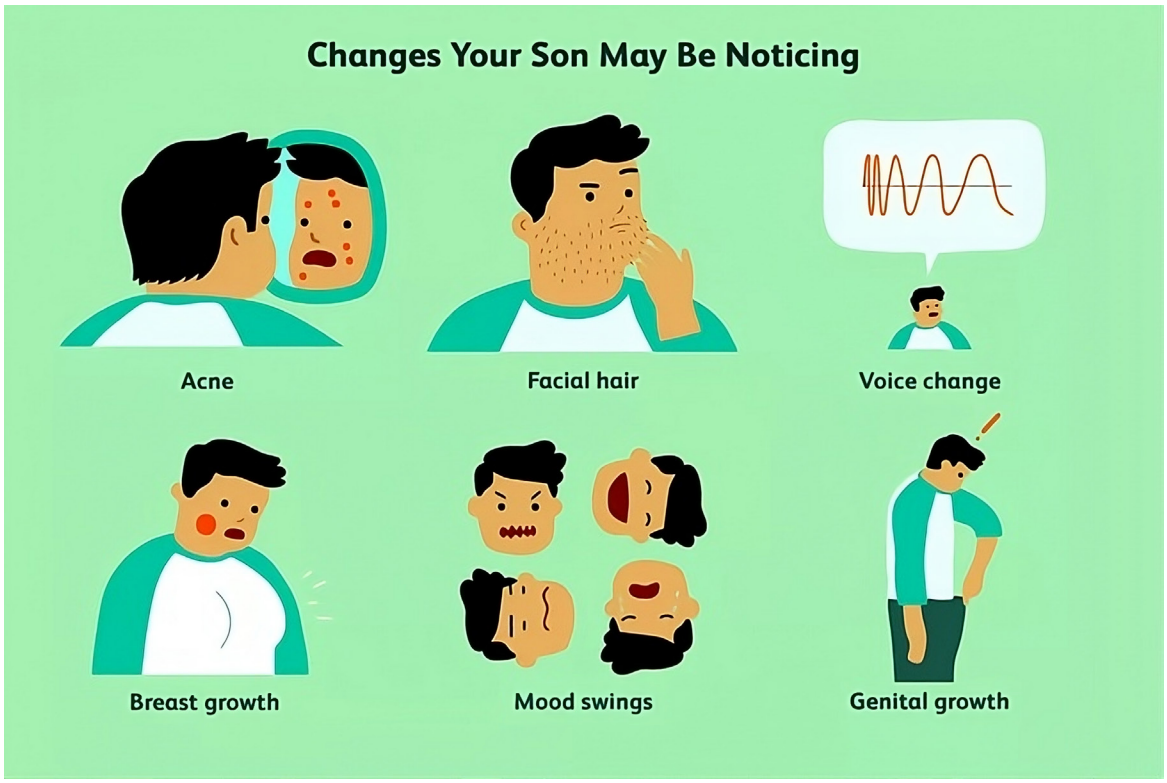
	When you start letting the air out, you should not hear any distinct sounds at first. You will then hear a thump, which will be followed by several other similar thumping sounds. The number on the dial when you heard the first thump is the patient's systolic blood pressure. The thumping sounds will eventually stop. The number on the dial when you heard the last sound is the patient's diastolic pressure. **Additional points** Although the recommendation is for multiple BP measurements and averaging them (complex computational task) to get the final BP value for patients, this may not be practical for busy primary care settings. Averaging BP readings is time-consuming and prone to errors in a busy clinic. A practical approach may be adopted below: If the first blood pressure (BP) is <140/90 mmHg, then no other blood pressure measurement is needed during that encounter. If the first BP is >140/90 mmHg, perform a second BP after 3-5 mins and use the second reading as the recorded BP for the encounter. If there is a large difference between the first and second readings (>5 mmHg), it is reasonable to do a third measurement and use the third BP as the recorded BP.

APPENDICES

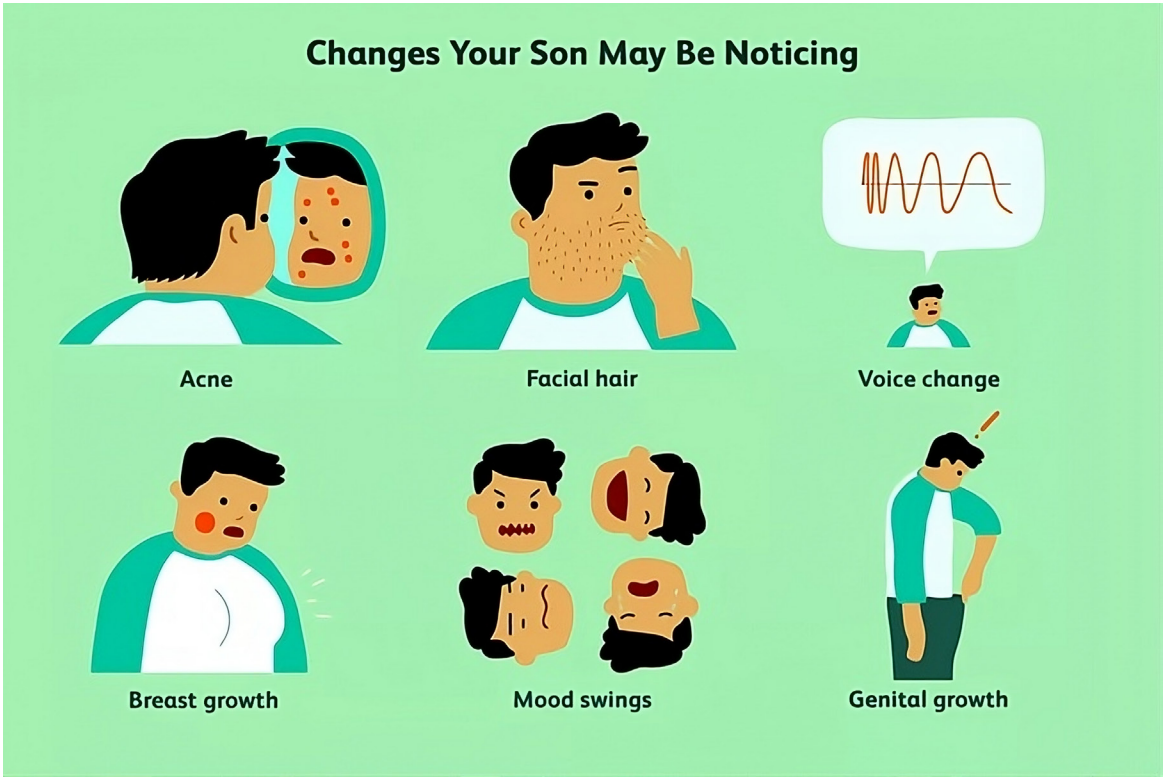
Appendix A: Integrated Mamagement Chart



Appendix B: Essential Care for Every Baby (ECEB) Chart



Appendix C: Good Attachment to the Breast



Appendix D: Essential Care for Every Baby in the First 90 Minutes of Life

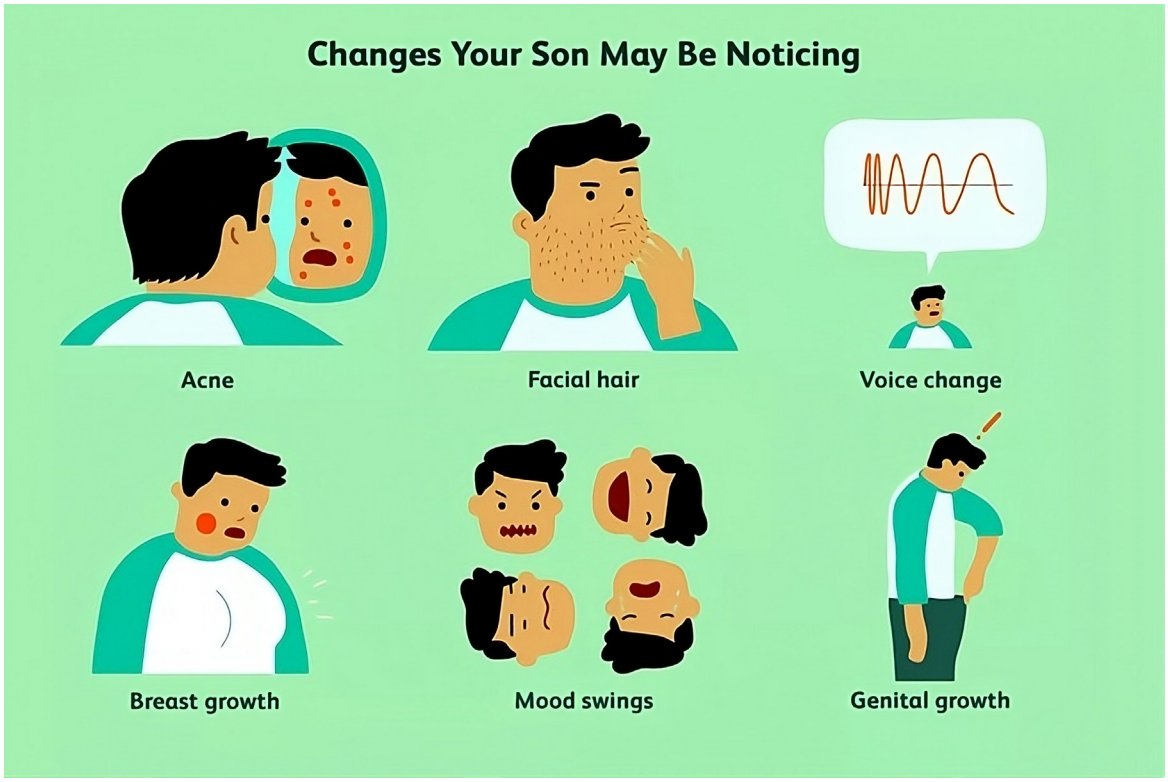


FIGURE 10. NEONATAL CONJUNCTIVITIS

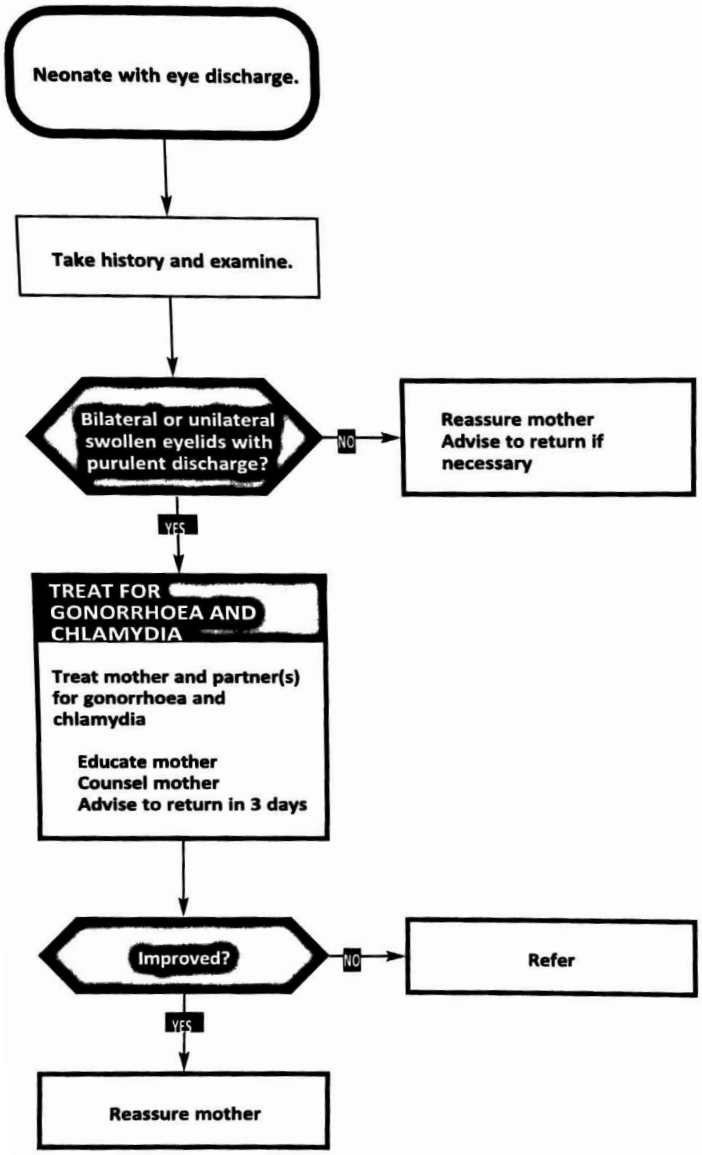
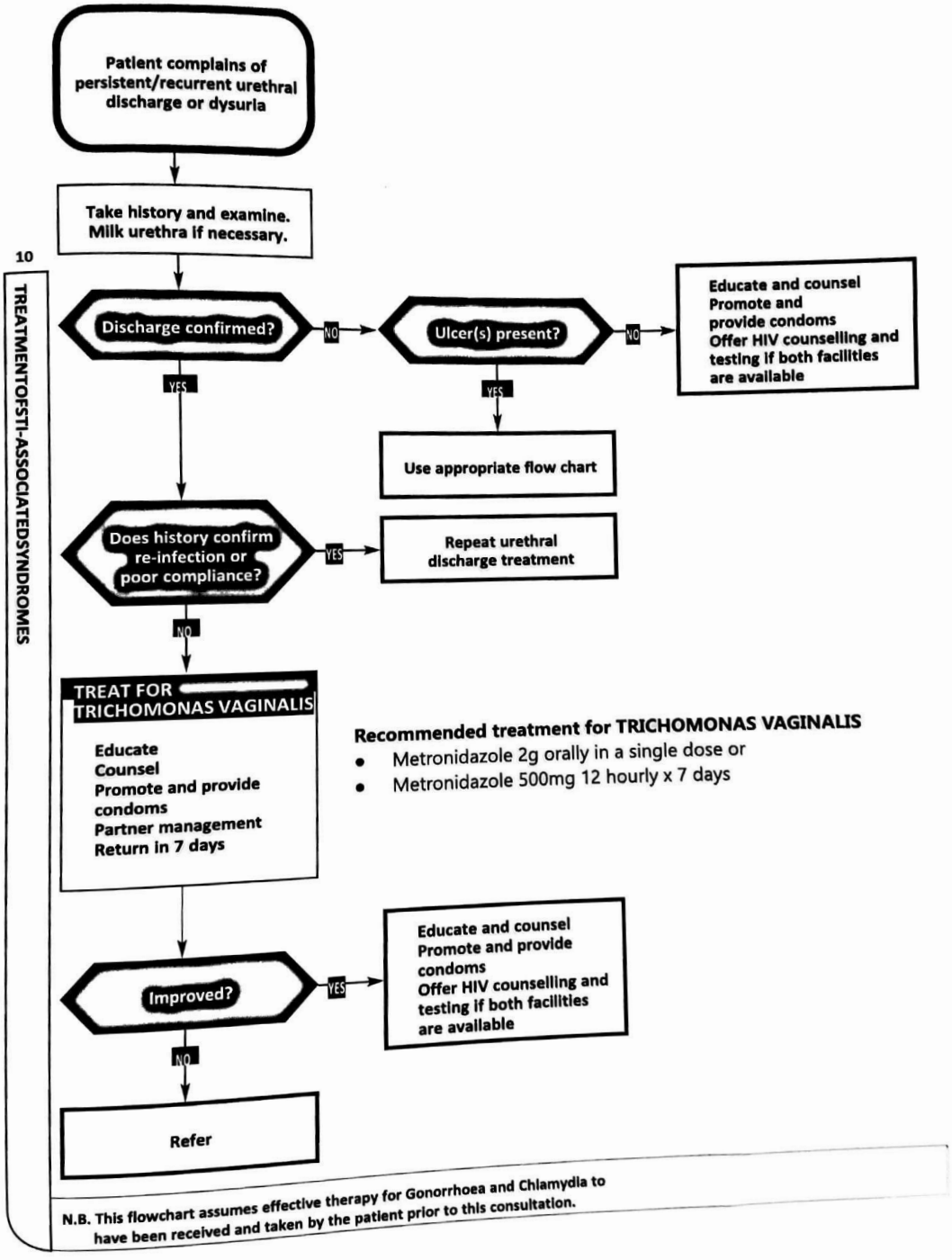


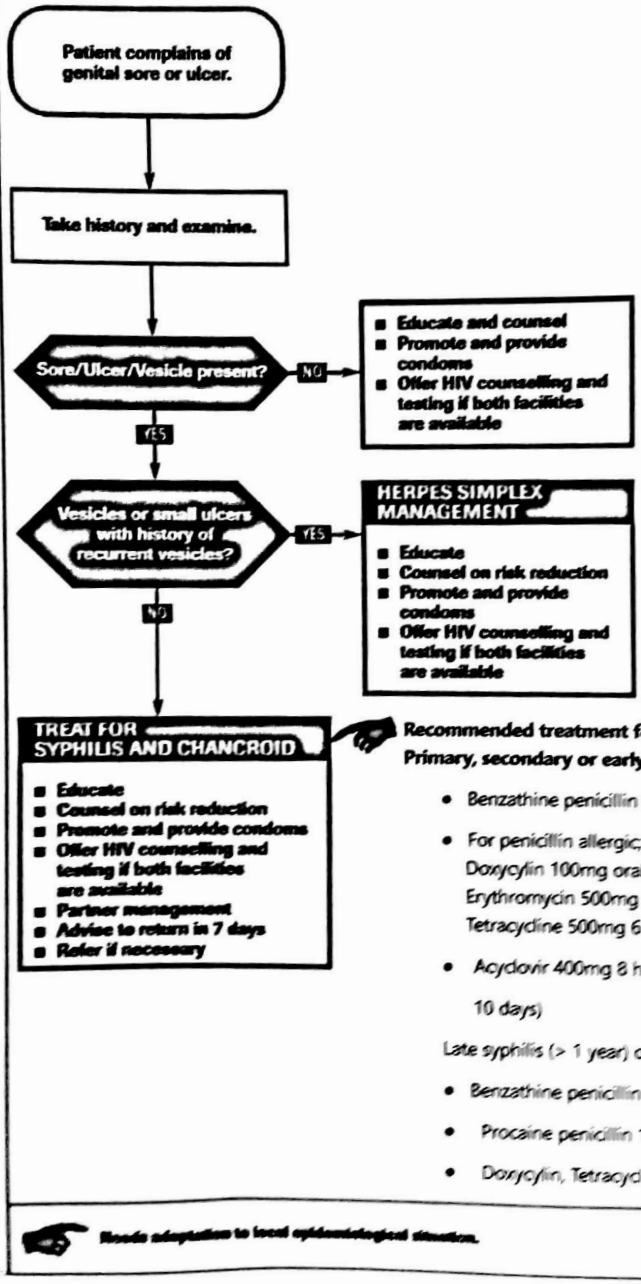
FIGURE 2. PERSISTENT/RECURRENT URETHRAL DISCHARGE IN MEN



GUIDELINES FOR THE MANAGEMENT OF SEXUALLY TRANSMITTED INFECTIONS

14 TREATMENT OF STI-ASSOCIATED SYNDROMES

FIGURE 3. GENITAL ULCERS



Recommended treatment for Syphilis

Primary, secondary or early syphilis (> 1 year)

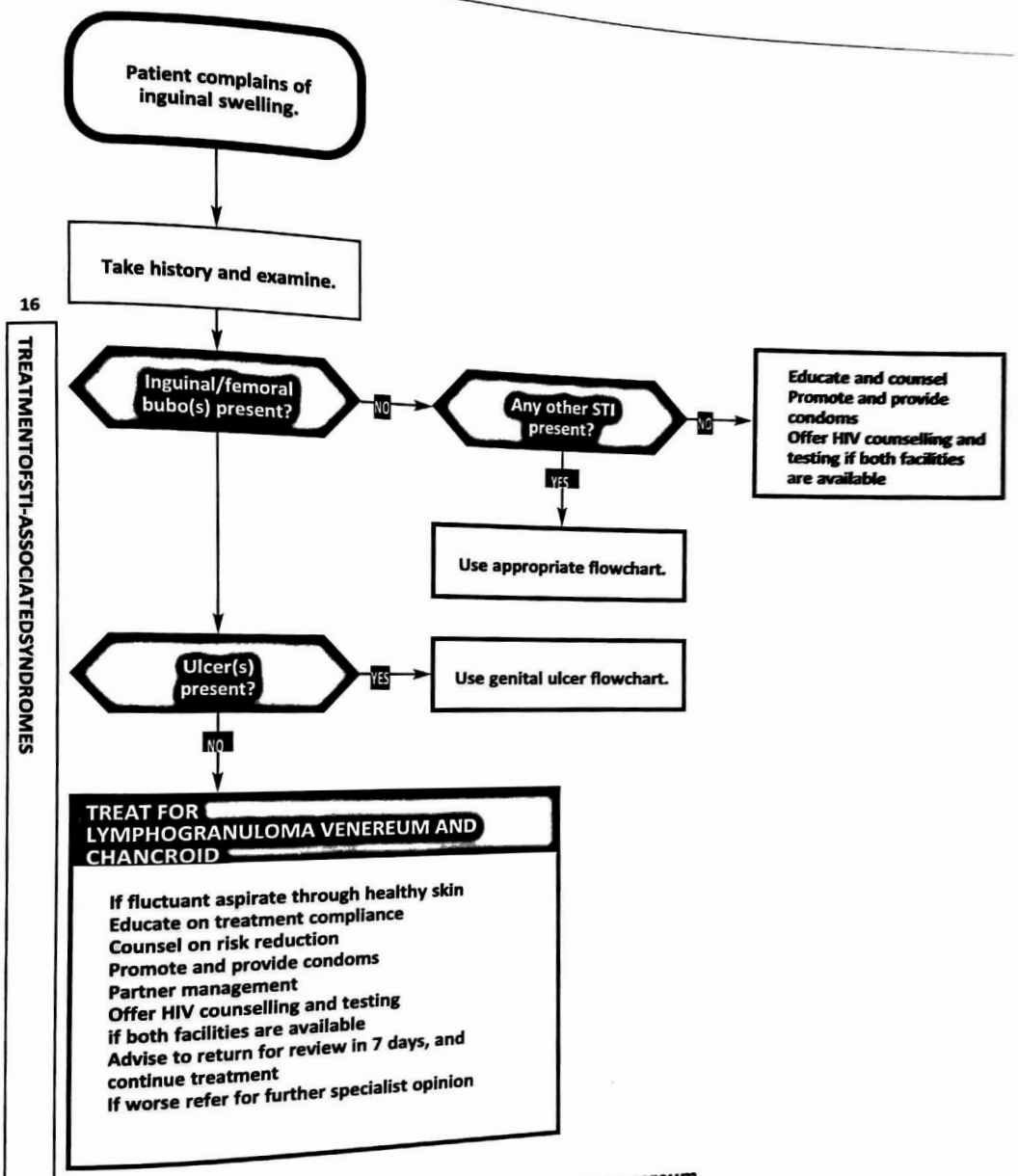
- Benzathine penicillin 2.4 million units IM X 1 dose
- For penicillin allergic;
  Doxycylin 100mg orally 12 hourly x 14 days Or
  Erythromycin 500mg orally 6hourly x 14 days Or
  Tetracycline 500mg 6 hourly x 14 days plus
- Acyclovir 400mg 8 hourly for 10 days (or 200mg 5 x daily for 10 days)

Late syphilis (> 1 year) or unknown

- Benzathine penicillin 2.4 million units IM weekly x 3 doses **Or**
- Procaine penicillin 1.2 million units IM daily x 20 days **Or**
- Doxycylin, Tetracycline or Erythromycin orally x 28 days

Needs adaptation to local epidemiological situation.

FIGURE 4. INGUINAL BUBO

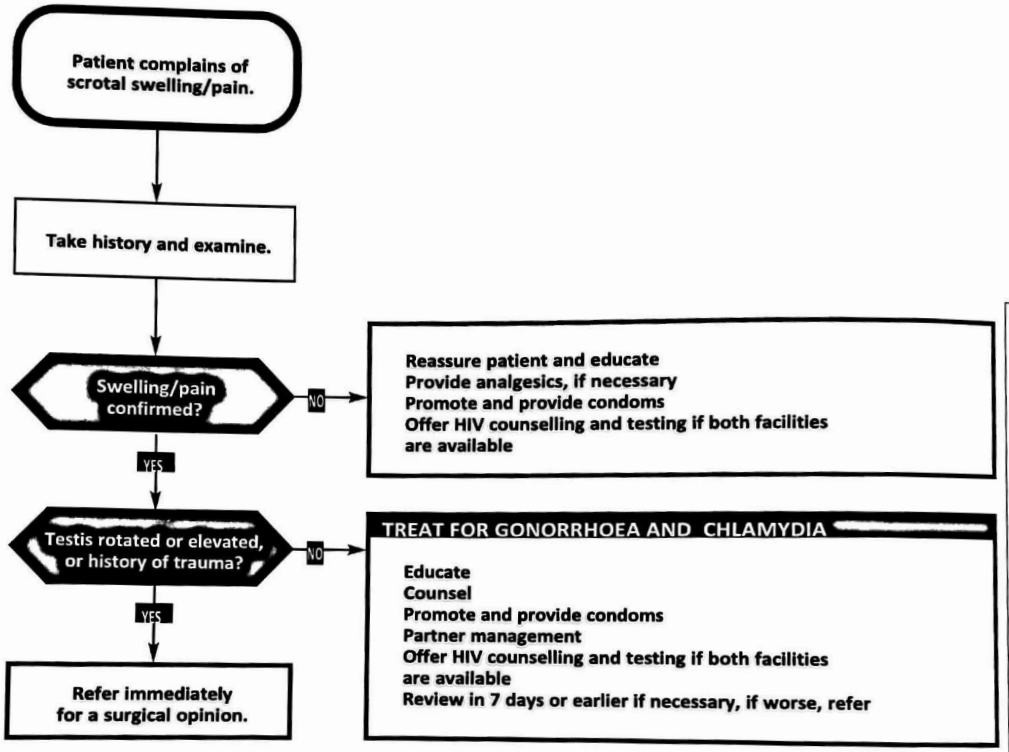


16 TREATMENTOFSTI-ASSOCIATEDSYNDROMES

Recommended treatment for Lymphogranuloma venereum

- Ciprofloxacin 500mg 12 hourly orally for 3 days **plus**
- Doxycylin 100mg orally 12 hourly x 14 days Or
- Erythromycin 500mg orally 6hourly x 14 days

FIGURE 5. SCROTAL SWELLING



Recommended treatment for Gonorrhea
Treatment options:

- Cefixime 400mg orally x 1 dose
- Ceftriaxone 250mg IM x 1 dose
- Spectinomycin 2 gram IM 1 dose

Always treat Chlamydia as well

Recommended treatment for Chlamydia

- Doxycylin 100mg orally 12 hourly x 7 days
  Or (alternatives)
- Erythromycin 500mg 6hourly x 7 days (safe for pregnant women)
- Tetracycline 500mg 6 hourly x 7 days

Always treat Gonorrhea as well

FIGURE 6. VAGINAL DISCHARGE

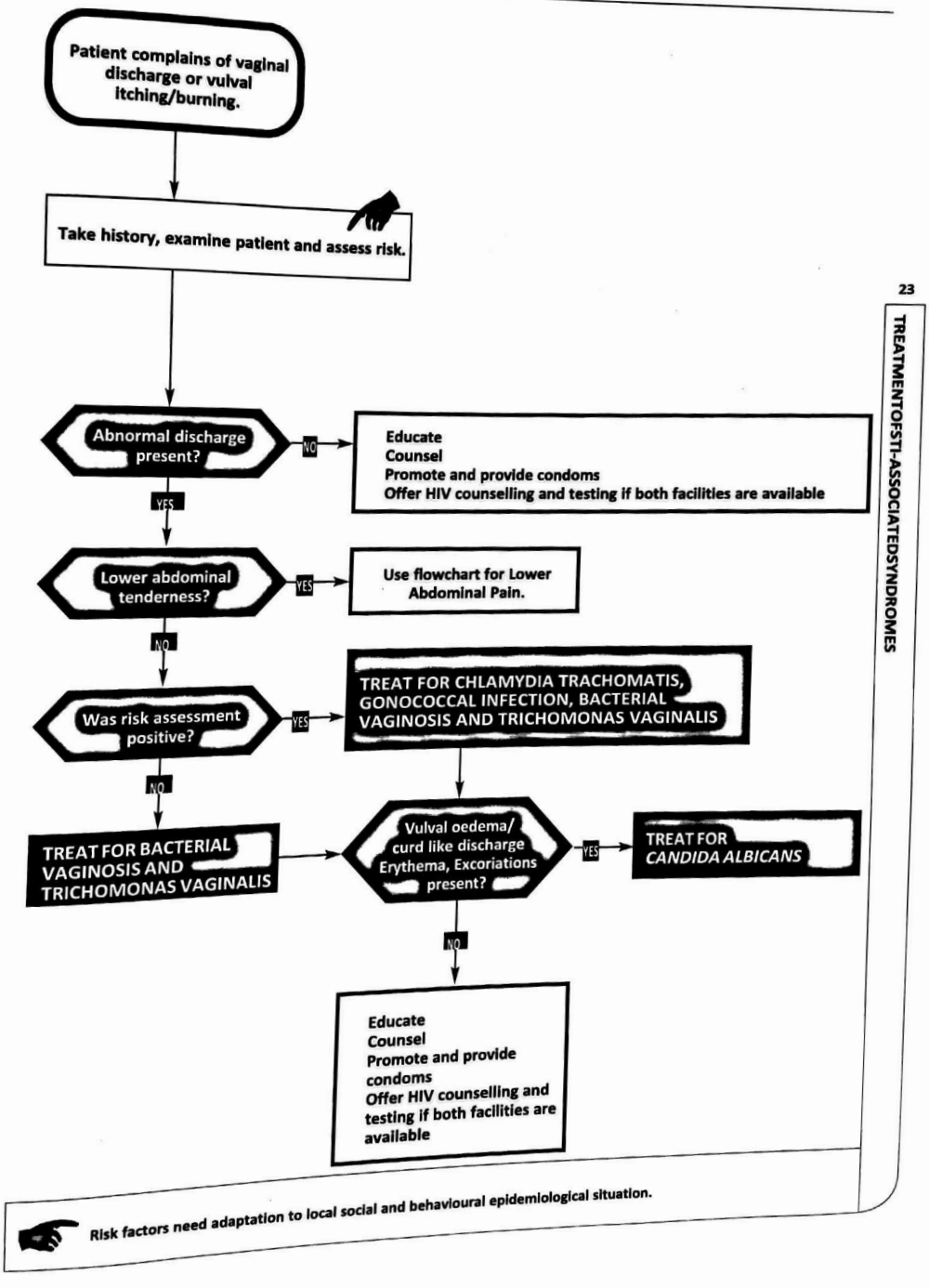


FIGURE 6. Vaginal discharge

Recommended treatment for Gonorrhea
Treatment options:

- Cefixime 400mg orally x 1 dose
- Ceftriaxone 250mg IM x 1 dose
- Spectinomycin 2 gram IM 1 dose

Always treat Chlamydia as well

Recommended treatment for Chlamydia

- Doxycylin 100mg orally 12 hourly x 7 days
  Or (alternatives)
- Erythromycin 500mg 6hourly x 7 days (safe for pregnant women)
- Tetracycline 500mg 6 hourly x 7 days

Always treat Gonorrhea as well

Recommended treatment for Bacterial vaginosis or Trichomonas

- Metronidazole 2g orally in a single dose or
- Metronidazole 500mg 12 hourly x 7 days

Recommended treatment for Candida infection

- Clotrimazole or miconazole topical x 3- 7 days
- Fluconazole 150mg oral x 1 dose
- Nystatin 100.000IU intravaginal daily x 14 days

FIGURE 7. VAGINAL DISCHARGE (SPECULUM AND BIMANUAL)

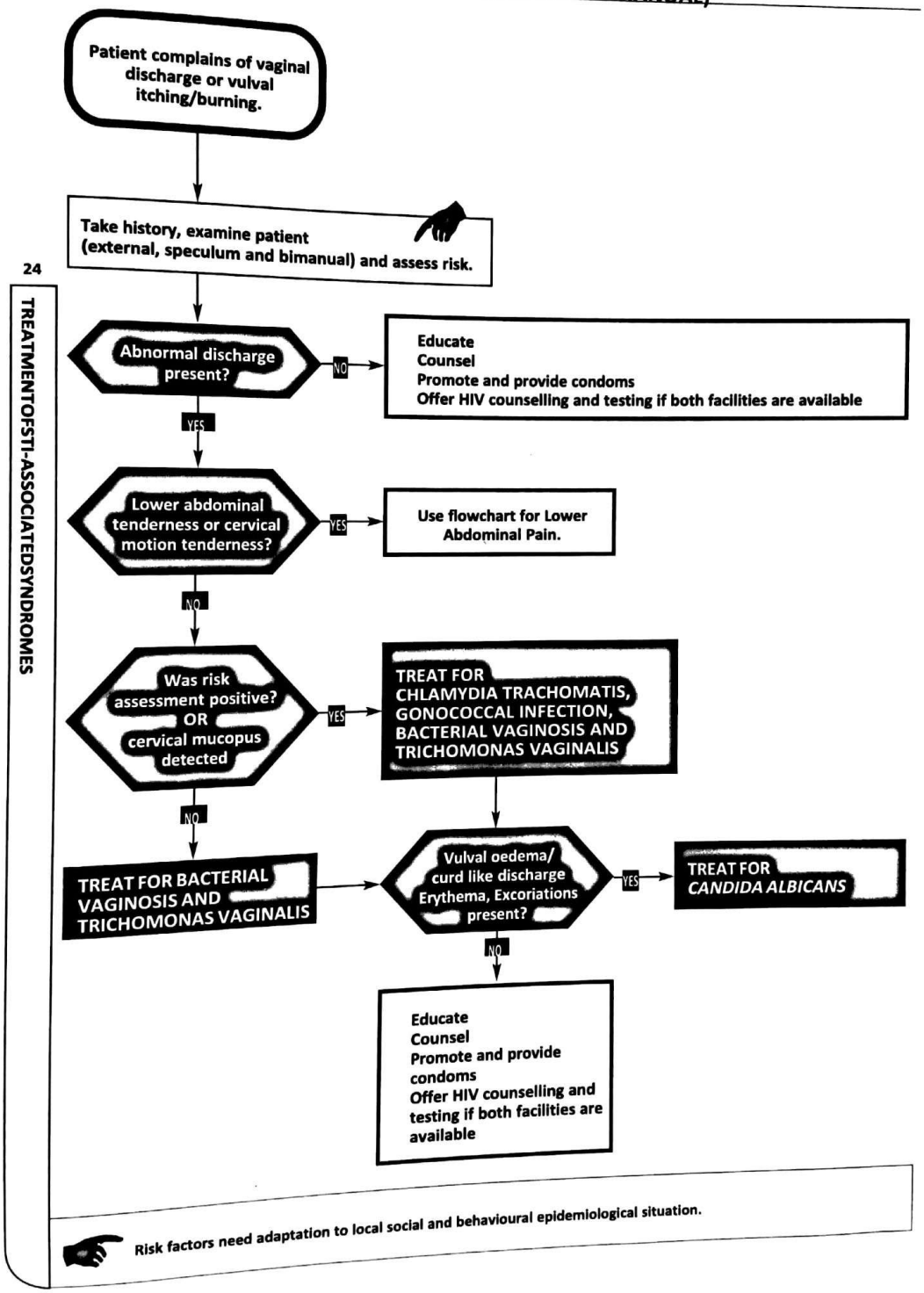
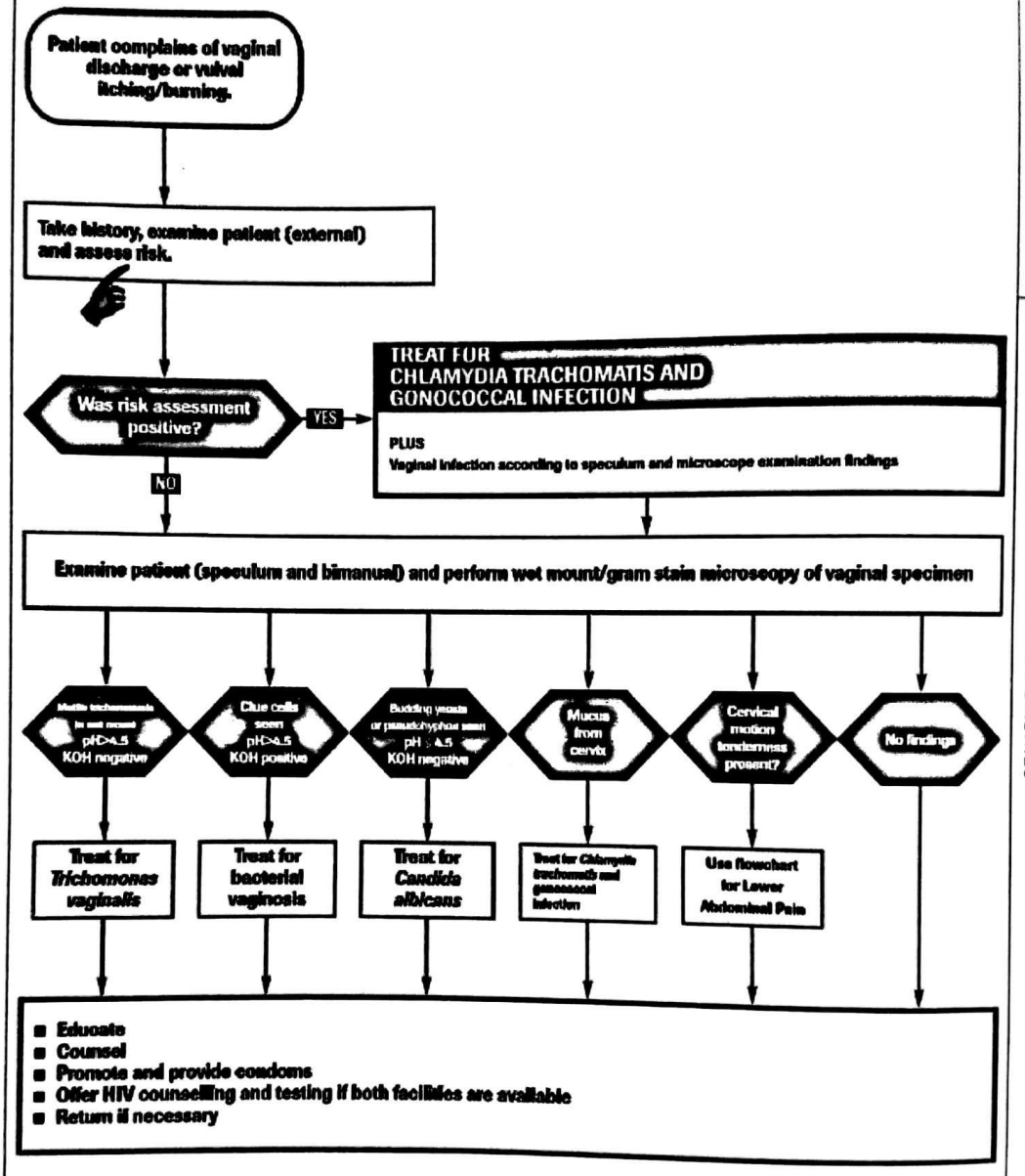


FIGURE 8. VAGINAL DISCHARGE (SPECULUM AND MICROSCOPE)

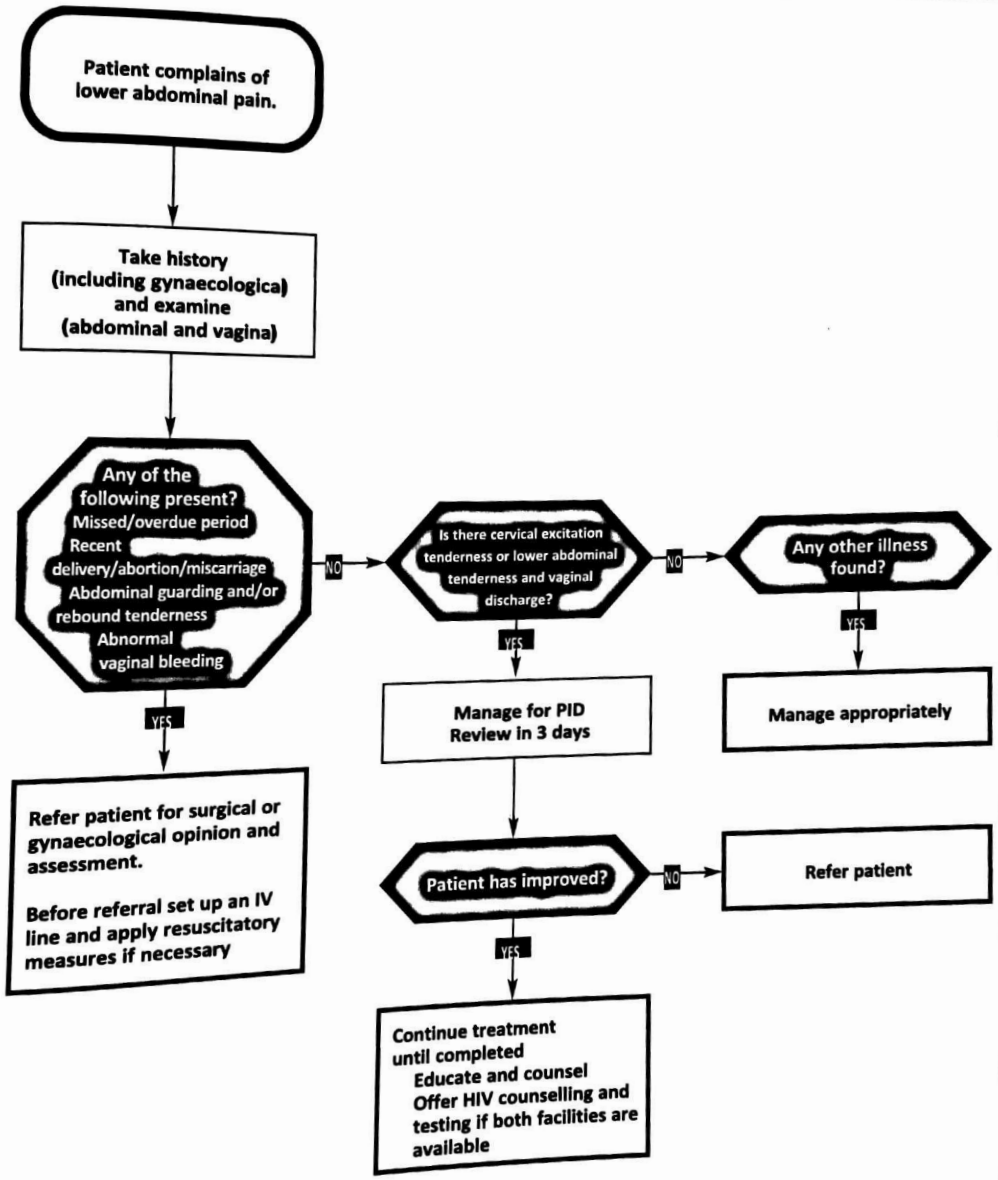


Notes:

1. KOH Test: 1 drop 10% KOH to reveal the amine odour (fishy)
2. Wet mount: smear on slide with 1 drop of saline and view at 400x

Risk factors need adaptation to local social and behavioural epidemiological situation.

FIGURE 9. LOWER ABDOMINAL PAIN



Recommended treatment for Pelvic inflammatory disease plus

- Ceftriaxone 250mg IM X 1 dose
- Doxycylin 100mg orally 12 hourly x 14 days
- Metronidazole 400mg 12 hourly x 14 days

Consider hospital admission for;

Severe illness, pregnancy & no improvement in 2-3 days

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Revised By:

COMMUNITY HEALTH PRACTITIONERS' REGISTRATION BOARD OF NIGERIA IN COLLABORATION WITH NPHCDA

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